

Competitive Programming Notebook

ICPC team reference • August 13, 2026

HOW TO FIND SOMETHING (in order of speed). 1. You know the routine's name → the alphabetical *Index of every routine* at the very back gives the page. 2. You know the topic → the running head on every page names the section (left) and the sub-topic (right); thumb until it matches. 3. You know neither → section **12 Problem Solving** maps a problem's *shape* and its *constraints* onto the technique, and docs/PLAYBOOK.md does the same at more length.

BEFORE YOU PASTE. Line 1 of every file says what it does; a `//` Needs: line names the templates it depends on; a `//` NOTE: line warns when another file declares the same symbol. Vertices and array positions are **0-based** unless the header says otherwise. Geometry: `P<11>` for every decision, `P<1d>` only when the answer is a real number.

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1 Miscellaneous

1.1 Template

```
#include <bits/stdc++.h>
using namespace std;

// #define int long long
#define ll long long
#define ld long double
#define fi first
#define se second
#define pb push_back
#define eb emplace_back
#define all(x) x.begin(), x.end()
#define rall(x) x.rbegin(), x.rend()
#define sz(x) (int)(x).size()
#define ones(n) __builtin_popcountll(n)
#define msb(n) (63 - __builtin_clzll(n))
#define lsb(n) __builtin_ctzll(n)
typedef vector<int> vi;
typedef pair<int, int> pii;

const int N = 2e5 + 5, M = 1e3 + 5, LOG = 31;
const int inf = 0x3f3f3f3f;
const ll llinf = 0x3f3f3f3f3f3f3f3f;
const int MOD = 1e9 + 7;
const ld eps = 1e-9, PI = acos(-1.0L); // ld, not double:
    ↪ geometry needs the digits

#ifdef LOCAL // compile with
    ↪ -DLOCAL
template <class T> void _p(const T& x);
void _p(const string& s) { cerr << "'" << s << "'"; }
void _p(char c) { cerr << '\\' << c << '\\'; }
template <class A, class B> void _p(const pair<A, B>& p) { cerr <<
    ↪ '('; _p(p.fi); cerr << ", "; _p(p.se); cerr << ')'; }
template <class T> void _p(const T& x) {
    if constexpr (is_arithmetic_v<T>) cerr << x;
    else { cerr << '{'; int f = 1; for (auto& e : x) { if (!f)
        ↪ cerr << ", "; f = 0; _p(e); } cerr << '}'; }
}
void _pr() { cerr << "\n"; }
template <class H, class... T> void _pr(const H& h, const T&... t)
    ↪ { _p(h); if (sizeof...(t)) cerr << " | "; _pr(t...); }
#define dbg(...) (cerr << "[" << #__VA_ARGS__ << "] = ",
    ↪ _pr(__VA_ARGS__))
#else
#define dbg(...)
#endif

void testCase() {
}

void preCompute() {
}

int32_t main() {
    ios_base::sync_with_stdio(false), cin.tie(nullptr),
    ↪ cout.tie(nullptr);
    cout << fixed << setprecision(10);
    preCompute();

    int tc = 1;
    cin >> tc; // DELETE THIS
    ↪ LINE for single-test problems
    for (int TC = 1; TC <= tc; TC++) {
        // cout << "Case #" << TC << ": ";
        testCase();
    }
}

/*

*/
```

1.2 fastIO

HAND-ROLLED FAST IO (fread/fwrite buffered). Use only when cin with sync off is measurably too

```
// slow: >1e6 numbers to read, or heavy output. Otherwise prefer
// the template's ios::sync_with_stdio.
const int MOD = 1e9 + 7;
const int BUF_SZ = 1 << 15;

inline namespace Input {
    char buf[BUF_SZ];
    int pos;
    int len;
    char next_char() {
        if (pos == len) {
            pos = 0;
            len = (int)fread(buf, 1, BUF_SZ, stdin);
            if (!len) { return EOF; }
        }
        return buf[pos++];
    }

    int read_int() {
        int x;
        char ch;
        int sgn = 1;
        while (!isdigit(ch = next_char())) {
            if (ch == '-') { sgn *= -1; }
        }
        x = ch - '0';
        while (isdigit(ch = next_char())) { x = x * 10 + (ch - '0'); }
        return x * sgn;
    }
} // namespace Input

inline namespace Output {
    char buf[BUF_SZ];
    int pos;

    void flush_out() {
        fwrite(buf, 1, pos, stdout);
        pos = 0;
    }

    void write_char(char c) {
        if (pos == BUF_SZ) { flush_out(); }
        buf[pos++] = c;
    }

    void write_int(int x) {
        static char num_buf[100];
        if (x < 0) {
            write_char('-');
            x *= -1;
        }
        int len = 0;
        for (; x >= 10; x /= 10) { num_buf[len++] = (char)('0' + (x % 10)); }
        write_char((char)('0' + x));
        while (len) { write_char(num_buf[--len]); }
        write_char('\n');
    }

    // auto-flush output when program exits
    void init_output() { assert(atexit(flush_out) == 0); }
} // namespace Output
```

1.3 grid

GRID DIRECTION VECTORS. First 4 entries = 4-neighbourhood (L,R,U,D), all 8 = king moves.

```
// Knight moves: dx {1,1,-1,-1,2,2,-2,-2}, dy
// {2,-2,2,-2,1,-1,1,-1}. Guard with 0<=x<n && 0<=y<m.
int dx[] = {0, 0, -1, 1, -1, -1, 1, 1};
int dy[] = {-1, 1, 0, 0, -1, 1, -1, 1};
char di[] = {'L', 'R', 'U', 'D'};
```

1.4 random

RANDOMNESS. Seed from the clock so tests cannot be prepared against you.

```
mt19937_64
    rng(chrono::steady_clock::now().time_since_epoch().count());
ll rnd(ll l, ll r) { return uniform_int_distribution<ll>(l,
    r)(rng); } // inclusive

vector<int> randPerm(int n) { // random
    // permutation of 0..n-1
    vector<int> v(n);
    iota(all(v), 0);
    shuffle(all(v), rng);
    return v;
}

// Shuffling the INPUT is a real algorithm, not just a utility:
// - minimum enclosing circle is O(n) only after a shuffle
// - quickselect / random pivots
// - defeats anti-hash and anti-quicksort tests
// - random rotation of points removes degenerate ties in a
    // geometry sweep
```

1.5 coordinateCompression

Coordinate Compression - O(N log N)

```
template<typename T>
struct Compress {
    vector<T> v;

    Compress(vector<T>& a) : v(a) {
        sort(all(v));
        v.erase(unique(all(v)), v.end());
        for (auto &x : a) {
            x = lower_bound(all(v), x) - v.begin();
        }
    }

    T operator[](int i) const {
        return v[i];
    }
};
```

1.6 grayCode

GRAY CODE - orders $0..2^n-1$ so consecutive values differ in exactly ONE bit. Use it to walk all

```
// subsets while changing one element at a time (incremental DP,
// XOR-basis sweeps, Tower of Hanoi).
// Gray Code Generator - O(2^N)
// g(n) = n ^ (n >> 1)
vector<int> grayCode(int n) {
    vector<int> res(1 << n);
    for (int i = 0; i < (1 << n); i++) {
        res[i] = i ^ (i >> 1);
    }
    return res;
}

int rev_g (int g) {
    int n = 0;
    for (; g; g >>= 1)
        n ^= g;
    return n;
}
```

1.7 customHashHashTable

UNORDERED_MAP THAT CANNOT BE HACKED. The default std::hash for integers is the identity, so an

```
// adversary can force every key into one bucket and turn O(1)
//   into O(n). splitmix64 + a clock seed
// fixes that. gp_hash_table is ~3x faster than unordered_map;
//   reserve() and max_load_factor help more.
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

// Custom Hash to prevent anti-hash tests in unordered_map /
//   gp_hash_table
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
        chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

// Usage: gp_hash_table<ll, int, custom_hash> table;
//         unordered_map<ll, int, custom_hash> safe_map;
```

1.8 pragmas

COMPILER PRAGMAS. O3 + unrolling, and AVX2 for auto-vectorised loops (bitset, simple array math).

```
// Put them at the very top, before any include. Risk: 'target'
//   crashes judges without those CPU
// features - never use avx2 on an unknown judge, and never on
//   Codeforces problems with old machines.
// GCC Optimizations
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

1.9 int128

__int128 Stream Operators Helper

```
typedef __int128_t int128;
typedef __uint128_t uint128;

ostream &operator<<(ostream &os, int128 n) {
    if (n == 0) return os << "0";
    if (n < 0) { os << "-"; n = -n; }
    string s;
    while (n > 0) { s += (char)('0' + (n % 10)); n /= 10; }
    reverse(all(s));
    return os << s;
}

istream &operator>>(istream &is, int128 &n) {
    string s; is >> s;
    n = 0; int i = 0, sign = 1;
    if (!s.empty() && s[0] == '-') { sign = -1; i = 1; }
    for (; i < (int)s.size(); i++) n = n * 10 + (s[i] - '0');
    n *= sign;
    return is;
}
```

1.10 SearchIdioms

The four searches. Getting the boundary right here saves more contest time than any template.

```
// BINARY SEARCH ON A PREDICATE. pred must be monotone: F F F T T
//   T.
// firstTrue returns the first x in [lo, hi] with pred(x), or hi+1
//   if none.
template <class F> ll firstTrue(ll lo, ll hi, F pred) {
    hi++;
    while (lo < hi) { ll m = lo + (hi - lo) / 2; pred(m) ? hi = m
    : lo = m + 1; }
    return lo;
}

// lastTrue returns the last x in [lo, hi] with pred(x), or lo-1
//   if none. T T T F F F
template <class F> ll lastTrue(ll lo, ll hi, F pred) {
    lo--;
    while (lo < hi) { ll m = lo + (hi - lo + 1) / 2; pred(m) ? lo
    = m : hi = m - 1; }
    return lo;
}

// Reals: fixed iteration count beats an epsilon loop (no infinite
//   loops, predictable time).
// 100 iterations halve the range 2^100 times - always enough.
// Returns the boundary: pred is false on [lo, x) and true on [x,
//   hi]. ('lo' and 'hi' converge.)
template <class F> ld binReal(ld lo, ld hi, F pred, int it = 100)
    {
        while (it--) { ld m = (lo + hi) / 2; pred(m) ? hi = m : lo =
        m; }
        return hi;
        // first TRUE, matching firstTrue
    }

// TERNARY SEARCH on a UNIMODAL function (strictly decreasing then
//   increasing => minimum).
// Integer version: stop when the window is tiny, then scan. Never
//   use 'l < r' alone.
template <class F> ll ternaryInt(ll lo, ll hi, F f) {
    // returns argmin
    while (hi - lo > 2) {
        ll m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        f(m1) <= f(m2) ? hi = m2 : lo = m1;
        // <= keeps the LEFTmost min
    }
    ll best = lo;
    for (ll x = lo; x <= hi; x++) if (f(x) < f(best)) best = x;
    return best;
}

template <class F> ld ternaryReal(ld lo, ld hi, F f, int it =
    200) {
    while (it--) { ld m1 = lo + (hi - lo) / 3, m2 = hi - (hi -
    lo) / 3; f(m1) < f(m2) ? hi = m2 : lo = m1; }
    return (lo + hi) / 2;
}

// If f is only WEAKLY unimodal (has flat regions), ternary search
//   is WRONG: one comparison
// f(m1) == f(m2) inside a plateau discards the side holding the
//   optimum. For any DISCRETELY
// CONVEX f (non-decreasing difference sequence, plateaus allowed)
//   use the derivative instead -
// this is strictly safer AND uses half the evaluations:
template <class F> ll convexArgmin(ll lo, ll hi, F f) {
    return firstTrue(lo, hi - 1, [&](ll x) { return f(x + 1) >=
    f(x); });
}

// NEWTON'S METHOD for roots of a smooth f (quadratic convergence;
//   needs a decent start).
template <class F, class DF> ld newton(ld x, F f, DF df, int it =
    60) {
    while (it--) { ld d = f(x) / df(x); x -= d; if (fabs1(d) <
    1e-15L * max((ld)1, fabs1(x))) break; }
    return x;
}

// Integer sqrt / cbrt with correction - NEVER trust sqrt() alone
//   near 1e18.
ll isqrt(ll n) { ll r = (ll)sqrt1((ld)n); while (r > 0 && r * r >
    n) r--; while ((r + 1) * (r + 1) <= n) r++; return r; }
ll icbrt(ll n) { ll r = (ll)cbrt1((ld)n); while (r > 0 && r * r *
    r > n) r--; while ((r + 1) * (r + 1) * (r + 1) <= n) r++;
    return r; }

// BINARY SEARCH ON THE ANSWER - the pattern, not the code:
// "minimise the maximum X" / "maximise the minimum X" / "can we
//   finish within T"
```

```
// -> guess the answer, write a GREEDY feasibility check, binary
//    search on it.
// PARALLEL BINARY SEARCH - when you need this for MANY queries at
//    once: bucket the queries by
// their current midpoint, sweep the shared structure once per
//    level, O(log) sweeps total.
```

1.11 BitTricks

BIT MANIPULATION CHEAT SHEET =====

```
/* ===== BIT MANIPULATION CHEAT SHEET
   =====
   BUILTINS (add ll suffix for 64-bit: __builtin_popcountll, ...)
   __builtin_popcount(x) #set bits      __builtin_ctz(x)
   => #trailing zeros (x != 0)
   __builtin_clz(x)      #leading zeros  __builtin_parity(x)
   => popcount & 1
   __lg(x) = 63 - clzll(x) = floor(log2 x)    (GCC only; write it
   => yourself elsewhere)
   lowest set bit: x & -x      clear it: x &= x - 1
   => set/flip: x != b, x ^= b
   is power of two: x && !(x & (x - 1))
   round UP to a power of two (x >= 1): x == 1 ? 1ULL : 2ULL <<
   => (63 - __builtin_clzll(x - 1))
   - clzll(0) is UNDEFINED, so x = 1 must be special-cased, and
   => the shift must be 64-bit

   SUBMASK ENUMERATION - iterate every subset of m:
   for (int s = m; ; s = (s - 1) & m) { ...; if (!s) break; }
   Over ALL masks this is O(3^n) total (each element is
   => in/out/neither).

   SUPERMASK enumeration (every s with m subset of s, within n bits):
   for (int s = m; s < (1 << n); s = (s + 1) | m) { ... }

   K-SUBSETS in increasing order (Gosper's hack):
   for (int s = (1 << k) - 1; s < (1 << n); ) { ...;
       int c = s & -s, r = s + c; s = ((r ^ s) >> 2) / c | r; }

   GRAY CODE g(i) = i ^ (i >> 1)      inverse: x ^= x >> 1
   => repeatedly (see 06 - grayCode.cpp)
   consecutive gray codes differ in exactly one bit - use for
   => "change one thing at a time".

   XOR FACTS
   x ^ x = 0, x ^ 0 = x, xor is its own inverse -> prefix xor
   => answers range xor
   xor of 0..n = [n, 1, n+1, 0][n % 4]
   a + b = (a ^ b) + 2*(a & b)      a | b = (a ^ b) + (a & b)      a
   => + b = (a | b) + (a & b)
   "OR of a range equals its SUM" <=> the values are pairwise
   => BIT-DISJOINT
   (so at most 30 non-zero values fit in any such window -> two
   => pointers)

   COUNTING PER BIT - the standard decomposition: handle each of the
   => 30/60 bits independently.
   sum over pairs of (a_i ^ a_j) = sum over bits b of (#with bit)
   => * (#without bit) * 2^b

   BITSET
   std::bitset<N> b; b._Find_first(), b._Find_next(i) (GCC) -
   => iterate set bits fast
   b <= k, b != other are O(N/64) - this is what makes bitset DP
   => and reachability work.

   =====

   => */

// Iterate all submasks of m (including m and 0).
template <class F> void forEachSubmask(int m, F f) {
    for (int s = m; s = (s - 1) & m) { f(s); if (!s) break; }
}

// All n-bit masks with exactly k bits set, in increasing order
//    (Gosper's hack).
vector<int> kSubsets(int n, int k) {
    vector<int> r;
    if (k == 0) return {0};
    for (int s = (1 << k) - 1; s < (1 << n); ) {
        r.push_back(s);
        int c = s & -s, x = s + c;
        s = ((x ^ s) >> 2) / c | x;
    }
    return r;
}

// Sum over all pairs (i<j) of (a_i XOR a_j), by handling each bit
//    independently.
ll pairXorSum(const vector<ll>& a, int B = 62) {
    ll n = a.size(), r = 0;
    for (int b = 0; b < B; b++) {
        ll c = 0;
        for (ll x : a) c += (x >> b) & 1;
        r += c * (n - c) % MOD * ((1LL << b) % MOD) % MOD;
    }
    return r % MOD;
}
```

1.12 StressTest

STRESS TESTING =====

```
/* ===== STRESS TESTING =====
The single highest-value 20 lines in this notebook. Use it the
    ↪ moment a solution WAs on a
test you cannot see. Write gen.cpp + brute.cpp, then loop until
    ↪ they disagree.

---- run.sh
    ↪ -----
g++ -O2 -o sol sol.cpp && g++ -O2 -o brute brute.cpp && g++ -O2
    ↪ -o gen gen.cpp
for i in $(seq 1 100000); do
    ./gen $i > in.txt
    ./sol < in.txt > out1.txt
    ./brute < in.txt > out2.txt
    if ! diff -qb out1.txt out2.txt > /dev/null; then echo "WA on
    ↪ test $i"; cat in.txt; break; fi
done
```

Windows cmd: replace the loop with 'for /L %%i in (1,1,100000) do
 ↪ (... fc out1.txt out2.txt)'

RULES THAT MAKE IT ACTUALLY WORK

1. Generate SMALL cases ($n \leq 8$, values ≤ 5). Bugs hide in tiny
 ↪ corner cases, not big ones.
2. Seed the generator from `argv[1]` so every iteration differs
 ↪ and is reproducible.
3. If the answer is not unique, do not diff - write a CHECKER
 ↪ that validates sol's output.
4. If there is no brute force, stress against yourself: compare
 ↪ two different solutions, or
 compare an $O(n^2)$ version of your own idea against the $O(n \log n)$ one.
5. For "find any valid X" problems, verify the constraints
 ↪ instead of comparing.

===== */

```
// ---- gen.cpp ----
// #include <bits/stdc++.h>
// using namespace std;
// int main(int argc, char** argv) {
//     mt19937 rng(atoi(argv[1]));
//     auto rnd = [&](int l, int r) { return
//     ↪ uniform_int_distribution<int>(l, r)(rng); };
//     int n = rnd(1, 8);
//     printf("%d\n", n);
//     for (int i = 0; i < n; i++) printf("%d ", rnd(1, 5));
//     puts("");
// }

// ---- in-process version: no files, no scripts. Put brute + fast
//     ↪ in one binary. ----
// Fastest way to start; switch to the shell loop only when you
//     ↪ need the failing input on disk.
template <class Gen, class Fast, class Brute>
void stress(Gen gen, Fast fast, Brute brute, int iters = 100000) {
    for (int it = 1; it <= iters; it++) {
        auto in = gen(it);
        auto a = fast(in), b = brute(in);
        if (!(a == b)) {
            printf("MISMATCH on iteration %d: fast=", it);
            cout << a << " brute=" << b << "\n";
            return;
            ↪ 'in' here too
        }
        puts("all tests passed");
    }
}
```

1.13 Fraction

Exact rational. Always normalised (gcd reduced, denominator > 0).

```
// Comparison uses __int128 so it is exact for |num|, |den| up to
    ↪ -9e18.
// Use whenever comparing ratios: slopes, "maximum average",
    ↪ scheduling exchange arguments.
struct Frac {
    ll n = 0, d = 1;
    Frac(ll n = 0, ll d = 1) : n(n), d(d) { norm(); }
    void norm() {
        if (d < 0) n = -n, d = -d;
        ll g = gcd(n < 0 ? -n : n, d);
        if (g) n /= g, d /= g;
        if (!d) d = 1;
        ↪ treat as 0/1; guard your input
    }
    Frac operator+(Frac o) const { return Frac(n * o.d + o.n * d,
    ↪ d * o.d); }
    Frac operator-(Frac o) const { return Frac(n * o.d - o.n * d,
    ↪ d * o.d); }
    Frac operator*(Frac o) const { return Frac(n * o.n, d * o.d);
    ↪ }
    Frac operator/(Frac o) const { return Frac(n * o.d, d * o.n);
    ↪ }
    bool operator<(Frac o) const { return (__int128)n * o.d <
    ↪ (__int128)o.n * d; }
    bool operator==(Frac o) const { return n == o.n && d == o.d; }
    bool operator<=(Frac o) const { return !(o < *this); }
    ld val() const { return (ld)n / d; }
    friend ostream& operator<<(ostream& os, Frac f) { return os
    ↪ << f.n << '/' << f.d; }
};

// OVERFLOW: + and * multiply denominators, so chains of
    ↪ operations blow up fast. If you only
// need COMPARISONS, keep the raw pair and cross-multiply in
    ↪ __int128 - do not normalise.
// For unbounded exactness use Python/BigInt, or the Stern-Brocot
    ↪ search (ContinuedFractions).
```

1.14 Inversions

INVERSIONS: #pairs $i < j$ with $a[i] > a[j]$. $O(n \log n)$ by merge sort (also sorts a).

```
ll inversions(vector<int>& a) {
    int n = a.size();
    if (n < 2) return 0;
    vector<int> buf(n);
    function<ll(int, int)> rec = [&](int l, int r) -> ll {
        ↪ // [l, r)
        if (r - l < 2) return 0;
        int m = (l + r) / 2;
        ll res = rec(l, m) + rec(m, r);
        int i = l, j = m, k = 1;
        while (i < m && j < r) {
            if (a[i] <= a[j]) buf[k++] = a[i++];
            else buf[k++] = a[j++], res += m - i;
        }
        ↪ // a[i..m) all beat a[j]
        while (i < m) buf[k++] = a[i++];
        while (j < r) buf[k++] = a[j++];
        copy(buf.begin() + 1, buf.begin() + r, a.begin() + 1);
        return res;
    };
    return rec(0, n);
}

// WHY IT KEEPS APPEARING
// - minimum number of ADJACENT SWAPS to sort = inversion count
// - minimum adjacent swaps to turn a into b: relabel b's
    ↪ positions, then count inversions
// - "how far is this permutation from sorted" / bubble-sort
    ↪ distance
// - inversions modulo 2 = permutation PARITY (= sign, used in
    ↪ determinants and 15-puzzle)
// BIT VERSION (needed when you must also support updates, or
    ↪ count online):
// compress values, sweep left to right, ans += (#inserted so
    ↪ far) - query(a[i]), then add a[i].
```

1.15 ExpressionParser

EXPRESSION PARSER (shunting-yard). Handles $+$ $-$ $*$ $/$ $\wedge$, unary minus, parentheses.
// Left-associative except $\wedge$. Extend by editing prec() / applyOp().

```

struct Parser {
    static int prec(char op) {
        switch (op) {
            case '+': case '-': return 1;
            case '*': case '/': case '%': return 2;
            case '^': return 3;
            case '~': return 4; // unary
            // minus (internal symbol)
            default: return -1;
        }
    }
    static bool rightAssoc(char op) { return op == '^' || op == '~'; }

    static ll apply(char op, ll a, ll b = 0) {
        switch (op) {
            case '+': return a + b;
            case '-': return a - b;
            case '*': return a * b;
            case '/': return a / b;
            case '%': return a % b;
            case '~': return -a;
            case '^': { ll r = 1; while (b > 0) { if (b & 1) r *= a; a *= a, b >>= 1; } return r; }
        }
        return 0;
    }
    static ll eval(const string& s) {
        vector<ll> val;
        vector<char> ops;
        bool wantOperand = true; // distinguishes unary from binary minus
        auto pop = [&]() {
            char op = ops.back(); ops.pop_back();
            if (op == '~') { ll a = val.back(); val.pop_back();
                val.push_back(apply(op, a)); }
            else { ll b = val.back(); val.pop_back(); ll a = val.back(); val.pop_back();
                val.push_back(apply(op, a, b)); }
        };
        for (size_t i = 0; i < s.size(); i++) {
            char c = s[i];
            if (isspace((unsigned char)c)) continue;
            if (isdigit((unsigned char)c)) {
                ll x = 0;
                while (i < s.size() && isdigit((unsigned char)s[i])) x = x * 10 + (s[i++] - '0');
                i--;
                val.push_back(x), wantOperand = false;
            } else if (c == '(') ops.push_back(c), wantOperand = true;
            else if (c == ')') {
                while (!ops.empty() && ops.back() != '(') pop();
                if (!ops.empty()) ops.pop_back();
                wantOperand = false;
            } else {
                if (c == '~' && wantOperand) c = '~'; // unary
                while (!ops.empty() && ops.back() != '(' && prec(ops.back()) > prec(c) || (prec(ops.back()) == prec(c) && !rightAssoc(c))) pop();
                ops.push_back(c), wantOperand = (c != '~') || true;
            }
        }
        while (!ops.empty()) pop();
        return val.empty() ? 0 : val.back();
    }
};
// To build an AST instead of evaluating, push node indices instead of values.
// For boolean/comparison operators just add them to prec() with lower precedence.

```

1.16 CycleFinding

CYCLE FINDING in a FUNCTIONAL GRAPH $x \rightarrow f(x)$. Every component is a "rho": a tail feeding
// into a cycle. Answers: where does x end up after k steps, cycle length, distance to cycle.

// Floyd (tortoise & hare), O(1) memory. Returns {start index of the cycle, cycle length}.

```

template <class F> pair<ll, ll> floydCycle(ll x0, F f) {
    ll t = f(x0), h = f(f(x0));
    while (t != h) t = f(t), h = f(f(h));
    ll mu = 0; // steps from x0 to the cycle entry
    t = x0;
    while (t != h) t = f(t), h = f(h), mu++;
    ll lam = 1; // cycle length
    h = f(t);
    while (t != h) h = f(h), lam++;
    return {mu, lam};
}
// f applied k times, using the rho structure (k can be 1e18).
template <class F> ll iterate(ll x0, ll k, F f) {
    auto [mu, lam] = floydCycle(x0, f);
    if (k < mu) { while (k--) x0 = f(x0); return x0; }
    for (ll i = 0; i < mu; i++) x0 = f(x0);
    ll rem = (k - mu) % lam;
    while (rem--) x0 = f(x0);
    return x0;
}
// WHOLE-GRAPH version when f is given as an array over 0..n-1: colour DFS, O(n).
// state: 0 unvisited, 1 on the current path, 2 done. Fills cycId/cycPos/distToCycle.
struct FuncGraph {
    int n;
    vector<int> f, state, cyc, pos, dist, cycLen; // cyc[v] = id of the cycle v reaches
    FuncGraph(vector<int> f) : n(f.size()), f(f), state(n, 0), cyc(n, -1), pos(n, -1), dist(n, 0) {
        for (int i = 0; i < n; i++) if (!state[i]) {
            vector<int> path;
            int v = i;
            while (!state[v]) state[v] = 1, path.push_back(v), v = f[v];
            if (state[v] == 1) { // found a NEW cycle, starting at v
                int id = cycLen.size(), len = 0;
                for (int u = v; u = f[u]; ) { cyc[u] = id, pos[u] = len++; if (f[u] == v) break; }
                cycLen.push_back(len);
            }
            for (int j = path.size() - 1; j >= 0; j--) {
                int u = path[j];
                state[u] = 2;
                if (cyc[u] == -1) cyc[u] = cyc[f[u]], dist[u] = dist[f[u]] + 1;
            }
        }
    }
    int kth(int v, ll k) { // f^k(v)
        while (k > 0 && dist[v] > 0) v = f[v], k--;
        if (!k) return v;
        int L = cycLen[cyc[v]], p = (pos[v] + k % L) % L, u = v;
        for (int i = 0; i < (p - pos[v] + L) % L; i++) u = f[u];
        return u;
    }
};
// USES: "apply the permutation k times", Pollard's rho, "where does the ball stop",
// detecting a repeat in a modular sequence, linked-list loop detection.

```


1.17 ParallelBinarySearch

PARALLEL BINARY SEARCH - binary search the answer for MANY queries at once, sweeping the

```
// shared structure once per level instead of once per query.
// PRECONDITION: test(i) must be MONOTONE in time - once it is
//   ↳ true it stays true. If updates
// can also undo things, parallel binary search does not apply;
//   ↳ use offline divide & conquer over
// the timeline with a rollback DSU (O2 - Data Structures/14)
//   ↳ instead.
// Total: O(log(maxT) * (cost of one full sweep)).
// Shape: "after which update does query q first become
//   ↳ satisfiable?"
// lo[q], hi[q] track each query's remaining range; every round,
//   ↳ bucket the queries by their
// midpoint, replay the updates in order, and test each query
//   ↳ when the sweep reaches its mid.
template <class Apply, class Reset, class Test>
vector<int> parallelBinarySearch(int q, int T, Apply applyUpdate,
//   ↳ Reset reset, Test test) {
//   ↳ applyUpdate(t): apply update t to the shared structure
//   ↳ reset(): clear the structure
//   ↳ test(i): is query i satisfied by the current structure?
vector<int> lo(q, 0), hi(q, T); // answer
//   ↳ in [lo, hi]; hi == T means "never"
while (true) {
vector<vector<int>> at(T + 1);
bool active = false;
for (int i = 0; i < q; i++) if (lo[i] < hi[i]) at[lo[i]
//   ↳ + hi[i] / 2].push_back(i), active = true;
if (!active) break;
reset();
for (int t = 0; t <= T; t++) {
if (t) applyUpdate(t - 1); // now
//   ↳ updates 0..t-1 are applied
for (int i : at[t]) { if (test(i)) hi[i] = t; else
//   ↳ lo[i] = t + 1; }
}
return lo; // lo[i]
//   ↳ == T => never satisfied
}
// CONCRETE USES
// - "for each query, the first moment a path/edge set connects u
//   ↳ and v" (structure = DSU)
// - k-th smallest in a range, offline (structure = BIT over
//   ↳ values, updates = insert a value)
// - "smallest capacity so that a flow/matching of size k exists"
// - anything where a single query's binary search would need its
//   ↳ own O(T) replay.
// If the structure supports ROLLBACK, prefer divide-and-conquer
//   ↳ over the timeline instead
// (see OfflineDynamicConnectivity) - it is one pass instead of
//   ↳ log passes.
```

1.18 MeetInTheMiddle

MEET IN THE MIDDLE - $n \leq 40$. Split into halves, enumerate $2^{(n/2)}$ subsets each side,

```
// then recombine by sorting + binary search / two pointers.
//   ↳ O(2(n/2) * n).

// All subset sums of a half, as a sorted vector.
vector<ll> subsetSums(const vector<ll>& a) {
int n = a.size();
vector<ll> s(1 << n, 0);
for (int m = 1; m < (1 << n); m++) {
int b = __builtin_ctz(m);
s[m] = s[m ^ (1 << b)] + a[b];
}
return s;
}

// #subsets with sum <= S. Works with NEGATIVE values too (no
//   ↳ pre-filter on the left half).
ll countAtMost(vector<ll> a, ll S) {
int n = a.size(), h = n / 2;
vector<ll> L(a.begin(), a.begin() + h), R(a.begin() + h,
//   ↳ a.end());
auto ls = subsetSums(L), rs = subsetSums(R);
sort(all(rs));
ll c = 0;
for (ll x : ls) c += upper_bound(all(rs), S - x) - rs.begin();
return c;
}

// Largest achievable sum <= S (the classic "closest subset sum
//   ↳ from below").
ll bestAtMost(vector<ll> a, ll S) {
int n = a.size(), h = n / 2;
vector<ll> L(a.begin(), a.begin() + h), R(a.begin() + h,
//   ↳ a.end());
auto ls = subsetSums(L), rs = subsetSums(R);
sort(all(rs));
ll best = LLONG_MIN;
for (ll x : ls) { // no 'x
//   ↳ <= S' filter: x may be > S
auto it = upper_bound(all(rs), S - x); // and
//   ↳ still pair with a negative right sum
if (it != rs.begin()) best = max(best, x + *prev(it));
}
return best;
}

// OTHER SHAPES
// - EXACT target: put one half in a hash map, look up (target -
//   ↳ x) from the other.
// - 4-SUM: split the four arrays 2+2, hash all pairwise sums.
// - "choose exactly k items": bucket each half's subsets BY
//   ↳ POPCOUNT, then pair bucket i with
// bucket k-i (sort each bucket once).
// - Optimising two objectives: sort one half by objective A and
//   ↳ take a prefix maximum of
// objective B, so each query is one binary search.
// If n <= 20 a plain 2n loop is simpler; meet in the middle
//   ↳ earns its keep from n = 30 upward.
```

1.19 Scheduling

```

SCHEDULING / EXCHANGE-ARGUMENT THEOREM CARD
=====

/* ===== SCHEDULING / EXCHANGE-ARGUMENT THEOREM CARD
   ↳ =====
Each of these turns a hard-looking problem into a sort. Prove any
↳ new one by the EXCHANGE
ARGUMENT: assume an optimal schedule, swap two ADJACENT jobs, and
↳ compare the two costs.

SMITH'S RULE (1 || sum w_j C_j : minimise the weighted sum of
↳ completion times)
Sort by p_j / w_j ASCENDING. Adjacent swap of h,k improves iff
↳ w_k*p_h - w_h*p_k > 0,
so ALWAYS compare by cross-multiplication (w_k*p_h vs w_h*p_k),
↳ never by float division.

MOORE-HODGSON (1 || sum U_j : minimise the NUMBER of late jobs),
↳ O(n log n)
Sort by deadline ascending; keep a max-heap of the processing
↳ times taken so far.
For each job: take it, push p_j, add p_j to the running time;
↳ if the running time exceeds
this job's deadline, pop the largest p and subtract it. The
↳ heap size is the answer
(the rejected jobs can go last in any order).

EARLIEST DEADLINE FIRST (preemptive, one machine)
If ANY schedule meets all deadlines, EDF does. Feasibility
↳ check for non-preemptive
unit-machine: sort by deadline and verify prefix sums of p_j
↳ never exceed d_j.

JOHNSON'S RULE (F2 || Cmax : two-machine flow shop, minimise
↳ makespan)
Set A = jobs with p1 < p2, sorted by p1 ASCENDING
Set B = jobs with p1 >= p2, sorted by p2 DESCENDING
Optimal order = A then B.

LAWLER (1 | prec | f_max : minimise the maximum penalty, with
↳ precedence), O(n^2)
Build the schedule BACK TO FRONT. With t = total remaining
↳ time, among the jobs that have
no unscheduled successor pick the one minimising f_j(t), and
↳ place it last.

CLASSIC GREEDY COMPANIONS
Interval scheduling (max #non-overlapping): sort by RIGHT
↳ endpoint, take greedily.
Interval point cover (stab all intervals): sort by right
↳ endpoint, place a point at it.
Minimum #machines for overlapping intervals: max overlap =
↳ sweep +1/-1 over endpoints.
Deadline scheduling with profits (each job unit length): sort
↳ by profit DESC, place each
job as late as possible before its deadline (DSU "next free
↳ slot" makes it near-linear).
Fractional knapsack: sort by value/weight DESC.
Minimise sum of |x - a_i|: x = the MEDIAN. Minimise sum of (x -
↳ a_i)^2: x = the MEAN.

=====
↳ */

// Moore-Hodgson: maximum number of jobs finished on time. jobs =
↳ {processing time, deadline}.
int maxOnTime(vector<pair<ll, ll>> jobs) {
    sort(all(jobs), [&](auto& a, auto& b) { return a.second <
↳ b.second; });
    priority_queue<ll> pq;
    ll t = 0;
    for (auto [p, d] : jobs) {
        t += p, pq.push(p);
        if (t > d) t -= pq.top(), pq.pop();
    }
    return pq.size();
}

// Johnson's rule: optimal two-machine flow-shop order (returns
↳ the job indices).
vector<int> johnson(const vector<pair<ll, ll>>& jobs) { // {p
↳ on machine 1, p on machine 2}
    vector<int> A, B;
    for (int i = 0; i < (int)jobs.size(); i++) (jobs[i].first <
↳ jobs[i].second ? A : B).push_back(i);
    sort(all(A), [&](int x, int y) { return jobs[x].first <
↳ jobs[y].first; });
    sort(all(B), [&](int x, int y) { return jobs[x].second >
↳ jobs[y].second; });

```

```

A.insert(A.end(), all(B));
return A;
}

```

1.20 Josephus

JOSEPHUS PROBLEM. n people in a circle 0..n-1, every k-th is removed; who survives?

```

// Recurrence: J(1) = 0, J(n) = (J(n-1) + k) mod n. O(n).
ll josephus(ll n, ll k) { ll r = 0; for (ll i = 2; i <= n; i++) r
↳ = (r + k) % i; return r; }
// k = 2 closed form: write n = 2^m + l with 0 <= l < 2^m, the
↳ survivor is 2*l (0-indexed).
ll josephus2(ll n) { return 2 * (n - (1LL << (63 -
↳ __builtin_clzll(n)))); }
// LARGE n, SMALL k: one full lap kills floor(n/k) people, so
↳ recurse on n - floor(n/k) and map the
// index back. O(k log n).
ll josephusFast(ll n, ll k) {
    if (n == 1) return 0;
    if (k == 1) return n - 1;
    if (k > n) return (josephusFast(n - 1, k) + k) % n;
    ll r = josephusFast(n - n / k, k) - n % k;
    return r < 0 ? r + n : r + r / (k - 1);
}

// ORDER OF ELIMINATION / the j-th person to die: keep a BIT over
↳ alive flags and binary-search the
// ((pos + k - 1) mod alive + 1)-th alive person each step. O(n
↳ log n) and handles "k changes".
// LAST TWO SURVIVORS: run the recurrence starting from J(2) = {0,
↳ 1} instead of J(1) = 0.

// OTHER ONE-LINE CLASSICS worth having on paper:
// GRAY CODE: g(i) = i ^ (i >> 1); inverse: x ^ = x >> 1, x
↳ >> 2, ... until 0.
// n-th PERMUTATION (factorial number system): digit i = k /
↳ (n-1-i)!, then k %= (n-1-i)!.
// 15-PUZZLE SOLVABLE iff (#inversions of the tiles, blank
↳ excluded) + (row of the blank counted
// from the BOTTOM, 1-based) is EVEN, for a 4x4 board. For an
↳ odd width w, solvable iff the
// inversion count is even. General w x h: inversions + (blank
↳ row from bottom) parity rule above
// when w is even, plain inversion parity when w is odd.
// TOWER OF HANOI: 2^n - 1 moves; on move t (1-based) move disk (t
↳ & -t) counted by trailing zeros,
// and for odd n the smallest disk cycles A->C->B->A, for even n
↳ A->B->C->A.
// BULLS AND COWS / stable-marriage style pairings: Gale-Shapley
↳ is O(n^2), always terminates, and
// is optimal for the proposing side.
// CATALAN-STYLE BALLOT COUNTING: paths from (0,0) to (n,m)
↳ staying strictly above y = x are
// (n-m)/(n+m) * C(n+m, m) (Bertrand's ballot theorem).

```


1.21 BigInt

BIG INTEGERS, non-negative, base 1e9. Enough for the usual "the answer does not fit in 64 bits"

```
// problems: exact factorials / Catalan numbers / determinants,
//   ↪ huge Fibonacci, exact comparisons.
// Add/sub  $O(n)$ , schoolbook multiply  $O(n*m)$  (fine to  $\sim 1e4$  digits),
//   ↪ divide by a small int  $O(n)$ .
// If you need big*big beyond that, convolve the limbs with FFT
//   ↪ and carry.
struct Big {
    static const ll B = 1000000000;
    vector<ll> d;
    // ↪ little-endian limbs, base 1e9
    Big(ll v = 0) { for (; v > 0; v /= B) d.push_back(v % B); }
    Big(const string& s) {
        for (int i = s.size(); i > 0; i -= 9)
            d.push_back(stoll(s.substr(max(0, i - 9), min(9,
// ↪ i)))));
        trim();
    }
    void trim() { while (!d.empty() && d.back() == 0)
// ↪ d.pop_back(); }
    bool operator<(const Big& o) const {
        if (d.size() != o.d.size()) return d.size() < o.d.size();
        for (int i = d.size() - 1; i >= 0; i--) if (d[i] !=
// ↪ o.d[i]) return d[i] < o.d[i];
        return false;
    }
    Big operator+(const Big& o) const {
        Big r; ll c = 0;
        for (size_t i = 0; i < max(d.size(), o.d.size()) || c;
// ↪ i++) {
            if (i < d.size()) c += d[i];
            if (i < o.d.size()) c += o.d[i];
            r.d.push_back(c % B), c /= B;
        }
        return r;
    }
    Big operator-(const Big& o) const {
// ↪ requires *this >= o
        Big r = *this; ll c = 0;
        for (size_t i = 0; i < r.d.size(); i++) {
            r.d[i] -= c + (i < o.d.size() ? o.d[i] : 0);
            if ((c = r.d[i] < 0)) r.d[i] += B;
        }
        r.trim();
        return r;
    }
    Big operator*(const Big& o) const {
        if (d.empty() || o.d.empty()) return Big();
        vector<lll> t(d.size() + o.d.size(), 0);
        for (size_t i = 0; i < d.size(); i++)
            for (size_t j = 0; j < o.d.size(); j++) t[i + j] +=
// ↪ (lll)d[i] * o.d[j];
        Big r; lll c = 0;
        for (size_t i = 0; i < t.size(); i++) c += t[i],
// ↪ r.d.push_back((ll)(c % B)), c /= B;
        for (; c; c /= B) r.d.push_back((ll)(c % B));
        r.trim();
        return r;
    }
    Big operator*(ll v) const { return *this * Big(v); }
    Big operator/(ll v) const {
// ↪ small divisor only
        Big r = *this; lll c = 0;
        for (int i = r.d.size() - 1; i >= 0; i--) c = c * B +
// ↪ r.d[i], r.d[i] = (ll)(c / v), c %= v;
        r.trim();
        return r;
    }
    ll operator%(ll v) const {
        lll c = 0;
        for (int i = d.size() - 1; i >= 0; i--) c = (c * B +
// ↪ d[i]) % v;
        return (ll)c;
    }
    string str() const {
        if (d.empty()) return "0";
        string s = to_string(d.back());
        for (int i = d.size() - 2; i >= 0; i--) {
            string t = to_string(d[i]);
            s += string(9 - t.size(), '0') + t;
        }
        return s;
    }
};
// SIGNED: keep a separate sign flag and dispatch +/- to add/sub
```

↪ on magnitudes; comparison first.
 // BASE 1e9 packs 9 decimal digits per limb, so printing is
 ↪ trivial - that is the whole point.
 // FAST DIVISION of two Bigs is long and almost never needed; if a
 ↪ problem really wants it, binary
 // search the quotient limb by limb, or move to Python-style
 ↪ thinking and avoid it.
 // Sqrt: binary search with the multiply above, or Newton on
 ↪ doubles then fix up by a few steps.

1.22 RankingAndDates

RANKING / UNRANKING COMBINATORIAL OBJECTS, and CALENDAR ARITHMETIC. Small, and each one costs

```
// 20 minutes to re-derive under pressure.

// --- FACTORIAL NUMBER SYSTEM: rank and unrank a PERMUTATION.
//  ↳  $O(n^2)$  as written; use an
// order-statistic tree or a BIT for  $O(n \log n)$  when  $n$  is large.
// order(p) = how many permutations of the same multiset are
//  ↳ lexicographically smaller.
ll permRank(vector<int> p) {
    //  ↳ p is a permutation of 0..n-1
    int n = p.size();
    ll r = 0, f = 1;
    for (int i = n - 1; i >= 0; i--) {
        int c = 0;
        for (int j = i + 1; j < n; j++) if (p[j] < p[i]) c++;
        //  ↳ the Lehmer code digit
        r += c * f, f *= (n - i);
    }
    return r;
}

vector<int> permUnrank(int n, ll k) {
    //  ↳ k-th permutation, 0-indexed
    vector<int> avail(n), p(n);
    iota(all(avail), 0);
    vector<ll> f(n + 1, 1);
    for (int i = 1; i <= n; i++) f[i] = f[i - 1] * i;
    for (int i = 0; i < n; i++) {
        ll d = k / f[n - 1 - i];
        k %= f[n - 1 - i];
        p[i] = avail[d], avail.erase(avail.begin() + d);
    }
    return p;
}

// --- JULIAN DAY NUMBER: date <-> integer, closed form, no loops,
//  ↳ correct for the Gregorian
// calendar including every leap rule. Date differences, "what
//  ↳ weekday", and "add 1000 days" all
// become integer arithmetic once you are here.
ll dateToInt(int y, int m, int d) {
    //  ↳ y-m-d Gregorian -> day number
    return 1461 * (y + 4800 + (m - 14) / 12) / 4
        + 367 * (m - 2 - (m - 14) / 12 * 12) / 12
        - 3 * ((y + 4900 + (m - 14) / 12) / 100) / 4 + d - 32075;
}

array<int, 3> intToDate(ll jd) {
    //  ↳ inverse: {y, m, d}
    ll x, n, i, j, dd;
    x = jd + 68569, n = 4 * x / 146097, x -= (146097 * n + 3) / 4;
    i = 4000 * (x + 1) / 1461001, x -= 1461 * i / 4 - 31;
    j = 80 * x / 2447, dd = x - 2447 * j / 80, x = j / 11;
    return {(int)(100 * (n - 49) + i + x), (int)(j + 2 - 12 * x),
        (int)dd};
}

int weekday(ll jd) { return (int)((jd + 1) % 7); }
//  ↳ // 0 = Sunday
// LEAP YEAR: y % 4 == 0 && (y % 100 != 0 || y % 400 == 0).
// ZELLER'S CONGRUENCE gives the weekday directly, but the day
//  ↳ number is strictly more useful.

// --- STERN-BROCOT SEARCH: the simplest fraction satisfying a
//  ↳ MONOTONE predicate. ---
// Descend the mediant tree, but take many steps in one direction
//  ↳ at once (double the step size)
// so a 1e18 denominator costs  $O(\log^2)$  instead of  $O(\text{answer})$ .
// USES: "smallest denominator q with property P", recovering p/q
//  ↳ from a decimal prefix, the
// simplest fraction strictly inside an interval, and best
//  ↳ rational approximation with a bound.
// ok() must be MONOTONE in the value p/q: false on small
//  ↳ fractions, true from some point on.
// Returns the SMALLEST such fraction with denominator <= N. Each
//  ↳ iteration takes as many steps
// as possible in one direction (found by doubling, then binary
//  ↳ search), so it is  $O(\log^2 N)$ 
// rather than  $O(\text{answer})$  - the difference between instant and
//  ↳ hopeless at  $N = 1e18$ .
```

```
template <class F>
pair<ll, ll> fracSearch(ll N, F ok) {
    ll lp = 0, lq = 1, rp = 1, rq = 0;
    //  ↳ // 0/1 (not ok) .. 1/0 (ok)
    for (;;) {
        ll moved = 0;
        auto walk = [&](ll& ap, ll& aq, ll& bp, ll& bq, bool want)
        //  ↳ // steps from a toward b
        {
            ll k = 0, s = 1;
            auto test = [&](ll t) {
                return aq + t * bq <= N && ok(ap + t * bp, aq + t
                //  ↳ // * bq) == want;
            };
            while (test(k + s)) k += s, s *= 2;
            for (s /= 2; s; s /= 2) if (test(k + s)) k += s;
            ap += k * bp, aq += k * bq, moved += k;
        };
        walk(lp, lq, rp, rq, false);
        //  ↳ // push the FALSE end up
        if (lq + rq > N) return {rp, rq};
        walk(rp, rq, lp, lq, true);
        //  ↳ // pull the TRUE end down
        if (lp + lq > N || !moved) return {rp, rq};
        //  ↳ // !moved: nothing left to refine
    }
}

// The Stern-Brocot tree IS the continued fraction expansion:
//  ↳ going left/right k times is one
// partial quotient. See 04 - Number Theory/05 - Rational for the
//  ↳ CF machinery and semiconvergents.
```

2 Data Structures (DS)

2.1 Segment Tree

2.1.1 SegmentTree

SEGMENT TREE, point update + range query, $O(\log n)$ each. Swap the merge in SEG to change the

```
// operation (sum/min/max/gcd/matrix); any associative monoid
// works. Size is rounded up to a power
// of two so the tree is perfect and indices are simple.
struct SEG {
    ll sum = 0;
    SEG() {}
    SEG(ll x){
        sum = x;
    }
};

struct segTree {
    vector<SEG> seg;
    int sz = 1,n;
    segTree(int nn){
        n = nn;
        while (sz < nn)
            sz *= 2;
        seg.assign(2 * sz , SEG());
    }

    SEG merge(SEG seg1, SEG seg2) {
        SEG ret;
        ret.sum = seg1.sum + seg2.sum;
        return ret;
    }

    void build(vector<int> &v,int x, int lx, int rx) {
        if (lx == rx) {
            seg[x] = SEG(v[lx]);
            return;
        }
        int mid = (lx + rx) / 2;
        int lf = 2 * x + 1, rt = 2 * x + 2;

        build(v,lf, lx, mid);
        build(v,rt, mid + 1, rx);
        seg[x] = merge(seg[lf], seg[rt]);
    }

    void build(vector<int> &v){
        build(v,0, 0, n-1);
    }

    void update(int i, ll val, int x, int lx, int rx) {
        if (lx == rx) {
            seg[x] = SEG(val);
            return;
        }
        int mid = (lx + rx) / 2,lf = 2*x+1,rt= 2*x+2;
        if(i <= mid)
            update(i, val, lf, lx, mid);
        else
            update(i, val, rt, mid + 1, rx);
        seg[x] = merge(seg[lf] , seg[rt]);
    }

    void update(int i, ll val) {
        update(i, val, 0, 0, n-1);
    }

    SEG query(int l, int r, int x, int lx, int rx) {
        if (l <= lx && r >= rx)
            return seg[x];
        if (l > rx || r < lx)
            return SEG();

        int mid = (lx + rx) / 2,lf = 2 * x + 1,rt = 2*x+2;

        return merge(query(l, r, lf, lx, mid) , query(l, r, rt,
        mid + 1, rx));
    }

    SEG query(int l, int r) {
        return query(l, r, 0, 0, n-1);
    }
};
```

2.1.2 LazySegmentTree

Lazy Segment Tree (range add, range sum) - build $O(n)$, update/query $O(\log n)$

```
// 1-based node indices, 0-based positions, ALL recursions on [0,
// n-1].
// To change the op: edit merge(), apply() and the identity in
// qry().
struct LazySeg {
    int n;
    vector<ll> t, lz;

    LazySeg(int n = 0) : n(n), t(4 * n, 0), lz(4 * n, 0) {}

    ll merge(ll a, ll b) { return a + b; }
    void apply(int v, int l, int r, ll x) { t[v] += x * (r - l +
    1); lz[v] += x; }

    void push(int v, int l, int r) {
        if (!lz[v]) return;
        int m = (l + r) / 2;
        apply(2 * v, l, m, lz[v]);
        apply(2 * v + 1, m + 1, r, lz[v]);
        lz[v] = 0;
    }

    void build(const vector<ll>& a, int v, int l, int r) {
        if (l == r) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(a, 2 * v, l, m), build(a, 2 * v + 1, m + 1, r);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }

    void build(const vector<ll>& a) { if (n) build(a, 1, 0, n -
    1); }

    void upd(int ql, int qr, ll x, int v, int l, int r) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) return apply(v, l, r, x);
        push(v, l, r);
        int m = (l + r) / 2;
        upd(ql, qr, x, 2 * v, l, m), upd(ql, qr, x, 2 * v + 1, m
        + 1, r);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }

    void upd(int l, int r, ll x) { upd(l, r, x, 1, 0, n - 1); }

    ll qry(int ql, int qr, int v, int l, int r) {
        if (qr < l || r < ql) return 0; //
        // identity
        if (ql <= l && r <= qr) return t[v];
        push(v, l, r);
        int m = (l + r) / 2;
        return merge(qry(ql, qr, 2 * v, l, m), qry(ql, qr, 2 * v
        + 1, m + 1, r));
    }

    ll qry(int l, int r) { return qry(l, r, 1, 0, n - 1); }
};
```

2.1.3 PersistentSegmentTree

PERSISTENT SEGMENT TREE - every update returns a NEW root and old versions stay queryable.

```
// O(log n) time and O(log n) new nodes per update. The standard
// use is offline k-th smallest in a
// range (build a version per prefix, then query version[r] minus
// version[l-1]).
struct Node {
    Node *l, *r;
    ll val = 0;

    Node(ll val) : l(NULL), r(NULL), val(val) {}

    Node() : l(NULL), r(NULL) {}

    Node(Node* l, Node* r) : l(l), r(r) {
        if (l != NULL) val += l->val;
        if (r != NULL) val += r->val;
    }

    void addChild() {
        l = new Node();
        r = new Node();
    }
};

struct PST {
    int n;

    PST(int n) : n(n + 1) {}

    Node merge(Node x, Node y) {
        Node ret;
        ret.val = x.val + y.val;
        return ret;
    }

    Node* update(Node* v, int i, ll val, int lx, int rx) {
        if (lx == rx) return new Node(v->val + val);
        int mid = (lx + rx) / 2;
        if (!v->l) v->addChild();
        if (i <= mid) return new Node(update(v->l, i, val, lx,
        // mid), v->r);
        return new Node(v->l, update(v->r, i, val, mid + 1, rx));
    }

    Node* update(Node* v, int i, ll val) { return update(v, i,
    // val, 0, n - 1); }

    Node query(Node* v, int l, int r, int lx, int rx) {
        if (l > rx || r < lx) return {};
        if (l <= lx && r >= rx) return *v;
        if (!v->l) v->addChild();
        int mid = (lx + rx) / 2;
        return merge(query(v->l, l, r, lx, mid), query(v->r, l,
    // r, mid + 1, rx));
    }

    Node query(Node* v, int l, int r) { return query(v, l, r, 0,
    // n - 1); }

    int getKth(Node* a, Node* b, int k, int lx, int rx) {
        if (lx == rx) return lx;
        if (!a->l) a->addChild();
        if (!b->l) b->addChild();
        int rem = b->l->val - a->l->val;
        int mid = (lx + rx) / 2;
        if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
        return getKth(a->r, b->r, k - rem, mid + 1, rx);
    }

    int getKth(Node* a, Node* b, int k) { return getKth(a, b, k,
    // 0, n - 1); }
};
```

2.1.4 SegmentTreeBeats

Rename one if you ever paste two of them together.

```
// NOTE: several segment-tree files here declare their own 'struct
// Node'.
// Rename one if you ever paste two of them together.
struct Node {
    ll mn;
    int mnCnt;
    ll secondMn;

    ll mx;
    int mxCnt;
    ll secondMx;

    ll sum;

    ll lazyAdd;
    ll lazyEqual;
    bool isLazyEqual;

    Node() {
        mx = -inf;
        secondMx = -inf;
        mxCnt = 0;

        mn = inf;
        secondMn = inf;
        mnCnt = 0;

        sum = 0;

        lazyAdd = 0;
        lazyEqual = 0;
        isLazyEqual = 0;
    };

    Node(int x) {
        mx = x;
        secondMx = -inf;
        mxCnt = 1;

        mn = x;
        secondMn = inf;
        mnCnt = 1;

        sum = x;

        lazyAdd = 0;
        lazyEqual = 0;
        isLazyEqual = 0;
    }
};

struct SegTree {
    int tree_size;
    vector<Node> seg_data;

    SegTree(int n) {
        tree_size = 1;
        while (tree_size < n) tree_size *= 2;
        seg_data.resize(2 * tree_size, Node());
    }

    Node merge(Node& lf, Node& ri) {
        Node ans = Node();

        ans.sum = lf.sum + ri.sum;

        ans.mx = max(lf.mx, ri.mx);
        ans.secondMx = max(lf.secondMx, ri.secondMx);
        ans.mxCnt = 0;
        if (lf.mx == ans.mx) ans.mxCnt += lf.mxCnt;
        else ans.secondMx = max(ans.secondMx, lf.mx);
        if (ri.mx == ans.mx) ans.mxCnt += ri.mxCnt;
        else ans.secondMx = max(ans.secondMx, ri.mx);

        ans.mn = min(lf.mn, ri.mn);
        ans.secondMn = min(lf.secondMn, ri.secondMn);
        ans.mnCnt = 0;
        if (lf.mn == ans.mn) ans.mnCnt += lf.mnCnt;
        else ans.secondMn = min(ans.secondMn, lf.mn);
        if (ri.mn == ans.mn) ans.mnCnt += ri.mnCnt;
        else ans.secondMn = min(ans.secondMn, ri.mn);

        return ans;
    }
};
```

```

void init(vector<int>& nums, int ni, int lx, int rx) {
    if (rx - lx == 1) {
        if (lx < nums.size())
            seg_data[ni] = Node(nums[lx]);
        return;
    }

    int mid = lx + (rx - lx) / 2;
    init(nums, 2 * ni + 1, lx, mid);
    init(nums, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 *
    ↪ ni + 2]);
}

void init(vector<int>& nums) {
    init(nums, 0, 0, tree_size);
}

void doPushEq(int ni, int lx, int rx, int x) {
    seg_data[ni].mn = seg_data[ni].mx =
    ↪ seg_data[ni].lazyEqual = x;
    seg_data[ni].isLazyEqual = 1;
    seg_data[ni].mnCnt = seg_data[ni].mxCnt = rx - lx;
    seg_data[ni].secondMx = -inf;
    seg_data[ni].secondMn = inf;

    seg_data[ni].sum = (ll)x * (rx - lx);
    seg_data[ni].lazyAdd = 0;
}

void doPushMinEq(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mn >= x) doPushEq(ni, lx, rx, x);
    else if (seg_data[ni].mx > x) {
        if (seg_data[ni].secondMn == seg_data[ni].mx)
            seg_data[ni].secondMn = x;
        seg_data[ni].sum -= (ll)(seg_data[ni].mx - x) *
    ↪ seg_data[ni].mxCnt;
        seg_data[ni].mx = x;
    }
}

void doPushMaxEq(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mx <= x) doPushEq(ni, lx, rx, x);
    else if (seg_data[ni].mn < x) {
        if (seg_data[ni].secondMx == seg_data[ni].mn)
            seg_data[ni].secondMx = x;

        seg_data[ni].sum += (ll)(x - seg_data[ni].mn) *
    ↪ seg_data[ni].mnCnt;
        seg_data[ni].mn = x;
    }
}

void doPushSum(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mn == seg_data[ni].mx)
        doPushEq(ni, lx, rx, seg_data[ni].mn + x);
    else {
        seg_data[ni].mx += x;
        if (seg_data[ni].secondMx != -inf)
            seg_data[ni].secondMx += x;

        seg_data[ni].mn += x;
        if (seg_data[ni].secondMn != inf)
            seg_data[ni].secondMn += x;

        seg_data[ni].sum += (ll)(rx - lx) * x;
        seg_data[ni].lazyAdd += x;
    }
}

void propagete(int ni, int lx, int rx) {
    if (rx - lx == 1) return;

    int mid = lx + (rx - lx) / 2;
    if (seg_data[ni].isLazyEqual) {
        doPushEq(2 * ni + 1, lx, mid, seg_data[ni].lazyEqual);
        doPushEq(2 * ni + 2, mid, rx, seg_data[ni].lazyEqual);
        seg_data[ni].lazyEqual = seg_data[ni].isLazyEqual = 0;
    }
    else {
        doPushSum(2 * ni + 1, lx, mid, seg_data[ni].lazyAdd);
        doPushSum(2 * ni + 2, mid, rx, seg_data[ni].lazyAdd);
        seg_data[ni].lazyAdd = 0;

        doPushMinEq(2 * ni + 1, lx, mid, seg_data[ni].mx);
        doPushMinEq(2 * ni + 2, mid, rx, seg_data[ni].mx);
    }
}

```

```

        doPushMaxEq(2 * ni + 1, lx, mid, seg_data[ni].mn);
        doPushMaxEq(2 * ni + 2, mid, rx, seg_data[ni].mn);
    }
}

void minimize(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r || seg_data[ni].mx <= x)
        return;
    if (lx >= l && rx <= r && seg_data[ni].secondMx < x) {
        doPushMinEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    minimize(l, r, x, 2 * ni + 1, lx, mid);
    minimize(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 *
    ↪ ni + 2]);
}

void minimize(int l, int r, int x) {
    minimize(l, r, x, 0, 0, tree_size);
}

void maximize(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r || seg_data[ni].mn >= x)
        return;
    if (lx >= l && rx <= r && seg_data[ni].secondMn > x) {
        doPushMaxEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    maximize(l, r, x, 2 * ni + 1, lx, mid);
    maximize(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 *
    ↪ ni + 2]);
}

void maximize(int l, int r, int x) {
    maximize(l, r, x, 0, 0, tree_size);
}

void add(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r)
        return;
    if (lx >= l && rx <= r) {
        doPushSum(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    add(l, r, x, 2 * ni + 1, lx, mid);
    add(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 *
    ↪ ni + 2]);
}

void add(int l, int r, int x) {
    add(l, r, x, 0, 0, tree_size);
}

void set(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r)
        return;
    if (lx >= l && rx <= r) {
        doPushEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    set(l, r, x, 2 * ni + 1, lx, mid);
    set(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 *
    ↪ ni + 2]);
}

```

```

    ↪ ni + 2]);
}

void set(int l, int r, int x) {
    set(l, r, x, 0, 0, tree_size);
}

Node get_range(int l, int r, int ni, int lx, int rx) {
    propagete(ni, lx, rx);

    if (lx >= l && rx <= r)
        return seg_data[ni];
    if (rx <= l || lx >= r)
        return Node();

    int mid = (lx + rx) / 2;

    Node lf = get_range(l, r, 2 * ni + 1, lx, mid);
    Node ri = get_range(l, r, 2 * ni + 2, mid, rx);
    return merge(lf, ri);
}

ll get_sum(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).sum;
}

int get_mx(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).mx;
}

int get_mn(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).mn;
}
};

```

2.1.5 DynamicSegTree

Dynamic (Implicit) Segment Tree - Space: $O(Q \log N)$ per query

// Allocates nodes on demand for ranges up to 1e9 or 1e18

```

template<typename T = ll>
struct DynamicSegTree {
    struct Node {
        T val = 0;
        int lc = -1, rc = -1;
    };

    vector<Node> tree;
    ll L, R;

    DynamicSegTree(ll L = 0, ll R = 1e9) : L(L), R(R) {
        add_node();
    }

    int add_node() {
        tree.pb(Node());
        return tree.size() - 1;
    }

    void update(int node, ll l, ll r, ll pos, T val) {
        tree[node].val += val;
        if (l == r) return;
        ll mid = l + (r - l) / 2;
        if (pos <= mid) {
            if (tree[node].lc == -1) {
                int nxt = add_node();
                tree[node].lc = nxt;
            }
            update(tree[node].lc, l, mid, pos, val);
        } else {
            if (tree[node].rc == -1) {
                int nxt = add_node();
                tree[node].rc = nxt;
            }
            update(tree[node].rc, mid + 1, r, pos, val);
        }
    }

    void update(ll pos, T val) {
        update(0, L, R, pos, val);
    }

    T query(int node, ll l, ll r, ll ql, ll qr) {
        if (node == -1 || ql > r || qr < l) return 0;
        if (ql <= l && r <= qr) return tree[node].val;
        ll mid = l + (r - l) / 2;
        return query(tree[node].lc, l, mid, ql, qr) +
    ↪ query(tree[node].rc, mid + 1, r, ql, qr);
    }

    T query(ll ql, ll qr) {
        return query(0, L, R, ql, qr);
    }
};

```


2.1.6 MergeSortTree

Merge Sort Tree - Space: $O(N \log N)$, Build: $O(N \log N)$, Query: $O(\log^2 N)$

```
// Stores sorted elements in each node to answer 2D range count
// queries
template<typename T = int>
struct MergeSortTree {
    int n;
    vector<vector<T>> tree;

    MergeSortTree(const vector<T>& a) : n(a.size()), tree(4 * n) {
        build(1, 0, n - 1, a);
    }

    void build(int node, int l, int r, const vector<T>& a) {
        if (l == r) {
            tree[node] = {a[l]};
            return;
        }
        int mid = (l + r) / 2;
        build(2 * node, l, mid, a);
        build(2 * node + 1, mid + 1, r, a);
        merge(all(tree[2 * node]), all(tree[2 * node + 1]),
        // back_inserter(tree[node]));

        // Returns number of elements <= k in index range [ql, qr]
        int query_count_le(int node, int l, int r, int ql, int qr, T
        // k) {
            if (ql > r || qr < l) return 0;
            if (ql <= l && r <= qr) {
                return upper_bound(all(tree[node]), k) -
                tree[node].begin();
            }
            int mid = (l + r) / 2;
            return query_count_le(2 * node, l, mid, ql, qr, k) +
            query_count_le(2 * node + 1, mid + 1, r, ql, qr, k);
        }

        int query_count_le(int ql, int qr, T k) {
            return query_count_le(1, 0, n - 1, ql, qr, k);
        }
    };
};
```

2.1.7 IterativeSegTree

Iterative (Non-Recursive) Segment Tree - Space: $O(N)$, Update: $O(\log N)$, Query: $O(\log N)$

```
// Fast, cache-friendly implementation for point update and range
// query
template<typename T = ll>
struct IterativeSegTree {
    int n;
    vector<T> tree;
    T neutral;
    function<T(T, T)> combine;

    IterativeSegTree(int n = 0, T neutral = 0, function<T(T, T)>
    // combine = [](T a, T b) { return a + b; })
        : n(n), tree(2 * n, neutral), neutral(neutral),
        // combine(combine) {}

    IterativeSegTree(const vector<T>& a, T neutral = 0,
    // function<T(T, T)> combine = [](T a, T b) { return a + b; })
        : n(a.size()), tree(2 * a.size(), neutral),
        // neutral(neutral), combine(combine) {
            for (int i = 0; i < n; i++) tree[n + i] = a[i];
            for (int i = n - 1; i > 0; i--) tree[i] = combine(tree[i
            // << 1], tree[i << 1 | 1]);
        }

    void update(int p, T val) {
        // combine(LEFT, RIGHT) - do NOT write combine(tree[p],
        // tree[p ^ 1]): when p is the right
        // child that reverses the arguments, which is wrong for
        // any non-commutative combine
        // (matrix product, affine composition, "merge two
        // structs").
        for (tree[p += n] = val; p > 1; p >>= 1)
            tree[p >> 1] = combine(tree[p & ~1], tree[p | 1]);
    }

    // Range query on [l, r] inclusive
    T query(int l, int r) {
        T resl = neutral, resr = neutral;
        for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
            if (l & 1) resl = combine(resl, tree[l++]);
            if (r & 1) resr = combine(tree[--r], resr);
        }
        return combine(resl, resr);
    }
};
```

2.1.8 SegTree2D

2D Segment Tree - Space: $O(N * M)$, Update: $O(\log N * \log M)$, Query: $O(\log N * \log M)$

```
// Supports point updates and 2D subgrid range sum queries
template<typename T = ll>
struct SegTree2D {
    int n, m;
    vector<vector<T>> tree;

    SegTree2D(int n = 0, int m = 0) : n(n), m(m), tree(4 * n,
        vector<T>(4 * m, 0)) {}

    void update_y(int vx, int lx, int rx, int vy, int ly, int ry,
        int x, int y, T val) {
        if (ly == ry) {
            if (lx == rx) tree[vx][vy] += val;
            else tree[vx][vy] = tree[2 * vx][vy] + tree[2 * vx +
        1][vy];
            return;
        }
        int my = (ly + ry) / 2;
        if (y <= my) update_y(vx, lx, rx, 2 * vy, ly, my, x, y,
            val);
        else update_y(vx, lx, rx, 2 * vy + 1, my + 1, ry, x, y,
            val);
        tree[vx][vy] = tree[vx][2 * vy] + tree[vx][2 * vy + 1];
    }

    void update_x(int vx, int lx, int rx, int x, int y, T val) {
        if (lx != rx) {
            int mx = (lx + rx) / 2;
            if (x <= mx) update_x(2 * vx, lx, mx, x, y, val);
            else update_x(2 * vx + 1, mx + 1, rx, x, y, val);
        }
        update_y(vx, lx, rx, 1, 0, m - 1, x, y, val);
    }

    void add(int x, int y, T val) {
        update_x(1, 0, n - 1, x, y, val);
    }

    T query_y(int vx, int vy, int ly, int ry, int qy1, int qy2) {
        if (qy1 > ry || qy2 < ly) return 0;
        if (qy1 <= ly && ry <= qy2) return tree[vx][vy];
        int my = (ly + ry) / 2;
        return query_y(vx, 2 * vy, ly, my, qy1, qy2) +
        query_y(vx, 2 * vy + 1, my + 1, ry, qy1, qy2);
    }

    T query_x(int vx, int lx, int rx, int qx1, int qx2, int qy1,
        int qy2) {
        if (qx1 > rx || qx2 < lx) return 0;
        if (qx1 <= lx && rx <= qx2) return query_y(vx, 1, 0, m -
        1, qy1, qy2);
        int mx = (lx + rx) / 2;
        return query_x(2 * vx, lx, mx, qx1, qx2, qy1, qy2) +
        query_x(2 * vx + 1, mx + 1, rx, qx1, qx2, qy1, qy2);
    }

    T query(int x1, int y1, int x2, int y2) {
        return query_x(1, 0, n - 1, x1, x2, y1, y2);
    }
};
```

2.1.9 MergeSortTreeFractional

MERGE SORT TREE WITH FRACTIONAL CASCADING - static "how many values $\leq x$ in $[l, r]$ " in $O(\log n)$

```
// instead of  $O(\log^2 n)$ : each node stores its sorted values plus
// pointers into the children, so the
// binary search happens ONCE at the root and is then followed
// down for free.
const int MAXN = 30005;
int a[MAXN];

struct Node {
    vector<int> vals;
    vector<int> to_left;
    vector<int> to_right;
} tree[4 * MAXN];

void build(int node, int start, int end) {
    if (start == end) {
        tree[node].vals.push_back(a[start]);
        tree[node].to_left.assign(2, 0);
        tree[node].to_right.assign(2, 0);
        return;
    }

    int mid = (start + end) / 2;
    int left_child = 2 * node;
    int right_child = 2 * node + 1;

    build(left_child, start, mid);
    build(right_child, mid + 1, end);

    int left_sz = tree[left_child].vals.size();
    int right_sz = tree[right_child].vals.size();
    int total_sz = left_sz + right_sz;

    tree[node].vals.resize(total_sz);
    tree[node].to_left.resize(total_sz + 1);
    tree[node].to_right.resize(total_sz + 1);

    int p1 = 0, p2 = 0;

    for (int i = 0; i < total_sz; i++) {
        tree[node].to_left[i] = p1;
        tree[node].to_right[i] = p2;

        if (p1 < left_sz && (p2 == right_sz ||
            tree[left_child].vals[p1] <= tree[right_child].vals[p2])) {
            tree[node].vals[i] = tree[left_child].vals[p1++];
        } else {
            tree[node].vals[i] = tree[right_child].vals[p2++];
        }
    }

    tree[node].to_left[total_sz] = p1;
    tree[node].to_right[total_sz] = p2;
}

int query(int node, int start, int end, int l, int r, int idx) {
    if (l > end || r < start) {
        return 0;
    }

    if (l <= start && end <= r) {
        return idx;
    }

    int mid = (start + end) / 2;
    int next_idx_left = tree[node].to_left[idx];
    int next_idx_right = tree[node].to_right[idx];

    return query(2 * node, start, mid, l, r, next_idx_left) +
        query(2 * node + 1, mid + 1, end, l, r,
        next_idx_right);
}
```

2.1.10 RangeMajorityCandidates

```

const int MAXN = 200005; // array bound
    ⇨ (Template.cpp calls this N)
int a[MAXN];
vector<int> pos[MAXN];
int K_LIMIT = 2;

struct Node {
    vector<pair<int, int>> cands;
};

Node tree_seg[4 * MAXN];

Node merge_nodes(const Node& left, const Node& right) {
    vector<pair<int, int>> combined;
    combined.reserve(left.cands.size() + right.cands.size());

    int i = 0, j = 0;
    while (i < left.cands.size() && j < right.cands.size()) {
        if (left.cands[i].first == right.cands[j].first) {
            combined.push_back({left.cands[i].first,
    ⇨ left.cands[i].second + right.cands[j].second});
            i++; j++;
        } else if (left.cands[i].first < right.cands[j].first) {
            combined.push_back(left.cands[i++]);
        } else {
            combined.push_back(right.cands[j++]);
        }
    }
    while (i < left.cands.size())
    ⇨ combined.push_back(left.cands[i++]);
    while (j < right.cands.size())
    ⇨ combined.push_back(right.cands[j++]);

    if (combined.size() <= K_LIMIT - 1) {
        return {combined};
    }

    vector<int> counts;
    counts.reserve(combined.size());
    for (const auto& p : combined) {
        counts.push_back(p.second);
    }

    nth_element(counts.begin(), counts.begin() + K_LIMIT - 1,
    ⇨ counts.end(), greater<int>());
    int penalty = counts[K_LIMIT - 1];

    vector<pair<int, int>> survived;
    survived.reserve(K_LIMIT - 1);
    for (const auto& p : combined) {
        if (p.second > penalty) {
            survived.push_back({p.first, p.second - penalty});
        }
    }

    return {survived};
}

void build(int node, int start, int end) {
    if (start == end) {
        tree_seg[node].cands.push_back({a[start], 1});
        return;
    }
    int mid = start + (end - start) / 2;
    build(2 * node, start, mid);
    build(2 * node + 1, mid + 1, end);
    tree_seg[node] = merge_nodes(tree_seg[2 * node], tree_seg[2 *
    ⇨ node + 1]);
}

Node query_candidates(int node, int start, int end, int l, int r)
    ⇨ {
    if (r < start || end < l) return {};
    if (l <= start && end <= r) return tree_seg[node];

    int mid = start + (end - start) / 2;
    Node left_res = query_candidates(2 * node, start, mid, l, r);
    Node right_res = query_candidates(2 * node + 1, mid + 1, end,
    ⇨ l, r);

    if (left_res.cands.empty()) return right_res;
    if (right_res.cands.empty()) return left_res;

    return merge_nodes(left_res, right_res);
}

```

```

// O(log N) fast static frequency count
int get_frequency(int val, int l, int r) {
    auto& v = pos[val];
    auto right_it = upper_bound(v.begin(), v.end(), r);
    auto left_it = lower_bound(v.begin(), v.end(), l);
    return distance(left_it, right_it);
}

void solve() {
    int n, q;
    cin >> n >> q;

    for (int i = 1; i <= n; i++) {
        cin >> a[i];
        pos[a[i]].push_back(i);
    }

    build(1, 1, n);

    while (q--) {
        int l, r;
        cin >> l >> r;

        int len = r - l + 1;
        Node res = query_candidates(1, 1, n, l, r);

        int max_freq = 0;
        for (const auto& p : res.cands) {
            int cand = p.first;
            int freq = get_frequency(cand, l, r);
            max_freq = max(max_freq, freq);
        }

        if (max_freq > (len + 1) / 2) {
            cout << 2 * max_freq - len << "\n";
        } else {
            cout << 1 << "\n";
        }
    }
}

```

2.1.11 SegTreeDescent

Segment Tree DESCENT - answer "first / last index satisfying P" in $O(\log n)$, not $O(\log^2 n)$.

```
// Here: max-segtree. first_ge(l, x) = smallest i >= l with a[i]
// >= x, or n if none.
// Swap the stored aggregate + the two 'good()' tests to get:
// first prefix-sum >= S,
// first zero, k-th set bit, "how far right can I extend". This is
// the workhorse behind
// greedy "extend as far as possible" and interval-cover problems.
struct SegMax {
    int n;
    vector<ll> t;

    SegMax(int n = 0) : n(n), t(4 * n, LLONG_MIN) {}

    void build(const vector<ll>& a, int v, int l, int r) {
        if (l == r) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(a, 2 * v, l, m), build(a, 2 * v + 1, m + 1, r);
        t[v] = max(t[2 * v], t[2 * v + 1]);
    }
    void build(const vector<ll>& a) { if (n) build(a, 1, 0, n - 1); }

    void set(int p, ll x, int v, int l, int r) {
        if (l == r) { t[v] = x; return; }
        int m = (l + r) / 2;
        p <= m ? set(p, x, 2 * v, l, m) : set(p, x, 2 * v + 1, m + 1, r);
        t[v] = max(t[2 * v], t[2 * v + 1]);
    }
    void set(int p, ll x) { set(p, x, 1, 0, n - 1); }

    ll qry(int ql, int qr, int v, int l, int r) {
        if (qr < l || r < ql) return LLONG_MIN;
        if (ql <= l && r <= qr) return t[v];
        int m = (l + r) / 2;
        return max(qry(ql, qr, 2 * v, l, m), qry(ql, qr, 2 * v + 1, m + 1, r));
    }
    ll qry(int l, int r) { return qry(l, r, 1, 0, n - 1); }

    // smallest i in [ql, n) with a[i] >= x (returns n if none)
    int first_ge(int ql, ll x, int v, int l, int r) {
        if (r < ql || t[v] < x) return n;
        if (l == r) return l;
        int m = (l + r) / 2, res = first_ge(ql, x, 2 * v, l, m);
        return res < n ? res : first_ge(ql, x, 2 * v + 1, m + 1, r);
    }
    int first_ge(int ql, ll x) { return first_ge(ql, x, 1, 0, n - 1); }

    // largest i in [0, qr] with a[i] >= x (returns -1 if none)
    int last_ge(int qr, ll x, int v, int l, int r) {
        if (l > qr || t[v] < x) return -1;
        if (l == r) return l;
        int m = (l + r) / 2, res = last_ge(qr, x, 2 * v, l, m);
        return res >= 0 ? res : last_ge(qr, x, 2 * v + 1, m + 1, r);
    }
    int last_ge(int qr, ll x) { return last_ge(qr, x, 1, 0, n - 1); }
};
```

2.2 Binary Indexed Tree (BIT)

2.2.1 FenwickTree

Fenwick Tree (Binary Indexed Tree) - 0-based indexing

// Space: $O(N)$, Time: $O(\log N)$ per update/query

```
template<typename T = ll>
struct FenwickTree {
    int n;
    vector<T> tree;

    FenwickTree(int n = 0) : n(n), tree(n + 1, 0) {}

    void add(int i, T delta) {
        for (i++; i <= n; i += i & -i) tree[i] += delta;
    }

    void set(int i, T val) {
        add(i, val - query(i, i));
    }

    T query(int i) {
        T sum = 0;
        for (i++; i > 0; i -= i & -i) sum += tree[i];
        return sum;
    }

    T query(int l, int r) {
        if (l > r) return 0;
        return query(r) - query(l - 1);
    }

    // Returns first 0-based index where prefix sum >= val
    int lower_bound(T val) {
        int pos = 0;
        for (int sz = 1 << (n ? __lg(n) : 0); sz > 0; sz >>= 1) {
            if (pos + sz <= n && tree[pos + sz] < val) {
                val -= tree[pos + sz];
                pos += sz;
            }
        }
        return pos;
    }
};
```

2.2.2 FenwickTree2D

2D Fenwick Tree - 0-based indexing

// Space: $O(N * M)$, Time: $O(\log N * \log M)$ per update/query

```
template<typename T = ll>
struct FenwickTree2D {
    int n, m;
    vector<vector<T>> tree;

    FenwickTree2D(int n = 0, int m = 0) : n(n), m(m), tree(n + 1,
        vector<T>(m + 1, 0)) {}

    void add(int x, int y, T val) {
        for (int i = x + 1; i <= n; i += i & -i) {
            for (int j = y + 1; j <= m; j += j & -j) {
                tree[i][j] += val;
            }
        }
    }

    void set(int x, int y, T val) {
        add(x, y, val - query(x, y, x, y));
    }

    T query(int x, int y) {
        T sum = 0;
        for (int i = min(x + 1, n); i > 0; i -= i & -i) {
            for (int j = min(y + 1, m); j > 0; j -= j & -j) {
                sum += tree[i][j];
            }
        }
        return sum;
    }

    // 2D Subgrid Range Sum from (x1, y1) to (x2, y2) inclusive
    T query(int x1, int y1, int x2, int y2) {
        if (x1 > x2 || y1 > y2) return 0;
        return query(x2, y2) - query(x1 - 1, y2) - query(x2, y1 - 1) + query(x1 - 1, y1 - 1);
    }
};
```

2.2.3 Offline2DFenwickTree

Offline 2D Fenwick Tree (Compressed 2D BIT)

```
// Space:  $O(N + Q \log N)$ , Time:  $O(\log^2 N)$  per query
template<typename T = ll>
struct BIT2D {
    int n;
    vector<vector<int>> vals;
    vector<vector<T>> bit;

    int ind(const vector<int>& v, int x) {
        return upper_bound(all(v), x) - v.begin() - 1;
    }

    // mx: max row index, todo: all (r, c) update coordinates that
    // will be called
    BIT2D(int mx, vector<array<int, 2>> todo) : n(mx), vals(mx +
    1), bit(mx + 1) {
        sort(all(todo), [](const auto& a, const auto& b) { return
        a[1] < b[1]; });

        for (int i = 0; i <= n; i++) vals[i].pb(INT_MIN);
        for (auto [r, c] : todo) {
            for (int x = r; x <= n; x |= x + 1) {
                if (vals[x].back() != c) vals[x].pb(c);
            }
        }
        for (int i = 0; i <= n; i++)
            bit[i].resize(vals[i].size(), 0);

        void add(int r, int c, T val) {
            for (; r <= n; r |= r + 1) {
                for (int i = ind(vals[r], c); i < (int)bit[r].size();
                i |= i + 1) {
                    bit[r][i] += val;
                }
            }
        }

        T query(int r, int c) {
            T res = 0;
            for (; r >= 0; r = (r & (r + 1)) - 1) {
                for (int i = ind(vals[r], c); i >= 0; i = (i & (i +
                1)) - 1) {
                    res += bit[r][i];
                }
            }
            return res;
        }

        T query(int r1, int c1, int r2, int c2) {
            return query(r2, c2) - query(r2, c1 - 1) - query(r1 - 1,
            c2) + query(r1 - 1, c1 - 1);
        }

        void reset() {
            for (auto& v : bit) fill(all(v), 0);
        }
};
```

2.2.4 BITRangeUpdateRangeQuery

BIT with RANGE UPDATE + RANGE QUERY (two BITs). 0-based, $O(\log n)$ per op.

```
//  $pre(i) = \sum_{j \leq i} a[j] = S1(i) * (i+1) - S2(i)$ , where a
//  $\rightarrow$  range-add of v on  $[l, r]$  does
//  $S1: +v$  at l,  $-v$  at r+1  $S2: +v*l$  at l,  $-v*(r+1)$  at r+1
template<typename T = ll>
struct BIT2 {
    int n;
    vector<T> b1, b2;

    BIT2(int n = 0) : n(n), b1(n + 1, 0), b2(n + 1, 0) {}
    void add(vector<T>& b, int i, T v) { for (i++; i <= n; i += i
    & -1) b[i] += v; }
    T sum(const vector<T>& b, int i) const { T s = 0; for (i++; i
    > 0; i -= i & -1) s += b[i]; return s; }

    void upd(int l, int r, T v) { // add v to  $[l, r]$ 
        add(b1, l, v), add(b1, r + 1, -v);
        add(b2, l, v * l), add(b2, r + 1, -v * (r + 1));
    }
    T pre(int i) const { return i < 0 ? 0 : sum(b1, i) * (i + 1)
    - sum(b2, i); }
    T qry(int l, int r) const { return pre(r) - pre(l - 1); }
};
```

2.3 Sparse Table

2.3.1 SparseTable

SPARSE TABLE - $O(1)$ range query after $O(N \log N)$ build, NO updates.

```
// The operation must be IDEMPOTENT (min, max, gcd, and, or)
//  $\rightarrow$  because the two covering blocks
// OVERLAP. For sum/product/xor - anything not idempotent - use 12
//  $\rightarrow$  - DisjointSparseTable.cpp
// (still  $O(1)$  query) or a Fenwick tree. Guard  $l \leq r$ :  $__lg(0)$  is
// undefined.
template<typename T, typename F = function<T(T, T)>>
struct SparseTable {
    int n;
    vector<vector<T>> mat;
    F func;

    SparseTable() = default;
    SparseTable(const vector<T>& a, F f = [](T x, T y) { return
    max(x, y); }) : n(a.size()), func(f) {
        int k = n ? __lg(n) + 1 : 0;
        mat.assign(k, a);
        for (int j = 1; j < k; j++) {
            for (int i = 0; i + (1 << j) <= n; i++) {
                mat[j][i] = func(mat[j - 1][i], mat[j - 1][i + (1
                << (j - 1))]);
            }
        }

        T query(int l, int r) const {
            int j = __lg(r - l + 1);
            return func(mat[j][l], mat[j][r - (1 << j) + 1]);
        }
};
```

2.3.2 SparseTable2D

2D Sparse Table - $O(N * M * \log N * \log M)$ build, $O(1)$ query

```
template<typename T = ll>
struct SparseTable2D {
    int n, m;
    // jmp[kr][kc][i][j] = max in 2^kr x 2^kc subgrid starting at
    // (i, j)
    vector<vector<vector<vector<T>>>> jmp;

    SparseTable2D(const vector<vector<T>>& grid) {
        n = grid.size();
        if (!n) return;
        m = grid[0].size();
        if (!m) return;

        int log_n = __lg(n) + 1, log_m = __lg(m) + 1;
        jmp.assign(log_n, vector<vector<vector<T>>>>(log_m,
        // vector<vector<T>>>(n, vector<T>(m))));

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                jmp[0][0][i][j] = grid[i][j];
            }
        }

        for (int kc = 1; kc < log_m; kc++) {
            for (int i = 0; i < n; i++) {
                for (int j = 0; j + (1 << kc) <= m; j++) {
                    jmp[0][kc][i][j] = max(jmp[0][kc - 1][i][j],
                    // jmp[0][kc - 1][i][j + (1 << (kc - 1))]);
                }
            }
        }

        for (int kr = 1; kr < log_n; kr++) {
            for (int kc = 0; kc < log_m; kc++) {
                for (int i = 0; i + (1 << kr) <= n; i++) {
                    for (int j = 0; j + (1 << kc) <= m; j++) {
                        jmp[kr][kc][i][j] = max(jmp[kr - 1][kc][i][j],
                        // jmp[kr - 1][kc][i + (1 << (kr - 1))][j]);
                    }
                }
            }
        }

        T query(int r1, int c1, int r2, int c2) const {
            int kr = __lg(r2 - r1 + 1), kc = __lg(c2 - c1 + 1);
            return max({jmp[kr][kc][r1][c1],
                jmp[kr][kc][r1][c2 - (1 << kc) + 1],
                jmp[kr][kc][r2 - (1 << kr) + 1][c1],
                jmp[kr][kc][r2 - (1 << kr) + 1][c2 - (1 <<
                // kc) + 1]});
        }
    };
};
```

2.3.3 SquareSparseTable

SQUARE SPARSE TABLE - $O(1)$ query for any SQUARE sub-grid, $O(N * M * \log(\min(N, M)))$ build.

```
// The 2D sparse table (02 - SparseTable2D.cpp) stores one table
// per (row-power, col-power) pair,
// so it costs log(N)*log(M) memory. If every query is a SQUARE -
// "best k x k block", "does some
// k x k window contain only ones", template matching on a grid -
// you only ever need the diagonal
// of that table, which is a factor of log cheaper in both time
// and memory.
// Query a k x k square: cover it with four overlapping 2^t x 2^t
// squares, t = floor(log2 k).
// The operation must be IDEMPOTENT (min/max/gcd/and/or) because
// those four squares overlap.
template<typename T = ll>
struct SquareSparseTable {
    int n, m;
    // jmp[k][i][j] = max of 2^k x 2^k square starting at (i, j)
    vector<vector<vector<T>>> jmp;

    SquareSparseTable(const vector<vector<T>>& grid) {
        n = grid.size();
        if (!n) return;
        m = grid[0].size();
        if (!m) return;

        int log_max = __lg(min(n, m)) + 1;
        jmp.assign(log_max, vector<vector<T>>>(n, vector<T>(m)));

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                jmp[0][i][j] = grid[i][j];
            }
        }

        for (int k = 1; k < log_max; k++) {
            int pw = 1 << (k - 1);
            for (int i = 0; i + (pw * 2) <= n; i++) {
                for (int j = 0; j + (pw * 2) <= m; j++) {
                    jmp[k][i][j] = max({jmp[k - 1][i][j],
                        jmp[k - 1][i][j + pw],
                        jmp[k - 1][i + pw][j],
                        jmp[k - 1][i + pw][j +
                        // pw]});
                }
            }
        }

        // Queries max in square subgrid from (r1, c1) to (r2, c2)
        T query(int r1, int c1, int r2, int c2) const {
            int k = __lg(r2 - r1 + 1), pw = 1 << k;
            return max({jmp[k][r1][c1],
                jmp[k][r1][c2 - pw + 1],
                jmp[k][r2 - pw + 1][c1],
                jmp[k][r2 - pw + 1][c2 - pw + 1]});
        }
    };
};
```

2.4 Trie

2.4.1 StringTrie

PREFIX TRIE over an alphabet. $O(L)$ per insert/query. Reach for it for prefix counting, autocomplete,

```
// "is any inserted word a prefix of this one", and as the
// skeleton Aho-Corasick is built on.
// Prefix Trie for Strings
// Time:  $O(L)$  per insert/query, Space:  $O(N * L * Alphabet)$ 
struct Trie {
    vector<vector<int>> nxt;
    vector<int> cnt, end_cnt;
    int alpha;

    Trie(int alpha = 26) : alpha(alpha) {
        add_node();
    }

    int add_node() {
        nxt.emplace_back(alpha, -1);
        cnt.emplace_back(0);
        end_cnt.emplace_back(0);
        return nxt.size() - 1;
    }

    void insert(const string& s, int val = 1) {
        int u = 0;
        cnt[u] += val;
        for (char c : s) {
            int ch = c - 'a';
            if (nxt[u][ch] == -1) nxt[u][ch] = add_node();
            u = nxt[u][ch];
            cnt[u] += val;
        }
        end_cnt[u] += val;
    }

    int count_prefix(const string& s) {
        int u = 0;
        for (char c : s) {
            int ch = c - 'a';
            if (nxt[u][ch] == -1) return 0;
            u = nxt[u][ch];
        }
        return cnt[u];
    }
};
```

2.4.2 BinaryTrie

Binary trie over $[0, 2^{(B+1)})$. insert / ERASE / query all $O(B)$.

```
// erase() is what lets you keep the trie equal to the current
// root->node path during a DFS
// ("min XOR with any ancestor"): add on entry, add(-1) on exit.
struct BTrie {
    static const int B = 30;
    vector<array<int, 2>> ch = {{-1, -1}};
    vector<int> cnt = {0};

    int node() { ch.push_back({-1, -1}); cnt.push_back(0); return
    ch.size() - 1; }
    void add(ll x, int d = 1) { // d = +1
        // insert, -1 erase
        int u = 0; cnt[0] += d;
        for (int i = B; i >= 0; i--) {
            int b = x >> i & 1;
            if (ch[u][b] < 0) ch[u][b] = node();
            u = ch[u][b], cnt[u] += d;
        }
    }
    int cn(int u, int b) const { return u < 0 || ch[u][b] < 0 ? 0
    : cnt[ch[u][b]]; }
    int size() const { return cnt[0]; }

    ll bestXor(ll x, bool mx) const { // mx=1
        // -> max x^y, mx=0 -> min x^y
        if (!cnt[0]) return -1;
        int u = 0; ll r = 0;
        for (int i = B; i >= 0; i--) {
            int b = (x >> i & 1) ^ mx;
            if (!cn(u, b)) b ^= 1;
            r |= (1ll) << i & b;
            u = ch[u][b];
        }
        return r;
    }
    ll maxXor(ll x) const { return bestXor(x, 1); }
    ll minXor(ll x) const { return bestXor(x, 0); }

    int cntLess(ll x, ll k) const { // #{y in
    trie : (x ^ y) < k}
        int u = 0, r = 0;
        for (int i = B; i >= 0 && u >= 0; i--) {
            int xb = x >> i & 1;
            if (k >> i & 1) r += cn(u, xb), u = ch[u][xb ^ 1];
            else u = ch[u][xb];
        }
        return r;
    }
};
```

2.4.3 PersistentTrie

Persistent Binary Trie - Space: $O(N * \text{BITS})$, Time: $O(\text{BITS})$ per insert/query

```
// Supports versioning and range max XOR queries between versions
↳ [l_ver, r_ver]
struct PersistentTrie {
    struct Node {
        int cnt = 0;
        array<int, 2> nxt = {-1, -1};
    };

    vector<Node> tree;
    vector<int> roots;
    int max_bit;

    PersistentTrie(int max_bit = 30) : max_bit(max_bit) {
        roots.pb(add_node());
    }

    int add_node(int old_node = -1) {
        if (old_node != -1) {
            tree.pb(tree[old_node]);
        } else {
            tree.pb(Node());
        }
        return tree.size() - 1;
    }

    // Inserts x starting from prev_root and returns the new
    // version root
    ↳ range [l_ver, r_ver]
    int insert(int prev_root, ll x) {
        int new_root = add_node(prev_root);
        int u = new_root, prev_u = prev_root;
        tree[u].cnt++;

        for (int i = max_bit; i >= 0; i--) {
            int b = (x >> i) & 1;
            int prev_child = (prev_u != -1) ? tree[prev_u].nxt[b]
            ↳ : -1;
            tree[u].nxt[b] = add_node(prev_child);
            u = tree[u].nxt[b];
            if (prev_u != -1) prev_u = tree[prev_u].nxt[b];
            tree[u].cnt++;
        }
        roots.pb(new_root);
        return new_root;
    }

    // Returns max (x ^ val) over elements inserted in version
    ↳ range [l_ver, r_ver]
    ll max_xor(int l_ver, int r_ver, ll x) const {
        int u_r = roots[r_ver], u_l = (l_ver > 0) ? roots[l_ver] -
        ↳ 1 : -1;
        ll res = 0;

        for (int i = max_bit; i >= 0; i--) {
            int b = (x >> i) & 1;
            int want = b ^ 1;

            int count_r = (u_r != -1 && tree[u_r].nxt[want] !=
            ↳ -1) ? tree[tree[u_r].nxt[want]].cnt : 0;
            int count_l = (u_l != -1 && tree[u_l].nxt[want] !=
            ↳ -1) ? tree[tree[u_l].nxt[want]].cnt : 0;

            if (count_r - count_l > 0) {
                res |= (1LL << i);
                u_r = tree[u_r].nxt[want];
                u_l = (u_l != -1 && tree[u_l].nxt[want] != -1) ?
            ↳ tree[u_l].nxt[want] : -1;
            } else {
                u_r = (u_r != -1 && tree[u_r].nxt[b] != -1) ?
            ↳ tree[u_r].nxt[b] : -1;
                u_l = (u_l != -1 && tree[u_l].nxt[b] != -1) ?
            ↳ tree[u_l].nxt[b] : -1;
            }
        }
        return res;
    }
};
```

2.5 Disjoint Set Union (DSU)

2.5.1 DSU

Disjoint Set Union (DSU) - Path Compression & Union by Size

```
// Time:  $O(\alpha(N))$  per operation
struct DSU {
    vector<int> par, sz;

    DSU(int n = 0) : par(n), sz(n, 1) {
        iota(all(par), 0);
    }

    int find(int x) {
        return par[x] == x ? x : par[x] = find(par[x]);
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool join(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) return false;
        if (sz[x] < sz[y]) swap(x, y);
        sz[x] += sz[y];
        par[y] = x;
        return true;
    }

    int size(int x) {
        return sz[find(x)];
    }
};
```

2.5.2 DSURollback

DSU with Rollback - No path compression to allow backtracking

```
// Time:  $O(\log N)$  per operation,  $O(1)$  rollback
struct DSURollback {
    vector<int> par, sz;
    vector<pair<int, int>> history; // {child_node,
    ↪ old_parent_size}

    DSURollback(int n = 0) : par(n), sz(n, 1) {
        iota(all(par), 0);
    }

    int find(int x) {
        return par[x] == x ? x : find(par[x]);
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool join(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) {
            history.eb(-1, -1);
            return false;
        }
        if (sz[x] < sz[y]) swap(x, y);
        history.eb(y, sz[x]);
        sz[x] += sz[y];
        par[y] = x;
        return true;
    }

    int size(int x) {
        return sz[find(x)];
    }

    void rollback() {
        if (history.empty()) return;
        auto [y, old_sz_x] = history.back();
        history.pop_back();
        if (y != -1) {
            sz[par[y]] = old_sz_x;
            par[y] = y;
        }
    }

    int get_state() const {
        return history.size();
    }

    void rollback_to(int state) {
        while ((int)history.size() > state) rollback();
    }
};
```

2.5.3 WeightedDSU

DSU with POTENTIALS: maintains $\text{pot}[v] = \text{value}(v) - \text{value}(\text{root})$, so you can answer

```
// "what is  $\text{value}(u) - \text{value}(v)$ " and detect contradictions.
// Group is  $(\mathbb{Z}, +)$  here; for parity/bipartiteness use XOR: change
    ↪ ALL FOUR of  $\text{pot}[x] += w \rightarrow \wedge$ ,
//  $\text{pu} - \text{pv} == w \rightarrow (\text{pu} \wedge \text{pv}) == w$ ,  $\text{pot}[\text{rv}] = \text{pu} - \text{pv} - w \rightarrow \text{pu} \wedge$ 
    ↪  $\text{pv} \wedge w$ , and DELETE the  $w = -w$  line
// (negation is not the inverse in  $\text{GF}(2)$ ), or  $\mathbb{Z}_k$  for mod- $k$ 
    ↪ relations.
struct WDSU {
    vector<int> par;
    vector<ll> pot; // potential
    ↪ relative to the parent
    vector<int> sz;

    WDSU(int n = 0) : par(n), pot(n, 0), sz(n, 1) {
        ↪ iota(all(par), 0);
        pair<int, ll> find(int x) { // {root,
            ↪ value(x) - value(root)}
            if (par[x] == x) return {x, 0};
            auto [r, p] = find(par[x]);
            par[x] = r, pot[x] += p; // path
            ↪ compression accumulates
            return {r, pot[x]};
        }
        // assert  $\text{value}(u) - \text{value}(v) == w$ . false = CONTRADICTION with
        ↪ what we already know.
        bool join(int u, int v, ll w) {
            auto [ru, pu] = find(u);
            auto [rv, pv] = find(v);
            if (ru == rv) return pu - pv == w;
            if (sz[ru] < sz[rv]) swap(ru, rv), swap(pu, pv), w = -w;
            par[rv] = ru, pot[rv] = pu - pv - w, sz[ru] += sz[rv];
            return true;
        }
        //  $\text{value}(u) - \text{value}(v)$ , or nullopt if they are not related yet
        bool diff(int u, int v, ll& out) {
            auto [ru, pu] = find(u);
            auto [rv, pv] = find(v);
            if (ru != rv) return false;
            out = pu - pv;
            return true;
        }
    };
    // BIPARTITENESS under edge additions: use XOR potentials with  $w =$ 
    ↪ 1 per edge; join() returning
    // false means an odd cycle appeared. Same structure answers "are
    ↪ u and v the same colour".
```

2.5.4 RangeParallelDSU

RANGE-PARALLEL DSU ("DSU on a sparse table") - merge two whole RANGES elementwise in $O(\log n)$.

```
// The operation is: "a[u..u+len-1] must equal a[v..v+len-1]
//   ↳ elementwise", i.e. union u+i with v+i
// for every i < len. Doing that naively is O(len) per constraint;
//   ↳ this does it in O(log n).
// IDEA: keep LOGN separate DSUs. par[k] answers "these two blocks
//   ↳ of length 2^k are identical".
// A range constraint of length len is covered by TWO blocks of
//   ↳ length 2^k (k = floor(log2 len)),
// exactly like a sparse-table query - so one union at level k,
//   ↳ twice. Afterwards, PUSH DOWN:
// if two blocks of length 2^k are equal then their two halves of
//   ↳ length 2^(k-1) are equal too.
// After the push-down, level 0 holds the true elementwise DSU.
// USES: "count the strings/arrays satisfying these
//   ↳ equal-substring constraints" (the answer is
// alphabet^(#components at level 0)), grid/string pattern
//   ↳ unification, and any problem that says
// "these two intervals must match".
struct RangeDSU {
    int n, LOG;
    vector<vector<int>>> par;
    RangeDSU(int n) : n(n), LOG(_lg(max(n, 1)) + 1), par(LOG,
    ↳ vector<int>(n)) {
        for (int k = 0; k < LOG; k++) iota(all(par[k]), 0);
    }
    int find(int k, int u) { return par[k][u] == u ? u :
    ↳ par[k][u] = find(k, par[k][u]); }
    void join(int k, int u, int v) {
        u = find(k, u), v = find(k, v);
        if (u != v) par[k][u] = v;
    }
    // a[u .. u+len-1] equals a[v .. v+len-1], elementwise. O(log
    ↳ n) amortised.
    void equalRanges(int u, int v, int len) {
        if (len <= 0) return;
        int k = _lg(len);
        join(k, u, v);
        join(k, u + len - (1 << k), v + len - (1 << k));
    }
    // Call ONCE after all the constraints. Then find(0, i) is the
    ↳ real elementwise component.
    void pushDown() {
        for (int k = LOG - 1; k > 0; k--) {
            int half = 1 << (k - 1);
            for (int i = 0; i + (1 << k) <= n; i++) {
                int j = find(k, i);
                if (j == i) continue;
                join(k - 1, i, j);
            }
            join(k - 1, i + half, j + half);
        }
    }
    int components() {
    ↳ // at level 0, after pushDown
        int c = 0;
        for (int i = 0; i < n; i++) if (find(0, i) == i) c++;
        return c;
    }
};
// The same "cover a range with two power-of-two blocks, then push
//   ↳ down" trick works for any
// idempotent relation, not just equality - it is the DSU analogue
//   ↳ of a sparse table.
// If instead you need to union a range of CONSECUTIVE elements to
//   ↳ each other ("a[l..r] all equal"),
// that is much simpler: a "next unvisited pointer" DSU where
//   ↳ find(i) jumps to the next index not
// yet merged, giving O(n alpha) total over all range-unions.
```

2.6 Square Root Decomposition

2.6.1 MoAlgorithm

MO'S ALGORITHM - answers m OFFLINE range queries in $O((n + m) \sqrt{n})$ when you can move an endpoint

```
// by one in O(1) (add/remove an element and maintain the answer).
//   ↳ Sort queries by block of l, then by
// r (alternating direction per block). Use it for distinct
//   ↳ counts, frequency-of-frequency, XOR pairs.
class MoArray {
private:
    struct Query {
        int l, r, idx;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd)
                return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    int n;
    int block_size;
    vector<int> a;

    // --- PROBLEM SPECIFIC STATE ---
    long long ans;
    vector<int> freq;

    inline void add(int idx) {
        if (freq[a[idx]] == 0) ans++;
        freq[a[idx]]++;
    }

    inline void remove(int idx) {
        freq[a[idx]]--;
        if (freq[a[idx]] == 0) ans--;
    }

    inline long long get_answer() const {
        return ans;
    }
    // -----

public:
    MoArray(const vector<int>& arr) : n(arr.size()), a(arr) {
        int max_val = 0;
        for (int x : a) {
            max_val = max(max_val, x);
        }
        freq.assign(max_val + 1, 0);
        ans = 0;
    }

    vector<long long> solve(const vector<pair<int, int>>&
    ↳ raw_queries) {
        int q = raw_queries.size();
        if (q == 0) return {};

        block_size = max(1, (int)(n / sqrt(q)));
        vector<Query> queries(q);

        for (int i = 0; i < q; ++i) {
            queries[i].l = raw_queries[i].first;
            queries[i].r = raw_queries[i].second;
            queries[i].idx = i;
            queries[i].bnd = queries[i].l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) add(--cur_l);
            while (cur_r < qry.r) add(++cur_r);
            while (cur_l < qry.l) remove(cur_l++);
            while (cur_r > qry.r) remove(cur_r--);
            answers[qry.idx] = get_answer();
        }

        return answers;
    }
};
```

2.6.2 MoOnSubtrees

MO'S ON SUBTREES - flatten the tree with an Euler tour so every subtree is a contiguous range,

// then run plain Mo's on it. Same $O((n + q) \sqrt{n})$. Use for
 ↳ "query over the subtree of v " when no
 // online structure fits (distinct colours in a subtree, k -th most
 ↳ frequent, etc.).

```
class MoSubtree {
private:
    struct Query {
        int l, r, idx;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd) return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    int n;
    int block_size;
    vector<vector<int>>> adj;
    vector<int> in, out, flat, clr;

    // --- PROBLEM SPECIFIC STATE ---
    int mx_freq;
    vector<int> freq;
    vector<long long> sum_freq;

    inline void add(int node_idx) {
        int c = clr[node_idx];
        sum_freq[freq[c]] -= c;
        freq[c]++;
        sum_freq[freq[c]] += c;
        if (freq[c] > mx_freq) mx_freq = freq[c];
    }

    inline void remove(int node_idx) {
        int c = clr[node_idx];
        sum_freq[freq[c]] -= c;
        freq[c]--;
        sum_freq[freq[c]] += c;
        while (mx_freq > 0 && sum_freq[mx_freq] == 0) mx_freq--;
    }

    inline long long get_answer() const {
        return sum_freq[mx_freq];
    }
    // -----

    void dfs(int u, int p) {
        in[u] = flat.size();
        flat.push_back(u);
        for (int v : adj[u]) {
            if (v == p) continue;
            dfs(v, u);
        }
        out[u] = flat.size() - 1;
    }

public:
    MoSubtree(int n, int max_val, const vector<int>& colors)
        : n(n), adj(n + 1), in(n + 1), out(n + 1), clr(colors) {
        mx_freq = 0;
        freq.assign(max_val + 1, 0);
        sum_freq.assign(n + 1, 0);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    vector<long long> solve(int root, const vector<int>&
  ↳ query_nodes) {
        flat.reserve(n);
        dfs(root, 0);

        int q = query_nodes.size();
        if (q == 0) return {};

        block_size = max(1, (int)(n / sqrt(q)));
        vector<Query> queries(q);

        for (int i = 0; i < q; ++i) {
            int u = query_nodes[i];
            queries[i].l = in[u];
```

```
            queries[i].r = out[u];
            queries[i].idx = i;
            queries[i].bnd = queries[i].l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) add(flat[--cur_l]);
            while (cur_r < qry.r) add(flat[++cur_r]);
            while (cur_l < qry.l) remove(flat[cur_l++]);
            while (cur_r > qry.r) remove(flat[cur_r--]);
            answers[qry.idx] = get_answer();
        }

        return answers;
    }
};
```

2.6.3 MoOnPaths

MO'S ON TREE PATHS - the Euler tour trick where a vertex appearing TWICE in the range cancels out,

// plus the LCA added back by hand. Answers path queries offline
 ⇒ in $O((n + q) \sqrt{n})$ with no HLD.

```
class MoTreePath {
private:
    int n, LOG, block_size;
    vector<vector<int>> adj, anc;
    vector<int> depth, in, out, flat;
    vector<long long> up, a;
    vector<bool> exist;

    // --- PROBLEM SPECIFIC STATE ---
    long long ans;
    vector<deque<int>> dep; // Tracks nodes by depth

    void add(int idx) {
        if (!dep[depth[idx]].empty()) {
            ans += 1ll * a[dep[depth[idx]].back()] * a[idx];
        }
        dep[depth[idx]].push_back(idx);
    }

    void remove(int idx) {
        if (dep[depth[idx]].size() == 2) {
            ans -= a[*dep[depth[idx]].begin()] *
        }
        if (dep[depth[idx]].front() == idx) {
            dep[depth[idx]].pop_front();
        } else {
            dep[depth[idx]].pop_back();
        }
    }

    void update(int idx) {
        int node = flat[idx];
        if (exist[node]) {
            remove(node);
        } else {
            add(node);
        }
        exist[node] = !exist[node];
    }

    inline long long get_answer() const {
        return ans;
    }

    // -----

    void dfs(int u, int p = -1) {
        if (p != -1) {
            anc[u][0] = p;
            up[u] = a[u] * a[u] + up[p];
        } else {
            depth[u] = 0;
            up[u] = a[u] * a[u];
        }
        in[u] = flat.size();
        flat.push_back(u);

        for (int v : adj[u]) {
            if (v == p) continue;
            depth[v] = depth[u] + 1;
            dfs(v, u);
        }

        out[u] = flat.size();
        flat.push_back(u); // Push again for Eulerian tour
    }

    void build_lca() {
        for (int k = 1; k < LOG; k++) {
            for (int i = 1; i <= n; i++) {
                anc[i][k] = anc[anc[i][k - 1]][k - 1];
            }
        }
    }

    int lift(int x, int k) {
        for (int bit = 0; bit < LOG; bit++) {
            if ((1 << bit) & k) x = anc[x][bit];
        }
        return x;
    }
};
```

```
int lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    u = lift(u, depth[u] - depth[v]);
    if (u == v) return u;
    for (int bit = LOG - 1; bit >= 0; bit--) {
        if (anc[u][bit] != anc[v][bit]) {
            u = anc[u][bit];
            v = anc[v][bit];
        }
    }
    return anc[u][0];
}

public:
    struct Query {
        int l, r, idx, x = -1;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd) return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    MoTreePath(int n_nodes, const vector<long long>& node_values)
        : n(n_nodes), a(node_values), LOG(log2(n_nodes) + 2) {

        adj.assign(n + 1, vector<int>());
        anc.assign(n + 1, vector<int>(LOG, 0));
        depth.assign(n + 1, 0);
        in.assign(n + 1, 0);
        out.assign(n + 1, 0);
        up.assign(n + 1, 0);
        exist.assign(n + 1, false);
        dep.assign(n + 1, deque<int>());
        ans = 0;
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    // PRIMARY ENTRY POINT: give it the paths as vertex pairs and
    // it builds the queries itself.
    // (in[], out[] and lca() are only valid AFTER the dfs, so you
    // cannot build them yourself.)
    vector<long long> solve(int root, const vector<pair<int,
    int>>& paths) {
        flat.clear();
        dfs(root);
        build_lca();
        vector<Query> queries;
        for (int i = 0; i < (int)paths.size(); i++) {
            int u = paths[i].first, v = paths[i].second;
            if (in[u] > in[v]) swap(u, v);
            int w = lca(u, v);
            // u inside v's subtree -> plain range; otherwise use
            // out[u]..in[v] and add back the LCA
            if (w == u) queries.push_back({in[u], in[v], i, -1,
            0});
            else queries.push_back({out[u], in[v], i, w, 0});
        }
        return solveRanges(queries);
    }

    vector<long long> solveRanges(vector<Query>& queries) {
        int q = queries.size();
        if (q == 0) return {};

        block_size = max(1, (int)(flat.size() / sqrt(q)));

        for (auto& qry : queries) {
            qry.bnd = qry.l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) update(--cur_l); // GROW
            while (cur_r < qry.r) update(++cur_r); // order
            // can pass through an invalid window
            while (cur_l < qry.l) update(cur_l++);
            while (cur_r > qry.r) update(cur_r--);
        }
    }
};
```

```

        answers[qry.idx] = get_answer() + (qry.x != -1 ?
    ↪ up[qry.x] : 0);
    }
    return answers;
}
};

```

2.6.4 BlockDecomposition

SQRT / BLOCK DECOMPOSITION. Split $[0, n)$ into blocks of $B \sim \sqrt{n}$ and keep one aggregate per

```

// block. Point update  $O(1..B)$ , range query  $O(\sqrt{n})$ . A
    ↪ BIT/segtree beats it for sum, so reach for
// this when the aggregate is NOT a group / has no lazy tag:
    ↪ block-sorted arrays, block frequency
// tables, "count values > x in  $[l, r]$ ", range-assign + weird
    ↪ query, or a per-block rebuild that is
// cheap. General shape below with sum; swap 'agg' and the two
    ↪ combine lines for anything else.
struct Blocks {
    int n, B;
    vector<ll> a, agg;
    ↪ agg[b] = aggregate of block b
    Blocks(const vector<ll>& v) : n(v.size()), B(max(1,
    ↪ (int)sqrt(n) + 1)), a(v),
        agg((n + B - 1) / B, 0) {
        for (int i = 0; i < n; i++) agg[i / B] += a[i];
    }
    void set(int i, ll v) { agg[i / B] += v - a[i], a[i] = v; }
    ↪ //  $O(1)$ ; for non-group aggregates

    ↪ rebuild the block instead,  $O(B)$ 
    ll query(int l, int r) {
    ↪ inclusive,  $O(n/B + B)$ 
        ll s = 0;
        int bl = l / B, br = r / B;
        if (bl == br) { for (int i = l; i <= r; i++) s += a[i];
    ↪ return s; }
        for (int i = l; i < (bl + 1) * B; i++) s += a[i];
        for (int b = bl + 1; b < br; b++) s += agg[b];
        for (int i = br * B; i <= r; i++) s += a[i];
        return s;
    }
};
// RANGE UPDATE, RANGE QUERY without lazy propagation: keep add[b]
    ↪ per block. A partial block is
// updated element-wise and its aggregate rebuilt; a full block
    ↪ only bumps add[b]. Query adds
// add[b] * (elements taken from b).
// BLOCK-SORTED ARRAY ("sqrt tree"): store each block also as a
    ↪ sorted copy. Then
// count of values  $\leq x$  in  $[l, r]$  = partial blocks scanned +
    ↪ binary search inside full blocks,
//  $O(\sqrt{n} \log n)$  per query,  $O(B \log B)$  per point update
    ↪ (re-sort one block). This is the
// offline-free alternative to a wavelet tree / merge-sort tree
    ↪ when updates are present.
// REBUILD TRICK: when an operation is expensive but rare (e.g.
    ↪ inserting into the middle), buffer
// changes and rebuild the whole decomposition every  $\sqrt{q}$ 
    ↪ operations  $\rightarrow O((n+q) \sqrt{q})$  total.
// CHOOSING B: query cost  $n/B + B*c$  is minimised at  $B = \sqrt{n*c}$ 
    ↪ where c is the per-element cost;
// if updates are much more frequent than queries, take bigger
    ↪ blocks, and vice versa.

```

2.6.5 MoWithUpdates

MO'S WITH UPDATES (3D Mo). Queries $\{l, r, t\}$ where $t = \# \text{updates applied before this query}$.

```

// Block size  $n^{2/3}$  gives  $O(n^{5/3})$  total. Use when the array
    ↪ changes between queries and no
// online structure fits.
struct MoUpd {
    struct Q { int l, r, t, idx; };
    struct U { int pos, oldV, newV; };

    int n, B;
    vector<int> a, rec;
    ↪ state at time 0; rec = while recording
    vector<Q> qs;
    vector<U> us;

    // ---- problem state: #distinct in the window ----
    vector<int> cnt;
    int cur = 0;
    void add(int v) { if (cnt[v]++ == 0) cur++; }
    void rem(int v) { if (--cnt[v] == 0) cur--; }
    int answer() const { return cur; }
    // -----

    MoUpd(vector<int> arr, int maxVal) : n(arr.size()), a(arr),
    ↪ rec(arr), cnt(maxVal + 1, 0) {
        B = max(1, (int)pow((double)max(n, 1), 2.0 / 3));
    }
    void addQuery(int l, int r) { qs.push_back({l, r,
    ↪ (int)us.size(), (int)qs.size()}); }
    void addUpdate(int pos, int newV) { us.push_back({pos,
    ↪ rec[pos], newV}); rec[pos] = newV; }

    void applyU(int i, int l, int r) {
    ↪ update i, forwards or backwards
        auto& u = us[i];
        if (l <= u.pos && u.pos <= r) rem(a[u.pos]), add(u.newV);
        a[u.pos] = u.newV;
        swap(u.oldV, u.newV);
    ↪ the operation its own inverse
    }

    vector<int> solve() {
        sort(all(qs), [&](const Q& x, const Q& y) {
            int bx = x.l / B, by = y.l / B;
            if (bx != by) return bx < by;
            int cx = x.r / B, cy = y.r / B;
            if (cx != cy) return (bx & 1) ? cx > cy : cx < cy;
            return (cx & 1) ? x.t > y.t : x.t < y.t;
        });
        vector<int> res(qs.size());
        int L = 0, R = -1, T = 0;
        for (auto& q : qs) {
            while (T < q.t) applyU(T++, L, R);
            while (T > q.t) applyU(--T, L, R);
            while (R < q.r) add(a[++R]);
            while (L > q.l) add(a[--L]);
            while (R > q.r) rem(a[R--]);
            while (L < q.l) rem(a[L++]);
            res[q.idx] = answer();
        }
        return res;
    }
};
// NOTE: 'a' stays at time 0; 'rec' tracks values while you record
    ↪ updates, so applyU() can
// step forwards AND backwards through time using the same
    ↪ swap-based inverse.

```


2.6.6 MoWithRollback

MO'S WITH ROLLBACK - for structures where ADD is easy but REMOVE is hard or impossible

```
// (running maximum, DSU, "best pair so far", knapsack-ish state).
// Per block of left endpoints: sort by r, keep a persistent right
//   ↳ part that only grows, and
// rebuild the short left part from scratch for every query,
//   ↳ rolling back afterwards.
// O((n + q) sqrt n) adds, ZERO removes.
struct MoRollback {
    struct Q { int l, r, idx; };
    int n, B;
    vector<ll> a;
    vector<Q> qs;

    // ---- problem state: max (value * count) seen; only add()
    // and a snapshot/rollback ----
    vector<int> cnt;
    ll best = 0;
    vector<pair<int, ll>> hist;           // {value
    // index, previous best}
    void add(int v, bool rec) {
        if (rec) hist.push_back({v, best});
        cnt[v]++;
        best = max(best, (ll)cnt[v] * v);
    }
    void rollback(size_t to) {
        while (hist.size() > to) {
            auto [v, b] = hist.back(); hist.pop_back();
            cnt[v]--; best = b;
        }
    }
    void clearAll() { fill(all(cnt), 0), best = 0, hist.clear(); }
    ll answer() const { return best; }
    //
    // -----

    MoRollback(vector<ll> arr, int maxVal) : n(arr.size()),
    // a(arr), cnt(maxVal + 1, 0) {
        B = max(1, (int)sqrtl((ld)max(1, n)));
    }
    void addQuery(int l, int r) { qs.push_back({l, r,
    // (int)qs.size()}); }

    vector<ll> solve() {
        sort(all(qs), [&](const Q& x, const Q& y) {
            int bx = x.l / B, by = y.l / B;
            return bx != by ? bx < by : x.r < y.r;
        });
        vector<ll> res(qs.size());
        size_t i = 0;
        while (i < qs.size()) {
            int b = qs[i].l / B, rEnd = min(n - 1, (b + 1) * B -
    // 1);
            size_t j = i;
            while (j < qs.size() && qs[j].l / B == b) j++;
            // pass 1: queries contained in this block - answer
    // each from a CLEAN state
            for (size_t k2 = i; k2 < j; k2++) if (qs[k2].r <=
    // rEnd) {
                clearAll();
                for (int k = qs[k2].l; k <= qs[k2].r; k++)
    // add(a[k], false);
                res[qs[k2].idx] = answer();
            }
            // pass 2: long queries - right part grows
    // monotonically, left part is rolled back
            clearAll();
            int R = rEnd;
            for (size_t k2 = i; k2 < j; k2++) if (qs[k2].r >
    // rEnd) {
                while (R < qs[k2].r) add(a[++R], false);
    // // grow-only, not recorded
                size_t snap = hist.size();
                for (int k = qs[k2].l; k <= rEnd; k++) add(a[k],
    // true); // short left part
                res[qs[k2].idx] = answer();
                rollback(snap);
            }
            i = j;
        }
        return res;
    }
};
```

2.7 Balanced Binary Search Tree (BBST)

2.7.1 Treap

TREAP (keyed) - a BST by key and a heap by random priority, so it is balanced in expectation.

// NOTE: 01 - Miscellaneous/04 - random.cpp declares the same
 ↳ 'rng'. Keep one copy.
 // O(log n) insert/erase/find, plus split and merge, which is what
 ↳ makes it better than std::set:
 // you can cut the tree at a key, operate on a whole range, and
 ↳ glue it back.
 mt19937_64
 ↳ rng(chrono::steady_clock::now().time_since_epoch().count());

```
template <typename T>
struct Treap {
    struct Node {
        T key, val, sum;
        T lazy = 0; // 1. Added lazy tag
        int sz = 1;
        uint64_t p = rng();
        Node *l = nullptr, *r = nullptr;

        Node(T k, T v = T()) : key(k), val(v), sum(v) {}
    } * root = nullptr;

    ~Treap() { destroy(root); }

    void destroy(Node* t) {
        if (!t) return;
        destroy(t->l);
        destroy(t->r);
        delete t;
    }

    int sz(Node* t) { return t ? t->sz : 0; }
    T sub_val(Node* t) { return t ? t->sum : 0; }

    // 2. The Push Function: Propagates pending updates to children
    void push(Node* t) {
        if (!t || t->lazy == 0) return;

        if (t->l) {
            t->l->key += t->lazy; // Shift the key (critical for
    // the DP solution)
            t->l->val += t->lazy; // Shift the value
            t->l->sum += t->lazy * sz(t->l); // Update subtree sum
            t->l->lazy += t->lazy; // Pass the tag down
        }
        if (t->r) {
            t->r->key += t->lazy;
            t->r->val += t->lazy;
            t->r->sum += t->lazy * sz(t->r);
            t->r->lazy += t->lazy;
        }
        t->lazy = 0; // Clear the tag on the current node
    }

    Node* update(Node* t) {
        if (t) {
            t->sz = 1 + sz(t->l) + sz(t->r);
            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
        }
        return t;
    }

    Node* merge(Node* l, Node* r) {
        push(l); // 3. Push before making routing decisions
        push(r);
        if (!l || !r) return l ? l : r;
        if (l->p > r->p) {
            l->r = merge(l->r, r);
            return update(l);
        }
        r->l = merge(l, r->l);
        return update(r);
    }

    pair<Node*, Node*> split(Node* t, T k) {
        if (!t) return {nullptr, nullptr};
        push(t); // 3. Push before evaluating keys for splits
        if (t->key <= k) {
            auto [l, r] = split(t->r, k);
            t->r = l;
            return {update(t), r};
        }
        auto [l, r] = split(t->l, k);
        t->l = r;
        return {l, update(t)};
    }
};
```

```

}

// --- Core Operations ---

// Range Update Method
void add_range(T l_val, T r_val, T add_val) {
    if (l_val > r_val) return;

    auto [left_mid, right] = split(root, r_val);
    auto [left, mid] = split(left_mid, l_val - 1);

    // Apply the lazy tag to the entire isolated segment
    if (mid) {
        mid->key += add_val;
        mid->val += add_val;
        mid->sum += add_val * sz(mid);
        mid->lazy += add_val;
    }

    root = merge(merge(left, mid), right);
}

bool search(T x) {
    Node* curr = root;
    while (curr) {
        push(curr); // Push during traversal
        if (curr->key == x) return true;
        curr = (x < curr->key) ? curr->l : curr->r;
    }
    return false;
}

void insert(T x, T val = T()) {
    auto [l, r] = split(root, x);
    auto [l2, r2] = split(l, x - 1);
    if (!r2) {
        r2 = new Node(x, val);
    } else {
        push(r2); // Push before modifying
        r2->val += val;
        update(r2);
    }
    root = merge(merge(l2, r2), r);
}

void erase(T x) {
    auto [l, r] = split(root, x);
    auto [l2, r2] = split(l, x - 1);
    destroy(r2);
    root = merge(l2, r);
}

// --- Utility Operations ---

Node* find_by_order(Node* t, int k) {
    if (!t) return nullptr;
    push(t); // Push before accessing children sizes
    int left_sz = sz(t->l);
    if (k < left_sz) return find_by_order(t->l, k);
    if (k == left_sz) return t;
    return find_by_order(t->r, k - left_sz - 1);
}

Node* find_by_order(int k) { return find_by_order(root, k); }

int order_of_key(Node* t, T k) {
    if (!t) return 0;
    push(t); // Push before comparing keys
    if (t->key >= k) return order_of_key(t->l, k);
    return sz(t->l) + 1 + order_of_key(t->r, k);
}

int order_of_key(T k) { return order_of_key(root, k); }

void print(Node* t) {
    if (!t) return;
    push(t); // Push before printing to get true values
    print(t->l);
    cout << t->key << " ";
    print(t->r);
}

void print() {
    print(root);
    cout << '\n';
}

};

```

2.7.2 ImplicitTreap

IMPLICIT TREAP - a treap keyed by POSITION (subtree size) instead of value, i.e. an array that

// NOTE: 01 - Miscellaneous/04 - random.cpp declares the same
 ↳ 'rng'. Keep one copy.
 // supports insert/erase in the middle, reverse a range, and move
 ↳ a block, all in $O(\log n)$.
 // This is the structure for "cut [l,r] out and paste it
 ↳ elsewhere" problems.
 mt19937_64
 ↳ rng(chrono::steady_clock::now().time_since_epoch().count());

```

template <typename T>
struct ImplicitTreap {
    struct Node {
        T val, sum;
        T lazy = 0; // Lazy tag for range additions
        bool rev = false; // Lazy tag for range reversals
        int sz = 1;
        uint64_t p = rng();
        Node *l = nullptr, *r = nullptr;

        Node(T v = T()) : val(v), sum(v) {}
    } *root = nullptr;

    // Default Constructor
    ImplicitTreap() = default;

    // Vector Constructor - Builds treap in  $O(N)$  time
    ImplicitTreap(const vector<T>& a) {
        build(a);
    }

    ~ImplicitTreap() { destroy(root); }

    void destroy(Node* t) {
        if (!t) return;
        destroy(t->l);
        destroy(t->r);
        delete t;
    }

    int sz(Node* t) { return t ? t->sz : 0; }
    T sub_val(Node* t) { return t ? t->sum : 0; }

    // Propagates pending lazy updates to children
    void push(Node* t) {
        if (!t) return;

        // 1. Process pending reversal
        if (t->rev) {
            swap(t->l, t->r);
            if (t->l) t->l->rev ^= true;
            if (t->r) t->r->rev ^= true;
            t->rev = false;
        }

        // 2. Process pending value addition
        if (t->lazy != 0) {
            if (t->l) {
                t->l->val += t->lazy;
                t->l->sum += t->lazy * sz(t->l);
                t->l->lazy += t->lazy;
            }
            if (t->r) {
                t->r->val += t->lazy;
                t->r->sum += t->lazy * sz(t->r);
                t->r->lazy += t->lazy;
            }
            t->lazy = 0; // Clear the tag
        }
    }

    Node* update(Node* t) {
        if (t) {
            t->sz = 1 + sz(t->l) + sz(t->r);
            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
        }
        return t;
    }

    // ---  $O(N)$  Linear Time Build ---
    void build(const vector<T>& a) {
        destroy(root);
        root = nullptr;
        if (a.empty()) return;
    }

```

```

vector<Node*> st;
for (const T& val : a) {
    Node* curr = new Node(val);
    Node* last = nullptr;

    // Maintain max-heap property on random priority 'p'
    while (!st.empty() && st.back()->p < curr->p) {
        last = st.back();
        st.pop_back();
    }

    curr->l = last;
    if (!st.empty()) {
        st.back()->r = curr;
    }
    st.push_back(curr);
}

// Post-order pass to compute subtree sizes and sums in
// O(N)
auto recompute = [&](auto& self, Node* t) -> void {
    if (!t) return;
    self(self, t->l);
    self(self, t->r);
    update(t);
};

root = st.front(); // Bottom of stack is the root
recompute(recompute, root);
}

// --- Dynamic Array Operations ---

Node* merge(Node* l, Node* r) {
    push(l);
    push(r);
    if (!l || !r) return l ? l : r;
    if (l->p > r->p) {
        l->r = merge(l->r, r);
        return update(l);
    }
    r->l = merge(l, r->l);
    return update(r);
}

pair<Node*, Node*> split(Node* t, int k) {
    if (!t) return {nullptr, nullptr};
    push(t);

    int left_sz = sz(t->l);
    if (left_sz < k) {
        auto [l, r] = split(t->r, k - left_sz - 1);
        t->r = l;
        return {update(t), r};
    } else {
        auto [l, r] = split(t->l, k);
        t->l = r;
        return {l, update(t)};
    }
}

int size() { return sz(root); }

void insert(int idx, T val) {
    auto [l, r] = split(root, idx);
    Node* node = new Node(val);
    root = merge(merge(l, node), r);
}

void erase(int idx) {
    if (idx < 0 || idx >= sz(root)) return;
    auto [left_mid, right] = split(root, idx + 1);
    auto [left, mid] = split(left_mid, idx);
    destroy(mid);
    root = merge(left, right);
}

void reverse_range(int l, int r) {
    if (l >= r || l < 0 || r >= sz(root)) return;

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    if (mid) mid->rev ^= true;

    root = merge(merge(left, mid), right);
}

```

```

void add_range(int l, int r, T add_val) {
    if (l > r || l < 0 || r >= sz(root)) return;

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    if (mid) {
        mid->val += add_val;
        mid->sum += add_val * sz(mid);
        mid->lazy += add_val;
    }

    root = merge(merge(left, mid), right);
}

T query_range(int l, int r) {
    if (l > r || l < 0 || r >= sz(root)) return T();

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    T ans = sub_val(mid);

    root = merge(merge(left, mid), right);
    return ans;
}

T get(int idx) {
    return query_range(idx, idx);
}

void print(Node* t) {
    if (!t) return;
    push(t);
    print(t->l);
    cout << t->val << " ";
    print(t->r);
}

void print() {
    print(root);
    cout << '\n';
}
};

```

2.8 Policy Based Data Structures

2.8.1 OrderedSet

Policy-Based Data Structure: ordered_set

// Time: $O(\log N)$ per operation

#include <ext/pb_ds/assoc_container.hpp>

#include <ext/pb_ds/tree_policy.hpp>

using namespace __gnu_pbds;

template<typename T>

using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,

tree_order_statistics_node_update>;

// Usage:

// st.find_by_order(k): Iterator to k-th element (0-based)

// st.order_of_key(x): Number of elements strictly smaller than x

2.9 Monotonic Data Structures

2.9.1 MonotonicStack

Monotonic Stack - Time: $O(N)$, Space: $O(N)$

```
// Computes Next/Previous Smaller and Greater element indices for
//    all positions
template<typename T = ll>
struct MonotonicStack {
    int n;
    vector<T> a;
    vector<int> prev_less, next_less;
    vector<int> prev_greater, next_greater;

    MonotonicStack(const vector<T>& arr) : n(arr.size()), a(arr),
        prev_less(n, -1), next_less(n, n),
        prev_greater(n, -1), next_greater(n, n) {

        vector<int> st;
        // Next Less & Previous Less
        for (int i = 0; i < n; i++) {
            while (!st.empty() && a[st.back()] >= a[i]) {
                next_less[st.back()] = i;
                st.pop_back();
            }
            if (!st.empty()) prev_less[i] = st.back();
            st.pb(i);
        }

        st.clear();
        // Next Greater & Previous Greater
        for (int i = 0; i < n; i++) {
            while (!st.empty() && a[st.back()] <= a[i]) {
                next_greater[st.back()] = i;
                st.pop_back();
            }
            if (!st.empty()) prev_greater[i] = st.back();
            st.pb(i);
        }
    }
};
```

2.9.2 TwoStackQueue

Monotonic Queue via Two Stacks (Sliding Window Min/Max)

```
// Space:  $O(N)$ , Time:  $O(1)$  amortized per push/pop/get
struct TwoStackQ {
    struct Node {
        ll mx = -llinf, mn = llinf, val;

        Node() : val(0) {}
        Node(ll x) : mx(x), mn(x), val(x) {}
    };

    stack<Node> a, b;

    int size() {
        return a.size() + b.size();
    }

    void mrg(Node &a, Node &b) {
        a.mn = min(a.mn, b.mn);
        a.mx = max(a.mx, b.mx);
    }

    // Pushes value into the queue
    void push(ll val) {
        auto nd = Node(val);
        if (!a.empty()) mrg(nd, a.top());
        a.push(nd);
    }

    // Transfers elements from stack a to stack b when b is empty
    void move() {
        while (!a.empty()) {
            auto nd = Node(a.top().val);
            if (!b.empty()) mrg(nd, b.top());
            b.push(nd);
            a.pop();
        }
    }

    // Returns min/max of all current elements in the queue in  $O(1)$ 
    Node get() {
        Node res;
        if (!b.empty()) mrg(res, b.top());
        if (!a.empty()) mrg(res, a.top());
        return res;
    }

    // Removes the oldest element from the queue
    void pop() {
        if (b.empty()) move();
        if (!b.empty()) b.pop();
    }
};
```

2.9.3 MonotonicQueue2D

2D Monotonic Queue (Sliding Window 2D Min/Max) - Time: $O(N * M)$, Space: $O(N * M)$

```
// Computes min in every K x L subgrid of an N x M matrix
template<typename T = ll>
struct MonotonicQueue2D {
    int n, m;

    // Returns a matrix res where res[i][j] is the min in subgrid
    // [i..i+K-1][j..j+L-1]
    vector<vector<T>> sliding_window_2d(const vector<vector<T>>&
    grid, int K, int L) {
        n = grid.size();
        m = grid[0].size();

        // 1. Sliding window min horizontally along each row
        vector<vector<T>> row_min(n, vector<T>(m - L + 1));
        for (int i = 0; i < n; i++) {
            deque<int> dq;
            for (int j = 0; j < m; j++) {
                while (!dq.empty() && grid[i][dq.back()] >=
                grid[i][j]) dq.pop_back();
                dq.pb(j);
                if (dq.front() <= j - L) dq.pop_front();
                if (j >= L - 1) row_min[i][j - L + 1] =
                grid[i][dq.front()];
            }

            // 2. Sliding window min vertically down each column of
            row_min
            vector<vector<T>> res(n - K + 1, vector<T>(m - L + 1));
            for (int j = 0; j < m - L + 1; j++) {
                deque<int> dq;
                for (int i = 0; i < n; i++) {
                    while (!dq.empty() && row_min[dq.back()][j] >=
                    row_min[i][j]) dq.pop_back();
                    dq.pb(i);
                    if (dq.front() <= i - K) dq.pop_front();
                    if (i >= K - 1) res[i - K + 1][j] =
                    row_min[dq.front()][j];
                }

                return res;
            }
        }
};
```

2.10 DynamicMedian

RUNNING MEDIAN / k-th SMALLEST under insertions, via two multisets kept balanced around the split.

```
// O(log n) per insert, O(1) to read the median and either side's
// sum. The pattern generalises to
// "maintain the sum of the k smallest" - which is what
// cost-to-move-to-median problems need.
multiset<ll> L, R;
ll sum_L = 0, sum_R = 0;
int k;

// Rebalances the multisets so L always has the lower half/median
// of CURRENT elements
void balance() {
    int total_size = sz(L) + sz(R);
    int k_L = (total_size + 1) / 2; // Fixed: Target size based on
    // actual current elements

    // If L is too big, move its largest element to R
    while (sz(L) > k_L) {
        auto it = prev(L.end());
        ll val = *it;

        sum_L -= val;
        sum_R += val;

        R.insert(val);
        L.erase(it);
    }

    // If L is too small, move the smallest element of R to L
    while (sz(L) < k_L) {
        auto it = R.begin();
        ll val = *it;

        sum_R -= val;
        sum_L += val;

        L.insert(val);
        R.erase(it);
    }
}

void add_val(ll val) {
    if (L.empty() || val <= *L.rbegin()) {
        L.insert(val);
        sum_L += val;
    } else {
        R.insert(val);
        sum_R += val;
    }
    balance();
}

void remove_val(ll val) {
    auto it = L.find(val);
    if (it != L.end()) {
        sum_L -= val;
        L.erase(it);
    } else {
        it = R.find(val);
        if (it == R.end()) return; // not
        // present in either half: do nothing
        sum_R -= val;
        R.erase(it);
    }
    balance();
}

ll get_cost() {
    // UB if empty -
    // guard the caller
    ll median = *L.rbegin();
    return sum_R - sum_L + median * (sz(L) - sz(R));
}
```

2.11 IntervalSetODT

INTERVAL SET / "Chtholly tree" (ODT). Keeps the array as maximal runs of equal value.

```
// assign(l, r, v) is amortised near-O(log n) WHEN the data has
// ↪ many range-assigns (random or
// adversarial-but-assign-heavy); each assign destroys the
// ↪ intervals it covers, so the total
// number of intervals created is O(q).
// THIS is the standard way to support range bitwise-AND / OR /
// ↪ MOD / assign together with
// range sum, because those operations rapidly make long stretches
// ↪ constant.
struct ODT {
    map<int, pair<int, ll>> mp; // l -> {r,
    // ↪ value}, covering [l, r]

    ODT(int n, ll v = 0) { mp[0] = {n - 1, v}; }

    // split so that an interval starts exactly at p; returns the
    // ↪ iterator to it
    auto split(int p) {
        auto it = prev(mp.upper_bound(p));
        if (it->first == p) return it;
        auto [r, v] = it->second;
        it->second.first = p - 1;
        return mp.emplace(p, make_pair(r, v)).first;
    }

    void assign(int l, int r, ll v) {
        auto itr = split(r + 1), itl = split(l);
        mp.erase(itl, itr);
        mp[l] = {r, v};
    }

    // Generic scan over [l, r]: f(l, r, value) is called once per
    // ↪ run. O(#runs touched).
    template <class F> void forEach(int l, int r, F f) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) f(it->first,
        // ↪ it->second.first, it->second.second);
    }

    ll sum(int l, int r) {
        ll s = 0;
        for (auto it = split(l), itl = split(r + 1); it != itl; ++it)
        // ↪ s += it->second.second;
        return s;
    }

    // RANGE ADD: touch each run once. Runs are not merged, so
    // ↪ adds alone do NOT keep the
    // amortised bound - this is only cheap when assigns keep
    // ↪ collapsing the structure.
    void add(int l, int r, ll x) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) it->second.second +=
        // ↪ x;
    }

    // RANGE AND / OR / MOD follow the same loop, applying the op
    // ↪ to each run's value.
    // They are fast because every value can only decrease O(log
    // ↪ V) times before it stops
    // changing, at which point neighbouring runs become equal and
    // ↪ can be merged.
    template <class Op> void apply(int l, int r, Op op) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) it->second.second =
        // ↪ op(it->second.second);
        merge(l, r);
    }

    void merge(int l, int r) { // glue
        // ↪ neighbouring runs of equal value
        auto it = split(l);
        while (it != mp.end() && it->first <= r) {
            auto nx = next(it);
            if (nx != mp.end() && nx->second.second ==
            // ↪ it->second.second &&
            // it->second.first + 1 == nx->first) {
            // it->second.first = nx->second.first;
            // mp.erase(nx);
            } else it = nx;
        }
    }
};

// Careful: worst case WITHOUT assigns is O(n) per operation. If
// ↪ the problem has only adds and
// queries, use a segment tree instead.
```

2.12 DisjointSparseTable

DISJOINT SPARSE TABLE - O(1) query for ANY associative operation (not just idempotent ones).

```
// Build O(n log n) time and memory. Use when a plain sparse table
// ↪ would double-count:
// sum, product, matrix product, hashing, "merge two structs",
// ↪ min-count pairs, ...
// (For min/max/gcd/or/and a normal sparse table is smaller - use
// ↪ that instead.)
template <typename T, class F = function<T(T, T)>>
struct DisjointST {
    int n, LOG;
    vector<vector<T>>> t;
    F op;

    DisjointST(const vector<T>& a, F op) : n(a.size()), op(op) {
        LOG = 1;
        while ((1 << LOG) < n) LOG++;
        t.assign(LOG + 1, a);
        for (int lvl = 1; lvl <= LOG; lvl++) {
            int half = 1 << lvl;
            for (int mid = half; mid < n + half; mid += 2 * half)
            // ↪ {
            // if (mid - 1 < n) { //
            // ↪ build leftwards from the split
            // t[lvl][mid - 1] = a[mid - 1];
            // for (int i = mid - 2; i >= mid - half && i >=
            // ↪ 0; i--)
            // t[lvl][i] = op(a[i], t[lvl][i + 1]);
            // }
            // if (mid < n) { //
            // ↪ and rightwards
            // t[lvl][mid] = a[mid];
            // for (int i = mid + 1; i < min(n, mid + half);
            // ↪ i++)
            // t[lvl][i] = op(t[lvl][i - 1], a[i]);
            // }
            // }
        }

        T query(int l, int r) const { //
            // ↪ inclusive [l, r]
            if (l == r) return t[0][l];
            int lvl = 63 - __builtin_clzll((unsigned long long)(l ^
            // ↪ r)); // highest differing bit
            return op(t[lvl][l], t[lvl][r]);
        }
    };

    // Every query splits at the unique power-of-two boundary between
    // ↪ l and r, and both halves are
    // already prefix/suffix-accumulated at that level - hence exactly
    // ↪ one op() per query.
```

2.13 WaveletTree

WAVELET TREE over values in $[lo, hi]$. Build $O(n \log V)$.

```
// kth(l, r, k)      : k-th SMALLEST in a[l..r]
//    ↪ O(log V)
// countLeq(l,r,x)   : how many a[i] <= x in a[l..r]
//    ↪ O(log V)
// countEq(l,r,x)    : how many a[i] == x in a[l..r]
//    ↪ O(log V)
// Online (no offline sorting, no persistence) - the practical
//    ↪ alternative to a merge-sort tree.
struct Wavelet {
    int lo, hi;
    Wavelet *l = nullptr, *r = nullptr;
    vector<int> b; // b[i] =
    ↪ #elements going left among first i

    // build on a[from, to) - the array IS reordered in place
    ↪ (stable partition by midpoint)
    Wavelet(vector<int>::iterator from, vector<int>::iterator to,
    ↪ int x, int y) : lo(x), hi(y) {
        if (from >= to) return;
        if (lo == hi) { b.assign(to - from + 1, 0); return; }
        int mid = lo + (hi - lo) / 2;
        auto f = [mid](int v) { return v <= mid; };
        b.reserve(to - from + 1);
        b.push_back(0);
        for (auto it = from; it != to; ++it) b.push_back(b.back()
    ↪ + f(*it));
        auto pivot = stable_partition(from, to, f);
        l = new Wavelet(from, pivot, lo, mid);
        r = new Wavelet(pivot, to, mid + 1, hi);
    }
    int kth(int L, int R, int k) { //
    ↪ 1-indexed k, inclusive [L, R] (1-based)
        if (L > R) return 0;
        if (lo == hi) return lo;
        int inLeft = b[R] - b[L - 1], lb = b[L - 1], rb = b[R];
        if (k <= inLeft) return l->kth(lb + 1, rb, k);
        return r->kth(L - lb, R - rb, k - inLeft);
    }
    int countLeq(int L, int R, int k) { // #{i in
    ↪ [L,R] : a[i] <= k}
        if (L > R || k < lo) return 0;
        if (hi <= k) return R - L + 1;
        int lb = b[L - 1], rb = b[R];
        return l->countLeq(lb + 1, rb, k) + r->countLeq(L - lb, R
    ↪ - rb, k);
    }
    int countEq(int L, int R, int k) {
        if (L > R || k < lo || k > hi) return 0;
        if (lo == hi) return R - L + 1;
        int lb = b[L - 1], rb = b[R], mid = lo + (hi - lo) / 2;
        if (k <= mid) return l->countEq(lb + 1, rb, k);
        return r->countEq(L - lb, R - rb, k);
    }
    ~Wavelet() { delete l; delete r; }
};
// Usage: compress values to [0, V), copy the array, then Wavelet
//    ↪ w(a.begin(), a.end(), 0, V-1);
// queries are 1-BASED on the original positions.
```

2.14 OfflineDynamicConnectivity

OFFLINE DYNAMIC CONNECTIVITY - edges appear and disappear over time; answer queries at each

```
// Needs: DSURollback (02 - DSU/02 - DSURollback.cpp).
// moment. Segment tree over the TIMELINE + DSU with rollback. O(q
//    ↪ log q * log n) - DSURollback has no path compression, so
//    ↪ find is log n, not alpha.
// The same skeleton answers ANY "structure supports add +
//    ↪ rollback, but not delete" problem:
// put each element on the time interval where it is alive, DFS
//    ↪ the segment tree, add on entry,
// roll back on exit. ("Deleting from a data structure in O(T log
//    ↪ n)")
struct DynConn {
    int n, T;
    vector<vector<pair<int, int>>> seg; // edges
    ↪ alive on each timeline node
    DSURollback d;
    vector<int> comps; // comps[t]
    ↪ = #components at time t

    DynConn(int n, int T) : n(n), T(T), seg(4 * max(1, T)), d(n),
    ↪ comps(T, 0) {}

    void addSeg(int v, int l, int r, int ql, int qr, pair<int,
    ↪ int> e) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) { seg[v].push_back(e); return; }
        int m = (l + r) / 2;
        addSeg(2 * v, l, m, ql, qr, e), addSeg(2 * v + 1, m + 1,
    ↪ r, ql, qr, e);
    }
    // edge (u,v) exists during times [tl, tr] (inclusive)
    void addEdge(int u, int v, int tl, int tr) { if (tl <= tr)
    ↪ addSeg(1, 0, T - 1, tl, tr, {u, v}); }

    void dfs(int v, int l, int r, int cur) {
        int snap = d.get_state();
        for (auto [a, b] : seg[v]) cur -= d.join(a, b);
        if (l == r) comps[l] = cur;
        else {
            int m = (l + r) / 2;
            dfs(2 * v, l, m, cur), dfs(2 * v + 1, m + 1, r, cur);
        }
        d.rollback_to(snap);
    }
    void solve() { if (T) dfs(1, 0, T - 1, n); }
};
// USAGE: collect every edge's [birth, death] interval first (an
//    ↪ edge that is never removed dies
// at T-1), then one solve() fills comps[]. To answer "are u and v
//    ↪ connected at time t", instead
// record the queries per leaf and test d.same(u, v) inside the l
//    ↪ == r branch.
```


2.15 CartesianTree

CARTESIAN TREE in O(n): heap-ordered by VALUE, BST-ordered by INDEX.

```
// The root of any range is its minimum, so "range minimum"
// ↳ structure becomes a TREE -
// which turns range problems into subtree problems (and enables
// ↳ divide & conquer on the min).
// Built with a monotonic stack; ties go left so the tree is
// ↳ unique.
struct CartTree {
    int n, root = -1;
    vector<int> l, r, par;

    CartTree(const vector<ll>& a) : n(a.size()), l(n, -1), r(n,
    ↳ -1), par(n, -1) {
        vector<int> st;
        for (int i = 0; i < n; i++) {
            int last = -1;
            while (!st.empty() && a[st.back()] > a[i]) last =
            ↳ st.back(), st.pop_back();
            l[i] = last;
            if (last != -1) par[last] = i;
            if (!st.empty()) r[st.back()] = i, par[i] = st.back();
            st.push_back(i);
        }
        root = st[0];
    }
};
// USES
// - "for every element, the maximal range where it is the
// ↳ minimum" = its subtree range,
// which is the contribution technique (sum over subarrays of
// ↳ the min) in tree form
// - largest rectangle in a histogram = best over nodes of value
// ↳ * subtreeSize
// - divide & conquer "solve(l, r) splits at the minimum" without
// ↳ recomputing RMQ
// - counting subarrays whose minimum is a given element
// - a Treap built on (index, random priority) is exactly a
// ↳ Cartesian tree - same code shape
```

2.16 SegTreeMerging

SEGMENT TREE MERGING - one dynamic segment tree per vertex over the VALUE domain, merged

```
// bottom-up along a tree. Total O(n log n) nodes touched over ALL
// ↳ merges (amortised), because
// each merge destroys as many nodes as it visits.
// Solves, per subtree and in one DFS: most frequent value, k-th
// ↳ value, count of a value,
// "sum of values in a range" - i.e. everything a merge-sort tree
// ↳ gives, but online per subtree.
struct SegMerge {
    struct Node { int l = 0, r = 0; ll sum = 0; int best = 0; ll
    ↳ bestCnt = 0; };
    vector<Node> t;
    int V;
    // value
    // ↳ domain is [0, V)

    SegMerge(int V, int reserve = 0) : V(V) { t.reserve(reserve),
    ↳ t.push_back(Node()); } // 0 = null
    int newNode() { t.push_back(Node()); return t.size() - 1; }

    void pull(int v) {
        int L = t[v].l, R = t[v].r;
        t[v].sum = t[L].sum + t[R].sum;
        if (t[L].bestCnt >= t[R].bestCnt) t[v].best = t[L].best,
        ↳ t[v].bestCnt = t[L].bestCnt;
        else t[v].best = t[R].best, t[v].bestCnt = t[R].bestCnt;
    }
    // Returns the (possibly new) node index. NEVER take an 'int&'
    ↳ into t here: newNode() can
    // reallocate the vector and invalidate the reference
    ↳ mid-recursion.
    int update(int v, int lo, int hi, int p, ll d) {
        if (!v) v = newNode();
        if (hi - lo == 1) { t[v].sum += d, t[v].best = lo,
        ↳ t[v].bestCnt = t[v].sum; return v; }
        int m = (lo + hi) / 2;
        if (p < m) { int c = update(t[v].l, lo, m, p, d); t[v].l
        ↳ = c; }
        else { int c = update(t[v].r, m, hi, p, d); t[v].r = c; }
        pull(v);
        return v;
    }
    int merge(int a, int b, int lo, int hi) { // destroys
    ↳ b; returns the merged root
        if (!a || !b) return a | b;
        if (hi - lo == 1) { t[a].sum += t[b].sum, t[a].bestCnt =
        ↳ t[a].sum; return a; }
        int m = (lo + hi) / 2;
        int L = merge(t[a].l, t[b].l, lo, m), R = merge(t[a].r,
        ↳ t[b].r, m, hi);
        t[a].l = L, t[a].r = R;
        pull(a);
        return a;
    }
    ll query(int v, int lo, int hi, int ql, int qr) { // sum of
    ↳ counts in [ql, qr)
        if (!v || qr <= lo || hi <= ql) return 0;
        if (ql <= lo && hi <= qr) return t[v].sum;
        int m = (lo + hi) / 2;
        return query(t[v].l, lo, m, ql, qr) + query(t[v].r, m,
        ↳ hi, ql, qr);
    }
    int kth(int v, int lo, int hi, ll k) { // k-th
    ↳ smallest (1-indexed), -1 if k too big
        if (!v || k > t[v].sum) return -1;
        if (hi - lo == 1) return lo;
        int m = (lo + hi) / 2;
        return k <= t[t[v].l].sum ? kth(t[v].l, lo, m, k) :
        ↳ kth(t[v].r, m, hi, k - t[t[v].l].sum);
    }
    // wrappers
    void update(int& root, int p, ll d) { root = update(root, 0,
    ↳ V, p, d); }
    int merge(int a, int b) { return merge(a, b, 0, V); }
    ll query(int root, int ql, int qr) { return query(root, 0, V,
    ↳ ql, qr); }
    int kth(int root, ll k) { return kth(root, 0, V, k); }
};
// TREE USAGE: root[v] starts with just v's own value; after
// ↳ recursing, merge every child's root
// into v's. Answer v's query right there. Never reuse a root
// ↳ after merging it away.
```

2.17 BitsetTricks

BITSET SPEEDUPS - the /64 family. All of these are "obviously too slow" DPs that fit anyway.

```
// TRANSITIVE CLOSURE (reachability) in  $O(n^3 / 64)$ .  $n$  up to ~5000.
template <int N>
void transitiveClosure(vector<bitset<N>>& g, int n) {
    for (int k = 0; k < n; k++)
        for (int i = 0; i < n; i++) if (g[i][k]) g[i] |= g[k];
}

// LCS in  $O(n*m/64)$  - bit-parallel (Allison-Dix). 'a' is the
// string whose length bounds the
// bitset width; row is a bitmask over positions of 'a'.
// x = row | match[b_i]
// row' = x & ((x - (row << 1 | 1)) ^ x)
// The "| 1" is the carry-in that makes the shift borrow correctly.
typedef unsigned long long u64;
int bitLCS(const string& a, const string& b) {
    int n = a.size(), W = (n + 63) / 64;
    if (!n || b.empty()) return 0;
    vector<vector<u64>> match(256, vector<u64>(W, 0));
    for (int i = 0; i < n; i++) match[(unsigned char)a[i]][i >> 6] |= 1ULL << (i & 63);
    vector<u64> row(W, 0), x(W), sh(W);
    for (char c : b) {
        const auto& M = match[(unsigned char)c];
        for (int i = 0; i < W; i++) x[i] = row[i] | M[i];
        u64 carry = 1; // sh =
        // (row << 1) | 1
        for (int i = 0; i < W; i++) { u64 nc = row[i] >> 63;
        sh[i] = (row[i] << 1) | carry; carry = nc; }
        u64 borrow = 0; // row = x
        // & ((x - sh) ^ x)
        for (int i = 0; i < W; i++) {
            u64 t = sh[i] + borrow;
            borrow = (t < sh[i]) || (x[i] < t); // t <
            // sh[i] catches the carry overflow
            u64 d = x[i] - t;
            row[i] = x[i] & (d ^ x[i]);
        }
    }
    int r = 0;
    for (u64 v : row) r += __builtin_popcountll(v);
    return r;
}

// SUBSET SUM / KNAPSACK FEASIBILITY: dp |= dp << w (see
// KnapsackFamily.cpp)
// BIPARTITE MATCHING on dense graphs: keep each left vertex's
// neighbourhood as a bitset and
// pick the next free right vertex with (adj[u] &
// free)._Find_first().
// COUNTING PAIRS with a property: bitset per value class, then
// popcount of ANDs.
// GCC EXTRAS: b._Find_first(), b._Find_next(i) iterate set bits
// in  $O(N/64)$  total.
// RULE OF THUMB: bitset turns  $1e9$  "simple" operations into  $\sim 1.5e7$ 
// word operations - budget
// about  $n*m/64 \leq 1e8$  for a comfortable pass.
```

2.18 VeniceAndSWAG

VENICE TECHNIQUE - a multiset with an $O(1)$ "add x to EVERY element".

```
// Keep a global offset; store values shifted by it. 15 lines, and
// it turns a whole family of
// "everyone loses X health, remove the dead" problems into a
// one-liner.
struct Venice {
    multiset<ll> s;
    ll water = 0; // global
    // offset

    void add(ll v) { s.insert(v + water); } // insert
    // the true value v
    void erase(ll v) { auto it = s.find(v + water); if (it !=
    // s.end()) s.erase(it); }
    void addAll(ll v) { water -= v; } //  $O(1)$ :
    // add v to every element
    ll getMin() const { return *s.begin() - water; }
    ll getMax() const { return *s.rbegin() - water; }
    int size() const { return s.size(); }
    // Pop everything whose TRUE value is  $\leq t$  (e.g. everyone who
    // died).
    template <class F> void popLE(ll t, F onPop) {
        while (!s.empty() && *s.begin() - water <= t) {
            onPop(*s.begin() - water); s.erase(s.begin()); }
    };
    // SAME IDEA elsewhere: a binary trie with a global XOR mask ("xor
    // every element by x"),
    // a BIT with a global additive offset, a lazy "add to all" on a
    // heap.

    // SWAG (Sliding Window Aggregation) for ANY monoid -
    // non-commutative and non-invertible ops
    // included: matrix product, affine composition, gcd, "merge two
    // structs".
    // Amortised  $O(1)$  push/pop/fold. This generalises 09/02 -
    // TwoStackQueue.cpp, which is hardwired
    // to min/max. Prefix-product-and-divide does NOT work without an
    // inverse - this does.
    template <class T, class Op>
    struct SWAG {
        vector<pair<T, T>> front_, back_; // {value,
        // aggregate from here to the bottom}
        T id;
        Op op;
        SWAG(T id, Op op) : id(id), op(op) {}

        T foldFront() const { return front_.empty() ? id :
        // front_.back().second; }
        T foldBack() const { return back_.empty() ? id :
        // back_.back().second; }
        T fold() const { return op(foldFront(), foldBack()); }
        // front first: order matters
        void push(T v) { back_.push_back({v, op(foldBack(), v)}); }
        void pop() {
            if (front_.empty())
                while (!back_.empty()) {
                    T v = back_.back().first; back_.pop_back();
                    front_.push_back({v, op(v, foldFront())});
                }
            // reversed accumulation
            if (!front_.empty()) front_.pop_back();
        }
        int size() const { return front_.size() + back_.size(); }
    };
    // USAGE: SWAG<ll, function<ll(ll,ll)>> q(0, [] (ll a, ll b){
    // return a + b; });
    // For non-commutative ops keep the argument order exactly as
    // written above, or the window
    // product comes out reversed.
```

2.19 MergeableHeap

MERGEABLE (MELDABLE) HEAP - a priority queue that also supports MERGING two heaps in $O(\log n)$,

```
// which std::priority_queue cannot do. This is a skew heap: merge
//   ↳ by walking the two right spines
// and swapping children, which keeps the amortised depth
//   ↳ logarithmic with no stored rank.
// Supports a lazy "add v to every key in this heap" tag, because
//   ↳ that is what the two algorithms
// that need a mergeable heap actually want.
// USES: Chu-Liu/Edmonds directed MST in  $O(E \log V)$  (each
//   ↳ contracted cycle merges its in-edge
// heaps and subtracts the chosen edge's weight from the whole
//   ↳ heap); Dijkstra-style k-shortest
// paths (Eppstein); "merge the two children's candidate sets" DP
//   ↳ on trees; leftist-heap solutions
// to scheduling problems (repeatedly merge and pop the largest,
//   ↳ e.g. minimise total lateness).
```

```
struct MHeap {
    struct Node { ll key, lz = 0; Node *l = 0, *r = 0; };
    static void down(Node* t) { //
        ↳ apply the lazy tag downwards
        if (!t || !t->lz) return;
        t->key += t->lz;
        if (t->l) t->l->lz += t->lz;
        if (t->r) t->r->lz += t->lz;
        t->lz = 0;
    }
    static Node* merge(Node* a, Node* b) { //
        ↳  $O(\log n)$  amortised, MIN-heap
        if (!a || !b) return a ? a : b;
        down(a), down(b);
        if (a->key > b->key) swap(a, b);
        a->r = merge(a->r, b);
        swap(a->l, a->r); //
        ↳ the "skew" step
        return a;
    }
    static Node* push(Node* t, ll key) { return merge(t, new
        ↳ Node[key]); }
    static ll top(Node* t) { return down(t), t->key; }
    static Node* pop(Node* t) { down(t); return merge(t->l,
        ↳ t->r); }
    static void addAll(Node* t, ll v) { if (t) t->lz += v; } //
        ↳  $O(1)$  add to every key
};
// TO GET A MAX-HEAP: negate the keys, or flip the comparison in
//   ↳ merge.
// LEFTIST HEAP is the same interface with a stored 'dist'
//   ↳ (null-path length) and the rule "keep the
// larger dist on the left"; it gives WORST-CASE  $O(\log n)$  instead
//   ↳ of amortised. Skew is shorter and
// fine unless you need the worst-case bound inside another
//   ↳ amortised argument.
// PAIRING HEAP has  $O(1)$  merge and decrease-key in practice and is
//   ↳ what you want if you need
// decrease-key (Dijkstra on very dense graphs); in contests a
//   ↳ binary heap with lazy deletion wins.
// LAZY DELETION is usually enough: push (key, id), and pop until
//   ↳ the top is still valid. That
// removes the need for decrease-key entirely and is 5 lines.
// SMALL-TO-LARGE with std::priority_queue also "merges" heaps:
//   ↳ repeatedly pop from the smaller and
// push into the larger,  $O(n \log^2 n)$  total. Shorter than this
//   ↳ file - use it unless the  $\log^2$  hurts.
```

2.20 PersistentAndUndo

PERSISTENCE AND UNDO - three ways to travel in a structure's history, and when each applies.

```
// ROLLBACK (05 - DSU/02): undo the last op. Cheapest. Needs a
//   ↳ stack discipline (LIFO).
// OFFLINE DELETION (14 - OfflineDynamicConnectivity): segment
//   ↳ tree over time + rollback.
// Handles arbitrary insert/delete, but you must know every
//   ↳ operation in advance.
// PERSISTENCE (this file): branch the history. The only one
//   ↳ that works ONLINE with deletions,
// and the only one that lets you query a past version after
//   ↳ moving on.
```

```
// --- PERSISTENT ARRAY: a segment tree over indices. get/set in
//   ↳  $O(\log n)$ ,  $O(\log n)$  new nodes. ---
struct PArray {
    struct Node { int l = 0, r = 0; ll v = 0; };
    vector<Node> t;
    int n;
    PArray(int n) : t(1), n(n) {} //
        ↳ node 0 is the shared all-zero
    int build(const vector<ll>& a, int lo, int hi) {
        int id = t.size();
        t.push_back({});
        if (lo == hi) { t[id].v = a[lo]; return id; }
        int m = (lo + hi) / 2, L = build(a, lo, m), R = build(a,
        ↳ m + 1, hi);
        t[id].l = L, t[id].r = R;
        return id;
    }
    int build(const vector<ll>& a) { return build(a, 0, n - 1); }
    ll get(int u, int i, int lo, int hi) {
        if (lo == hi) return t[u].v;
        int m = (lo + hi) / 2;
        return i <= m ? get(t[u].l, i, lo, m) : get(t[u].r, i, m
        ↳ + 1, hi);
    }
    ll get(int u, int i) { return get(u, i, 0, n - 1); }
    int set(int u, int i, ll v, int lo, int hi) {
        int id = t.size();
        t.push_back(t[u]);
        if (lo == hi) { t[id].v = v; return id; }
        int m = (lo + hi) / 2;
        if (i <= m) t[id].l = set(t[u].l, i, v, lo, m);
        else t[id].r = set(t[u].r, i, v, m + 1, hi);
        return id;
    }
    int set(int u, int i, ll v) { return set(u, i, v, 0, n - 1); }
};
// --- PERSISTENT DSU on top of it.  $O(\log^2 n)$  per operation. ---
// The catch that makes it work: NO PATH COMPRESSION (that would
//   ↳ mutate an old version), so you
// MUST union by size/rank to keep the height  $O(\log n)$ .
struct PDSU {
    PArray par, sz;
    PDSU(int n) : par(n), sz(n) {}
    pair<int, int> init(int n) { //
        ↳ returns {parRoot, szRoot}
        vector<ll> p(n), s(n, 1);
        iota(all(p), 0);
        return {par.build(p), sz.build(s)};
    }
    int find(int pr, int x) { ll p = par.get(pr, x); return p ==
        ↳ x ? x : find(pr, p); }
    pair<int, int> join(int pr, int sr, int a, int b) {
        a = find(pr, a), b = find(pr, b);
        if (a == b) return {pr, sr};
        if (sz.get(sr, a) < sz.get(sr, b)) swap(a, b);
        return {par.set(pr, b, a), sz.set(sr, a, sz.get(sr, a) +
        ↳ sz.get(sr, b))};
    }
};
/* THE OTHER TWO TIME-TRAVEL TRICKS, as recipes
QUEUE UNDO. You have a structure that supports only "undo the
//   ↳ last op" (rollback DSU), but the
deletions arrive in FIFO order (a sliding window). Keep the
//   ↳ undo stack partitioned so the
OLDEST element is near the top; when it is not, pop a prefix
//   ↳ and re-push it in the other order,
always rebuilding the SMALLER half.  $O(\log n)$  amortised extra.
//   ↳ This is the online alternative to
the offline segment-tree-on-time trick, and it is what "sliding
//   ↳ window connectivity" needs.
STATIC TO DYNAMIC (the logarithmic method). Any STATIC structure
//   ↳ with  $O(B(n))$  build and  $O(Q(n))$ 
query becomes insert-only: keep  $O(\log n)$  instances of sizes
```

↪ $2^0, 2^1, 2^2, \dots$; inserting is
 binary-counter carrying, so you rebuild an instance of size 2^k
 ↪ only every 2^k insertions.
 Insert becomes $O(B(n)/n * \log n)$ amortised and query $O(Q(n) \log n)$ (ask all instances).
 Use it to make a convex hull / a wavelet tree / an Aho-Corasick
 ↪ automaton accept insertions.
 PARTIALLY vs FULLY PERSISTENT: partially = you may query any past
 ↪ version but only modify the
 latest (much cheaper - a version-stamped array often suffices);
 ↪ fully = you may modify any
 version, branching the history into a tree. The code above is
 ↪ fully persistent.
 WHEN NOT TO BOTHER: if the problem is offline and only needs
 ↪ "delete", the segment tree over the
 timeline with a rollback DSU is shorter, faster, and has a
 ↪ smaller constant than any of this. */

3 Graph Theory

3.1 Graph Traversal

3.1.1 TopoSort

Topological Sort (Kahn's Algorithm) - $O(V + E)$

```

struct TopoSort {
    int n;
    vector<vector<int>> adj;
    vector<int> deg, order;

    TopoSort(int n = 0) : n(n), adj(n + 1), deg(n + 1, 0) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        deg[v]++;
    }

    bool solve() {
        queue<int> q;
        for (int i = 1; i <= n; i++) {
            if (!deg[i]) q.push(i);
        }
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            order.pb(u);
            for (int v : adj[u]) {
                if (--deg[v]) q.push(v);
            }
        }
        return (int)order.size() == n; // false if graph has cycles
    }
};
  
```

3.1.2 BFS_DFS

BFS / DFS / 0-1 BFS / multi-source. 0-based. The bread and butter - keep them short.

```

// Unweighted shortest path + parent for reconstruction.
pair<vector<int>, vector<int>> bfs(int n, const
    ↪ vector<vector<int>>& adj, vector<int> src) {
    vector<int> d(n, -1), par(n, -1);
    queue<int> q;
    for (int s : src) d[s] = 0, q.push(s); // MULTI-SOURCE:
    ↪ pass every source at once
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int v : adj[u]) if (d[v] < 0) d[v] = d[u] + 1,
    ↪ par[v] = u, q.push(v);
    }
    return {d, par};
}

vector<int> pathTo(int t, const vector<int>& par) {
    vector<int> p;
    for (int v = t; v >= 0; v = par[v]) p.push_back(v);
    reverse(all(p));
    return p;
}

// 0-1 BFS: edge weights are only 0 or 1 -> deque instead of a
    ↪ heap,  $O(V + E)$ .
// Use for "turning costs 1", "breaking one wall is free", "flip
    ↪ at most k edges".
vector<ll> bfs01(int n, const vector<vector<pair<int, int>>>&
    ↪ adj, int s) {
    vector<ll> d(n, llinf);
    deque<int> q;
    d[s] = 0, q.push_back(s);
    while (!q.empty()) {
        int u = q.front(); q.pop_front();
        for (auto [v, w] : adj[u]) if (d[u] + w < d[v]) {
            d[v] = d[u] + w;
            w ? q.push_back(v) : q.push_front(v);
        }
    }
    return d;
}

// Iterative DFS with entry/exit times (avoids stack overflow at n
    ↪ = 2e5).
void dfsIter(int n, const vector<vector<int>>& adj, int root,
    ↪ vector<int>& tin, vector<int>& tout) {
    tin.assign(n, -1), tout.assign(n, -1);
    vector<int> it(n, 0), st = {root};
    int timer = 0;
    tin[root] = timer++;
    while (!st.empty()) {
        int u = st.back();
        if (it[u] < (int)adj[u].size()) {
            int v = adj[u][it[u]++];
            if (tin[v] < 0) tin[v] = timer++, st.push_back(v);
        } else tout[u] = timer++, st.pop_back();
    }
}

// GRID: 4- and 8-neighbour offsets (see 03 - grid.cpp). Encode a
    ↪ cell as r*m+c and reuse the
// graph routines above; add extra state dimensions (mask, parity,
    ↪ keys) by widening the index.
// CYCLE DETECTION, directed: colour DFS (0 white, 1 grey, 2
    ↪ black); a grey->grey edge closes a
// cycle - walk the recursion stack back to extract it.
    ↪ Undirected: any edge to an already-visited
// non-parent vertex closes one.
  
```

3.2 Shortest Paths

3.2.1 Dijkstra

Dijkstra's Shortest Path Algorithm - $O((V + E) \log V)$

```

struct Dijkstra {
    int n;
    vector<vector<pair<int, ll>>> adj;
    vector<ll> dist;
    vector<int> par;

    Dijkstra(int n = 0) : n(n), adj(n + 1), dist(n + 1, llinf),
    ↪ par(n + 1, -1) {}

    void add_edge(int u, int v, ll w, bool directed = false) {
        adj[u].eb(v, w);
        if (!directed) adj[v].eb(u, w);
    }

    void solve(int src) {
        fill(all(dist), llinf);
        fill(all(par), -1);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
    ↪ greater<>> pq;

        dist[src] = 0;
        pq.emplace(0, src);

        while (!pq.empty()) {
            auto [d, u] = pq.top();
            pq.pop();
            if (d > dist[u]) continue;

            for (auto& [v, w] : adj[u]) {
                if (dist[u] + w < dist[v]) {
                    dist[v] = dist[u] + w;
                    par[v] = u;
                    pq.emplace(dist[v], v);
                }
            }
        }
    };

```

3.2.2 FloydWarshall

Floyd-Warshall All-Pairs Shortest Path - $O(V^3)$

```

struct FloydWarshall {
    int n;
    vector<vector<ll>> dist;

    FloydWarshall(int n = 0) : n(n), dist(n + 1, vector<ll>(n +
    ↪ 1, llinf)) {
        for (int i = 0; i <= n; i++) dist[i][i] = 0;
    }

    void add_edge(int u, int v, ll w, bool directed = false) {
        dist[u][v] = min(dist[u][v], w);
        if (!directed) dist[v][u] = min(dist[v][u], w);
    }

    void solve() {
        for (int k = 1; k <= n; k++) {
            for (int i = 1; i <= n; i++) {
                for (int j = 1; j <= n; j++) {
                    if (dist[i][k] < llinf && dist[k][j] < llinf)
    ↪ {
                        dist[i][j] = min(dist[i][j], dist[i][k] +
    ↪ dist[k][j]);
                    }
                }
            }
        }
    };

```

3.2.3 BellmanFord

If you paste both, keep only ONE copy of it.

```

// NOTE: 08 - Spanning Trees/01 - MST.cpp declares an identical
    ↪ 'struct Edge'.
// If you paste both, keep only ONE copy of it.
// Bellman-Ford Shortest Path with Negative Cycle Detection -  $O(V$ 
    ↪  $* E)$ 
struct Edge {
    int u, v;
    ll w;
};

struct BellmanFord {
    int n;
    vector<Edge> edges;
    vector<ll> dist;
    vector<int> par;

    BellmanFord(int n = 0) : n(n), dist(n + 1, llinf), par(n + 1,
    ↪ -1) {}

    void add_edge(int u, int v, ll w) {
        edges.pb({u, v, w});
    }

    // Returns false if a negative cycle is reachable from src
    bool solve(int src) {
        fill(all(dist), llinf);
        fill(all(par), -1);
        dist[src] = 0;

        for (int i = 0; i < n - 1; i++) {
            bool relaxed = false;
            for (const auto& e : edges) {
                if (dist[e.u] < llinf && dist[e.u] + e.w <
    ↪ dist[e.v]) {
                    dist[e.v] = dist[e.u] + e.w;
                    par[e.v] = e.u;
                    relaxed = true;
                }
            }
            if (!relaxed) break;
        }

        for (const auto& e : edges) {
            if (dist[e.u] < llinf && dist[e.u] + e.w < dist[e.v])
    ↪ {
                return false; // Negative cycle detected
            }
        }
        return true;
    };

```

3.2.4 DominatorTree

```

/*
=====
CATALOGUE: Lengauer-Tarjan Dominator Tree ( $O((N + M) \log N)$ )
=====
DESCRIPTION:
Finds the "bottlenecks" in any directed graph. A node 'u'
→ dominates 'v' if
EVERY path from the root to 'v' is forced to go through 'u'.

USAGE:
1. Build graph: Create a 2D vector 'adj' (1-based) containing
→ the directed edges.
2. Initialize: DominatorTree dt(n, root, adj);
(Automatically builds the tree on instantiation)

EXTRACTING ANSWERS:
- dt.dom[u]: The Immediate Dominator of 'u' (the closest
→ bottleneck to 'u').
- dt.id[u]: If dt.id[u] == 0 after building, 'u' is
→ UNREACHABLE from the root.
=====
*/
struct DominatorTree {
    int T = 0, n;
    vector<vector<int>> rg, bucket, adj;
    vector<int> dsu, par, sdom, idom, dom, label, id, rev;

    DominatorTree(int _n, int r, vector<vector<int>>& adj):
        n(_n + 1), rg(n), bucket(n), adj(adj), dsu(n),
        par(n), sdom(n), idom(n), dom(n), label(n), id(n), rev(n)
    → { dfs(r); build(r); }

    int find(int u, int x = 0) {
        if (u == dsu[u]) return x ? -1 : u;
        int v = find(dsu[u], x + 1);
        if (v < 0) return u;
        if (sdom[label[dsu[u]]] < sdom[label[u]]) label[u] =
    → label[dsu[u]];
        dsu[u] = v;
        return x ? v : label[u];
    }

    void dfs(int u) {
        id[u] = ++T, rev[T] = u;
        label[T] = sdom[T] = dsu[T] = T;
        for (int& w : adj[u]) {
            if (!id[w]) {
                dfs(w);
                par[id[w]] = id[u];
            }
            rg[id[w]].push_back(id[u]);
        }
    }

    void build(int r) {
        // FIXED: Loop MUST go down to i >= 1 to process the
    → root's bucket!
        for (int i = T; i >= 1; i--) {
            for (int& u : rg[i]) sdom[i] = min(sdom[i],
    → sdom[find(u)]);
            if (i > 1) bucket[sdom[i]].push_back(i);

            for (int& w : bucket[i]) {
                int v = find(w);
                idom[w] = sdom[v] == sdom[w] ? sdom[w] : v;
            }

            if (i > 1) dsu[i] = par[i];
        }

        for (int i = 2; i <= T; i++) {
            if (idom[i] != sdom[i]) idom[i] = idom[idom[i]];
        }

        for (int u = 1; u <= T; ++u) {
            dom[rev[u]] = rev[idom[u]];
        }
    }
};

void testCase() {
    int n, m, s;
    cin >> n >> m >> s;
    s++; // Convert to 1-based indexing

    vector<vector<int>> adj(n + 1);

```

```

    for (int i = 0; i < m; i++) {
        int u, v;
        cin >> u >> v;
        adj[u + 1].push_back(v + 1);
    }

    // Automatically builds upon instantiation
    DominatorTree dt(n, s, adj);

    for (int i = 1; i <= n; i++) {
        if (i == s) {
            // pS is S
            cout << s - 1 << (i == n ? " " : " ");
        } else if (!dt.id[i]) {
            // If we can not reach i from S, print -1
            cout << -1 << (i == n ? " " : " ");
        } else {
            // Print the 0-based parent (which is now guaranteed
    → to be calculated)
            cout << dt.dom[i] - 1 << (i == n ? " " : " ");
        }
    }
    cout << "\n";
}

```


3.2.5 ShortestPathEdgeModify

SHORTEST PATH WITH ONE EDGE CHANGED, answered OFFLINE in $O(1)$ per query after $O((n+m) \log n)$.

```
// Query: "if edge id had weight x instead, what is dist(s,t)?" -
//   ↳ each query is independent.
// IDEA. Let ds/dt be the distance arrays from s and to t, P the
//   ↳ shortest path, base = ds[t].
// * Edge NOT on P: the new path either ignores it (base) or
//   ↳ crosses it once:
//   min(base, ds[u] + x + dt[v], ds[v] + x + dt[u]).
// * Edge ON P at position i: either still use it (base - w +
//   ↳ x), or avoid it entirely, and
//   ↳ "the best s->t path avoiding path-edge i" is precomputed
//   ↳ for every i at once:
//   ↳ for a non-path edge (u,v,w), the path ds[u] -> u -> v ->
//   ↳ dt[v] skips exactly the path edges
//   ↳ in the window [L[u], R[v]], where L[x] = last path vertex
//   ↳ on some shortest s->x path and
//   ↳ R[x] = first path vertex on some shortest x->t path. Sweep
//   ↳ i from 0 to k-1 with a heap.
struct SPEdgeModify {
    struct Edge { int u, v; ll w; };
    int n, k; //
    ↳ k = #edges on the shortest path
    ll base;
    vector<Edge> E;
    vector<ll> ds, dt, best; //
    ↳ best[i] = s->t avoiding path edge i
    vector<int> pos; //
    ↳ pos[e] = index on P, or -1
    SPEdgeModify(int n, vector<Edge> E, int s, int t) : n(n),
    ↳ E(E) {
        int m = E.size();
        vector<vector<pair<int, int>>> adj(n);
        for (int i = 0; i < m; i++)
            adj[E[i].u].push_back({E[i].v, i}),
            adj[E[i].v].push_back({E[i].u, i});
        vector<int> par(n, -1), pare(n, -1), ordS, ordT;
        auto dij = [&](int src, vector<ll>& d, vector<int>& ord,
        ↳ bool tree) {
            d.assign(n, llinf);
            priority_queue<pair<ll, int>, vector<pair<ll, int>>,
            ↳ greater<>> pq;
            d[src] = 0, pq.push({0, src});
            while (!pq.empty()) {
                auto [dd, u] = pq.top(); pq.pop();
                if (dd > d[u]) continue;
                ord.push_back(u);
                for (auto [v, id] : adj[u]) if (d[u] + E[id].w <
                ↳ d[v]) {
                    d[v] = d[u] + E[id].w;
                    if (tree) par[v] = u, pare[v] = id;
                    pq.push({d[v], v});
                }
            }
            dij(s, ds, ordS, true), dij(t, dt, ordT, false);
            base = ds[t];
            vector<int> pv, pe; //
            ↳ the path, s ... t
            for (int c = t; c != s; c = par[c]) pv.push_back(c),
            ↳ pe.push_back(pare[c]);
            pv.push_back(s), reverse(all(pv)), reverse(all(pe));
            k = pe.size();
            pos.assign(m, -1);
            vector<char> onP(n, 0);
            for (int x : pv) onP[x] = 1;
            for (int i = 0; i < k; i++) pos[pe[i]] = i;
            vector<int> L(n, 0), R(n, k);
            for (int i = 0; i <= k; i++) L[pv[i]] = R[pv[i]] = i;
            for (int u : ordS) for (auto [v, id] : adj[u])
            ↳ // relaxed in shortest-path order
                if (!onP[v] && ds[v] == ds[u] + E[id].w) L[v] =
            ↳ max(L[v], L[u]);
            for (int u : ordT) for (auto [v, id] : adj[u])
                if (!onP[v] && dt[v] == dt[u] + E[id].w) R[v] =
            ↳ min(R[v], R[u]);
            vector<vector<pair<ll, int>>> ev(max(k, 1));
            ↳ // ev[start] = {cost, end}
            for (int i = 0; i < m; i++) {
                if (pos[i] >= 0) continue;
                auto [u, v, w] = E[i];
                if (L[u] < R[v]) ev[L[u]].push_back({ds[u] + w +
                ↳ dt[v], R[v]});
                if (L[v] < R[u]) ev[L[v]].push_back({ds[v] + w +
                ↳ dt[u], R[u]});
            }
        }
```

```
best.assign(max(k, 1), llinf);
priority_queue<pair<ll, int>, vector<pair<ll, int>>,
↳ greater<>> sw;
for (int i = 0; i < k; i++) {
    for (auto& e : ev[i]) sw.push(e);
    while (!sw.empty() && sw.top().second <= i) sw.pop();
    if (!sw.empty()) best[i] = sw.top().first;
}
}
ll query(int id, ll x) const { //
    ↳ edge id gets weight x
    auto [u, v, w] = E[id];
    if (pos[id] < 0) return min({base, ds[u] + x + dt[v],
    ↳ ds[v] + x + dt[u]});
    return min(base - w + x, best[pos[id]]);
};
// RELATED, same precomputation:
// * "Does edge e lie on SOME shortest path?" <=> ds[u] + w +
//   ↳ dt[v] == base (either direction).
// * "Does e lie on EVERY shortest path?" <=> removing it
//   ↳ increases the distance; the sweep
//   ↳ above answers this for path edges (best[i] > base means
//   ↳ path edge i is essential).
// * REPLACEMENT PATHS (delete one edge, ask the new distance) =
//   ↳ query(id, +infinity).
// * SECOND SHORTEST WALK (may reuse edges): min over edges of
//   ↳ ds[u] + w + dt[v] strictly > base.
// * If instead the whole graph changes per query, you need a
//   ↳ fresh Dijkstra - this trick only
//   ↳ works because s, t and every other weight stay fixed.
```


3.2.6 CyclesAndConstraints

MINIMUM MEAN CYCLE, and SYSTEMS OF DIFFERENCE CONSTRAINTS - the two shortest-path tools that

```
// are not about shortest paths.

// --- KARP'S MINIMUM MEAN CYCLE,  $O(V \cdot E)$ . Returns the smallest
//   possible (cycle weight)/(length).
//  $dp[k][v]$  = min weight of a walk of EXACTLY  $k$  edges ending at  $v$ 
//   (from a virtual source that
//   reaches everything). Karp: the answer is min over  $v$  of max
//   over  $k < n$  of  $(dp[n][v] - dp[k][v]) / (n - k)$ .
// Reach for it whenever the objective is an AVERAGE around a
//   cycle: "minimum cost per unit time",
// "is there a cycle with negative average", profit-per-day loops.
ld minMeanCycle(int n, const vector<array<ll, 3>>& e) {
    e = {u, v, w}, directed
    vector<vector<ll>> dp(n + 1, vector<ll>(n, llinf));
    fill(all(dp[0]), 0);
    // virtual source reaches every v
    for (int k = 1; k <= n; k++)
        for (auto [u, v, w] : e)
            if (dp[k - 1][u] < llinf) dp[k][v] = min(dp[k][v],
    dp[k - 1][u] + w);
    ld best = 1e18;
    for (int v = 0; v < n; v++) {
        if (dp[n][v] == llinf) continue;
        ld cur = -1e18;
        for (int k = 0; k < n; k++)
            if (dp[k][v] < llinf) cur = max(cur, (ld)(dp[n][v] -
    dp[k][v]) / (n - k));
        best = min(best, cur);
    }
    return best;
    // 1e18 if the graph is acyclic
}

// MINIMUM COST-TO-TIME RATIO CYCLE (minimise sum(cost)/sum(time),
//   time > 0): binary search x and
// test for a negative cycle with weights cost - x*time. Same
//   skeleton, one Bellman-Ford per step.
// MAXIMUM mean cycle: negate the weights.

// --- SYSTEM OF DIFFERENCE CONSTRAINTS. Every constraint  $x_j -$ 
//    $x_i \leq w$  becomes an edge  $i \rightarrow j$  of
// weight w; add a super-source with a 0-edge to every vertex and
//   run Bellman-Ford.
// FEASIBLE iff there is no negative cycle. The distances are then
//   a solution, and they are the
// POINTWISE MAXIMUM one (every other solution is  $\leq$  it
//   componentwise, after fixing  $x_{\text{source}} = 0$ ).
// TRANSLATIONS you will need:  $x_j - x_i \geq w$  is  $x_i - x_j \leq$ 
//    $-w$ .  $x_j - x_i = w$  is both.
//  $x_i \leq c$  is  $x_i - x_0 \leq c$  against the source. Strict " $<$ " on
//   integers is " $\leq w - 1$ ".
// USES: scheduling with precedence and deadlines, "arrange these
//   values so all the gaps hold",
// consistency of a set of relative measurements, and the dual of
//   a min-cost-flow.
bool differenceConstraints(int n, const vector<array<ll, 3>>& c,
    vector<ll>& x) {
    vector<array<ll, 3>> e = c;
    c = {i, j, w} meaning  $x_j - x_i \leq w$ 
    for (int v = 0; v < n; v++) e.push_back({(ll)n, (ll)v, 0});
    // super-source n
    x.assign(n + 1, 0);
    for (int it = 0; it <= n; it++) {
        bool any = false;
        for (auto [u, v, w] : e)
            if (x[u] + w < x[v]) x[v] = x[u] + w, any = true;
        if (!any) { x.resize(n); return true; }
    }
    return false;
    // negative cycle: infeasible
}

// --- BELLMAN-FORD EXTRAS worth remembering
// NEGATIVE CYCLE RECOVERY: after n rounds, take a vertex that
//   still relaxed, walk n parent
// pointers back (this lands you ON the cycle - the relaxed vertex
//   may only be reachable from it),
// then follow parents until a vertex repeats.
// SPFA is Bellman-Ford with a queue; it is  $O(VE)$  worst case and
//   killable, but the SLF heuristic
// (push to the front if the new distance beats the front) makes
//   it fast in practice.
// ONLY SHORTEST PATHS WITH AT MOST K EDGES: run exactly k rounds,
//   updating from a SNAPSHOT of the
// previous round (otherwise one round can propagate along many
//   edges).
```

```
// JOHNSON'S APSP for sparse graphs with negative edges:
//   Bellman-Ford from a super-source gives
// potentials  $h[]$ , reweight  $w'(u, v) = w + h[u] - h[v] \geq 0$ , then
//   run n Dijkstras.  $O(VE + V E \log V)$ .
```

3.2.7 SegmentTreeGraph

SEGMENT-TREE GRAPH - add "u -> every vertex in [l, r]" (or the reverse) with $O(\log n)$ edges

```
// instead of  $O(n)$ . Then run Dijkstra/BFS as usual. Total  $O((n + q$ 
//    $\log n) \log n)$ .
// Two trees: OUT (parent -> children, cost 0) to broadcast INTO a
//   range,
//   IN (children -> parent, cost 0) to collect FROM a
//   range.
// Leaves of both trees are the original vertices.
struct SegGraph {
    int n, cnt;
    vector<int> outId, inId; // node ids
    // of the two trees
    vector<vector<pair<int, ll>>> g;

    SegGraph(int n) : n(n) {
        cnt = n;
        outId.assign(4 * n, -1), inId.assign(4 * n, -1);
        g.assign(n + 8 * n, {});
        buildOut(1, 0, n - 1), buildIn(1, 0, n - 1);
    }
    int node() { return cnt++; }
    void buildOut(int v, int l, int r) {
        if (l == r) { outId[v] = l; return; }
        outId[v] = node();
        int m = (l + r) / 2;
        buildOut(2 * v, l, m), buildOut(2 * v + 1, m + 1, r);
        g[outId[v]].push_back({outId[2 * v], 0});
        g[outId[v]].push_back({outId[2 * v + 1], 0});
    }
    void buildIn(int v, int l, int r) {
        if (l == r) { inId[v] = l; return; }
        inId[v] = node();
        int m = (l + r) / 2;
        buildIn(2 * v, l, m), buildIn(2 * v + 1, m + 1, r);
        g[inId[2 * v]].push_back({inId[v], 0});
        g[inId[2 * v + 1]].push_back({inId[v], 0});
    }
    void addRange(int v, int l, int r, int ql, int qr, int u, ll
    w, bool toRange) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) {
            if (toRange) g[u].push_back({outId[v], w});
            // u -> everything in [ql, qr]
            else g[inId[v]].push_back({u, w});
            // everything in [ql, qr] -> u
            return;
        }
        int m = (l + r) / 2;
        addRange(2 * v, l, m, ql, qr, u, w, toRange);
        addRange(2 * v + 1, m + 1, r, ql, qr, u, w, toRange);
    }
    void edgeToRange(int u, int l, int r, ll w) { addRange(1, 0,
    n - 1, l, r, u, w, true); }
    void edgeFromRange(int l, int r, int u, ll w) { addRange(1,
    0, n - 1, l, r, u, w, false); }
    void edge(int u, int v, ll w) { g[u].push_back({v, w}); }

    vector<ll> dijkstra(int s) {
        vector<ll> d(cnt, llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        greater<>> pq;
        d[s] = 0, pq.push({0, s});
        while (!pq.empty()) {
            auto [dd, u] = pq.top(); pq.pop();
            if (dd > d[u]) continue;
            for (auto [v, w] : g[u]) if (dd + w < d[v]) d[v] = dd
            + w, pq.push({d[v], v});
        }
        d.resize(n); // only the
        // original vertices matter
        return d;
    }
};
// The same construction gives 2-SAT range clauses and "range ->
//   range" edges (route both
//   through one fresh auxiliary node).
```

3.3 Graph Connectivity

3.3.1 LowLink

LowLink: BRIDGES + ARTICULATION POINTS + 2-EDGE-CONNECTED COMPONENTS - $O(V + E)$

```
// Skips the parent by EDGE ID, so MULTI-EDGES are handled
    ↪ correctly
// (a doubled edge is never a bridge). 0-based vertices.
// comp[u] = id of u's 2-edge-connected component; contracting
    ↪ those gives the BRIDGE TREE,
// on which "number of bridges between u and v" is just a tree
    ↪ path length.
struct LowLink {
    int n, timer = 0, nc = 0;
    vector<vector<pair<int, int>>> adj;           // {to,
    ↪ edge id}
    vector<int> tin, low, comp;
    vector<bool> isCut;
    vector<pair<int, int>> bridges;
    vector<int> st;

    LowLink(int n = 0) : n(n), adj(n, vector<pair<int, int>>()), tin(n, -1), low(n),
    ↪ comp(n, -1), isCut(n, false) {}
    void add_edge(int u, int v, int id) { adj[u].push_back({v,
    ↪ id}), adj[v].push_back({u, id}); }

    void dfs(int u, int pe) {
        tin[u] = low[u] = timer++;
        st.push_back(u);
        int kids = 0;
        for (auto [v, id] : adj[u]) {
            if (id == pe) continue;           // skip
    ↪ the edge we came in on, not the vertex
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {
                dfs(v, id), kids++;
                low[u] = min(low[u], low[v]);
                if (low[v] > tin[u]) bridges.push_back({u, v});
                if (low[v] >= tin[u] && pe != -1) isCut[u] = true;
            }
        }
        if (pe == -1 && kids > 1) isCut[u] = true;
        if (low[u] == tin[u]) {               // u is
    ↪ the top of a 2-edge-connected comp
            while (true) {
                int v = st.back(); st.pop_back();
                comp[v] = nc;
                if (v == u) break;
            }
            nc++;
        }
    }
    void solve() { for (int i = 0; i < n; i++) if (tin[i] == -1)
    ↪ dfs(i, -1); }

    // Bridge tree: nc nodes, one edge per bridge.
    vector<vector<int>> bridgeTree() {
        vector<vector<int>> t(nc);
        for (auto [u, v] : bridges)
    ↪ t[comp[u]].push_back(comp[v]), t[comp[v]].push_back(comp[u]);
        return t;
    }
};
```

3.3.2 SCC

Strongly Connected Components (Tarjan's Algorithm) - $O(V + E)$

```
// IMPORTANT: component ids come out in REVERSE topological order
    ↪ - for every edge u->v with
    ↪ comp[u] != comp[v] we have comp[u] > comp[v]. Iterate ids
    ↪ DOWNWARD to walk the condensation
// forwards. build_dag() may emit duplicate arcs (fine for
    ↪ reachability, wrong for counting).
struct SCC {
    int n, timer = 0, num_comps = 0;
    vector<vector<int>> adj;
    vector<int> tin, low, comp;
    vector<bool> in_st;
    stack<int> st;
    vector<vector<int>> comps;

    SCC(int n = 0) : n(n), adj(n, vector<int>()), tin(n, -1), low(n, -1),
    ↪ comp(n, -1), in_st(n, false) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
    }

    void dfs(int u) {
        tin[u] = low[u] = ++timer;
        st.push(u);
        in_st[u] = true;

        for (int v : adj[u]) {
            if (!tin[v]) {
                dfs(v);
                low[u] = min(low[u], low[v]);
            } else if (in_st[v]) {
                low[u] = min(low[u], tin[v]);
            }
        }

        if (low[u] == tin[u]) {
            comps.pb({});
            while (true) {
                int v = st.top();
                st.pop();
                in_st[v] = false;
                comp[v] = num_comps;
                comps.back().pb(v);
                if (u == v) break;
            }
            num_comps++;
        }
    }

    void solve() {
        for (int i = 1; i <= n; i++) {
            if (!tin[i]) dfs(i);
        }
    }

    vector<vector<int>> build_dag() {
        vector<vector<int>> dag(num_comps);
        for (int u = 1; u <= n; u++) {
            for (int v : adj[u]) {
                if (comp[u] != comp[v]) dag[comp[u]].pb(comp[v]);
            }
        }
        return dag;
    }
};
```

3.3.3 BlockCutTree

BLOCK-CUT TREE (vertex biconnected components). $O(V + E)$.

```
// Builds a forest whose nodes are: the original CUT VERTICES,
//   ↳ plus one node per BICONNECTED
// COMPONENT ("block"). Every cut vertex is joined to each block
//   ↳ containing it.
// Answers "is there a path u -> v avoiding vertex w", "which
//   ↳ blocks does a path cross",
// "articulation structure of the graph" - all become tree queries.
struct BlockCut {
    int n, timer = 0;
    vector<vector<int>> adj, comps, tree;           // comps[i]
    ↳ = vertices of block i
    vector<int> tin, low, stk, id;                 // id[v] =
    ↳ tree node of original vertex v
    vector<bool> isCut;

    BlockCut(int n = 0) : n(n), adj(n), tin(n, -1), low(n), id(n,
    ↳ -1), isCut(n, false) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
    ↳ adj[v].push_back(u); }

    void dfs(int u, int p) {
        tin[u] = low[u] = timer++;
        stk.push_back(u);
        int kids = 0;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; }      // skip ONE
            ↳ parent edge (multi-edge safe)
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {
                dfs(v, u), kids++;
                low[u] = min(low[u], low[v]);
                if (low[v] >= tin[u]) {              // u closes
                    ↳ a block
                    isCut[u] = (tin[u] > 0 || kids > 1);
                    comps.push_back({u});
                    while (true) {
                        int w = stk.back(); stk.pop_back();
                        comps.back().push_back(w);
                        if (w == v) break;
                    }
                }
            }
        }
        void build() {
            for (int v = 0; v < n; v++) if (adj[v].empty())
            ↳ comps.push_back({v}); // isolated vertex
            for (int i = 0; i < n; i++) if (tin[i] == -1) { timer =
            ↳ 0; dfs(i, -1); stk.clear(); }
            tree.assign(comps.size(), {});
            for (int v = 0; v < n; v++) if (isCut[v]) id[v] =
            ↳ tree.size(), tree.push_back({});
            for (int i = 0; i < (int)comps.size(); i++)
                for (int v : comps[i]) {
                    if (!isCut[v]) id[v] = i;
                    else tree[i].push_back(id[v]),
                    ↳ tree[id[v]].push_back(i);
                }
        }
    };

    // A vertex that is NOT a cut vertex lies in exactly one block ->
    ↳ id[v] is that block.
    // An edge (u,v) always lies in exactly one block. Leaves of the
    ↳ tree are blocks; a graph is
    // biconnected iff the tree has a single node.
```

3.3.4 TriangleAndCycleCounting

COUNTING SMALL SUBGRAPHS in $O(m \sqrt{m})$ - the "degree ordering" trick. Orient every edge from

```
// the lower (degree, index) endpoint to the higher one. In that
//   ↳ DAG every out-degree is  $O(\sqrt{m})$ ,
// because a vertex with out-degree d has d neighbours of degree
//   ↳  $\geq d$ , so  $d^2 \leq 2m$ .
// Undirected simple graph, 0-indexed, edges given as pairs.

// --- TRIANGLES: enumerate each once.  $O(m \sqrt{m})$ . ---
ll countTriangles(int n, const vector<pair<int, int>>& es) {
    vector<int> deg(n, 0);
    for (auto [u, v] : es) deg[u]++, deg[v]++;
    auto better = [&](int a, int b) { return deg[a] != deg[b] ?
    ↳ deg[a] < deg[b] : a < b; };
    vector<vector<int>> out(n);
    for (auto [u, v] : es) better(u, v) ? out[u].push_back(v) :
    ↳ out[v].push_back(u);
    vector<char> mark(n, 0);
    ll cnt = 0;
    for (int u = 0; u < n; u++) {
        for (int v : out[u]) mark[v] = 1;
        for (int v : out[u]) for (int w : out[v]) cnt += mark[w];
        for (int v : out[u]) mark[v] = 0;
    }
    return cnt;
    ↳ each triangle counted exactly once

// LISTING them: inside the inner loop, (u,v,w) with mark[w] IS a
//   ↳ triangle.
// PER-VERTEX / PER-EDGE triangle counts: add 1 to each of u,v,w
//   ↳ (or to each of the three edges)
// at that moment - this is what "local clustering coefficient"
//   ↳ and many CF problems want.

// --- 4-CYCLES:  $O(m \sqrt{m})$  with the same orientation. ---
// For each u, walk two steps (u -> v -> w) in the oriented graph
//   ↳ and count how many times each w
// is reached; a pair of distinct paths u->w gives a 4-cycle. Sum
//   ↳  $C(cnt[w], 2)$ .
ll countFourCycles(int n, const vector<pair<int, int>>& es) {
    vector<int> deg(n, 0);
    for (auto [u, v] : es) deg[u]++, deg[v]++;
    auto better = [&](int a, int b) { return deg[a] != deg[b] ?
    ↳ deg[a] < deg[b] : a < b; };
    vector<vector<int>> adj(n), out(n);
    for (auto [u, v] : es) {
        adj[u].push_back(v), adj[v].push_back(u);
        better(u, v) ? out[u].push_back(v) : out[v].push_back(u);
    }
    vector<ll> cnt(n, 0);
    ll r = 0;
    for (int u = 0; u < n; u++) {
        for (int v : adj[u]) for (int w : out[v]) if (better(u,
        ↳ w)) r += cnt[w]++;
        for (int v : adj[u]) for (int w : out[v]) if (better(u,
        ↳ w)) cnt[w] = 0;
    }
    return r;

// WHY IT IS  $O(m \sqrt{m})$ : the inner loop runs sum over v of deg(v)
//   ↳ * outdeg(v)  $\leq O(m \sqrt{m})$ .
//
// OTHER COUNTING FACTS WORTH HAVING:
// * #triangles =  $\text{trace}(A^3) / 6$  for the adjacency matrix A;
//   ↳ with bitsets that is  $O(n^3 / 64)$  and
//   ↳ is often the shorter answer when  $n \leq 2000$  (also gives
//   ↳ per-pair common-neighbour counts).
// * #paths of length 2 = sum over v of  $C(\text{deg } v, 2)$ . Subtract 3
//   ↳ * #triangles for "paths that are
//   ↳ not part of a triangle".
// * A graph is TRIANGLE-FREE  $\Rightarrow m \leq n^2/4$  (Mantel / Turan
//   ↳ for  $K_3$ ). More generally Turan:
//   ↳ a  $K_{r+1}$ -free graph has at most  $(1 - 1/r) n^2 / 2$  edges.
// * #closed walks of length k =  $\text{trace}(A^k)$  - use matrix power
//   ↳ for k up to  $1e18$  on small n.
// * COUNTING CYCLES OF LENGTH  $\leq 5$  is polynomial by similar
//   ↳ tricks; length  $\geq 6$  is as hard as
//   ↳ matrix multiplication in general. For counting ALL simple
//   ↳ cycles use inclusion-exclusion
//   ↳ over subsets ( $2^n$ ) or DP over bitmask start-vertex.
// * GIRTH (shortest cycle) of an unweighted graph: BFS from
//   ↳ every vertex,  $O(nm)$ . For the
//   ↳ shortest cycle THROUGH a given edge, delete it and run BFS.
```

3.4 Satisfiability

3.4.1 TwoSat

2-SAT via Kosaraju - $O(n + m)$. Self-contained, 0-based variables.

```
// Literal encoding: 2*i = (x_i false), 2*i+1 = (x_i true).
// Satisfiable iff x and !x are never in the same SCC; then take
//   ↳ the literal whose SCC is LATER
// in topological order.
struct TwoSat {
    int n;
    vector<vector<int>> adj, radj;
    vector<int> ord, comp;
    vector<bool> used, val;

    TwoSat(int n = 0) : n(n), adj(2 * n), radj(2 * n) {}
    int lit(int i, bool t) { return 2 * i + t; }
    int add_var() { n++; adj.resize(2 * n), radj.resize(2 * n);
    ↳ return n - 1; }

    void imply(int a, int b) {                                     //
    ↳ literal a => literal b (and contrapositive)
        adj[a].push_back(b), radj[b].push_back(a);
        adj[b ^ 1].push_back(a ^ 1), radj[a ^ 1].push_back(b ^ 1);
    }
    void either(int a, int b) { imply(a ^ 1, b); }               // (a OR
    ↳ b)
    void force(int a) { imply(a ^ 1, a); }                       // a must
    ↳ be true
    void nand_(int a, int b) { either(a ^ 1, b ^ 1); }           // not
    ↳ both
    void eq(int a, int b) { imply(a, b), imply(b, a); } // a <=> b
    void xor_(int a, int b) { eq(a, b ^ 1); }

    // AT MOST ONE of 'lits' is true, with O(k) edges instead of
    ↳ O(k^2) (prefix/"lightbulb" trick):
    // aux_i means "someone at index <= i was chosen".
    void at_most_one(const vector<int>& lits) {
        int prev = -1;
        for (int c : lits) {
            int cur = lit(add_var(), true);
            imply(c, cur);
            if (prev != -1) imply(prev, cur), imply(c, prev ^ 1);
            prev = cur;
        }
    }

    void dfs1(int u) { used[u] = true; for (int v : adj[u]) if
    ↳ (!used[v]) dfs1(v); ord.push_back(u); }
    void dfs2(int u, int c) { comp[u] = c; for (int v : radj[u])
    ↳ if (comp[v] < 0) dfs2(v, c); }

    bool solve() {
        used.assign(2 * n, false), comp.assign(2 * n, -1),
    ↳ val.assign(n, false), ord.clear();
        for (int i = 0; i < 2 * n; i++) if (!used[i]) dfs1(i);
        int c = 0;
        for (int i = 2 * n - 1; i >= 0; i--) if (comp[ord[i]] <
    ↳ 0) dfs2(ord[i], c++);
        for (int i = 0; i < n; i++) {
            if (comp[2 * i] == comp[2 * i + 1]) return false;
            val[i] = comp[2 * i + 1] > comp[2 * i];
        }
        return true;                                             // read
    ↳ val[0..base_n-1]; aux vars are padding
    }
};

// RANGE CLAUSE ("if x then ALL of [l,r]") in O(log n) edges:
//   ↳ build a segment tree over the
// variables, give every internal node an aux variable, add parent
//   ↳ => both children, then
// imply(x, node) for the O(log n) covering nodes. Mirror the tree
//   ↳ (child => parent) for
// "if any of [l,r] then x".
```

3.5 Eulerian Path

3.5.1 EulerianPath

Hierholzer's Algorithm for Eulerian Path / Circuit - $O(V + E)$

// Works for both Directed and Undirected graphs

```
struct EulerianPath {
    int n, m;
    vector<vector<pair<int, int>>> adj; // {neighbor, edge_id}
    vector<bool> used_edge;
    vector<int> in_deg, out_deg, path;

    EulerianPath(int n = 0, int m = 0) : n(n), m(m), adj(n + 1),
    ↳ used_edge(m, false), in_deg(n + 1, 0), out_deg(n + 1, 0) {}

    void add_edge(int u, int v, int edge_id, bool directed =
    ↳ true) {
        adj[u].pb({v, edge_id});
        out_deg[u]++;
        in_deg[v]++;
        if (!directed) {
            adj[v].pb({u, edge_id});
            out_deg[v]++;
            in_deg[u]++;
        }
    }

    // Finds Eulerian path starting at start_node. Returns false
    ↳ if graph is not Eulerian
    bool solve(int start_node) {
        vector<int> ptr(n + 1, 0);
        stack<int> st;
        st.push(start_node);

        while (!st.empty()) {
            int u = st.top();
            if (ptr[u] < adj[u].size()) {
                auto [v, id] = adj[u][ptr[u]++];
                if (!used_edge[id]) {
                    used_edge[id] = true;
                    st.push(v);
                }
            } else {
                path.pb(u);
                st.pop();
            }
        }
        reverse(all(path));

        // Verify that every edge was traversed
        for (bool used : used_edge) {
            if (!used) return false;
        }
        return true;
    }
};
```

3.6 Flow and Min Cut

3.6.1 Dinic

DINIC MAX FLOW. $O(V^2 E)$ in general, $O(E \sqrt{V})$ on unit-capacity graphs, $O(V^2 E)$ worst case but

```
// fast in practice. After maxflow, vertices reachable from s in
//   ↳ the residual graph form the min cut.
// Scaling (start with big capacities) helps on dense graphs with
//   ↳ large capacities.
// DINIC MAX FLOW.  $O(V^2 E)$  general,  $O(E \sqrt{E})$  with unit
//   ↳ capacities,  $O(E \sqrt{V})$  for bipartite
// matching. This is the SCALING variant (big capacities first),
//   ↳ so it stays fast on wide ranges.
// addEdge(a, b, cap, rcap): pass rcap = cap for an UNDIRECTED
//   ↳ edge. Edge::flow() recovers the flow
// on each arc. After calc(s,t), leftOfMinCut(v) is true exactly
//   ↳ for the SOURCE side of a minimum
// cut - the cut edges are those from a left vertex to a right
//   ↳ vertex. 0-indexed.
struct Dinic {
    struct Edge {
        int to, rev;
        ll c, oc;
        ll flow() { return max(oc - c, 0LL); } // if you need flows
    };

    vector<int> lvl, ptr, q;
    vector<vector<Edge>> > adj;

    Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {}

    void addEdge(int a, int b, ll c, ll rcap = 0) {
        adj[a].push_back({b, (int) adj[b].size(), c, c});
        adj[b].push_back({a, (int) adj[a].size() - 1, rcap,
↳ rcap});
    }

    ll dfs(int v, int t, ll f) {
        if (v == t || !f) return f;
        for (int &i = ptr[v]; i < (int) adj[v].size(); i++) {
            Edge &e = adj[v][i];
            if (lvl[e.to] == lvl[v] + 1)
                if (ll p = dfs(e.to, t, min(f, e.c))) {
                    e.c -= p, adj[e.to][e.rev].c += p;
                    return p;
                }
        }
        return 0;
    }

    ll calc(int s, int t) {
        ll flow = 0;
        q[0] = s;
        for (int L = 0; L < 31; L++)
            do {
                // 'int L=30' maybe faster for random data
                lvl = ptr = vector<int>((int) q.size());
                int qi = 0, qe = lvl[s] = 1;
                while (qi < qe && !lvl[t]) {
                    int v = q[qi++];
                    for (Edge e: adj[v])
                        if (!lvl[e.to] && e.c >> (30 - L))
                            q[qi++] = e.to, lvl[e.to] = lvl[v] +
↳ 1;
                }
                while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
            } while (lvl[t]);
        return flow;
    }

    bool leftOfMinCut(int a) { return lvl[a] != 0; }
};
```

3.6.2 MCMF

MIN-COST MAX-FLOW with JOHNSON POTENTIALS (Dijkstra instead of SPFA on every augmentation).

```
//  $O(F * E \log V)$  where  $F$  = number of augmentations. Much faster
//   ↳ than the SPFA version and the
// one to use by default. Negative edge costs are allowed at
//   ↳ construction PROVIDED there is no negative-cost CYCLE:
// run one Bellman-Ford
// to initialise the potentials (setPotentialsBF), after that
//   ↳ every reduced cost is  $\geq 0$ .
struct MCMF {
    struct E { int to, rev; ll cap, cost, flow = 0; };
    int n;
    vector<vector<E>> g;
    vector<ll> pot, dist;
    vector<int> pv, pe;

    MCMF(int n) : n(n), g(n), pot(n, 0), dist(n), pv(n), pe(n) {}
    void addEdge(int u, int v, ll cap, ll cost) {
        g[u].push_back({v, (int) g[v].size(), cap, cost});
        g[v].push_back({u, (int) g[u].size() - 1, 0, -cost});
    }
    void setPotentialsBF(int s) { // only
↳ needed if some cost < 0
        fill(all(pot), llinf);
        pot[s] = 0;
        for (int it = 0; it < n; it++) {
            bool ch = false;
            for (int u = 0; u < n; u++) if (pot[u] < llinf)
                for (auto& e : g[u]) if (e.cap - e.flow > 0 &&
↳ pot[u] + e.cost < pot[e.to])
                    pot[e.to] = pot[u] + e.cost, ch = true;
            if (!ch) break;
        }
        for (auto& x : pot) if (x == llinf) x = 0;
    }
    bool dijkstra(int s, int t) {
        fill(all(dist), llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
↳ greater<>> > pq;
        dist[s] = 0, pq.push({0, s});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int i = 0; i < (int) g[u].size(); i++) {
                auto& e = g[u][i];
                if (e.cap - e.flow <= 0) continue;
                ll nd = d + e.cost + pot[u] - pot[e.to]; //
↳ reduced cost, always  $\geq 0$ 
                if (nd < dist[e.to]) dist[e.to] = nd, pv[e.to] =
↳ u, pe[e.to] = i, pq.push({nd, e.to});
            }
        }
        return dist[t] < llinf;
    }
    // Returns {flow, cost}. Pass maxf to cap the flow (e.g. for
↳ min-cost k-flow).
    pair<ll, ll> run(int s, int t, ll maxf = llinf) {
        ll flow = 0, cost = 0;
        while (flow < maxf && dijkstra(s, t)) {
            for (int i = 0; i < n; i++) if (dist[i] < llinf)
↳ pot[i] += dist[i];
            ll aug = maxf - flow;
            for (int v = t; v != s; v = pv[v]) aug = min(aug,
↳ g[pv[v]][pe[v]].cap - g[pv[v]][pe[v]].flow);
            for (int v = t; v != s; v = pv[v]) {
                auto& e = g[pv[v]][pe[v]];
                e.flow += aug, g[v][e.rev].flow -= aug;
                cost += aug * e.cost;
            }
            flow += aug;
        }
        return {flow, cost};
    }
    // MIN-COST flow of exactly K units: run(s, t, K) and check
↳ flow == K.
    // MIN-COST (not max) flow: stop as soon as dist[t] + pot[t] -
↳ pot[s] > 0 (no profitable path).
};
// MODELLING: assignment on a SPARSE graph; for a dense/complete
↳ one use 07 - Matching/03 - Hungarian * transportation * "k
↳ disjoint paths of minimum total cost" *
// min-cost bipartite matching * scheduling with penalties * MCMF
↳ is also how you solve
// "choose k items with a matroid-ish constraint minimising cost".
```


3.6.3 FlowWithLowerBounds

FLows WITH LOWER BOUNDS (circulation with demands). Every edge needs $d(e) \leq f(e) \leq c(e)$.

```
// Build an auxiliary network, run one max flow; feasible iff
//   ↳ every edge out of s' saturates.
//   ↳ excess(v) = (sum of d over edges INTO v) - (sum of d over
//   ↳ edges OUT of v)
//   ↳ excess(v) > 0 -> edge s' -> v of capacity excess(v); < 0 ->
//   ↳ edge v -> t' of capacity -excess
//   ↳ every original edge (u,v) keeps capacity c - d
//   ↳ For an s-t flow, add t->s with capacity INF first, which turns
//   ↳ it into a circulation.
// Needs Dinic (01 - Dinic.cpp) with addEdge(u, v, cap) and
//   ↳ calc(s, t).
struct LowerBoundFlow {
    int n, S, T; // S, T are
    ↳ the auxiliary super source/sink
    Dinic din;
    vector<ll> excess, low;
    vector<array<int, 2>> eid; // per
    ↳ original edge: {u, index in din.adj[u]}

    LowerBoundFlow(int n) : n(n), S(n), T(n + 1), din(n + 2),
    ↳ excess(n + 2, 0) {}
    void addEdge(int u, int v, ll lo, ll hi) {
        low.push_back(lo);
        eid.push_back({u, (int)din.adj[u].size()});
        din.addEdge(u, v, hi - lo);
        excess[v] += lo, excess[u] -= lo;
    }
    bool feasible() { //
    ↳ circulation only (no s, t)
        ll need = 0;
        for (int v = 0; v < n; v++) {
            if (excess[v] > 0) din.addEdge(S, v, excess[v]), need
    ↳ += excess[v];
            else if (excess[v] < 0) din.addEdge(v, T, -excess[v]);
        }
        return din.calc(S, T) == need;
    }
    // Min / max s-t flow subject to the bounds - see the RECIPES
    ↳ block below for the ONLY
    // correct order of operations. You must DELETE the t->s edge
    ↳ (zero both residual directions)
    // before the second max flow, otherwise Dinic's first
    ↳ augmenting path runs s -> t backwards
    // along that arc and CANCELS the feasible flow instead of
    ↳ extending it.
    // feasible() appends the S/T edges every time it is called -
    ↳ call it exactly once.
    ll flowOn(int i) { return low[i] +
    ↳ din.adj[eid[i][0]][eid[i][1]].flow(); }
};

/* RECIPES
"each edge must be used at least once" -> d(e) = 1
"each vertex must be visited between L and R" -> split v into
↳ v_in -> v_out with [L, R]
"exact degree subgraph" -> d = c =
↳ required degree on the split edge
MAX s-t FLOW WITH BOUNDS: add t->s with capacity INF, run
↳ feasible(); then delete that edge
and run one more max flow from s to t on the residual graph.
↳ The answer is the flow that
was on t->s plus the extra augmentation.
MIN s-t FLOW WITH BOUNDS: same, but augment from t to s to
↳ cancel as much as possible. */
```

3.6.4 StoerWagner

STOER-WAGNER GLOBAL MIN CUT - minimum cut over ALL vertex pairs, undirected weighted, $O(V^3)$.

```
// No source/sink. Returns {cut weight, one side of the cut}.
// Key lemma: in a maximum-adjacency order, if s is the
//   ↳ second-to-last and t the last vertex
// added, then the minimum s-t cut is exactly {t}. Merge s and t
//   ↳ and repeat n-1 times.
pair<ll, vector<int>> globalMinCut(vector<vector<ll>> w) {
    int n = w.size();
    if (n < 2) return {0, {}};
    vector<char> gone(n, 0);
    vector<vector<int>> grp(n);
    for (int i = 0; i < n; i++) grp[i] = {i};
    ll bestW = llinf;
    vector<int> best;
    for (int phase = 0; phase + 1 < n; phase++) {
        vector<ll> ww(n, 0);
        vector<char> added(n, 0);
        int prev = -1, last = -1;
        for (int it = 0; it < n - phase; it++) { //
    ↳ maximum-adjacency order
            int sel = -1;
            for (int j = 0; j < n; j++)
                if (!gone[j] && !added[j] && (sel < 0 || ww[j] >
    ↳ ww[sel])) sel = j;
            added[sel] = 1, prev = last, last = sel;
            for (int j = 0; j < n; j++) if (!gone[j] &&
    ↳ !added[j]) ww[j] += w[sel][j];
        }
        ll cut = 0; //
    ↳ cut-of-the-phase separates {last}
        for (int j = 0; j < n; j++) if (!gone[j] && j != last)
    ↳ cut += w[last][j];
        if (cut < bestW) bestW = cut, best = grp[last];
        gone[last] = 1; // merge
    ↳ last into prev
        grp[prev].insert(grp[prev].end(), all(grp[last]));
        for (int j = 0; j < n; j++) if (!gone[j] && j != prev)
            w[prev][j] += w[last][j], w[j][prev] = w[prev][j];
    }
    return {bestW, best};
}

// USES: "split the network into two non-empty parts as cheaply as
//   ↳ possible", edge connectivity
// of a weighted undirected graph, "minimum number of edges to
//   ↳ delete to disconnect the graph"
// (unit weights). Doing this with n-1 max-flows is both slower
//   ↳ and longer to write.
// KARGER (randomized contraction) finds a min cut with
//   ↳ probability  $\geq 1/C(n,2)$  per run; the
// useful corollary is that a graph has AT MOST  $C(n,2)$  distinct
//   ↳ minimum cuts.
```

3.6.5 MinCutModelling

MIN-CUT MODELLING - the reason flow appears at ICPC. The code is never the hard part; choosing

```
// Needs: Dinic (01 - Dinic.cpp).
// what the vertices and the infinite capacities mean is. Rule of
  ↳ thumb: if the objective contains
// an "infinity", you are CUTTING, not flowing.

// --- MAXIMUM WEIGHT CLOSURE / PROJECT SELECTION ---
// Choose a set S closed under the dependency arcs (if v in S and
  ↳ v -> u is a dependency, u in S)
// maximising the total weight. Build: s -> v with capacity w(v)
  ↳ for every w(v) > 0;
// v -> t with capacity -w(v) for every w(v) < 0; u -> v with INF
  ↳ for every dependency u needs v.
// Answer = (sum of positive weights) - maxflow. S = the source
  ↳ side of the min cut.
// USES: "each project earns p but needs these machines that cost
  ↳ c"; "pick cells maximising
// profit but a chosen cell forces its neighbours";
  ↳ densest-subgraph style selections.
11 maxClosure(int n, const vector<ll>& w, const vector<pair<int,
  ↳ int>>& need, vector<int>* chosen = nullptr) {
    Dinic din(n + 2);
    int s = n, t = n + 1;
    ll pos = 0;
    for (int v = 0; v < n; v++)
        if (w[v] > 0) pos += w[v], din.addEdge(s, v, w[v]);
        else if (w[v] < 0) din.addEdge(v, t, -w[v]);
    for (auto [u, v] : need) din.addEdge(u, v, llinf); //
    ↳ choosing u forces choosing v
    ll cut = din.calc(s, t);
    if (chosen) for (int v = 0; v < n; v++) if
    ↳ (din.leftOfMinCut(v)) chosen->push_back(v);
    return pos - cut;
}
/* THE OTHER STANDARD REDUCTIONS
TWO LABELS WITH PAIR PENALTIES ("cats or dogs", image
  ↳ segmentation)
Every vertex gets label A or B. Cost a_v for A, b_v for B, and
  ↳ c_uv extra if u and v differ.
Build s -> v with b_v, v -> t with a_v, and u -> v with c_uv
  ↳ in both directions.
The min cut is the minimum total cost; the source side takes
  ↳ label A.
PRECONDITION: the pair penalty must be SUBMODULAR (cost(A,A) +
  ↳ cost(B,B) <= cost(A,B) +
  ↳ cost(B,A)). A penalty that rewards disagreement cannot be
  ↳ modelled by a cut - it is NP-hard.

VERTEX CAPACITIES / VERTEX-DISJOINT PATHS
Split v into v_in -> v_out with the vertex capacity.
  ↳ Vertex-disjoint s-t paths use capacity 1.

BIPARTITE FAMILY (see 07 - Matching and 10 - Theorems)
max matching = max flow with unit capacities
min vertex cover = max matching (Konig) -> recover
  ↳ via the alternating-BFS set
max independent set = n - max matching
min path cover of a DAG (vertex-disjoint) = n - max matching of
  ↳ the split graph

"MUST BE ON THE SOURCE SIDE" -> add s -> v with INF. "MUST BE
  ↳ ON THE SINK SIDE" -> v -> t INF.
That is how you force decisions inside an otherwise free min cut.

EDGE / VERTEX DISJOINT PATHS = Menger. Recover the paths by
  ↳ decomposing the integral flow.

LOWER BOUNDS ON EDGES -> 03 - FlowWithLowerBounds.cpp.

MAXIMUM DENSITY SUBGRAPH (maximise |E(S)| / |S|): binary search
  ↳ lambda, then min cut with
s -> v capacity m, v -> t capacity m + 2*lambda - deg(v), u ->
  ↳ v capacity 1. Feasible iff
the min cut is < n*m. O(log(n^2) * maxflow) since the density
  ↳ is a ratio of integers <= n^2.

MINIMUM CUT WITHOUT s AND t (global) -> 04 - StoerWagner.cpp,
  ↳ O(V^3), or n-1 max flows.

ALL-PAIRS MIN CUT -> the Gomory-Hu tree: n-1 max flows build a
  ↳ weighted tree in which the min
u-v cut equals the MINIMUM EDGE on the u-v tree path. Use it
  ↳ when many pairs are asked.

COMPLEXITY TABLE - read this at minute 3, not minute 90
```

Dinic	$O(V^2 E)$ general / $O(E \sqrt{E})$ unit caps	
↳ $O(E \sqrt{V})$ bipartite		
Edmonds-Karp	$O(V E^2)$	-- do not
Push-relabel / HLPP	$O(V^2 \sqrt{E})$	-- dense
↳ graphs, $n \sim 1000$, $m \sim n^2/2$		
MCMF (SSP + potentials)	$O(F * E \log V)$	-- F = total
↳ flow, so bound F first		
Hungarian	$O(n^2 m)$	-- dense
↳ assignment, $n \leq \sim 1000$		
Hopcroft-Karp	$O(E \sqrt{V})$	-- bipartite
↳ matching, $n \geq$ few thousand		
Kuhn	$O(V E)$	-- shorter,
↳ fine to ~ 2000 vertices		
Blossom (general)	$O(V^3)$	
Stoer-Wagner	$O(V^3)$	
Gomory-Hu	$O(V * \text{maxflow})$	

If the flow value itself can be huge (capacities up to $1e9$), an
 ↳ $O(F \dots)$ bound is not a bound -
 scale the capacities or use a strongly polynomial algorithm. */

3.6.6 GomoryHu

GOMORY-HU TREE - ALL-PAIRS MINIMUM CUT from only n-1 max-flow runs.

```
// Needs: Dinic (01 - Dinic.cpp).
// The result is a weighted tree on the same vertex set with the
  ↳ property that, for EVERY pair
// (u, v), the minimum u-v cut in the original graph equals the
  ↳ MINIMUM EDGE WEIGHT on the u-v
// path in the tree. (There are at most n-1 distinct min-cut
  ↳ values in any graph - that is the
// whole content of the theorem.)
// Reach for it when a problem asks about min cuts between many
  ↳ pairs: "sum/max/k-th of all
// pairwise min cuts", "order the vertices to maximise the sum of
  ↳ consecutive cuts", "how many
// pairs have min cut >= k". Undirected graphs only.
// Complexity: (n-1) x Dinic. The tree is returned as parent +
  ↳ weight arrays (par[0] = -1).
struct GomoryHu {
    int n;
    vector<array<ll, 3>> edges; //
    ↳ {u, v, capacity}, undirected
    vector<int> par;
    vector<ll> w;
    GomoryHu(int n) : n(n), par(n, 0), w(n, 0) {}
    void addEdge(int u, int v, ll c) { edges.push_back({(ll)u,
    ↳ (ll)v, c}); }
    void build() {
        par.assign(n, 0), w.assign(n, 0), par[0] = -1;
        for (Dinic din(n);
            for (auto [u, v, c] : edges) din.addEdge(u, v, c, c);
        ↳ // undirected: both directions
        w[i] = din.calc(i, par[i]);
        for (int j = i + 1; j < n; j++)
        ↳ // re-parent the same-side vertices
            if (par[j] == par[i] && din.leftOfMinCut(j))
        par[j] = i;
    }
    ll minCut(int u, int v) const { //
        ↳ min edge on the tree path
        ll best = llinf;
        vector<int> du(n, 0);
        for (int x = u; x != -1; x = par[x]) du[x] = 1;
        vector<int> pv;
        int m = v;
        while (!du[m]) pv.push_back(m), m = par[m];
        ↳ // m = the meeting vertex
        for (int x = u; x != m; x = par[x]) best = min(best,
        ↳ w[x]);
        for (int x : pv) best = min(best, w[x]);
        return best;
    }
};
// EQUIVALENT-FLOW TREE (Gusfield's variant) is the same n-1 flows
  ↳ without the re-parenting step;
// it gets the min-cut VALUES right but not the actual cuts. The
  ↳ version above is the real
// Gomory-Hu, so the tree edges are genuine cuts.
// GLOBAL MIN CUT is the minimum edge of the whole tree - but
  ↳ Stoer-Wagner (04) is  $O(V^3)$  and
// needs no flow at all, so prefer it when that is all you want.
// DIRECTED graphs have no Gomory-Hu tree; you genuinely need
  ↳  $O(n^2)$  flows there.
```

3.7 Matching

3.7.1 HopcroftKarp

HOPCROFT-KARP - maximum bipartite matching in $O(E \sqrt{V})$, the fast alternative to Kuhn's $O(VE)$.

```
// Use it when n is above a few thousand; below that Kuhn is
// shorter and fast enough.
// HOPCROFT-KARP - maximum bipartite matching in  $O(E \sqrt{V})$ , the
// fast alternative to Kuhn's
//  $O(VE)$ . Use above a few thousand vertices; below that Kuhn ( $O2$ )
// is shorter and fast enough.
// INDEXING TRAP: leftMatch has size m and is indexed by a RIGHT
// vertex; rightMatch has size n and
// is indexed by a LEFT vertex. Reading the names the obvious way
// is the default way to get this
// wrong. For the minimum vertex cover / maximum independent set
// (Konig), see 02 - KuhnAndKonig.
struct HopcroftKarp {
    vector<int> leftMatch, rightMatch, dist, cur;
    vector<vector<int>> > a;
    int n, m;

    HopcroftKarp() {}

    HopcroftKarp(int n, int m) {
        this->n = n;
        this->m = m;
        a = vector<vector<int>> >(n);
        leftMatch = vector<int>(m, -1);
        rightMatch = vector<int>(n, -1);
        dist = vector<int>(n, -1);
        cur = vector<int>(n, -1);
    }

    void addEdge(int x, int y) {
        a[x].push_back(y);
    }

    int bfs() {
        int found = 0;
        queue<int> q;
        for (int i = 0; i < n; i++)
            if (rightMatch[i] < 0) dist[i] = 0, q.push(i);
            else dist[i] = -1;

        while (!q.empty()) {
            int x = q.front();
            q.pop();
            for (int i = 0; i < int(a[x].size()); i++) {
                int y = a[x][i];
                if (leftMatch[y] < 0) found = 1;
                else if (dist[leftMatch[y]] < 0)
                    dist[leftMatch[y]] = dist[x] + 1,
                    q.push(leftMatch[y]);
            }
        }

        return found;
    }

    int dfs(int x) {
        for (; cur[x] < int(a[x].size()); cur[x]++) {
            int y = a[x][cur[x]];
            if (leftMatch[y] < 0 || (dist[leftMatch[y]] ==
                dist[x] + 1 && dfs(leftMatch[y]))) {
                leftMatch[y] = x;
                rightMatch[x] = y;
                return 1;
            }
        }
        return 0;
    }

    int maxMatching() {
        int match = 0;
        while (bfs()) {
            for (int i = 0; i < n; i++) cur[i] = 0;
            for (int i = 0; i < n; i++)
                if (rightMatch[i] < 0) match += dfs(i);
        }
        return match;
    }
};
```

3.7.2 KuhnAndKonig

Kuhn's bipartite matching, $O(V^2E)$ but tiny and easy to modify. Left size n, right size m.

```
// In practice fast; randomise the adjacency order + greedy
// pre-match if you need more speed.
struct Kuhn {
    int n, m;
    vector<vector<int>> adj; // left -> right
    vector<int> mt; // mt[right] =
    // matched left, or -1
    vector<bool> used;

    Kuhn(int n = 0, int m = 0) : n(n), m(m), adj(n), mt(m, -1) {}
    void add_edge(int u, int v) { adj[u].push_back(v); }

    bool try_(int u) {
        for (int v : adj[u]) if (!used[v]) {
            used[v] = true;
            if (mt[v] < 0 || try_(mt[v])) { mt[v] = u; return
            true; }
        }
        return false;
    }

    int maxMatching() {
        int r = 0;
        vector<int> ord(n);
        iota(all(ord), 0);
        shuffle(all(ord), rng);
        for (int u : ord) { used.assign(m, false); r += try_(u); }
        return r;
    }

    // KONIG: minimum VERTEX COVER = maximum matching. Recover it:
    // alternate from every UNMATCHED left vertex; let Z =
    // reached set.
    // cover = (left \ Z) union (right and Z); max independent
    // set = complement of the cover.
    pair<vector<int>, vector<int>> minVertexCover() {
        vector<int> matchL(n, -1);
        for (int v = 0; v < m; v++) if (mt[v] >= 0) matchL[mt[v]]
            = v;
        vector<bool> zl(n, false), zr(m, false);
        vector<int> st;
        for (int u = 0; u < n; u++) if (matchL[u] < 0) zl[u] =
            true, st.push_back(u);
        while (!st.empty()) {
            int u = st.back(); st.pop_back();
            for (int v : adj[u]) if (!zr[v]) {
                zr[v] = true;
                if (mt[v] >= 0 && !zl[mt[v]]) zl[mt[v]] = true,
                st.push_back(mt[v]);
            }
        }
        vector<int> L, R;
        for (int u = 0; u < n; u++) if (!zl[u]) L.push_back(u);
        for (int v = 0; v < m; v++) if (zr[v]) R.push_back(v);
        return {L, R}; // |L| + |R| ==
    }

    int maxMatching() {
};

/* THEOREMS you actually quote (bipartite graphs):
Konig: max matching = min vertex cover
max independent set = V - min vertex cover
MIN PATH COVER of a DAG (VERTEX-DISJOINT paths covering all
vertices)
= n - max matching of the split graph (u_out -> v_in
for every edge)
CAREFUL: this is the vertex-disjoint cover. To get
Dilworth (FINITE poset)'s width (minimum
number of CHAINS, i.e. paths allowed to share
vertices) run it on the
TRANSITIVE CLOSURE of the DAG first.
Hall: a perfect matching on the left exists iff  $|N(S)| \geq$ 
 $|S|$  for every subset S of the LEFT part X
Dilworth: min chain cover of a poset = longest antichain (build
the DAG, TRANSITIVELY CLOSE
it, then use the path cover above)
Mirsky: the dual - longest chain = minimum number of
antichains partitioning the poset
Gallai:  $\alpha + \tau = n$  always;  $\nu + \rho = n$  only if there
are NO ISOLATED VERTICES
(alpha = max independent set, tau = min vertex cover,
nu = max matching,
rho = min edge cover)
Hall defect:  $\nu(G) = |X| - \max$  over W subset of X of  $(|W| -$ 
 $|N(W)|)$ 
Tutte: G has a perfect matching iff  $\text{odd}(G - U) \leq |U|$  for
every U
```

Tutte-Berge: $nu(G) = (1/2) * \min \text{ over } U \text{ of } (|V| + |U| - \text{odd}(G \rightarrow -U))$
 Menger: $\min \text{ edge cut}(s,t) = \max \text{ edge-disjoint } s\text{-}t \text{ paths};$
 $\min \text{ vertex cut}(s,t) = \max \text{ internally-vertex-disjoint } s\text{-}t \text{ paths } (s,t \text{ non-adjacent}) *$

3.7.3 Hungarian

HUNGARIAN ALGORITHM - minimum-cost perfect matching on a DENSE bipartite graph, $O(n^2 m)$.

```
// a is 1-indexed: a[1..n][1..m] with n <= m. Returns the cost;
// ans[j] = row matched to column j.
// For MAXIMUM cost, negate the matrix.
pair<ll, vector<int>> hungarian(const vector<vector<ll>>& a) {
    int n = a.size() - 1, m = a[0].size() - 1;
    vector<ll> u(n + 1, 0), v(m + 1, 0);
    vector<int> p(m + 1, 0), way(m + 1, 0);
    for (int i = 1; i <= n; i++) {
        p[0] = i;
        int j0 = 0;
        vector<ll> minv(m + 1, llinf);
        vector<char> used(m + 1, false);
        do {
            used[j0] = true;
            int i0 = p[j0], j1 = 0;
            ll delta = llinf;
            for (int j = 1; j <= m; j++) if (!used[j]) {
                ll cur = a[i0][j] - u[i0] - v[j];
                if (cur < minv[j]) minv[j] = cur, way[j] = j0;
                if (minv[j] < delta) delta = minv[j], j1 = j;
            }
            for (int j = 0; j <= m; j++)
                if (used[j]) u[p[j]] += delta, v[j] -= delta;
            else minv[j] -= delta;
            j0 = j1;
        } while (p[j0] != 0);
        do { int j1 = way[j0]; p[j0] = p[j1], j0 = j1; } while
        (j0);
    }
    vector<int> ans(m + 1);
    for (int j = 1; j <= m; j++) ans[j] = p[j];
    return {-v[0], ans}; // -v[0] is
    // the optimal total cost
}
// Use it for: assignment problems, "match every task to a worker
//   -> minimising total time",
// minimum-cost perfect matching in a complete bipartite graph.
// FORBIDDEN PAIRS: use a large finite cost (1e15), NOT llinf -
//   -> the reduced cost a[i][j]-u[i]-v[j]
// would overflow. ans[j] == 0 means column j is unmatched
//   -> (possible only when n < m).
// WHICH ONE: on a DENSE / complete graph this beats MCMF - n^2*m
//   -> here versus n augmentations of
// Dijkstra over n^2 arcs (n^3 log n) there. Use MCMF only when
//   -> the graph is genuinely SPARSE.
```

3.7.4 Blossom

MAXIMUM MATCHING IN A GENERAL (NON-BIPARTITE) GRAPH - Edmonds' blossom algorithm, $O(V^3)$.

```
// Everything else in this folder assumes bipartite. Reach for
//   -> this the moment the two sides are
// not fixed: "pair up people who are compatible", "tile a board
//   -> where the colouring argument
// fails", "maximum set of disjoint edges" on an arbitrary graph.
// The trick it exists for: an ODD cycle can be traversed in two
//   -> directions, so an alternating
// search can enter and leave it on the same side. Edmonds
//   -> CONTRACTS such a cycle (a "blossom")
// into a single vertex, searches the contracted graph, and
//   -> expands afterwards.
// Certificate: the answer equals the Tutte-Berge bound (see 10 -
//   -> Theorems/01 - GraphTheorems).
struct Blossom {
    int n;
    vector<vector<char>> g; //
    // adjacency MATRIX
    vector<int> match, p, base;
    vector<char> used, inBlossom;
    Blossom(int n) : n(n), g(n, vector<char>(n, 0)), match(n,
    // -1), p(n), base(n),
    // used(n), inBlossom(n) {}
    void addEdge(int u, int v) { g[u][v] = g[v][u] = 1; }
    int lca(int a, int b) {
        vector<char> seen(n, 0);
        for (;;) { a = base[a], seen[a] = 1; if (match[a] < 0)
        // break; a = p[match[a]]; }
        for (;;) { b = base[b]; if (seen[b]) return b; b =
        // p[match[b]]; }
    }
    void mark(int v, int b, int child) { //
    // walk v up to the blossom base
        while (base[v] != b) {
            inBlossom[base[v]] = inBlossom[base[match[v]]] = 1;
            p[v] = child, child = match[v], v = p[match[v]];
        }
    }
    bool augment(int root) { //
    // one BFS for an augmenting path
        fill(all(used), 0), fill(all(p), -1);
        iota(all(base), 0);
        used[root] = 1;
        queue<int> q({root});
        while (!q.empty()) {
            int v = q.front(); q.pop();
            for (int to = 0; to < n; to++) {
                if (!g[v][to] || base[v] == base[to] || match[v]
                // == to) continue;
                if (to == root || (match[to] >= 0 && p[match[to]]
                // >= 0)) { // odd cycle -> contract
                    int b = lca(v, to);
                    fill(all(inBlossom), 0);
                    mark(v, b, to), mark(to, b, v);
                    for (int i = 0; i < n; i++) if
                    // (inBlossom[base[i]]) {
                        base[i] = b;
                        if (!used[i]) used[i] = 1, q.push(i);
                    }
                } else if (p[to] < 0) {
                    p[to] = v;
                    if (match[to] < 0) {
                    // free vertex: flip the path
                        for (int u = to, pv, nv; u >= 0; u = nv)
                            pv = p[u], nv = match[pv], match[u] =
                    // pv, match[pv] = u;
                            return true;
                    }
                    used[match[to]] = 1, q.push(match[to]);
                }
            }
        }
        return false;
    }
    int solve() {
        int r = 0;
        for (int v = 0; v < n; v++) //
        // greedy start: fewer BFS rounds
            if (match[v] < 0)
                for (int u = 0; u < n; u++)
                    if (g[v][u] && match[u] < 0) { match[v] = u,
                    // match[u] = v, r++; break; }
                for (int v = 0; v < n; v++) if (match[v] < 0) r +=
                    // augment(v);
                    return r; //
    }
```

```

    ↪ match[v] = partner, or -1
}
};
// PERFECT MATCHING EXISTS iff solve() * 2 == n. Tutte's theorem
    ↪ gives the certificate when it
// does not: some U with odd(G - U) > |U|.
// MINIMUM EDGE COVER (no isolated vertices) = n - maximum
    ↪ matching (Gallai).
// MAXIMUM INDEPENDENT SET is still NP-hard here - Konig's alpha =
    ↪ n - nu is BIPARTITE ONLY.
// WEIGHTED general matching needs the dual-variable version (~150
    ↪ lines); if the graph is small
// (n <= 20) a bitmask DP over matched vertices is far shorter and
    ↪ usually enough.
// CHINESE POSTMAN (undirected): make every degree even by
    ↪ duplicating a minimum-weight perfect
// matching on the ODD-degree vertices, using all-pairs shortest
    ↪ paths as the edge weights, then
// take an Eulerian circuit. The MIXED version (some edges
    ↪ directed) is NP-hard.
// THE MATCHING GAME (a token moves along edges to unvisited
    ↪ vertices; stuck loses): the first
// player wins iff the start vertex is covered by EVERY maximum
    ↪ matching, i.e. iff deleting it
// decreases the matching size. With this file that works on a
    ↪ general graph, not just bipartite.

```

3.8 Spanning Trees

3.8.1 MST

MINIMUM SPANNING TREE. Kruskal $O(m \log m)$ - use on sparse graphs / when you also want the

```

// Needs: DSU (02 - Data Structures/05 - DSU/01).
// KRUSKAL RECONSTRUCTION TREE. Prim  $O(n^2)$  - use on dense or
    ↪ IMPLICIT complete graphs.
struct Edge { int u, v; ll w; bool operator<(const Edge& o) const
    ↪ { return w < o.w; } };

ll kruskal(int n, vector<Edge> e, vector<Edge>* used = nullptr) {
    sort(all(e));
    DSU d(n);
    ll tot = 0;
    for (auto& x : e) if (d.join(x.u, x.v)) { tot += x.w; if
    ↪ (used) used->push_back(x); }
    return tot; // check
    ↪ d.size(0) == n for connectivity
}
// Dense / implicit graphs: cost(i,j) given by a function.  $O(n^2)$ ,
    ↪ no edge list needed.
template <class F>
ll prim(int n, F cost) {
    vector<ll> best(n, llinf);
    vector<bool> in(n, false);
    best[0] = 0;
    ll tot = 0;
    for (int it = 0; it < n; it++) {
        int u = -1;
        for (int i = 0; i < n; i++) if (!in[i] && (u < 0 ||
    ↪ best[i] < best[u])) u = i;
        in[u] = true, tot += best[u];
        for (int v = 0; v < n; v++) if (!in[v]) best[v] =
    ↪ min(best[v], cost(u, v));
    }
    return tot;
}
// KRUSKAL RECONSTRUCTION TREE: binary tree with 2n-1 nodes;
    ↪ internal node = merging edge weight.
// Then min possible MAX edge on a path u->v = val[lca(u,v)], and
    ↪ "all vertices reachable using
// edges of weight <= w" is exactly a subtree. Turns "binary
    ↪ search the threshold + DSU" into  $O(1)$ .
struct KRT {
    int n, cnt;
    vector<ll> val;
    vector<array<int, 2>> ch;
    KRT(int n, vector<Edge> e) : n(n), cnt(n), val(2 * n, 0),
    ↪ ch(2 * n, {-1, -1}) {
        sort(all(e));
        DSU d(2 * n);
        vector<int> top(2 * n);
        iota(all(top), 0);
        for (auto& x : e) {
            int a = top[d.find(x.u)], b = top[d.find(x.v)];
            if (a == b) continue;
            val[cnt] = x.w, ch[cnt] = {a, b};
            d.join(x.u, x.v);
            top[d.find(x.u)] = cnt++;
        }
    } // root = cnt-1
    ↪ (if connected); then build LCA on it
};
// SECOND-BEST MST: build the MST, then for each non-tree edge
    ↪ (u,v,w) the answer candidate is
// mst - maxEdgeOnPath(u,v) + w. Get maxEdgeOnPath with binary
    ↪ lifting or the KRT.

```

3.8.2 DirectedMSTandExtras

DIRECTED MST (Chu-Liu / Edmonds) - minimum spanning ARBORESCENCE rooted at r , $O(V \cdot E)$.

```
// Needs: struct Edge {int u,v; ll w;} and DSU (both from 01 -
// MST.cpp).
// Every non-root vertex takes its cheapest incoming edge; if that
// creates a cycle, contract it,
// subtract the chosen weights, and repeat. Returns -1 if some
// vertex is unreachable from r.
ll directedMST(int n, int r, vector<Edge> e) { // Edge =
    // {u, v, w} meaning u -> v
    ll total = 0;
    vector<int> id(n), par(n);
    while (true) {
        vector<ll> best(n, llinf);
        vector<int> from(n, -1);
        for (auto& x : e) if (x.u != x.v && x.w < best[x.v])
            best[x.v] = x.w, from[x.v] = x.u;
        best[r] = 0;
        for (int i = 0; i < n; i++) if (i != r && from[i] < 0)
            return -1; // unreachable
        int cnt = 0;
        fill(all(id), -1), fill(all(par), -1);
        for (int i = 0; i < n; i++) {
            if (i != r) total += best[i];
            int v = i;
            while (v != r && par[v] == -1 && id[v] == -1) par[v]
                = i, v = from[v];
            if (v != r && id[v] == -1 && par[v] == i) { // found
                // a new cycle
                for (int u = from[v]; u != v; u = from[u]) id[u]
                    = cnt;
                id[v] = cnt++;
            }
            if (!cnt) break; // no
        } // cycle -> done
        for (int i = 0; i < n; i++) if (id[i] == -1) id[i] =
            cnt++;
        for (auto& x : e) {
            int u = x.u, v = x.v;
            x.u = id[u], x.v = id[v];
            if (x.u != x.v) x.w -= best[v]; //
        } // reduced cost inside the contraction
        n = cnt, r = id[r];
    }
    return total;
}

// SECOND-BEST MST,  $O(E \log V + E \cdot \log V)$ . The second-best tree
// differs from the MST by exactly
// ONE edge swap. For each non-tree edge  $(u,v,w)$  the candidate is
//  $mst - \maxEdgeOnPath(u,v) + w$ .
// You must track BOTH the maximum and the strict second maximum
// on the path, because if the
// maximum equals  $w$  the swap is not a real change - use the second
// maximum instead.
// With binary lifting:  $up[k][v]$ ,  $mx1[k][v]$ ,  $mx2[k][v]$  merged
// pairwise while lifting.
// (The Kruskal Reconstruction Tree in 01 - MST.cpp gives
//  $\maxEdgeOnPath$  directly as  $val[lca]$ .)

// BORUVKA - MST in  $O(E \log V)$  by phases; each phase every
// component picks its cheapest outgoing
// edge and they are all merged at once. Use it when the edge set
// is IMPLICIT:
// XOR-MST : cheapest cross edge per component via a binary
// trie
// Manhattan MST: cheapest cross edge per octant (see
// Transforms.cpp)
ll boruvka(int n, const vector<Edge>& e) {
    DSU d(n);
    ll total = 0;
    int comps = n;
    while (comps > 1) {
        vector<int> best(n, -1);
        for (int i = 0; i < (int)e.size(); i++) {
            int a = d.find(e[i].u), b = d.find(e[i].v);
            if (a == b) continue;
            if (best[a] < 0 || e[i].w < e[best[a]].w) best[a] = i;
            if (best[b] < 0 || e[i].w < e[best[b]].w) best[b] = i;
        }
        bool any = false;
        for (int i = 0; i < n; i++) if (best[i] >= 0) {
            auto& x = e[best[i]];
            if (d.join(x.u, x.v)) total += x.w, comps--, any =

```

```
    true;
    }
    if (!any) break;
    //
    // disconnected
    }
    return comps == 1 ? total : -1;
}
```

3.8.3 SecondBestMST

SECOND-BEST SPANNING TREE - cheapest spanning tree whose weight is STRICTLY greater than the MST.

```
// Needs: Edge, kruskal (01 - MST.cpp).
// It differs from the MST in exactly one edge: add a non-tree
    ↳ edge (u,v,w), drop the heaviest edge
// on the tree path u->v whose weight is < w. By the cycle
    ↳ property every edge on that path has
// weight <= w, so "heaviest edge < w" is either the path maximum
    ↳ (if it is < w) or the second
// LARGEST DISTINCT value on the path. Binary lifting carries
    ↳ both. O(m log n).
11 secondBestMST(int n, const vector<Edge>& e) {
    vector<Edge> used;
    11 mst = kruskal(n, e, &used);
    if ((int)used.size() != n - 1) return -1; //
    ↳ disconnected
    vector<vector<pair<int, ll>>> g(n);
    for (auto& x : used) g[x.u].push_back({x.v, x.w}),
    ↳ g[x.v].push_back({x.u, x.w});
    typedef array<ll, 2> T; //
    ↳ {largest, 2nd largest DISTINCT}
    const T NONE = {LLONG_MIN, LLONG_MIN};
    auto join = [&](T a, T b) {
        array<ll, 4> v = {a[0], a[1], b[0], b[1]};
        sort(v.rbegin(), v.rend());
        T r = {v[0], LLONG_MIN};
        for (ll x : v) if (x < v[0]) { r[1] = x; break; }
        return r;
    };
    int L = 1;
    while ((1 << L) < n) L++;
    vector<vector<int>>> up(L + 1, vector<int>(n, 0));
    vector<vector<T>>> mx(L + 1, vector<T>(n, NONE));
    vector<int> dep(n, 0), st{0};
    vector<char> vis(n, 0);
    vis[0] = 1;
    while (!st.empty()) { //
        ↳ iterative DFS from 0
        int u = st.back(); st.pop_back();
        for (auto [v, w] : g[u]) if (!vis[v])
            vis[v] = 1, dep[v] = dep[u] + 1, up[0][v] = u,
        ↳ mx[0][v] = {w, LLONG_MIN}, st.push_back(v);
        }
        for (int k = 1; k <= L; k++)
            for (int v = 0; v < n; v++) {
                int m = up[k - 1][v];
                up[k][v] = up[k - 1][m], mx[k][v] = join(mx[k -
        ↳ 1][v], mx[k - 1][m]);
            }
        auto path = [&](int u, int v) { //
        ↳ top-2 distinct weights on u->v
            T r = NONE;
            if (dep[u] < dep[v]) swap(u, v);
            for (int k = L; k >= 0; k--) if (dep[up[k][u]] >= dep[v])
        ↳ r = join(r, mx[k][u]), u = up[k][u];
            if (u == v) return r;
            for (int k = L; k >= 0; k--) if (up[k][u] != up[k][v])
                r = join(join(r, mx[k][u]), mx[k][v]), u = up[k][u],
        ↳ v = up[k][v];
            return join(join(r, mx[0][u]), mx[0][v]);
        };
        11 best = LLONG_MAX;
        for (auto& x : e) { //
        ↳ tree edges give p = {w, MIN} and
            T p = path(x.u, x.v); //
        ↳ fall through to the MIN branch
            11 rem = p[0] < x.w ? p[0] : p[1];
            if (rem != LLONG_MIN) best = min(best, mst - rem + x.w);
        }
        return best == LLONG_MAX ? -1 : best;
        ↳ // -1: no second tree exists
    }
    // K-TH BEST SPANNING TREE: repeat the swap argument with a
    ↳ priority queue over (tree, forced-in /
    // forced-out edge sets) - Gabow's algorithm, O(k m log n). Rare;
    ↳ the second-best case is the one
    // that actually shows up.
    // MINIMUM BOTTLENECK SPANNING TREE = any MST (minimising the
    ↳ maximum edge is implied by Kruskal).
    // MST OF A GRAPH PLUS ONE NEW EDGE: add it, find the cycle, drop
    ↳ the cycle maximum. O(n).
    // MINIMUM SPANNING TREE WITH DEGREE CONSTRAINT at one vertex r:
    ↳ build the MST of G - r, then add
    // the cheapest edge from r into each component, then improve by
    ↳ swaps while degree < k.
```

3.9 Exact Exponential

3.9.1 CliqueColouringMIS

EXACT ALGORITHMS FOR NP-HARD GRAPH PROBLEMS ON SMALL n . The constraint IS the hint:

```
// n <= 20-24 -> plain subset DP,  $O(2^n * n)$ 
// n <= 40-50 -> meet in the middle, or Bron-Kerbosch (fast in
//   practice on sparse graphs)
// n <= 100+ -> only if the graph is bipartite/perfect/a tree
//   - then it is polynomial (Konig)
// adj[] is an ADJACENCY BITMASK: bit j of adj[i] is set iff i-j
//   is an edge. No self loops.

// --- MAXIMUM CLIQUE, Bron-Kerbosch with PIVOTING. Worst case
//    $O(3^{(n/3)})$ , very fast in practice. ---
// R = clique built so far, P = candidates, X = already-explored
//   vertices (avoids duplicates).
// The pivot u in P/X is chosen to maximise |P & adj[u]|; only P \
//   adj[u] needs branching, because
// any maximal clique either omits u or contains a non-neighbour
//   of u.
int bestClique;
unsigned bestSet;
void bk(unsigned R, unsigned P, unsigned X, const
    vector<unsigned>& adj) {
    if (!P && !X) {
        if (__builtin_popcount(R) > bestClique) bestClique =
        __builtin_popcount(R), bestSet = R;
        return;
    }
    int u = __builtin_ctz(P | X), best = -1;
    for (unsigned t = P | X; t; t &= t - 1) {
        int v = __builtin_ctz(t), c = __builtin_popcount(P &
        adj[v]);
        if (c > best) best = c, u = v;
    }
    for (unsigned cand = P & ~adj[u]; cand; cand &= cand - 1) {
        int v = __builtin_ctz(cand);
        bk(R | 1u << v, P & adj[v], X & adj[v], adj);
        P &= ~(1u << v), X |= 1u << v;
    }
}

int maxClique(const vector<unsigned>& adj) {
    // n <= 32
    bestClique = 0, bestSet = 0;
    bk(0, (1u << adj.size()) - 1, 0, adj);
    return bestClique;
    // the clique itself is in bestSet
}

// MAXIMUM INDEPENDENT SET = maximum clique of the COMPLEMENT
//   graph. Just flip adj and mask off i.
// MINIMUM VERTEX COVER = n - MIS. On BIPARTITE graphs both are
//   polynomial: Konig says
// min vertex cover = maximum matching, and MIS = n - matching
//   (see 07 - Matching).

// --- CHROMATIC NUMBER,  $O(2^n * n)$  by inclusion-exclusion
//   (Bjorklund-Husfeldt-Koivisto). ---
// Let ind[S] = 1 if S is independent. The number of ways to cover
//   V with k ORDERED independent
//   sets is  $\sum_S (-1)^{n-|S|} * I(S)^k$ , where  $I(S) =$ 
//   #independent subsets of S. The chromatic
//   number is the smallest k making that non-zero. Compute I over
//   subsets with one SOS pass.
int chromatic(const vector<unsigned>& adj) {
    // n <= 20 with this modulus
    int n = adj.size(), N = 1 << n;
    if (!n) return 0;
    const ll M = (1LL << 61) - 1;
    // any large prime works
    vector<ll> I(N), pw(N, 1);
    I[0] = 1;
    for (int S = 1; S < N; S++) {
        // I[S] = #independent subsets of S
        int v = __builtin_ctz(S), rest = S ^ 1 << v;
        // (v in it, or not)
        I[S] = (I[rest] + I[rest & ~adj[v]]) % M;
    }
    for (int k = 1; k <= n; k++) {
        ll tot = 0;
        for (int S = 0; S < N; S++) {
            pw[S] = (__int128)pw[S] * I[S] % M;
        }
        // I(S)^k
        ll t = (n - __builtin_popcount(S)) & 1 ? M - pw[S] :
        pw[S];
        tot = (tot + t) % M;
    }
    if (tot % M) return k;
}
```

```
return n;
}

// The count is done MODULO a prime, so a zero could in principle
// be a false negative; with a
// 61-bit modulus that never happens on contest inputs. Check
// bipartiteness first - k <= 2 is  $O(n+m)$ .

// --- MINIMUM STEINER TREE (connect a set of k terminals in a
//   weighted graph),  $O(3^k n + 2^k n^2)$ .
// dp[S][v] = cheapest tree containing terminal set S and touching
//   vertex v.
// merge: dp[S][v] = min over submasks T of dp[T][v] +
//   dp[S\T][v]
// grow: dp[S][v] = min over edges (u,v) of dp[S][u] + w
//   (run Dijkstra per S)
ll steinerTree(int n, const vector<vector<pair<int, ll>>& g,
    const vector<int>& term) {
    int k = term.size();
    if (!k) return 0;
    vector<vector<ll>> dp(1 << k, vector<ll>(n, llinf));
    for (int i = 0; i < k; i++) dp[1 << i][term[i]] = 0;
    for (int S = 1; S < 1 << k; S++) {
        for (int T = (S - 1) & S; T; T = (T - 1) & S)
            for (int v = 0; v < n; v++)
                if (dp[T][v] < llinf && dp[S ^ T][v] < llinf)
                    dp[S][v] = min(dp[S][v], dp[T][v] + dp[S ^
                    T][v]);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        greater<>> pq;
        for (int v = 0; v < n; v++) if (dp[S][v] < llinf)
            pq.push({dp[S][v], v});
        while (!pq.empty()) {
            // Dijkstra to relax the tree outward
            auto [d, u] = pq.top(); pq.pop();
            if (d > dp[S][u]) continue;
            for (auto [v, w] : g[u]) if (dp[S][u] + w < dp[S][v])
                dp[S][v] = dp[S][u] + w, pq.push({dp[S][v], v});
        }
    }
    return *min_element(all(dp[(1 << k) - 1]));
}

// STEINER TREE IN A GRID / with k up to ~10 is the usual contest
// size. If k == 2 it is a shortest
// path; if k == n it is the MST; the DP interpolates between them.
// RELATED EXACT-EXPONENTIAL TOOLS:
// * HAMILTONIAN PATH / TSP: dp[mask][v],  $O(2^n n^2)$ . See 08 -
//   DP/02 - BitmaskDP.
// * SET COVER: dp over covered elements,  $O(2^n * \#sets)$ , or IE
//   like the chromatic number.
// * GRAPH COLOURING with k fixed: try to 3-colour by 2-SAT? No
//   - 3-colouring is NP-hard; use
//   the IE formula above, or branch on the highest-degree
//   vertex with memoisation.
// * MAXIMUM INDEPENDENT SET for n <= 40-50: meet in the middle
//   - split the vertices in half,
//   enumerate independent subsets of the left half, and for
//   each store the best independent
//   subset of the right half compatible with its neighbourhood
//   (SOS over the right mask).
// * COUNTING INDEPENDENT SETS / colourings on a graph of small
//   TREewidth: DP over a tree
//   decomposition,  $O(2^{tw} * n)$  - usually a grid problem in
//   disguise (see broken-profile DP).
```

3.10 Theorems

3.10.1 GraphTheorems

GRAPH THEOREM SHEET =====

```
/* ===== GRAPH THEOREM SHEET =====
Every hypothesis here matters. A theorem quoted without its
  ↳ precondition is a wrong answer.
```

MATCHING AND COVERING

KONIG (BIPARTITE only): $\max \text{ matching} = \min \text{ vertex cover}$.
 Constructive: Z = vertices reachable by alternating paths
 ↳ from the UNMATCHED left vertices;
 the cover is $(X \setminus Z) \cup (Y \cap Z)$.
 GALLAI: for EVERY graph on n vertices, $\alpha + \tau = n$
 ↳ (independent set + vertex cover).
 for every graph WITH NO ISOLATED VERTEX, $\nu + \rho = n$
 ↳ (matching + edge cover).
 Bipartite + Konig gives $\alpha = n - \nu$.
 HALL (bipartite): a matching saturating every vertex of the
 ↳ LEFT part X exists iff
 $|N(S)| \geq |S|$ for every S contained in X . (Not every S of V
 ↳ - the side matters.)
 ORE'S DEFECT FORM: $\nu(G) = |X| - \max \text{ over } S \text{ contained in } X \text{ of}$
 $(|S| - |N(S)|)$.
 HALL FOR REGULAR BIPARTITE: every k -regular bipartite graph (k
 ↳ ≥ 1) has a perfect matching,
 so its edges split into exactly k perfect matchings.
 ↳ Equivalently KONIG'S EDGE COLOURING
 THEOREM: $\chi'(G) = \Delta(G)$ for every bipartite MULTIGRAPH.
 ↳ (Timetabling, "split into
 Delta rounds", Latin-square completion.)
 BERGE: a matching is maximum iff there is no augmenting path.
 ↳ (Why Kuhn / HK / Blossom stop.)
 TUTTE (general graphs): G has a PERFECT matching iff $\text{odd}(G - U)$
 ↳ $\leq |U|$ for every U contained
 in V , where $\text{odd}(H)$ = the number of components of H with an
 ↳ ODD number of vertices.
 U = empty forces n even.
 TUTTE-BERGE: $\nu(G) = (1/2) * \min \text{ over } U \text{ of } (|V| + |U| - \text{odd}(G$
 ↳ $- U))$.
 DILWORTH (FINITE poset): min number of chains covering P = size
 ↳ of the maximum antichain.
 MIRSKY (dual): min number of antichains partitioning P = length
 ↳ of the longest chain.
 SPERNER: the largest antichain in the subset lattice of $[n]$ has
 ↳ size $C(n, \lfloor n/2 \rfloor)$.
 PATH COVERS - the distinction that most notebooks get wrong:
 min VERTEX-DISJOINT path cover of a DAG = $n - (\max \text{ matching}$
 ↳ of the split graph, no closure).
 min chain cover for Dilworth = $n - (\max \text{ matching of the}$
 ↳ TRANSITIVE CLOSURE's split graph).

CONNECTIVITY AND FLOW

MENGER: for NON-ADJACENT s, t : min s - t VERTEX cut = $\max$
 ↳ internally-vertex-disjoint s - t paths.
 For any $s \neq t$: min s - t EDGE cut = $\max \text{ edge-disjoint } s$ - t
 ↳ paths.
 WHITNEY: $\kappa(G) \leq \lambda(G) \leq \Delta(G)$ (vertex conn $\leq$
 ↳ edge conn $\leq$ min degree).
 A graph on ≥ 3 vertices is 2-connected iff every two
 ↳ vertices lie on a common cycle.
 MAX-FLOW MIN-CUT, with INTEGRALITY: integer capacities $\Rightarrow$ some
 ↳ maximum flow is integral (and
 Dinic produces one). Corollary used constantly: a
 ↳ 0/1-capacity $\max$ flow decomposes into that
 many edge-disjoint s - t paths.
 ROBBINS: an undirected graph has a STRONG orientation iff it is
 ↳ connected and BRIDGELESS.
 Constructive in $O(V+E)$: DFS, orient tree edges away from the
 ↳ root and back edges toward it.
 ESWARAN-TARJAN (augmentation):
 directed - min arcs to make a digraph strongly connected =
 ↳ $\max(\#\text{sources}, \#\text{sinks})$ of the
 condensation (0 if the condensation is a single node).
 undirected - for a CONNECTED graph, min edges to make it
 ↳ 2-edge-connected = $\text{ceil}(L/2)$ where
 L is the number of leaves of the bridge tree (0 if the
 ↳ bridge tree is one node).

DEGREE SEQUENCES

ERDOS-GALLAI: $d_1 \geq \dots \geq d_n \geq 0$ is the degree sequence of a
 ↳ SIMPLE graph iff $\sum d_i$ is
 even and for every k in $[1, n]$:
 $\sum_{i \leq k} d_i \leq k(k-1) + \sum_{i > k} \min(d_i, k)$.
 Only k up to the largest index with $d_k \geq k$ needs checking
 ↳ (the Durfee square), so it is
 $O(n)$ after sorting with prefix sums.
 HAVEL-HAKIMI: graphical iff $d_1 \leq n-1$ and, after deleting d_1
 ↳ and subtracting 1 from the next

d_1 entries (re-sorted), the rest is graphical. Gives a
 ↳ CONSTRUCTION, which Erdos-Gallai
 does not. $O(n^2)$ naive.
 LANDAU (tournaments): $s_1 \leq \dots \leq s_n$ is a score sequence iff
 ↳ $\sum_{i \leq k} s_i \geq C(k, 2)$ for
 every k , WITH EQUALITY at $k = n$.

TOURNAMENTS

REDEI: every finite tournament has a Hamiltonian PATH
 ↳ (insertion sort, $O(n \log n)$).
 CAMION: every STRONGLY CONNECTED tournament has a Hamiltonian
 ↳ CYCLE.
 MOON: every strongly connected tournament is vertex-pancyclic -
 ↳ for every vertex v and every
 k in $[3, n]$ there is a k -cycle through v .

COUNTING

MATRIX-TREE (Kirchhoff), undirected: #spanning trees (or the
 ↳ sum over trees of the product of
 weights) = ANY cofactor of $L = D - A$, loops excluded.
 MATRIX-TREE, DIRECTED (Tutte), with $A_{ij} = \#\text{arcs } i \rightarrow j$:
 $L_{\text{out}} = D_{\text{out}} - A$, delete row AND column $r \rightarrow$ arborescences
 ↳ CONVERGING TO r (in-trees).
 $L_{\text{in}} = D_{\text{in}} - A$, delete row AND column $r \rightarrow$ arborescences
 ↳ DIVERGING FROM r (out-trees).
 These two are the pair everyone swaps. Verify on the 2-vertex
 ↳ graph $1 \rightarrow 2$.
 BEST THEOREM: for a CONNECTED digraph with $\text{indeg}(v) = \text{outdeg}(v)$
 ↳ everywhere, the number of
 Eulerian circuits up to starting point is $\text{ec}(G) = t_w(G) * \prod_v (\text{outdeg}(v) - 1)!$,
 where t_w is the number of arborescences CONVERGING TO any
 ↳ fixed w (the L_{out} form above);
 t_w is the same for every w in such a graph. Multiply by
 ↳ $\text{outdeg}(v)$ to fix a start vertex.
 CAYLEY: $n^{(n-2)}$ labelled trees; with prescribed degrees, $(n-2)!$
 ↳ $/ \prod (d_i - 1)!$;
 labelled forests with k trees whose roots are k specified
 ↳ vertices: $k * n^{(n-k-1)}$.

EULERIAN CONDITIONS (the check EulerianPath.cpp needs before it

↳ starts)
 UNDIRECTED: a circuit exists iff every degree is even and all
 ↳ EDGES lie in one component.
 A trail exists iff exactly 0 or 2 vertices have odd degree
 ↳ (start at an odd one), same
 connectivity condition.
 DIRECTED: a circuit exists iff $\text{indeg} = \text{outdeg}$ at every vertex
 ↳ and all vertices of nonzero
 degree lie in one component of the underlying undirected
 ↳ graph. A trail exists iff exactly
 one vertex has out - in = 1 (the start), exactly one has in -
 ↳ out = 1 (the end), the rest
 are balanced, same connectivity condition.

COLOURING

BROOKS: for connected G , $\chi(G) \leq \Delta(G)$ UNLESS G is
 ↳ complete or an odd cycle, where
 $\chi = \Delta + 1$. Greedy on any order gives $\Delta + 1$; greedy on
 ↳ a degeneracy order gives $d+1$.
 VIZING (SIMPLE graphs): $\Delta \leq \chi' \leq \Delta + 1$.
 ↳ Multigraphs: $\chi' \leq \Delta + \mu$, and
 Shannon's $\chi' \leq \text{floor}(3 \Delta / 2)$. Deciding class 1 vs
 ↳ class 2 is NP-complete - do not
 look for a polynomial algorithm mid-contest. Bipartite is the
 ↳ exception (Konig above).
 GALLAI-HASSE-ROY-VITAVER: $\chi(G) = 1 + \min \text{ over orientations of}$
 ↳ (longest directed path length).
 Equivalently every orientation contains a directed path on
 ↳ $\chi(G)$ vertices.

EXTREMAL

TURAN: a K_{r+1} -free graph has $e \leq (1 - 1/r) n^2 / 2$, with
 ↳ equality only for the balanced
 complete r -partite Turan graph. $r = 2$ is MANTEL:
 ↳ triangle-free $\Rightarrow e \leq \text{floor}(n^2/4)$.
 Every graph has a bipartite subgraph with $\geq m/2$ edges (greedy
 ↳ $\max$ -cut).
 A graph is bipartite iff it has no odd cycle.
 RAMSEY, the known small values: $R(3,3)=6$, $R(3,4)=9$, $R(3,5)=14$,
 ↳ $R(3,6)=18$, $R(3,7)=23$,
 $R(3,8)=28$, $R(3,9)=36$, $R(4,4)=18$, $R(4,5)=25$. $R(4,6)$ in $[36,40]$
 ↳ and $R(5,5)$ in $[43,46]$ are
 OPEN. Bound: $R(s,t) \leq C(s+t-2, s-1)$.
 NASH-WILLIAMS ARBORICITY: $a(G) = \max \text{ over subgraphs } S \text{ with } \geq 2$
 ↳ vertices of
 $\text{ceil}(m_S / (n_S - 1))$. Consequences: degeneracy d satisfies

$\hookrightarrow a \leq d \leq 2a - 1$; $m \leq a(n-1)$;
 triangle enumeration is $O(m \cdot a)$.
 TUTTE / NASH-WILLIAMS TREE PACKING: G has t edge-disjoint
 $\hookrightarrow$ spanning trees iff every partition
 of V into k non-empty parts has at least $t(k-1)$ crossing
 $\hookrightarrow$ edges.
 PETERSEN: every BRIDGELESS CUBIC graph has a perfect matching.
 $\hookrightarrow$ Every $2k$ -regular graph splits
 into k edge-disjoint 2-factors, so it has an Eulerian circuit
 $\hookrightarrow$ and an orientation with
 $\text{indeg} = \text{outdeg} = k$.

PLANARITY

EULER: connected plane graph $V - E + F = 2$; with c components,
 $\hookrightarrow V - E + F = 1 + c$.
 Simple planar with $V \geq 3$: $E \leq 3V - 6$. Triangle-free (girth $\geq$
 $\hookrightarrow 4$): $E \leq 2V - 4$.
 Every planar graph has a vertex of degree ≤ 5 , i.e. it is
 $\hookrightarrow 5$ -degenerate.
 KURATOWSKI / WAGNER: planar iff no K_5 and no $K_{3,3}$
 $\hookrightarrow$ subdivision (minor).
 FOUR COLOUR: every planar graph is 4-colourable - but there is
 $\hookrightarrow$ no usable algorithm, so never
 plan around it. FIVE COLOUR is constructive in $O(n)$ from the
 $\hookrightarrow$ degree- ≤ 5 vertex.
 K_5 , $K_{3,3}$ and the Petersen graph are non-planar. In a planar
 $\hookrightarrow$ graph min cut = shortest path
 in the dual.

HAMILTONICITY (sufficient conditions only - the general problem
 $\hookrightarrow$ is NP-complete)

DIRAC: $n \geq 3$ and min degree $\geq n/2 \Rightarrow$ Hamiltonian.
 ORE: $\deg(u) + \deg(v) \geq n$ for every non-adjacent pair $\Rightarrow$
 $\hookrightarrow$ Hamiltonian.
 Both are constructive by rotation-extension.

TREE FACTS

JORDAN: every tree has 1 or 2 CENTRES; if 2 they are adjacent;
 $\hookrightarrow$ they are the middle of any
 diameter path.
 CENTROID: every tree has 1 or 2 centroids; c is a centroid iff
 $\hookrightarrow$ every component of $T - c$ has
 $\leq n/2$ vertices iff c minimises the sum of distances. If 2
 $\hookrightarrow$ they are adjacent and n is even.
 That $\leq n/2$ bound is exactly what makes centroid
 $\hookrightarrow$ decomposition $O(\log n)$ deep.
 DIAMETER: the greedy two-BFS is valid on trees and on
 $\hookrightarrow$ positively weighted trees, NOT on
 general graphs. Every longest path passes through a centre.
 $\hookrightarrow$ Joining two trees by an edge
 gives diameter $\max(d_1, d_2, r_1 + r_2 + 1)$, minimised by joining
 $\hookrightarrow$ centre to centre.
 A tree is bipartite, has ≥ 2 leaves, and sum of degrees =
 $\hookrightarrow 2(n-1)$.

WHAT IS NP-HARD, AND WHAT ONLY LOOKS IT

NP-hard: max clique, chromatic number (≥ 3 colours),
 $\hookrightarrow$ Hamiltonian path/cycle, TSP, MIXED
 Chinese postman, vertex cover / independent set on general
 $\hookrightarrow$ graphs, max cut (general),
 Steiner tree with unbounded terminals, longest path,
 $\hookrightarrow$ dominating set, feedback vertex/arc
 set, edge-colouring class decision, bandwidth.
 Polynomial despite appearances: vertex cover and independent
 $\hookrightarrow$ set on BIPARTITE graphs (Konig)
 and on trees / interval / chordal graphs; 2-colouring; 2-SAT;
 $\hookrightarrow$ max cut on PLANAR graphs;
 min mean cycle; min-cost flow; directed MST; maximum matching
 $\hookrightarrow$ in a GENERAL graph (Blossom);
 min path cover of a DAG; Steiner tree with $k \leq -15$
 $\hookrightarrow$ terminals; TSP with $n \leq -20$.

===== */

3.11 Special Structures

3.11.1 StableChordalComplement

THREE GRAPH TOOLS THAT ARE NOT ABOUT PATHS OR FLOW.

```

// --- GALE-SHAPLEY STABLE MARRIAGE,  $O(n^2)$ . ---
// n proposers, n receivers, each with a full preference ranking.
   $\hookrightarrow$  A matching is STABLE if no pair
// mutually prefer each other to their assigned partners. One
   $\hookrightarrow$  always exists, and this finds the
// PROPOSER-OPTIMAL one: every proposer gets the best partner they
   $\hookrightarrow$  could have in ANY stable
// matching, and simultaneously every receiver gets their worst.
   $\hookrightarrow$  (Swap the roles to flip that.)
// pref[i] = the proposers' preference lists; rank[j][i] =
   $\hookrightarrow$  receiver j's ranking of proposer i.
vector<int> stableMarriage(const vector<vector<int>>& pref, const
   $\hookrightarrow$  vector<vector<int>>& rank) {
    int n = pref.size();
    vector<int> match(n, -1), nxt(n, 0);
     $\hookrightarrow$  match[receiver] = proposer
    vector<int> free_(n);
    iota(all(free_), 0);
    while (!free_.empty()) {
        int m = free_.back(); free_.pop_back();
        int w = pref[m][nxt[m]++];
         $\hookrightarrow$  propose down the list
        if (match[w] < 0) match[w] = m;
        else if (rank[w][m] < rank[w][match[w]])
         $\hookrightarrow$  free_.push_back(match[w]), match[w] = m;
        else free_.push_back(m);
    }
    return match;
}

// HOSPITALS/RESIDENTS (one side has capacity c) is the same loop
   $\hookrightarrow$  with a heap of accepted
// proposers per hospital. STABLE ROOMMATES (one pool, no sides)
   $\hookrightarrow$  may have NO stable matching -
// Irving's algorithm decides it, and it is a different, longer
   $\hookrightarrow$  algorithm.

// --- TRAVERSAL OF THE COMPLEMENT GRAPH in  $O(n + m)$ , never
   $\hookrightarrow$  materialising the  $O(n^2)$  edges. ---
// Keep the not-yet-visited vertices in a linked structure; from
   $\hookrightarrow$  u, walk that set and step to
// every w that is NOT a neighbour of u. Each vertex leaves the
   $\hookrightarrow$  set once, and each REAL edge is
// inspected once, so the total is  $O(n + m)$ .
// Reach for it whenever the input describes the NON-edges:
   $\hookrightarrow$  "connected components of the graph of
// pairs that do NOT conflict", complement colouring, complement
   $\hookrightarrow$  BFS distances.
vector<int> complementComponents(int n, const
   $\hookrightarrow$  vector<vector<int>>& adj) {
    vector<char> isAdj(n, 0);
    set<int> unvis;
    for (int i = 0; i < n; i++) unvis.insert(i);
    vector<int> comp(n, -1);
    int c = 0;
    while (!unvis.empty()) {
        int s = *unvis.begin();
        unvis.erase(unvis.begin());
        vector<int> st{s};
        comp[s] = c;
        while (!st.empty()) {
            int u = st.back(); st.pop_back();
            for (int v : adj[u]) isAdj[v] = 1;
            vector<int> take;
            for (int w : unvis) if (!isAdj[w]) take.push_back(w);
            for (int w : take) unvis.erase(w), comp[w] = c,
             $\hookrightarrow$  st.push_back(w);
            for (int v : adj[u]) isAdj[v] = 0;
        }
        c++;
    }
    return comp;
   $\hookrightarrow$  comp[v] = component id
}

// The complement of a DISCONNECTED graph is connected, so a
   $\hookrightarrow$  complement graph almost always has
// very few components - that is why this is fast in practice as
   $\hookrightarrow$  well as in theory.

// --- CHORDAL GRAPHS: maximum cardinality search gives a PERFECT
   $\hookrightarrow$  ELIMINATION ORDERING. ---
// A graph is CHORDAL if every cycle of length  $\geq 4$  has a chord.
   $\hookrightarrow$  Equivalently it has a PEO: an
// order in which every vertex's LATER neighbours form a clique.
// Why you care: on a chordal graph, MAXIMUM CLIQUE, CHROMATIC

```

Number Theory

```
    ↪ NUMBER, MAXIMUM INDEPENDENT SET
// and MINIMUM CLIQUE COVER are all POLYNOMIAL - the four problems
    ↪ that are NP-hard in general.
// Interval graphs and trees are chordal, so "these intervals
    ↪ conflict" problems land here.
vector<int> mcsOrder(int n, const vector<vector<int>>& adj) {
    ↪ // reverse of a PEO if chordal
    vector<int> w(n, 0), ord;
    vector<char> used(n, 0);
    for (int it = 0; it < n; it++) {
        int b = -1;
        for (int v = 0; v < n; v++) if (!used[v] && (b < 0 ||
    ↪ w[v] > w[b])) b = v;
        used[b] = 1, ord.push_back(b);
        for (int u : adj[b]) if (!used[u]) w[u]++;
    }
    reverse(all(ord));
    return ord;
    ↪ this IS a PEO iff chordal
}

bool isChordal(int n, const vector<vector<int>>& adj) {
    auto ord = mcsOrder(n, adj);
    vector<int> pos(n);
    for (int i = 0; i < n; i++) pos[ord[i]] = i;
    vector<char> mark(n, 0);
    for (int i = 0; i < n; i++) {
        ↪ v's later neighbours must be
        int v = ord[i], par = -1;
        ↪ a clique; it is enough to check
        for (int u : adj[v]) if (pos[u] > i && (par < 0 || pos[u]
    ↪ < pos[par])) par = u;
        if (par < 0) continue;
        for (int u : adj[par]) mark[u] = 1;
        for (int u : adj[v]) if (pos[u] > i && u != par &&
    ↪ !mark[u]) {
            for (int x : adj[par]) mark[x] = 0;
            return false;
        }
        for (int u : adj[par]) mark[u] = 0;
    }
    return true;
}

// GIVEN A PEO: greedy colouring in PEO order uses exactly omega
    ↪ colours (so chi = omega); the
// maximum clique is the largest "v plus its later neighbours";
    ↪ the maximum independent set is the
// greedy "take v, delete its later neighbours" over the PEO.
```

4 Number Theory

4.1 Euclidean Algorithm

4.1.1 ExtendedEuclid

EXTENDED EUCLID - returns $g = \gcd(a, b)$ and x, y with $a*x + b*y = g$. This is the one routine that

```
// gives you modular inverses for a NON-prime modulus, solves
    ↪ linear Diophantine equations, and
// underpins CRT.  $|x| \leq b/(2g)$  and  $|y| \leq a/(2g)$ , so no overflow
    ↪ for 64-bit inputs.
long long extGCD(long long a, long long b, long long &x, long long
    ↪ &y) {
    if (b == 0) {
        x = 1; y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

long long modInverse(long long a, long long m) {
    long long x, y;
    long long g = extGCD(a, m, x, y);
    if (g != 1) return -1; // Inverse doesn't exist
    return (x % m + m) % m;
}
```

4.1.2 LinearDiophantine

Linear Diophantine Equations (2 Variables & N Variables)

```
// 1. Solves  $a * x + b * y = c$  (Particular solution & GCD check)
// 2. Solves  $a_1*x_1 + a_2*x_2 + \dots + a_n*x_n = c$  for N variables
struct LinearDiophantine {
    static ll ext_gcd(ll a, ll b, ll &x, ll &y) {
        if (b == 0) {
            x = 1; y = 0;
            return a;
        }
        ll x1, y1;
        ll d = ext_gcd(b, a % b, x1, y1);
        x = y1;
        y = x1 - y1 * (a / b);
        return d;
    }

    ↪ // Solves  $a * x + b * y = c$ . Returns false if no solution
    ↪ exists
    static bool solve_2var(ll a, ll b, ll c, ll &x, ll &y, ll &g)
    ↪ {
        if (a == 0 && b == 0) {
            if (c == 0) { x = y = g = 0; return true; }
            return false;
        }
        g = ext_gcd(abs(a), abs(b), x, y);
        if (c % g != 0) return false;
        x *= c / g;
        y *= c / g;
        if (a < 0) x = -x;
        if (b < 0) y = -y;
        return true;
    }

    ↪ // Solves  $a_1*x_1 + a_2*x_2 + \dots + a_n*x_n = c$  for N
    ↪ variables.
    ↪ Returns true if a solution exists and populates x vector
    static bool solve_nvar(const vector<ll>& a, ll c, vector<ll>&
    ↪ x) {
        int n = a.size();
        if (n == 0) return c == 0;
        x.assign(n, 0);

        vector<ll> prefix_g(n);
        prefix_g[0] = abs(a[0]);
        for (int i = 1; i < n; i++) prefix_g[i] =
    ↪ std::gcd(prefix_g[i - 1], a[i]);

        if (prefix_g.back() == 0) return c == 0;
        if (c % prefix_g.back() != 0) return false;

        ll cur_c = c;
        for (int i = n - 1; i > 0; i--) {
            ll x_prev, y_curr, g;
            solve_2var(prefix_g[i - 1], a[i], cur_c, x_prev,
    ↪ y_curr, g);
            x[i] = y_curr;
            cur_c = prefix_g[i - 1] * x_prev;
        }
        x[0] = cur_c / a[0];
        return true;
    }
};
```

4.1.3 FloorSum

EUCLIDEAN-LIKE ALGORITHM (AtCoder Library 'floor_sum').

```
// floorSum(n, m, a, b) = sum_{i=0}^{n-1} floor((a*i + b) / m)
//    in O(log m).
// This is THE tool for "sum over a lattice line": counting
//    lattice points under a line, sums of
// floor/mod expressions, and anything of the shape sum floor((a i
//    + b) / m) that a loop cannot do
// because n is up to 1e9 or more. Works like the Euclidean
//    algorithm: reduce a mod m, then SWAP
// the roles of the axes (count by rows instead of by columns) and
//    recurse.
// Requires n >= 0, m >= 1, a >= 0, b >= 0. Overflow: a*n + b must
//    fit in ll (use __int128 if not).
ll floorSum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m) ans += (n - 1) * n / 2 * (a / m), a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    ll ymax = (a * n + b) / m;
    if (ymax == 0) return ans;
    ll xmax = ymax * m - b;
    ans += ymax * (n - (xmax + a - 1) / a);
    return ans + floorSum(ymax, a, m, (a - xmax % a) % a);
}
// NEGATIVE a or b: shift them into range first.
// a < 0: let a2 = a mod m in [0,m); floorSum(n,m,a,b) =
//    floorSum(n,m,a2,b) - (a2-a)/m * (n-1)n/2
// b < 0: let b2 = b mod m in [0,m); subtract n * (b2-b)/m.
//    (Same trick, easier: use __int128.)
//
// WHAT IT SOLVES, directly:
// * #lattice points (x,y) with 1 <= x <= n and 1 <= y <= (a x +
//    b)/m    -> floorSum itself
// * sum_{i=0}^{n-1} ((a i + b) mod m) = (a * n(n-1)/2 + b*n) -
//    m * floorSum(n, m, a, b)
// * sum_{i=1}^n floor(n / i) in O(sqrt n) instead - use
//    DIVISOR BLOCKS (below), not this.
// * "how many multiples of q lie in [l, r]" = floor(r/q) -
//    floor((l-1)/q); the basic case.
//
// THE OTHER TWO SUMS of the same family (needed together often
//    enough to be worth knowing they
// exist): S1 = sum i * floor((a i + b)/m) and S2 = sum floor((a i
//    + b)/m)^2 obey the same
// recursion with a 3-tuple state - the "universal Euclidean
//    algorithm" generalises
// this to any monoid: you supply the two operators U ("move
//    right") and R ("move up") and it
// composes the word for the line in O(log) monoid multiplications.
//
// DIVISOR BLOCKS - the other floor trick, O(sqrt n):
// floor(n / i) is constant on maximal blocks; for a block
//    starting at l the block ends at
//    r = n / (n / l). Use it for sum_{i=1}^n floor(n/i), sum of
//    tau/sigma, Du sieve, Mobius sums.
//    for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); /*
//    value n/l on [l, r] */ }
// Two-argument version (blocks constant for BOTH n/i and m/i):
//    r = min(n/(n/l), m/(m/l)).
```

4.2 Primes and Divisors

4.2.1 Sieve

Linear Sieve & Smallest Prime Factor (SPF) - Time: O(N), Space: O(N)

```
// Computes primes list, SPF array, primality test, and O(log N)
//    prime factorization
struct Sieve {
    int n;
    vector<int> primes;
    vector<int> spf;

    Sieve(int n = 1e7) : n(n), spf(n + 1, 0) {
        for (int i = 2; i <= n; i++) {
            if (spf[i] == 0) {
                spf[i] = i;
                primes.pb(i);
            }
            for (int p : primes) {
                if (p > spf[i] || (ll)i * p > n) break;
                spf[i * p] = p;
            }
        }

        bool is_prime(int x) const {
            return x >= 2 && x <= n && spf[x] == x;
        }

        // O(log N) prime factorization using precomputed SPF array
        vector<int> get_prime_factors(int x) const {
            vector<int> factors;
            while (x > 1) {
                factors.pb(spf[x]);
                x /= spf[x];
            }
            return factors;
        }
    };
};
```

4.2.2 Factorization

Trial Division Prime Factorization - Time: O(sqrt(N))

```
template <typename T = ll>
struct Factorization {
    static vector<T> get_prime_factors(T n) {
        vector<T> factors;
        for (T i = 2; i * i <= n; i++) {
            while (n % i == 0) {
                factors.pb(i);
                n /= i;
            }
        }
        if (n > 1) factors.pb(n);
        return factors;
    }
};
```

4.2.3 Divisors

Trial Division Divisor Generation - Time: O(sqrt(N))

```
template <typename T = ll>
struct Divisors {
    static vector<T> get_divisors(T n) {
        vector<T> divs;
        for (T i = 1; i * i <= n; i++) {
            if (n % i == 0) {
                divs.pb(i);
                if (n / i != i) divs.pb(n / i);
            }
        }
        return divs;
    }
};
```


4.2.4 Sieve1e9

Fast Bitset Sieve up to $N = 10^9$ - Time: $\sim 0.3s$ for 10^9 , Space: $N / 16$ bytes ($\sim 60MB$)

```
// Stores odd numbers only (index i represents number 2*i + 1)
// WARNING: the bitset is ~60 MB - the instance MUST be
    ↪ global/static, never a local.
template <int MAXPR = (int)1e9>
struct Sieve1e9 {
    bitset<MAXPR / 2> is_prime_odd;

    Sieve1e9(int n = MAXPR) {
        is_prime_odd.set();
        is_prime_odd[0] = 0; // 1 is not prime
        for (int i = 1; (2 * i + 1) * (2 * i + 1) <= n; i++) {
            if (is_prime_odd[i]) {
                int p = 2 * i + 1;
                // bound is (n-1)/2, NOT n/2: n/2 writes one index
                ↪ past the end of the bitset
                for (int j = 2 * i * (i + 1); j <= (n - 1) / 2; j
                    ↪ += p) {
                    is_prime_odd[j] = 0;
                }
            }
        }

        bool is_prime(int x) const {
            if (x == 2) return true;
            if (x < 2 || x % 2 == 0) return false;
            return is_prime_odd[x / 2];
        }
    };
};
```

4.2.5 SegmentedSieve

Segmented Sieve for Range $[L, R]$ - Time: $O(\sqrt{R}) + (R - L) \log \log R$, Space: $O(\sqrt{R}) + R - L$

// Computes all prime numbers in range $[L, R]$ where $R <= 1e12$ and
 ↪ $(R - L) <= 1e7$

```
struct SegmentedSieve {
    static vector<ll> get_primes_in_range(ll L, ll R) {
        ll limit = sqrtl((ld)R);
        while (limit * limit > R) limit--;
        while ((limit + 1) * (limit + 1) <= R) limit++;
        vector<bool> is_prime_small(limit + 1, true);
        vector<ll> primes;
        for (ll i = 2; i <= limit; i++) {
            if (is_prime_small[i]) {
                primes.pb(i);
                for (ll j = i * i; j <= limit; j += i)
                    ↪ is_prime_small[j] = false;
            }
        }

        vector<bool> is_prime_range(R - L + 1, true);
        for (ll p : primes) {
            ll start = max(p * p, (L + p - 1) / p * p);
            for (ll j = start; j <= R; j += p) {
                is_prime_range[j - L] = false;
            }
        }
        for (ll x = L; x <= min(R, 1LL); x++) is_prime_range[x -
            ↪ L] = false; // 0 and 1

        vector<ll> result;
        for (ll i = 0; i <= R - L; i++) {
            if (is_prime_range[i]) result.pb(L + i);
        }
        return result;
    };
};
```

4.2.6 MillerRabinPollard

Primality + factorization for n up to $1e18$. isPrime $O(\log^3 n)$, factor $O(n^{1/4})$.

```
// THE gate for any problem with a_i up to 1e18 - divisors, phi,
    ↪ tau, sigma all follow from factor().
typedef unsigned long long u64;
u64 mulmod(u64 a, u64 b, u64 m) { return (__int128)a * b % m; }
u64 powmod(u64 a, u64 e, u64 m) {
    u64 r = 1; a %= m;
    for (; e >= 1; a = mulmod(a, a, m)) if (e & 1) r =
        ↪ mulmod(r, a, m);
    return r;
}

// Deterministic for all n < 3.3e24 with these 7 bases
    ↪ (Miller-Rabin).
bool isPrime(u64 n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    u64 d = n - 1; int s = __builtin_ctzll(d); d >>= s;
    for (u64 a : {2, 325, 9375, 28178, 450775, 9780504,
        ↪ 1795265022}) {
        u64 p = powmod(a % n, d, n); int i = s;
        while (p != 1 && p != n - 1 && a % n && i--) p =
            ↪ mulmod(p, p, n);
        if (p != n - 1 && i != s) return false;
    }
    return true;
}

u64 pollard(u64 n) { //
    ↪ returns a non-trivial divisor (Brent)
    auto f = [n](u64 x) { return mulmod(x, x, n) + 1; };
    u64 x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    while (t++ % 40 || gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = mulmod(prd, x > y ? x - y : y - x, n))) prd = q;
        x = f(x), y = f(f(y));
    }
    return gcd(prd, n);
}

vector<u64> factor(u64 n) { //
    ↪ unsorted multiset of primes
    if (n <= 1) return {}; // guard:
    ↪ factor(0) would loop forever
    if (isPrime(n)) return {n};
    u64 x = pollard(n);
    auto l = factor(x), r = factor(n / x);
    l.insert(l.end(), all(r));
    return l;
}

// {prime, exponent} pairs, sorted
vector<pair<u64, int>> factorize(u64 n) {
    auto f = factor(n); sort(all(f));
    vector<pair<u64, int>> r;
    for (u64 p : f) (!r.empty() && r.back().first == p) ?
        ↪ r.back().second++ : (r.push_back({p, 1}), 0);
    return r;
}

vector<u64> divisors(u64 n) { // all
    ↪ divisors, unsorted
    vector<u64> d = {1};
    for (auto [p, e] : factorize(n)) {
        int sz = d.size();
        for (u64 pe = 1, k = 0; k < (u64)e; k++) {
            pe *= p;
            for (int i = 0; i < sz; i++) d.push_back(d[i] * pe);
        }
    }
    return d;
}

u64 phi(u64 n) { u64 r = n; for (auto [p, e] : factorize(n)) r -=
    ↪ r / p; return r; } // Euler totient
u64 tau(u64 n) { u64 r = 1; for (auto [p, e] : factorize(n)) r *=
    ↪ e + 1; return r; } // #divisors
u64 sigma(u64 n) {
    ↪ // sum of divisors
    u64 r = 1;
    for (auto [p, e] : factorize(n)) { u64 t = 1, pe = 1; for
        ↪ (int i = 0; i < e; i++) t += (pe *= p); r *= t; }
    return r;
}
```

4.2.7 Multiplicative Sieve

Linear sieve computing ANY multiplicative function for all $i \leq n$ in $O(N)$.

```
// Shipped: spf, phi, mu, tau (#divisors), sigma (sum of divisors).
// To add your own f: you need f(p), f(p^k) and f(a*b)=f(a)f(b)
// for coprime a,b.
struct MultSieve {
    int n;
    vector<int> spf, primes, phi, mu, tau, cnt; // cnt[i]
    // = exponent of spf[i] in i
    vector<ll> sigma, pw; // pw[i]
    // = spf[i]^cnt[i]

    MultSieve(int n) : n(n), spf(n + 1), phi(n + 1), mu(n + 1),
    // tau(n + 1), cnt(n + 1),
    // sigma(n + 1), pw(n + 1) {
        phi[1] = mu[1] = tau[1] = sigma[1] = pw[1] = 1;
        for (int i = 2; i <= n; i++) {
            if (!spf[i]) {
                spf[i] = i, primes.push_back(i);
                phi[i] = i - 1, mu[i] = -1, tau[i] = 2, cnt[i] = 1;
                pw[i] = i, sigma[i] = i + 1LL;
            }
            for (int p : primes) {
                if (p > spf[i] || (ll)i * p > n) break;
                int j = i * p;
                spf[j] = p;
                if (p == spf[i]) { // p
                    // divides i: same prime, exponent grows
                    cnt[j] = cnt[i] + 1, pw[j] = pw[i] * p;
                    phi[j] = phi[i] * p;
                    mu[j] = 0;
                    tau[j] = tau[i] * (cnt[i] + 1) * (cnt[j] + 1);
                    sigma[j] = sigma[i] * ((pw[i] * p - 1) / (p - 1)) * ((pw[j] * p - 1) / (p - 1));
                } else { //
                    // coprime: multiply
                    cnt[j] = 1, pw[j] = p;
                    phi[j] = phi[i] * (p - 1);
                    mu[j] = -mu[i];
                    tau[j] = tau[i] * 2;
                    sigma[j] = sigma[i] * (p + 1);
                }
            }
        }
    }

    vector<int> factors(int x) const { // O(log
    // x) via spf
        vector<int> f;
        while (x > 1) f.push_back(spf[x]), x /= spf[x];
        return f;
    }
};
```

4.3 Modular Arithmetic

4.3.1 CRT

```
// NOTE: extGCD below is the same one as in 01 - Euclidean
// Algorithm. Paste only one copy.
long long extGCD(long long a, long long b, long long &x, long long
// &y) {
    if (b == 0) {
        x = 1; y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

// Solves x = a_i (mod m_i). Returns {x, lcm(m_1, ..., m_k)}.
pair<long long, long long> CRT(const vector<long long>& a, const
// vector<long long>& m) {
    long long res = a[0], lcm = m[0];
    for (int i = 1; i < a.size(); i++) {
        long long x, y;
        long long g = extGCD(lcm, m[i], x, y);
        if ((a[i] - res) % g != 0) return {-1, -1}; // No solution

        long long step = m[i] / g;
        long long k = ((a[i] - res) / g) % step * (x % step) %
    // step;
        res = (long long)((__int128)k * lcm + res) %
    // ((__int128)lcm * step);
        lcm *= step;
        res = (res % lcm + lcm) % lcm;
    }
    return {res, lcm};
}
```


4.3.2 DiscreteLogAndRoots

Discrete log, modular square root, primitive root, discrete k-th root, multiplicative order.

```
// Needs: powmod (02 - Primes and Divisors/06 -
    ↳ MillerRabinPollard.cpp).
// All need mulmod/powmod from MillerRabinPollard.

// BSGS: smallest  $x \geq 0$  with  $a^x = b \pmod{m}$ . Works for ANY  $m$ 
    ↳ (handles  $\gcd(a, m) \neq 1$ ).  $O(\sqrt{m})$ .
ll discreteLog(ll a, ll b, ll m) {
    a %= m, b %= m;
    ll k = 1, add = 0, g;
    while ((g = gcd(a, m)) > 1) { // strip
        ↳ common factors
        if (b == k) return add;
        if (b % g) return -1;
        b /= g, m /= g, add++, k = k * (a / g) % m;
    }
    ll n = (ll)sqrtl((ld)m) + 1, an = 1;
    for (ll i = 0; i < n; i++) an = an * a % m;
    unordered_map<ll, ll> vals;
    for (ll q = 0, cur = b; q <= n; q++) vals[cur] = q, cur = cur
    ↳ * a % m;
    for (ll p = 1, cur = k; p <= n; p++) {
        cur = cur * an % m;
        if (vals.count(cur)) { ll ans = n * p - vals[cur] + add;
        ↳ if (ans >= 0) return ans; }
    }
    return -1;
}

// Tonelli-Shanks:  $x$  with  $x^2 = a \pmod{p}$ ,  $p$  odd prime. Returns -1
    ↳ if  $a$  is not a QR.
ll sqrtMod(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    if (powmod(a, (p - 1) / 2, p) != 1) return -1; //
    ↳ Euler's criterion
    if (p % 4 == 3) return powmod(a, (p + 1) / 4, p);
    ll s = p - 1, n = 2; int r = 0;
    while (s % 2 == 0) s /= 2, r++;
    while (powmod(n, (p - 1) / 2, p) != p - 1) n++; // find a
    ↳ non-residue
    ll x = powmod(a, (s + 1) / 2, p), b = powmod(a, s, p), g =
    ↳ powmod(n, s, p);
    for (int m = r; m = r) {
        ll t = b;
        for (r = 0; r < m && t != 1; r++) t = t * t % p;
        if (r == 0) return x;
        ll gs = powmod(g, 1LL << (m - r - 1), p);
        g = gs * gs % p, x = x * gs % p, b = b * g % p;
    }
}

// Smallest primitive root of a PRIME  $p$  (generator of  $\mathbb{Z}_p^*$ ).
ll primitiveRoot(ll p) {
    if (p == 2) return 1;
    auto f = factorize(p - 1);
    for (ll g = 2; g < p; g++) {
        bool ok = true;
        for (auto [q, e] : f) if (powmod(g, (p - 1) / q, p) == 1)
        ↳ { ok = false; break; }
        if (ok) return g;
    }
    return -1;
}

// Multiplicative order of  $a$  mod  $m$  (smallest  $e > 0$  with  $a^e = 1$ ).
    ↳ Requires  $\gcd(a, m) = 1$ .
ll multOrder(ll a, ll m) {
    ll e = phi(m), r = e;
    for (auto [q, k] : factorize(e))
        while (r % q == 0 && powmod(a, r / q, m) == 1) r /= q;
    return r;
}

// All  $x$  with  $x^k = a \pmod{p}$ ,  $p$  prime. Reduce to a discrete log
    ↳ in base  $g$ .
vector<ll> discreteRoot(ll k, ll a, ll p) {
    if (a == 0) return {0};
    ll g = primitiveRoot(p), y = discreteLog(powmod(g, k, p), a,
    ↳ p);
    if (y < 0) return {};
    ll x0 = powmod(g, y, p), d = gcd(k, p - 1), step = powmod(g,
    ↳ (p - 1) / d, p);
    vector<ll> r;
    for (ll i = 0, cur = x0; i < d; i++, cur = cur * step % p)
    ↳ r.push_back(cur);
}
```

```
sort(all(r)), r.erase(unique(all(r)), r.end());
return r;
}
```

4.3.3 LucasAndBinomialMod

$C(n, r) \pmod{m}$ for HUGE n . Pick by what m is:

```
// Needs: Mint (06 - Math/01), and factorize (02 - Primes and
    ↳ Divisors/06) for the last section.
//  $m$  prime and small  $\rightarrow$  lucas
//  $m = p^e \rightarrow$  binomPrimePower (Andrew
    ↳ Granville / Wilson generalisation)
//  $m$  arbitrary (squarefree-ish)  $\rightarrow$  binomAnyMod (CRT over prime
    ↳ powers)

// LUCAS:  $p$  prime, needs fact/invFact built up to  $p$ .  $C(n, r) = \prod$ 
    ↳  $C(n_i, r_i)$  over base- $p$  digits.
Mint lucas(ll n, ll r) {
    if (r < 0 || r > n) return 0;
    Mint res = 1;
    while (n || r) {
        ll a = n % MOD, b = r % MOD;
        if (b > a) return 0;
        res *= fact[a] * invFact[b] * invFact[a - b];
        n /= MOD, r /= MOD;
    }
    return res;
}

// --- arbitrary modulus ---
ll pw(ll a, ll e, ll m) { ll r = 1; a %= m; for (; e; e >>= 1, a
    ↳ = a * a % m) if (e & 1) r = r * a % m; return r; }
ll inv_mod(ll a, ll m) { //
    ↳ requires  $\gcd(a, m) = 1$ 
    ll g = m, x = 0, x1 = 1, a1 = a % m;
    while (a1) { ll q = g / a1; g -= q * a1, swap(g, a1); x -= q
    ↳ * x1, swap(x, x1); }
    return (x % m + m) % m;
}

//  $n!$  with all factors of  $p$  removed, mod  $p^e$ .  $O(p^e + \log n)$ .
ll factNoP(ll n, ll p, ll pe) {
    if (n == 0) return 1;
    vector<ll> f(pe);
    f[0] = 1;
    for (ll i = 1; i < pe; i++) f[i] = (i % p ? f[i - 1] * i :
    ↳ f[i - 1]) % pe;
    ll r = 1;
    for (ll m = n; m; m /= p) {
        r = r * pw(f[pe - 1], m / pe, pe) % pe;
        r = r * f[m % pe] % pe;
    }
    return r;
}

//  $C(n, r) \pmod{p^e}$ 
ll binomPrimePower(ll n, ll r, ll p, ll pe) {
    if (r < 0 || r > n) return 0;
    ll k = 0; //
    ↳ exponent of  $p$  in  $C(n, r)$  (Kummer's theorem:
    for (ll m = n / p, a = r / p, b = (n - r) / p; m; m /= p, a
    ↳ /= p, b /= p) k += m - a - b;
    if (k >= 60 || pw(p, k, pe * p) % pe == 0) { ll t = pw(p, k,
    ↳ pe); if (t == 0) return 0; }
    ll num = factNoP(n, p, pe), d1 = factNoP(r, p, pe), d2 =
    ↳ factNoP(n - r, p, pe);
    ll res = num * inv_mod(d1, pe) % pe * inv_mod(d2, pe) % pe;
    return res * pw(p, k, pe) % pe;
}

//  $C(n, r) \pmod{m}$  for arbitrary  $m$ : factor  $m$ , solve mod each  $p^e$ ,
    ↳ CRT.
ll binomAnyMod(ll n, ll r, ll m) {
    ll res = 0, M = m;
    for (auto [p, e] : factorize(m)) {
        ll pe = 1; for (int i = 0; i < e; i++) pe *= p;
        ll c = binomPrimePower(n, r, p, pe), t = M / pe;
        res = (res + c % pe * t % M * inv_mod(t % pe, pe)) % M;
    }
    return res;
}

// KUMMER'S THEOREM: the exponent of prime  $p$  in  $C(n, r)$  = number of
    ↳ CARRIES when adding
//  $r$  and  $n-r$  in base  $p$ . (Used above to pull out
    ↳ the  $p$ -part.)
// WILSON:  $(p-1)! = -1 \pmod{p}$  for prime  $p$ .
```

4.3.4 CongruencesAndTowers

Linear congruences and power towers. Needs powmod / phi / gcd.

// Needs: powmod, phi (04 - NT/02 - Primes and Divisors/06 -
 ↳ MillerRabinPollard.cpp).

```
// Solve a*x = b (mod m). Returns all solutions in [0, m) (there
↳ are gcd(a,m) of them, or none).
vector<ll> linCongruence(ll a, ll b, ll m) {
    a = ((a % m) + m) % m, b = ((b % m) + m) % m;
    ll g = gcd(a, m);
    if (b % g) return {};
    ll m2 = m / g, a2 = a / g, b2 = b / g;
    // a2 is invertible mod m2
    ll x0 = 0, x1 = 1, g2 = m2, a3 = a2 % m2;
    while (a3) { ll q = g2 / a3; g2 -= q * a3, swap(g2, a3); x0
↳ -= q * x1, swap(x0, x1); }
    ll inv = ((x0 % m2) + m2) % m2, x = b2 % m2 * inv % m2;
    vector<ll> r;
    for (ll i = 0; i < g; i++) r.push_back((x + i * m2) % m);
    return r;
}

// GENERALISED EULER: a^k = a^(k mod phi(m) + phi(m)) (mod m)
↳ whenever k >= log2(m).
// This is what makes power towers computable even when gcd(a, m)
↳ != 1.
ll capTower(const vector<ll>& a, int i, ll CAP = 64) { //
↳ min(true tower value, CAP)
    if (i == (int)a.size()) return 1;
    if (a[i] == 0) return capTower(a, i + 1, CAP) == 0 ? 1 : 0;
    ↳ // 0^0 = 1, 0^(k>0) = 0
    if (a[i] == 1) return 1;
    ll e = capTower(a, i + 1, CAP), r = 1;
    for (ll k = 0; k < e; k++) { r *= a[i]; if (r >= CAP) return
↳ CAP; }
    return r;
}

// a[i] ^ a[i+1] ^ ... ^ a[last] (mod m), right-associative.
ll powTower(const vector<ll>& a, int i, ll m) {
    if (m == 1) return 0;
    if (i == (int)a.size()) return 1 % m;
    ll ph = phi(m), t = capTower(a, i + 1);
    ll e = t < 64 ? t : powTower(a, i + 1, ph); // exact
    ↳ when tiny, +phi when large
    return powmod(a[i] % m, e, m);
}

// Extended CRT for NON-coprime moduli: x = a1 (mod m1), x = a2
↳ (mod m2).
// Returns {x, lcm} or {-1, -1}. Chain it to merge many
↳ congruences.
pair<ll, ll> crt2(ll a1, ll m1, ll a2, ll m2) {
    ll g = gcd(m1, m2), l = m1 / g * m2;
    if ((a2 - a1) % g) return {-1, -1};
    ll m2g = m2 / g;
    ll x0 = 0, x1 = 1, gg = m2g, aa = (m1 / g) % m2g;
    while (aa) { ll q = gg / aa; gg -= q * aa, swap(gg, aa); x0
↳ -= q * x1, swap(x0, x1); }
    ll inv = ((x0 % m2g) + m2g) % m2g;
    __int128 k = (__int128)((a2 - a1) / g % m2g) * inv % m2g;
    ll x = (ll)((__int128)k * m1 + a1) % l;
    return {(x % l + l) % l, l};
}
```

4.3.5 Garner

GARNER'S ALGORITHM - reconstruct x from residues modulo PAIRWISE COPRIME m_i, in mixed-radix

```
// form, so you can emit x modulo a target MOD without ever
↳ building prod(m_i).
// This is what CRT cannot do once the product overflows 64 bits.
// x = x1 + x2*m1 + x3*m1*m2 + ... , 0 <= xi < mi
ll garner(vector<ll> a, vector<ll> m, ll MODOUT) {
    int k = a.size();
    vector<ll> x(k);
    auto inv = [](ll a, ll M) { // requires
↳ gcd(a, M) = 1
        ll g = M, X = 0, x1 = 1, A = ((a % M) + M) % M;
        while (A) { ll q = g / A; g -= q * A, swap(g, A); X -= q
↳ * x1, swap(X, x1); }
        return (X % M + M) % M;
    };
    for (int i = 0; i < k; i++) {
        x[i] = ((a[i] % m[i]) + m[i]) % m[i];
        ll mul = 1;
        for (int j = 0; j < i; j++) {
            x[i] = (x[i] - x[j] % m[i] * mul) % m[i];
            mul = mul * (m[j] % m[i]) % m[i];
        }
        x[i] = (x[i] % m[i] + m[i]) % m[i] * inv(mul, m[i]) %
↳ m[i];
    }
    ll res = 0, mul = 1;
    for (int i = 0; i < k; i++) {
        res = (res + x[i] % MODOUT * mul) % MODOUT;
        mul = mul * (m[i] % MODOUT) % MODOUT;
    }
    return (res % MODOUT + MODOUT) % MODOUT;
}

// MAIN USE: arbitrary-modulus convolution. Run NTT under three
↳ NTT primes, then Garner the
// three residues down to the target modulus.
// NTT PRIMES: 998244353 = 119*2^23 + 1 (g=3) 167772161 =
↳ 5*2^25 + 1 (g=3)
// 469762049 = 7*2^26 + 1 (g=3) 1004535809 =
↳ 479*2^21 + 1 (g=3)
// 754974721 = 45*2^24 + 1 (g=11) <- note: NOT
↳ 9*2^23+1, a common misprint
// The 3-prime product 5.95e25 covers n * (MOD-1)^2 for any MOD <
↳ 2^30 and n up to ~5e7.
```

4.4 Theorems

4.4.1 NumberTheoryFormulas

NUMBER THEORY - FORMULA SHEET =====

```

/* ===== NUMBER THEORY - FORMULA SHEET
   =====
MULTIPLICATIVE FUNCTIONS  (f(ab)=f(a)f(b) for gcd(a,b)=1)
    tau(n)=#divisors=prod(e_i+1)    sigma(n)=sum of divisors=prod
    (p^(e_i+1)-1)/(p-1)
    phi(n)=n*prod(1-1/p)            mu(n)= 0 if NOT SQUAREFREE (some
    p^2 | n), else (-1)^#primes
    sum_{d|n} phi(d) = n            sum_{d|n} mu(d) = [n==1]
    (this is the key identity)
    sum_{d|n} mu(d)/d = phi(n)/n    sum_{d|n} mu(d)*tau(n/d) = 1

DIRICHLET CONVOLUTION  (f*g)(n) = sum_{d|n} f(d) g(n/d)
    1*1 = tau    Id*1 = sigma    phi*1 = Id    mu*1 =
    (=> eps([n==1])
    mu is the inverse of 1, so f = g*1 <=> g = f*mu.

MOBIUS INVERSION
    g(n) = sum_{d|n} f(d)          <=> f(n) = sum_{d|n} mu(d) g(n/d)
    g(n) = sum_{n/d, d<=N} f(d)    <=> f(n) = sum_{n/d, d<=N}
    mu(d/n) g(d)                  (the "upward" form)
    Standard use: [gcd(i,j)==1] = sum_{d | gcd(i,j)} mu(d).

GCD / LCM SUMS  (all O(N log N) or better with a mu/phi sieve)
    #coprime pairs (i,j)<=n        = sum_{d=1..n} mu(d) *
    floor(n/d)^2
    sum_{i,j<=n} gcd(i,j)          = sum_{d=1..n} phi(d) *
    floor(n/d)^2
    sum_{d|n} d*phi(n/d)           = sum_{i=1..n} gcd(i,n)
    sum_{i=1..n} lcm(i,n)           = n/2 * (1 + sum_{d|n} d*phi(d))
    sum_{i,j<=n} lcm(i,j)           = sum_g g * sum_{a,b<=n/g,
    gcd(a,b)=1} a*b
    general: sum_{i,j} f(gcd(i,j)) = sum_d (f*mu)(d) * floor(n/d)^2

DIVISOR BLOCKING - evaluates sum_{i=1..n} F(floor(n/i)) in
    O(sqrt n):
    for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); } //
    floor(n/i) == n/l on [l, r]
    Gives sum tau(i), sum sigma(i), sum floor(n/i), and Min_25 /
    Lucy-style prefix sums.

MODULAR
    Fermat:  a^(p-1) = 1 (mod p), p prime, p !| a    => a^-1 =
    a^(p-2)
    Euler:   a^phi(m) = 1 (mod m) if gcd(a,m)=1
    GENERALISED EULER (a and m NOT coprime), for n >= log2(m):
    a^n = a^(n mod phi(m) + phi(m)) (mod m)    <-
    power towers
    Wilson:  (p-1)! = -1 (mod p) iff p prime
    CRT:     x = a_i (mod m_i) has a unique solution mod lcm(m_i)
    iff a_i = a_j (mod gcd(m_i, m_j))
    Legendre: exponent of p in n! = sum_{i=1} floor(n / p^i) =
    (n - s_p(n)) / (p - 1)
    Kummer:  exponent of p in C(n,r) = #carries when adding r and
    (n-r) in base p
    LTE (p odd, p | a-b, p !| a,b):  v_p(a^n - b^n) = v_p(a-b) +
    v_p(n)
    p=2 (a, b both ODD):
    n odd : v_2(a^n - b^n) = v_2(a-b)    <- the case
    everyone forgets
    n even : v_2(a^n - b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1
    n odd : v_2(a^n + b^n) = v_2(a+b)
    n even : v_2(a^n + b^n) = 1
    LTE for SUMS (p odd, p | a+b, p !| a,b):
    n odd -> v_p(a^n + b^n) = v_p(a+b) + v_p(n)
    n even -> p does not divide a^n + b^n at all

QUADRATIC RESIDUES (p odd prime)
    Euler's criterion: a^((p-1)/2) = +1 if a is a QR, -1 if not
    #QRs = (p-1)/2.  x^2=a has 0 or 2 roots.  Use Tonelli-Shanks.
    -1 is a QR iff p = 1 (mod 4); 2 is a QR iff p = +-1 (mod 8)

PRIMITIVE ROOTS exist exactly for m = 1, 2, 4, p^k, 2p^k (p odd
    prime). Count = phi(phi(m)).

REPRESENTATIONS
    n is a sum of two squares iff every prime p = 3 (mod 4) has an
    EVEN exponent
    n is a sum of three squares iff n != 4^a (8b+7)    every n is
    a sum of four squares
    Primitive Pythagorean triples: (m^2-k^2, 2mk, m^2+k^2), m>k>0,
    gcd(m,k)=1, m-k odd

SIZE FACTS (for choosing an algorithm)
    max tau(n): 1344 for n<=1e9, 6720 for n<=1e12, 103680 for
    n<=1e18

```

```

sum_{i<=n} tau(i) ~ n ln n ; sum_{i<=n} phi(i) ~ 3n^2/pi^2
pi(n) ~ n/ln n ; n-th prime ~ n ln n ; Bertrand: a prime
    always lies in (n, 2n)
gaps: for n<=1e18 the max prime gap is < 1500
Pisano period pi(m) | ... : F(n) mod m repeats with period <= 6m
=====
    */

```

4.4.2 MobiusInversion

Working code for the identities on the formula sheet. Needs MultSieve (mu, phi).

```

// Needs: MultSieve (04 - NT/02 - Primes and Divisors/07 -
    MultiplicativeSieve.cpp) for mu/phi.

// #pairs (i,j) with 1<=i,j<=n and gcd(i,j)=1    O(n)
ll coprimePairs(int n, const MultSieve& S) {
    ll r = 0;
    for (int d = 1; d <= n; d++) r += (ll)S.mu[d] * (n / d) * (n
    / d);
    return r;
}
// sum over all pairs of gcd(i,j)    O(n)
ll gcdSum(int n, const MultSieve& S) {
    ll r = 0;
    for (int d = 1; d <= n; d++) r += (ll)S.phi[d] * (n / d) * (n
    / d);
    return r;
}
// General: sum_{i,j<=n} f(gcd(i,j)) = sum_d (f*mu)(d) *
    floor(n/d)^2
// Build g = f*mu once by sieving over multiples: for d: for
    m=d, 2d, ...: g[m] += mu[m/d]*f[d]
vector<ll> dirichletMu(const vector<ll>& f, const MultSieve& S) {
    // g = f * mu    O(n log n)
    int n = f.size() - 1;
    vector<ll> g(n + 1, 0);
    for (int d = 1; d <= n; d++) if (f[d])
        for (int m = d; m <= n; m += d) g[m] += (ll)S.mu[m / d] *
        f[d];
    return g;
}

// DIVISOR BLOCKING: sum_{i=1..n} floor(n/i) in O(sqrt n). Same
    skeleton for sum tau / sum sigma.
ll sumFloorDiv(ll n) {
    ll r = 0;
    for (ll l = 1, rr; l <= n; l = rr + 1) { rr = n / (n / l); r
    += (n / l) * (rr - l + 1); }
    return r;
}
// sum_{i=1..n} tau(i) = sum_{i=1..n} floor(n/i)    (count
    multiples)
// sum_{i=1..n} sigma(i) = sum_{i=1..n} i * floor(n/i)

// COUNT MULTIPLES / INCLUSION-EXCLUSION OVER DIVISORS:
// "how many i<=n are divisible by at least one of p1..pk" -> IE
    over the 2^k squarefree products,
// sign = mu of the product. For k up to ~20 enumerate submasks;
    beyond that, sieve mu.

// sum_{i=1..n} gcd(i, n) = sum_{d|n} d * phi(n/d)
ll gcdSumWithN(ll n) {
    ll r = 0;
    for (auto d : divisors(n)) r += (ll)d * phi(n / d);
    return r;
}
// sum_{i=1..n} lcm(i, n) = n/2 * (1 + sum_{d|n} d*phi(d))
ll lcmSumWithN(ll n) {
    ll s = 1;
    for (auto d : divisors(n)) s += (ll)d * phi(d);
    return n / 2 * s + (n % 2 ? s / 2 : 0);
}

```

4.4.3 PrimeCounting

LUCY_HEDGEHOG: $\pi(n) = \#\text{primes} \leq n$, and the SUM of primes $\leq n$, in $O(n^{3/4})$ time and

```
// O(sqrt n) memory. Handles n up to ~1e12 in well under a second;
// 1e13 in a few seconds.
// Idea: S(v, p) = (count or sum) of the integers in [2, v] left
// after sieving by all primes < p.
// Sieving by p only touches v >= p^2, so the whole table lives on
// the O(sqrt n) distinct
// values of floor(n / i).
ll primePi(ll n) {
    if (n < 2) return 0;
    ll r = (ll)sqrtl((ld)n);
    while (r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    vector<ll> S(r + 1), L(r + 1); // S[v]
    for v <= r, L[i] for v = n/i
    for (ll i = 1; i <= r; i++) S[i] = i - 1, L[i] = n / i - 1;
    for (ll p = 2; p <= r; p++) {
        if (S[p] == S[p - 1]) continue; // p is
        not prime
        ll sp = S[p - 1], p2 = p * p;
        for (ll i = 1; i <= min(r, n / p2); i++)
            L[i] -= (i * p <= r ? L[i * p] : S[n / (i * p)]) - sp;
        for (ll v = r; v >= p2; v--) S[v] -= S[v / p] - sp;
    }
    return L[1];
}
// Sum of all primes <= n. Same recurrence with f(x) = x(x+1)/2 -
// 1 and a factor of p.
ll primeSum(ll n) {
    if (n < 2) return 0;
    ll r = (ll)sqrtl((ld)n);
    while (r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    // __int128: n(n+1)/2 wraps for n > 4.29e9, and the ANSWER
    // (-n^2/(2 ln n)) wraps at n ~ 3e10.
    auto f = [](ll x) { return x % 2 ? (1ll)(x + 1) / 2 * x - 1 :
    (1ll)x / 2 * (x + 1) - 1; };
    vector<ll> S(r + 1), L(r + 1);
    for (ll i = 1; i <= r; i++) S[i] = f(i), L[i] = f(n / i);
    for (ll p = 2; p <= r; p++) {
        if (S[p] == S[p - 1]) continue;
        ll sp = S[p - 1], p2 = p * p;
        for (ll i = 1; i <= min(r, n / p2); i++)
            L[i] -= p * ((i * p <= r ? L[i * p] : S[n / (i * p)])
        - sp);
        for (ll v = r; v >= p2; v--) S[v] -= p * (S[v / p] - sp);
    }
    return L[1];
}
// SAME SKELETON computes the prefix sum of ANY
// completely-multiplicative f (change the two
// initialisations and the multiplier). For general multiplicative
// f use the Min_25 sieve, which
// runs this as its first phase and then re-adds the prime-power
// contributions.
// APPROXIMATIONS for sanity checks: pi(n) ~ n/ln n, the k-th
// prime ~ k ln k.
```

4.4.4 Representations

Representing integers: sums of two squares, Pythagorean triples, Legendre symbol.

```
// Needs: powmod, factorize (02 - Primes and Divisors/06 -
// MillerRabinPollard.cpp).
// Needs factorize / sqrtMod / isPrime.

int legendre(ll a, ll p) { // 1
    QR, -1 non-residue, 0 if p | a
    a %= p; if (a < 0) a += p;
    if (!a) return 0;
    return powmod(a, (p - 1) / 2, p) == 1 ? 1 : -1;
}
// n is a sum of two squares <=> every prime p = 3 (mod 4)
// appears to an EVEN power.
bool isSumOfTwoSquares(ll n) {
    if (n <= 0) return n == 0;
    for (auto [p, e] : factorize(n)) if (p % 4 == 3 && (e & 1))
        return false;
    return true;
}
// One representation n = x^2 + y^2 (Cornacchia). Returns {-1, -1}
// if none exists.
pair<ll, ll> sumOfTwoSquares(ll n) {
    if (n == 0) return {0, 0};
    ll t = n, pw2 = 1;
    while (t % 4 == 0) t /= 4, pw2 *= 2; // pull
    out squares of 2
    if (!isSumOfTwoSquares(t)) return {-1, -1};
    // Gaussian-integer product over the prime factorisation
    ll xr = 1, yr = 0; //
    current value as xr + yr*i
    for (auto [p, e] : factorize(t)) {
        if (p == 2) { for (int i = 0; i < e; i++) { ll nx = xr -
        yr, ny = xr + yr; xr = nx, yr = ny; } continue; }
        if (p % 4 == 3) { ll k = 1; for (int i = 0; i < e / 2;
        i++) k *= p; xr *= k, yr *= k; continue; }
        ll r = sqrtMod(p - 1, p); // r^2
        = -1 (mod p)
        ll a = p, b = r;
        while (b * b > p) { ll c = a % b; a = b, b = c; } //
        Cornacchia descent: p = b^2 + c^2
        ll c = (ll)sqrtl((ld)(p - b * b));
        while (c * c < p - b * b) c++;
        while (c * c > p - b * b) c--;
        for (int i = 0; i < e; i++) { ll nx = xr * b - yr * c, ny
        = xr * c + yr * b; xr = nx, yr = ny; }
        return {llabs(xr) * pw2, llabs(yr) * pw2};
    }
}
// #representations of n as an ORDERED pair of squares (with
// signs) = 4 * (d_1(n) - d_3(n)),
// where d_i counts divisors = i (mod 4). [Jacobi's two-square
// theorem]

// PRIMITIVE Pythagorean triples with hypotenuse <= L: (m^2-k^2,
// 2mk, m^2+k^2),
// m > k > 0, gcd(m,k) = 1, m-k odd. Every triple is a multiple of
// exactly one primitive triple.
vector<array<ll, 3>> pythagorean(ll L) {
    vector<array<ll, 3>> r;
    for (ll m = 2; m * m <= L; m++)
        for (ll k = 1 + (m % 2); k < m; k += 2) // m-k
        must be ODD => opposite parity
            if (gcd(m, k) == 1 && m * m + k * k <= L)
                r.push_back({m * m - k * k, 2 * m * k, m * m + k
                * k});
    return r;
}
// n is a sum of THREE squares <=> n != 4^a * (8b + 7). Every
// n is a sum of FOUR squares.
```

4.4.5 DuSieve

DU SIEVE (Dirichlet prefix sums) - prefix sums of a multiplicative f in SUBLINEAR time.

```
// Pick g with an easy prefix sum G and an easy (f*g) prefix sum;
// then
// g(1)*S(n) = sum_{i=1..n} (f*g)(i) - sum_{i=2..n} g(i) *
// S(floor(n/i))
// Sieve S(x) directly for x <= LIM (take LIM ~ n^(2/3)); recurse
// with memoisation above that.
// Total O(n^(2/3)) with the sieve, O(n^(3/4)) without.
// mu * 1 = eps -> M(n) = 1 - sum_{i=2..n} M(n/i)
// phi * 1 = Id -> Phi(n) = n(n+1)/2 - sum_{i=2..n} Phi(n/i)
struct DuSieve {
    ll LIM;
    vector<ll> mu, phi;
    // sums for x <= LIM
    unordered_map<ll, ll> memoM, memoP;

    DuSieve(ll n) {
        LIM = max((ll)1000, (ll)powl((ll)n, 2.0L / 3));
        vector<int> spf(LIM + 1, 0), primes, m(LIM + 1, 0), p(LIM
    + 1, 0);
        m[1] = p[1] = 1;
        for (ll i = 2; i <= LIM; i++) {
            if (!spf[i]) spf[i] = i, primes.push_back(i), m[i] =
    -1, p[i] = i - 1;
            for (int q : primes) {
                if (q > spf[i] || i * q > LIM) break;
                spf[i * q] = q;
                if (q == spf[i]) m[i * q] = 0, p[i * q] = p[i] *
    q;
                else m[i * q] = -m[i], p[i * q] = p[i] * (q - 1);
            }
        }
        mu.assign(LIM + 1, 0), phi.assign(LIM + 1, 0);
        for (ll i = 1; i <= LIM; i++) mu[i] = mu[i - 1] + m[i],
    phi[i] = phi[i - 1] + p[i];
    }
    ll M(ll n) {
        //
        sum_{i=1..n} mu(i) (Mertens)
        if (n <= LIM) return mu[n];
        auto it = memoM.find(n);
        if (it != memoM.end()) return it->second;
        ll r = 1;
        for (ll l = 2, rr; l <= n; l = rr + 1) { rr = n / (n /
    l); r -= (rr - l + 1) * M(n / l); }
        return memoM[n] = r;
    }
    ll P(ll n) {
        //
        sum_{i=1..n} phi(i)
        if (n <= LIM) return phi[n];
        auto it = memoP.find(n);
        if (it != memoP.end()) return it->second;
        ll r = n % 2 ? (n + 1) / 2 * n : n / 2 * (n + 1);
        for (ll l = 2, rr; l <= n; l = rr + 1) { rr = n / (n /
    l); r -= (rr - l + 1) * P(n / l); }
        return memoP[n] = r;
    }
};
// USES: squarefree count up to n = sum_{d<=sqrt n} mu(d) *
// floor(n/d^2);
// #coprime pairs up to n = 2*Phi(n) - 1; sum of gcd/lcm
// over pairs at n ~ 1e10;
// any "sum over 1..n of a multiplicative function" that a
// linear sieve cannot reach.
```

4.4.6 MoreFormulaCards

NUMBER THEORY, SECOND FORMULA SHEET =====

```
/* ===== NUMBER THEORY, SECOND FORMULA SHEET =====
Everything here was machine-verified against brute force / sympy.

QUADRATIC RECIPROCITY. p, q distinct ODD primes.
(p/q)(q/p) = (-1)^((p-1)/2 * (q-1)/2)
=> (p/q) = (q/p), UNLESS p = q = 3 (mod 4), where (p/q) =
-> -(q/p)
SUPPLEMENTS: (-1/p) = 1 iff p = 1 (mod 4); (2/p) = 1 iff p =
-> +-1 (mod 8);
(3/p) = 1 iff p = +-1 (mod 12)
EULER'S CRITERION (p odd prime AND p !| a): a^((p-1)/2) =
-> (a/p) (mod p).
If p | a the value is 0, not +-1 - state the hypothesis.
#k-th ROOTS of a != 0 mod p: d = gcd(k, p-1); the count is d
-> if a^((p-1)/d) = 1, else 0.
JACOBI SYMBOL (a/n) for ODD n > 0, n NOT necessarily prime -
-> O(log n), no factoring: */
int jacobi(ll a, ll n) {
    //
    -> n odd, n > 0
    int r = 1;
    a = ((a % n) + n) % n;
    //
    -> (a/n) is periodic in a mod n
    while (a) {
        while (a % 2 == 0) { a /= 2; if (n % 8 == 3 || n % 8 ==
    -> 5) r = -r; }
        swap(a, n);
        if (a % 4 == 3 && n % 4 == 3) r = -r;
        a %= n;
    }
    return n == 1 ? r : 0;
}
/* CAUTION: (a/n) = 1 does NOT prove a is a QR mod a composite n.
-> (a/n) = -1 DOES prove it is
not. Use it as a cheap filter, and to find a non-residue fast
-> for Tonelli-Shanks.

FIBONACCI (F_0 = 0, F_1 = 1)
FAST DOUBLING, O(log n), no matrices:
F(2k) = F(k) * (2 F(k+1) - F(k)), F(2k+1) = F(k)^2 +
-> F(k+1)^2
gcd(F_m, F_n) = F_gcd(m,n) => F_m | F_n iff m | n (m >= 3)
Cassini: F_{n-1} F_{n+1} - F_n^2 = (-1)^n
d'Ocagne: F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}
sum_{i<=n} F_i = F_{n+2} - 1 sum_{i<=n} F_i^2 = F_{2n+1}
-> F_{n+1}
F_92 is the last Fibonacci number fitting in a signed 64-bit
-> integer.
PISANO PERIOD pi(m) = the period of F(n) mod m
pi(1)=1, pi(2)=3, and pi(m) is EVEN for every m > 2
pi(lcm(a,b)) = lcm(pi(a), pi(b)) pi(p^k) = p^(k-1) *
-> pi(p)
pi(m) <= 6m, with EQUALITY exactly at m = 2 * 5^k (verified:
-> m = 10, 50, 250)
p = +-1 (mod 5) => pi(p) | p-1; p = +-2 (mod 5) => pi(p) |
-> 2(p+1)
ZECKENDORF: every n >= 1 is a UNIQUE sum of NON-CONSECUTIVE
-> Fibonacci numbers, found
greedily from the top. The basis of "Fibonacci base" digit DP
-> and of Wythoff's game.

PELL: x^2 - D y^2 = 1, D > 0 non-square. Always infinitely many
-> solutions.
sqrt(D) = [a0; a1, ..., a_L] is periodic with a_L = 2 a0 and a
-> palindromic middle.
FUNDAMENTAL SOLUTION = the convergent h_k / k_k with
k = L - 1 if the period L is EVEN, k = 2L - 1 if L
-> is ODD.
ALL solutions: x_m + y_m sqrt(D) = (x1 + y1 sqrt(D))^m, i.e.
x_{m+1} = x1 x_m + D y1 y_m, y_{m+1} = x1 y_m + y1 x_m
NEGATIVE PELL x^2 - D y^2 = -1 is solvable IFF the period L is
-> ODD; then the fundamental
solution is the convergent at k = L - 1.
SIZE WARNING: solutions explode. D = 61 gives x = 1766319049,
-> but D = 661 gives a
39-digit x. 'll' is useless past D ~ 150 - use bignum, or work
-> modulo the answer.
CF MACHINERY (a0 = floor(sqrt D); m = 0, d = 1, a = a0), all
-> exact integers:
m' = d*a - m, d' = (D - m'^2) / d, a' = floor((a0 +
-> m') / d')
BEST RATIONAL APPROXIMATION: the closest p/q to x with q <= N
-> is a convergent OR a
semiconvergent (t h1 + h2)/(t k1 + k2) with 1 <= t <= a_next.
-> Check both.
FAREY MEDIANT: for neighbours a/b < c/d (bc - ad = 1),
```


→ $(a+c)/(b+d)$ is the unique fraction with the SMALLEST denominator strictly between them.

SUMS OF SQUARES

TWO: $n = x^2 + y^2$ possible IFF every prime $p = 3 \pmod{4}$ has → an EVEN exponent.

$r_2(n) = 4 (d_1(n) - d_3(n))$, $d_i = \#$ divisors congruent → to $i \pmod{4}$.

THREE: $n = x^2 + y^2 + z^2$ possible IFF n is NOT of the form → $4^a (8b + 7)$.

FOUR: ALWAYS possible (Lagrange). $r_4(n) = 8 * \text{sum over } d \mid n$ → with $4 \nmid d$ of d .

PRIMITIVE PYTHAGOREAN TRIPLES: $(m^2 - k^2, 2mk, m^2 + k^2)$, $m >$ → $k > 0$, $\gcd(m, k) = 1$, $m - k$ odd.

GAUSSIAN INTEGERS $Z[i]$: the norm $N(a+bi) = a^2 + b^2$ is → multiplicative. Primes of $Z[i]$:

$1+i$ (over 2); $p = 1 \pmod{4}$ SPLITS as $p_i * \text{conj}(p_i)$; $p = 3$ → $\pmod{4}$ stays prime.

$\gcd$ by rounded division: $q = \text{round}(a \text{ conj}(b) / N(b))$, $r = a -$ → qb , then Euclid on norms.

FROBENIUS / CHICKEN McNUGGET. $a, b \geq 2$, $\gcd(a, b) = 1$:

largest non-representable $g(a, b) = ab - a - b$;

→ $\#$ non-representable $= (a-1)(b-1)/2$.

For $k \geq 3$ coins there is NO closed form: run Dijkstra on

→ residues mod a_1 ($d[r] =$ the smallest representable value congruent to r), then $g = \max_r$ → $d[r] - a_1$. $O(a_1 k \log a_1)$.

BEATTY / RAYLEIGH. $r, s > 1$ IRRATIONAL with $1/r + 1/s = 1 \Rightarrow$

→ $\text{floor}(n r)$ and $\text{floor}(n s)$, $n \geq 1$, PARTITION the positive integers exactly. (Wythoff's

→ losing positions are exactly $(\text{floor}(n \phi), \text{floor}(n \phi^2))$.)

THREE-DISTANCE (Steinhaus). The points $\{a\}, \{2a\}, \dots, \{na\}$ cut

→ $[0, 1)$ into intervals of AT

MOST 3 distinct lengths, and when there are 3 the largest is → the sum of the other two.

Use: $\min$ over $k \leq n$ of $(k*a \bmod m)$ in $O(\log)$ via the continued → fraction of a/m .

DETERMINISTIC MILLER-RABIN BASE SETS (all machine-verified)

$n < 2^{32}$: $\{2, 7, 61\}$

→ (valid to 4759123141)

$n < 3.2e18$: the first 12 primes $\{2..37\}$

$n < 3.317e24 > 2^{64}$: $\{2, 325, 9375, 28178, 450775, 9780504,$

→ $1795265022\}$

$\{2, 3, 5, 7\}$ is NOT enough even for 32 bits: 3215031751 =

→ $151*751*28351$ passes all four.

Skip a base a when n divides a , or small primes are reported

→ composite.

NTT PRIMES (form, primality and primitive root all

→ machine-verified)

p = $c * 2^k + 1$ g $\max$ transform

→ length

998244353 = $119 * 2^{23} + 1$ 3 2^{23} ← the

→ default

167772161 = $5 * 2^{25} + 1$ 3 2^{25}

469762049 = $7 * 2^{26} + 1$ 3 2^{26}

754974721 = $45 * 2^{24} + 1$ 11 2^{24} ← NOT

→ $9*2^{23}+1$ (common misprint)

1004535809 = $479 * 2^{21} + 1$ 3 2^{21}

2013265921 = $15 * 2^{27} + 1$ 31 2^{27}

1224736769 = $73 * 2^{24} + 1$ 3 2^{24}

3221225473 = $3 * 2^{30} + 1$ 5 2^{30}

9223372036737335297 = $549755813881 * 2^{24} + 1$, $g = 3$

→ (64-bit, needs __int128 mulmod)

THE 3-PRIME TRIPLE for an arbitrary modulus: 167772161 *

→ $469762049 * 754974721 = 5.95e25$.

THE BINDING CONSTRAINT IS LENGTH, NOT MAGNITUDE: 754974721 caps

→ the transform at 2^{24} , so

the input length is capped at 2^{23} . At that length the

→ magnitude headroom is still $6x$.

64-BIT CEILINGS - memorise these

$a*b$ fits in signed ll iff $a, b < 3037000499$ (=

→ $\text{floor}(\sqrt{2^{63} - 1})$)

$n(n+1)/2$ fits in ll iff $n \leq 4294967295$

sum of primes $\leq n$ fits iff $n < 3e10$ (the sum is

→ $\sim n^2 / (2 \ln n)$)

sum of $\phi(i)$, $i \leq n$, fits iff $n \leq 5508509576$ (the sum is

→ $\sim 0.3040 n^2$)

$\max \tau(n)$: 1344 for $n \leq 1e9$, 6720 for $1e12$, 103680 for $1e18$

NEED __int128 FOR: any modmul with modulus $> 3.04e9$ (Pollard,

→ BSGS, Tonelli, CRT, Garner);

floorSum with $n > 1e9$; Cornacchia's $b*b > p$ test;

→ prime/totient SUMS at $n \geq 1e10$.

FFT PRECISION, the real criterion (not $|result| < 1e15$):

safe iff $(\text{sum } a_i^2 + \text{sum } b_i^2) * \log_2(N) < 9e14$.

Splitting at $\sqrt{M}$ additionally needs $M^{1.5} < 9.2e18$, i.e.

→ $M < 4.4e12$, with the

inputs already reduced into $[0, M)$.

FWHT: the forward transform inflates magnitudes by up to N , so

→ the pointwise product is

bounded by $(N * \max|a|)^2$. Under a modulus the XOR inverse

→ must MULTIPLY BY $\text{inv}(N) \bmod p$ -

an integer ' $x \neq n$ ' is exact over Z and WRONG mod p .

===== */

4.5 Rational

4.5.1 ContinuedFractions

Continued fractions, convergents, best rational approximation, Stern-Brocot search.

```
// x = [a0; a1, a2, ...]. For a rational p/q this is just the
// ↪ Euclidean algorithm's quotients.
vector<ll> cfrac(ll p, ll q) {
    vector<ll> a;
    // FLOOR division: C++ truncates toward zero, so cfrac(-7,2)
    // ↪ would give [-3,-2] instead of
    // the correct [-4,2] and every convergent would be wrong.
    while (q) { ll d = p / q - ((p % q) && ((p < 0) ^ (q < 0)));
        ↪ a.push_back(d); ll r = p - d * q; p = q, q = r; }
    return a;
}
// Convergents h[i]/k[i] of a continued fraction: h = a*h1 + h2, k
// ↪ = a*k1 + k2.
// Every convergent is a BEST approximation: no fraction with a
// ↪ smaller denominator is closer.
vector<pair<ll, ll>> convergents(const vector<ll>& a) {
    vector<pair<ll, ll>> c;
    ll h1 = 1, h2 = 0, k1 = 0, k2 = 1;
    for (ll x : a) {
        ll h = x * h1 + h2, k = x * k1 + k2;
        c.push_back({h, k});
        h2 = h1, h1 = h, k2 = k1, k1 = k;
    }
    return c;
}
// Best rational approximation to p/q with denominator <= N.
// The answer is a convergent OR a SEMICONVERGENT
// ↪ (t*h1+h2)/(t*k1+k2) - you must compare both.
pair<ll, ll> bestApprox(ll p, ll q, ll N) {
    ll h1 = 1, h2 = 0, k1 = 0, k2 = 1, P = p, Q = q;
    pair<ll, ll> best = {p >= 0 ? 0 : -((-p + q - 1) / q), 1};
    ↪ // 0/1 (or floor) is legal
    auto better = [&](pair<ll, ll> a, pair<ll, ll> b) {
        ↪ // is a closer to p/q than b?
        __int128 da = (__int128)llabs(p * a.second - a.first * q)
        ↪ * b.second;
        __int128 db = (__int128)llabs(p * b.second - b.first * q)
        ↪ * a.second;
        return da != db ? da < db : a.second < b.second;
    };
    while (Q) {
        ll a = P / Q, r = P % Q;
        if (k1 && a * k1 + k2 > N) {
            ↪ // cannot take the full step
            ll t = (N - k2) / k1;
            if (t > 0) { pair<ll, ll> c = {t * h1 + h2, t * k1 +
            ↪ k2}; if (better(c, best)) best = c; }
            break;
        }
        ll h = a * h1 + h2, k = a * k1 + k2;
        if (k > N) break;
        h2 = h1, h1 = h, k2 = k1, k1 = k;
        if (better({h1, k1}, best)) best = {h1, k1};
        P = Q, Q = r;
    }
    return best;
}
// STERN-BROCOT SEARCH: find the simplest fraction p/q with f(p/q)
// ↪ true, when f is monotone.
// Walk the mediant tree; take exponential jumps in each direction
// ↪ so it is O(log^2).
// Classic uses: "smallest denominator between two reals",
// ↪ recovering a fraction from a decimal,
// and answering "is the answer rational with small denominator"
// ↪ style problems.
// FAREY / mediant fact: if a/b < c/d are neighbours (bc - ad = 1)
// ↪ then their mediant
// (a+c)/(b+d) is the unique simplest fraction strictly between
// ↪ them.
```

4.6 Sequences

4.6.1 FibonacciAndPisano

FIBONACCI, FAST. The notebook had the identities on a formula sheet and no routine; this is it.

```
// F(0)=0, F(1)=1. Fast doubling is O(log n) with two
// ↪ multiplications per bit and NO matrices -
// shorter and ~2x faster than 2x2 matrix exponentiation.
// F(2k) = F(k) * (2*F(k+1) - F(k))
// F(2k+1) = F(k)^2 + F(k+1)^2
pair<ll, ll> fibPair(ll n, ll mod) {
    ↪ returns {F(n), F(n+1)} mod
    if (!n) return {0, 1 % mod};
    auto [a, b] = fibPair(n >> 1, mod);
    ll c = a * ((2 * b % mod - a % mod + mod) % mod) % mod;
    ↪ // F(2k)
    ll d = (a * a + b * b) % mod;
    ↪ // F(2k+1)
    return (n & 1) ? make_pair(d, (c + d) % mod) : make_pair(c,
    ↪ d);
}
ll fib(ll n, ll mod) { return fibPair(n, mod).first; }
// LUCAS NUMBERS L(n) = F(n-1) + F(n+1) = 2F(n+1) - F(n).
// NEGATIVE index: F(-n) = (-1)^(n+1) F(n).
// The mod must be < -3e9 for the ll products above, or use
// ↪ __int128 / Mint.

// --- PISANO PERIOD pi(m): the period of F(n) mod m. Needed to
// ↪ reduce a huge index. ---
// pi(1) = 1, pi(2) = 3, and pi(m) is EVEN for every m > 2.
// pi(lcm(a,b)) = lcm(pi(a), pi(b)) pi(p^k) = p^(k-1) * pi(p)
// pi(m) <= 6m, with equality exactly at m = 2 * 5^k.
// p = +1 (mod 5) => pi(p) divides p-1; p = +-2 (mod 5) =>
// ↪ pi(p) divides 2(p+1).
// So: factor m, get pi for each prime power by testing the
// ↪ divisors of that bound, then lcm.
ll pisanoPrime(ll p) {
    ↪ p prime
    if (p == 2) return 3;
    if (p == 5) return 20;
    ll cand = (p % 5 == 1 || p % 5 == 4) ? p - 1 : 2 * (p + 1);
    vector<ll> divs;
    for (ll d = 1; d * d <= cand; d++)
        if (cand % d == 0) divs.push_back(d), divs.push_back(cand
    ↪ / d);
    sort(all(divs));
    for (ll d : divs) {
        auto [a, b] = fibPair(d, p);
        if (a == 0 && b == 1 % p) return d;
    }
    return cand;
}
// Full pi(m): factorise m (Pollard rho), take pisanoPrime(p) *
// ↪ p^(e-1) per prime power, lcm them.
/* MORE FIBONACCI FACTS worth the page
gcd(F_m, F_n) = F_gcd(m,n) => F_m | F_n iff m | n (m
↪ >= 3)
Cassini: F_{n-1} F_{n+1} - F_n^2 = (-1)^n
Catalan: F_n^2 - F_{n-r} F_{n+r} = (-1)^(n-r) F_r^2
d'Ocagne: F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}
sum_{i<=n} F_i = F_{n+2} - 1 sum_{i<=n} F_i^2 = F_n
↪ F_{n+1}
sum_{i<=n} F_{2i-1} = F_{2n} sum_{i<=n} F_{2i} =
↪ F_{2n+1} - 1
F_n = (phi^n - psi^n)/sqrt(5); F_92 is the last one fitting in a
↪ signed 64-bit integer.
ZECKENDORF: every n >= 1 is a UNIQUE sum of NON-CONSECUTIVE
↪ Fibonacci numbers, found greedily
from the largest. This is the "Fibonacci base", and it is
↪ what makes Fibonacci-nim work.
LCM of F_1..F_n and "which F are divisible by k" both follow
↪ from the gcd identity.
Fibonacci mod p when 5 is a QR mod p: sqrt(5) exists, so the
↪ closed form works in F_p directly.
When it is not, work in F_p[x]/(x^2-5) (a "Phi field") - two
↪ components, same O(log n). */

// --- TRIBONACCI / any linear recurrence: matrix exponentiation
// ↪ (06 - Math/03), or Kitamasa /
// Bostan-Mori for a long recurrence. Berlekamp-Massey RECOVERS
// ↪ the recurrence from ~2k terms.
// --- LUCAS SEQUENCES U_n, V_n with U_{n+1} = P U_n - Q U_{n-1}
// ↪ generalise all of this and have
// the same doubling identities; they are what Baillie-PSW
// ↪ primality is built on.
```

5.1 StarsAndBars

```
Stars and Bars (Number of Solutions to  $x_1 + x_2 + \dots + x_k = n$ ) - O(1)
// 1. Non-negative integer solutions ( $x_i \geq 0$ ):  $C(n+k-1, k-1)$ 
// 2. Positive integer solutions ( $x_i \geq 1$ ):  $C(n-1, k-1)$ 
struct StarsAndBars {
    // Requires precomputed Combination C(n, k)
    static ll count_non_negative(ll n, ll k, function<ll(ll, ll)> nCr) {
        if (n < 0 || k <= 0) return 0;
        return nCr(n+k-1, k-1);
    }

    static ll count_positive(ll n, ll k, function<ll(ll, ll)> nCr) {
        if (n < k || k <= 0) return 0;
        return nCr(n-1, k-1);
    }
};
```

5.2 BurnsidePolya

```
Burnside / Polya - count colourings up to symmetry. #orbits = (1/|G|) sum_g |fix(g)|
// Needs: Mint (06 - Math/01); phi (02 - Primes and Divisors/06)
// for the necklace formulas.
// Needs Mint and phi() (single-value Euler totient).

// Necklaces: n beads in a circle, k colours, ROTATIONS only.
// A rotation by r has gcd(r,n) cycles -> k^gcd(r,n) fixed
// colourings; group by d = gcd.
Mint necklaces(ll n, ll k) {
    Mint s = 0;
    for (ll d = 1; d * d <= n; d++) if (n % d == 0) {
        s += Mint(phi(n / d)) * Mint(k).pow(d);
        if (d != n / d) s += Mint(phi(d)) * Mint(k).pow(n / d);
    }
    return s * Mint(n).inv();
}

// Bracelets: rotations AND reflections (dihedral group, |G| = 2n).
Mint bracelets(ll n, ll k) {
    Mint s = necklaces(n, k) * Mint(n); // = sum
    // over the n rotations
    if (n & 1) s += Mint(n) * Mint(k).pow((n+1)/2);
    else s += Mint(n/2) * (Mint(k).pow(n/2) + Mint(k).pow(n/2 + 1));
    return s * Mint(2*n).inv();
}

// GENERAL RECIPE: for each group element g, count its CYCLES c(g)
// on the objects being
// coloured; it fixes k^c(g) colourings. Answer = (1/|G|) * sum_g k^c(g).
// square grid n x n under the 4 rotations: cycles = n^2,
// ceil(n^2/4)... (work them out per problem)
// cube faces under 24 rotations: cycle counts 6,3,3,4,2,2,...
// -> (k^6 + 3k^4 + 12k^3 + 8k^2)/24
// POLYA (colours have weights): replace k^c by prod over cycles
// of (sum of x_i^len).
Mint burnside(const vector<ll>& cyclesPerElement, ll k) {
    Mint s = 0;
    for (ll c : cyclesPerElement) s += Mint(k).pow(c);
    return s * Mint((ll)cyclesPerElement.size()).inv();
}
```

5.3 CombinatoricsFormulas

```
COMBINATORICS - FORMULA SHEET =====
/* ===== COMBINATORICS - FORMULA SHEET
   =====
BASICS C(n,k)=n!/(k!(n-k)!) P(n,k)=n!/(n-k)! multinomial
   n!/(k1!..km!)
STARS AND BARS: x1+...+xk = n, xi>=0 -> C(n+k-1, k-1); xi>=1
   -> C(n-1, k-1)
   with upper bounds xi<=c: IE over which
   variables overflow
   Compositions of n into k parts C(n-1,k-1); into any parts
   -> 2^(n-1)

BINOMIAL IDENTITIES (memorise these - recognising one is worth
   30 minutes)
Pascal C(n,k) = C(n-1,k-1) + C(n-1,k)
Symmetry C(n,k) = C(n,n-k)
Absorption k*C(n,k) = n*C(n-1,k-1)
Hockey stick sum_{i=r..n} C(i,r) = C(n+1, r+1)
Vandermonde sum_k C(m,k) C(n,p-k) = C(m+n, p)
sum_k C(n,k) = 2^n sum_k (-1)^k C(n,k) = [n=0]
sum_k k*C(n,k) = n*2^(n-1) sum_k C(n,k)^2 = C(2n,n)
sum_k C(k,r) x^k ... upper negation: C(-n,k) = (-1)^k C(n+k-1, k)
sum_{k} C(n,k) C(k,m) = C(n,m) 2^(n-m)
Binomial inversion: f(n)=sum_k C(n,k) g(k) <=> g(n)=sum_k
   (-1)^(n-k) C(n,k) f(k)

INCLUSION-EXCLUSION
|A1 union ... union An| = sum |Ai| - sum |Ai n Aj| + ...
#surjections [n]->[k] = sum_{i=0..k} (-1)^i C(k,i) (k-i)^n = k!
   * S2(n,k)
"exactly m" from "at least m": E(m) = sum_{j>=m} (-1)^(j-m)
   C(j,m) A(j)
derangements D(n) = n! sum_{k=0..n} (-1)^k / k! =
   (n-1)(D(n-1)+D(n-2)), D0=1, D1=0
permutations with exactly k fixed points = C(n,k) * D(n-k)

CATALAN Cat(n) = C(2n,n)/(n+1) = C(2n,n) - C(2n,n+1), Cat(n+1)
   = sum Cat(i)Cat(n-i)
1,1,2,5,14,42,132,429,1430,4862 - counts: balanced bracket
   strings of length 2n;
   binary trees with n nodes; triangulations of an (n+2)-gon;
   monotone lattice paths under
   the diagonal; stack-sortable permutations; non-crossing
   partitions/chords.
BALLOT / CYCLE LEMMA: paths from 0 to k in n steps staying > 0
   = (k/n) C(n, (n+k)/2)
REFLECTION: #paths (0,0)->(a,b) touching the line y=x+c =
   #paths to the reflected point

STIRLING
2nd kind S2(n,k) = #partitions of n LABELLED items into k
   non-empty UNLABELLED blocks
S2(n,k) = k*S2(n-1,k) + S2(n-1,k-1); S2(n,k) = (1/k!) sum_i
   (-1)^i C(k,i) (k-i)^n
x^n = sum_k S2(n,k) * x_falling(k) surjections = k!
S2(n,k)
1st kind (unsigned) c(n,k) = #permutations of n with exactly k
   cycles
c(n,k) = c(n-1,k-1) + (n-1) c(n-1,k); sum_k c(n,k) x^k =
   x(x+1)..(x+n-1)
BELL B(n) = sum_k S2(n,k) = #set partitions. B(n+1) = sum_k
   C(n,k) B(k). Bell triangle.

PARTITIONS (unordered, into positive parts)
p(n) via pentagonal number theorem, O(sqrt n):
p(n) = sum_{k>=1} (-1)^(k+1) [ p(n - k(3k-1)/2) + p(n -
   k(3k+1)/2) ]
partitions into <= k parts = partitions into parts <= k
   (conjugate)

TWELVEFOLD WAY (n balls into k boxes)
balls boxes any injective surjective
dist. dist. k^n k!/(k-n)! k! S2(n,k)
ident. dist. C(n+k-1, n) C(k,n) C(n-1, n-k)
dist. ident. sum_{i<=k} S2(n,i) [n<=k] S2(n,k)
ident. ident. p_k(n) [n<=k] p(n into
   exactly k parts)

BURNSIDE / POLYA #orbits = (1/|G|) sum_{g in G} |fix(g)|
necklaces (rotations), n beads, k colours: (1/n) sum_{d|n}
   phi(n/d) k^d
bracelets (+ reflections): (necklace sum + reflection sum) /
   (2n),
   reflection sum = n*k^((n+1)/2) if n odd, (n/2)(k^(n/2) +
   k^(n/2+1)) if n even
```

TREES / GRAPHS

Cayley: $n^{(n-2)}$ labelled trees on n vertices; with given
 $\hookrightarrow$ degrees d_i : $(n-2)! / \prod (d_i-1)!$
 Prufer code $\leftrightarrow$ labelled tree, bijective
 Matrix-Tree: #spanning trees = any cofactor of the Laplacian (D
 $\hookrightarrow -A$)
 LGV: #non-intersecting path systems = $\det(M)$ with $M_{ij} =$
 $\hookrightarrow$ #paths $a_i \rightarrow b_j$

GENERATING FUNCTIONS

OGF $1/(1-x) = \sum x^n$ $1/(1-x)^k = \sum C(n+k-1,$
 $\hookrightarrow k-1) x^n$
 $x/(1-x-x^2) = \text{Fibonacci}$ $(1-\sqrt{1-4x})/(2x) = \text{Catalan}$
 $\text{prod } 1/(1-x^i) = \text{partitions}$ $\text{prod } (1+x^i) = \text{distinct-part}$
 $\hookrightarrow \text{partitions}$
 EGF $e^x = \sum x^n/n!$ EGF of set partitions =
 $\hookrightarrow e^x(e^x - 1)$
 EXPONENTIAL FORMULA: if $C(x)$ is the EGF of connected
 $\hookrightarrow$ structures, $e^C(x)$ is the EGF of all
 Coefficient extraction: $[x^n] A(x)B(x) = \sum_k a_k b_{n-k}$ (a
 $\hookrightarrow$ convolution $\rightarrow$ NTT)
 Lagrange inversion, for $F(0)=0, F'(0) \neq 0$: $[x^n] G(F^{-1})(x) =$
 $\hookrightarrow (1/n)[x^{n-1}] G'(x)(x/F(x))^n$

MISC

hook length: #standard Young tableaux of shape $\lambda = n! /$
 $\hookrightarrow \text{prod}(\text{hook lengths})$
 Erdos-Ko-Rado, Dilworth (min chain cover = max antichain),
 $\hookrightarrow$ Sperner (max antichain $C(n, n/2)$)

=====

$\hookrightarrow *$

5.4 CatalanStirlingBell

Catalan, Stirling (both kinds), Bell, derangements, partitions. Needs Mint + buildFact().

```
Mint catalan(int n) { return C(2 * n, n) * inv_[n + 1]; }
// paths from 0 to k in n steps (+1) that stay strictly positive
// - ballot / cycle lemma
Mint ballot(int n, int k) {
    if (n == 0) return k == 0; // the
    // empty path
    if (k <= 0 || k > n || (n + k) % 2) return 0; // k <= 0
    // was unguarded: Mint(-2) wraps
    return Mint(k) * inv_[n] * C(n, (n + k) / 2);
}

Mint derange(int n) { // D(n) =
    // (n-1)(D(n-1)+D(n-2))
    Mint a = 1, b = 0; // D0=1,
    // D1=0
    for (int i = 2; i <= n; i++) { Mint c = Mint(i - 1) * (a +
    // b); a = b, b = c; }
    return n ? b : a;
}

// permutations of n with exactly k fixed points
Mint fixedPoints(int n, int k) { return C(n, k) * derange(n - k);
    // }

// Stirling 2nd kind, single value: S2(n,k) = (1/k!) sum_i (-1)^i
// C(k,i) (k-i)^n O(k log n)
Mint stirling2(ll n, int k) {
    Mint r = 0;
    for (int i = 0; i <= k; i++) {
        Mint t = C(k, i) * Mint(k - i).pow(n);
        (i & 1) ? r -= t : r += t;
    }
    return r * invFact[k];
}

Mint surjections(ll n, int k) { return stirling2(n, k) * fact[k];
    // }

// Whole ROW of Stirling 2nd (all k for fixed n) in O(n^2), or O(n
// log n) as a convolution of
// a_i = (-1)^i/i! and b_i = i^n/i! (NTT).
vector<Mint> stirling2Row(int n) {
    vector<Mint> a(n + 1), b(n + 1), r(n + 1, 0);
    for (int i = 0; i <= n; i++) a[i] = (i & 1 ? Mint(0) -
    // invFact[i] : invFact[i]), b[i] = Mint(i).pow(n) * invFact[i];
    for (int k = 0; k <= n; k++) for (int i = 0; i <= k; i++)
    // r[k] += a[i] * b[k - i];
    return r;
}

// Stirling 1st kind (unsigned), whole row: prod_{i=0..n-1} (x +
// i) O(n^2)
vector<Mint> stirling1Row(int n) {
    vector<Mint> p = {1};
    for (int i = 0; i < n; i++) {
        vector<Mint> q(p.size() + 1, 0);
        for (size_t j = 0; j < p.size(); j++) q[j + 1] += p[j],
    // q[j] += p[j] * Mint(i);
        p = q;
    }
    return p; // p[k] =
    // c(n, k)
}

// Bell numbers B(0..n) via the Bell triangle, O(n^2)
vector<Mint> bell(int n) {
    vector<Mint> B(n + 1), row = {1};
    B[0] = 1;
    for (int i = 1; i <= n; i++) {
        vector<Mint> nr = {row.back()};
        for (Mint x : row) nr.push_back(nr.back() + x);
        row = nr, B[i] = row[0];
    }
    return B;
}

// Integer partitions p(0..n) via the pentagonal number theorem,
// O(n sqrt n)
vector<Mint> partitions(int n) {
    vector<Mint> p(n + 1, 0);
    p[0] = 1;
    for (int i = 1; i <= n; i++)
        for (int k = 1;; k++) {
            ll g1 = (11)k * (3 * k - 1) / 2, g2 = (11)k * (3 * k
    // + 1) / 2;
            if (g1 > i) break;
            Mint t = p[i - g1] + (g2 <= i ? p[i - g2] : Mint(0));
            (k & 1) ? p[i] += t : p[i] -= t;
        }
}
```

```

    return p;
}

```

5.5 InclusionExclusion

INCLUSION-EXCLUSION recipes. Needs Mint + buildFact().

```

// x1+...+xk = n with 0 <= xi <= ub[i]. IE over which variables
// overflow their bound.
// O(2^k) - fine for k <= 20. With equal bounds use the closed
// form below instead.
Mint boundedStarsBars(ll n, const vector<ll>& ub) {
    int k = ub.size();
    Mint r = 0;
    for (int m = 0; m < (1 << k); m++) {
        ll rem = n;
        for (int i = 0; i < k; i++) if (m >> i & 1) rem -= ub[i]
        + 1;
        if (rem < 0) continue;
        Mint t = C(rem + k - 1, k - 1);
        (__builtin_popcount(m) & 1) ? r -= t : r += t;
    }
    return r;
}
// Same with a COMMON bound c: sum_j (-1)^j C(k,j) C(n - j(c+1) +
// k-1, k-1). O(k)
Mint boundedStarsBarsEqual(ll n, ll k, ll c) {
    Mint r = 0;
    for (ll j = 0; j <= k && j * (c + 1) <= n; j++) {
        Mint t = C(k, j) * C(n - j * (c + 1) + k - 1, k - 1);
        (j & 1) ? r -= t : r += t;
    }
    return r;
}
// #surjections [n] -> [k] (equivalently k! * S2(n,k))
Mint surj(ll n, ll k) {
    Mint r = 0;
    for (ll i = 0; i <= k; i++) {
        Mint t = C(k, i) * Mint(k - i).pow(n);
        (i & 1) ? r -= t : r += t;
    }
    return r;
}
// "AT LEAST" -> "EXACTLY" (binomial / Mobius inversion over the
// subset lattice):
// if A(j) = #configurations having at least a chosen set of j
// properties (summed over sets),
// then E(m) = sum_{j=m}^k (-1)^(j-m) C(j,m) A(j) counts those
// with EXACTLY m.
Mint exactlyFromAtLeast(const vector<Mint>& A, int m) {
    Mint r = 0;
    for (int j = m; j < (int)A.size(); j++) {
        Mint t = C(j, m) * A[j];
        ((j - m) & 1) ? r -= t : r += t;
    }
    return r;
}
// BINOMIAL INVERSION: f(n) = sum_k C(n,k) g(k) <=> g(n) =
// sum_k (-1)^(n-k) C(n,k) f(k)
vector<Mint> binomialInvert(const vector<Mint>& f) {
    int n = f.size();
    vector<Mint> g(n, 0);
    for (int i = 0; i < n; i++)
        for (int k = 0; k <= i; k++) {
            Mint t = C(i, k) * f[k];
            ((i - k) & 1) ? g[i] -= t : g[i] += t;
        }
    return g;
}
// COUNTING COPRIME PAIRS / "gcd is exactly g": count multiples
// then Mobius-invert downwards:
// exact[g] = cntMultiples[g] - sum_{g | d, d > g} exact[d]
// (iterate g downwards)
vector<ll> exactGcdCounts(const vector<ll>& cntMultiples) {
    int n = cntMultiples.size() - 1;
    vector<ll> e(n + 1);
    for (int g = n; g >= 1; g--) {
        e[g] = cntMultiples[g];
        for (int d = 2 * g; d <= n; d += g) e[g] -= e[d];
    }
    return e;
}

```

5.6 TreeCounting

COUNTING TREES AND GRAPHS.

```

// Needs: Mint (06 - Math/01) and determinant (06 - Math/04 -
// Linear Algebra/02).

// PRUFER CODE: bijection between labelled trees on n>=2 vertices
// and sequences in [0,n)^(n-2).
// Immediate corollaries: Cayley n^(n-2) labelled trees; with
// prescribed degrees d_i the count is
// (n-2)! / prod (d_i - 1)! (because vertex i appears exactly d_i
// - 1 times in the code).
vector<int> toPrufer(const vector<vector<int>>& adj) { //
    // repeatedly strip the SMALLEST leaf
    int n = adj.size();
    if (n <= 2) return {};
    vector<int> deg(n), code;
    vector<bool> gone(n, false);
    priority_queue<int, vector<int>, greater<int>> q;
    for (int i = 0; i < n; i++) if ((deg[i] = adj[i].size()) ==
    1) q.push(i);
    for (int it = 0; it < n - 2; it++) {
        int leaf = q.top(); q.pop();
        gone[leaf] = true;
        int nb = -1;
        for (int v : adj[leaf]) if (!gone[v]) nb = v;
        code.push_back(nb);
        if (--deg[nb] == 1) q.push(nb);
    }
    return code;
}
vector<pair<int, int>> fromPrufer(vector<int> code) {
    int n = code.size() + 2;
    vector<int> deg(n, 1);
    for (int x : code) deg[x]++;
    priority_queue<int, vector<int>, greater<int>> q;
    for (int i = 0; i < n; i++) if (deg[i] == 1) q.push(i);
    vector<pair<int, int>> e;
    for (int x : code) {
        int leaf = q.top(); q.pop();
        e.push_back({leaf, x});
        if (--deg[x] == 1) q.push(x);
    }
    int a = q.top(); q.pop();
    e.push_back({a, q.top()});
    return e;
}

// MATRIX-TREE THEOREM: #spanning trees of a graph = any cofactor
// of the Laplacian L = D - A.
// Multigraphs work (put the multiplicity in A). Needs
// determinant() from GaussianElimination.
Mint spanningTrees(int n, const vector<pair<int, int>>& edges) {
    if (n == 1) return 1;
    vector<vector<Mint>> L(n, vector<Mint>(n, 0));
    for (auto [u, v] : edges) {
        L[u][u] += 1, L[v][v] += 1;
        L[u][v] -= 1, L[v][u] -= 1;
    }
    vector<vector<Mint>> M(n - 1, vector<Mint>(n - 1)); //
    // delete row/col 0
    for (int i = 1; i < n; i++) for (int j = 1; j < n; j++) M[i -
    1][j - 1] = L[i][j];
    return determinant(M);
}

// DIRECTED version (Tutte): #arborescences rooted at r = cofactor
// of (Dout - A) deleting row/col r
// for in-trees, or (Din - A) for out-trees.
// WEIGHTED version: put edge weights in A and the weighted degree
// in D -> sum over spanning
// trees of the product of weights (useful mod p for "sum over all
// spanning trees" problems).

// EXPONENTIAL FORMULA: if C(x) is the EGF of CONNECTED structures
// then e^(C(x)) is the EGF of
// all structures. Concretely, with g_n = #labelled graphs on n
// nodes = 2^C(n,2):
// c_n = g_n - sum_{k=1}^{n-1} C(n-1, k-1) * c_k * g_{n-k}
// (connected labelled graphs)
vector<Mint> connectedLabelledGraphs(int n) {
    vector<Mint> g(n + 1, c(n + 1, 0));
    for (int i = 0; i <= n; i++) g[i] = Mint(2).pow((ll)i * (i -
    1) / 2);
    for (int i = 1; i <= n; i++) {
        c[i] = g[i];
        for (int k = 1; k < i; k++) c[i] -= C(i - 1, k - 1) *
        c[k] * g[i - k];
    }
}

```



```

    return c;
}

```

5.7 SubsetConvolution

TRANSFORMS OVER THE SUBSET LATTICE. n bits, arrays of size 2^n .

```

// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).

// ZETA (sum over subsets) and its inverse MOBIUS.  $O(n 2^n)$ . This
// is SOS DP.
void zetaSub(vector<Mint>& f, int n) { // f[m]
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (m >> i & 1) f[m]
            += f[m ^ (1 << i)];
}
void mobiusSub(vector<Mint>& f, int n) { //
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (m >> i & 1) f[m]
            -= f[m ^ (1 << i)];
}
void zetaSuper(vector<Mint>& f, int n) { // f[m]
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (!(m >> i & 1))
            f[m] += f[m | (1 << i)];
}
void mobiusSuper(vector<Mint>& f, int n) {
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (!(m >> i & 1))
            f[m] -= f[m | (1 << i)];
}
// OR-convolution:  $h[m] = \sum_{a|b=m} f[a] g[b]$  -> zetaSub,
// multiply, mobiusSub
// AND-convolution:  $h[m] = \sum_{a\&b=m} f[a] g[b]$  ->
// zetaSuper, multiply, mobiusSuper
vector<Mint> orConv(vector<Mint> f, vector<Mint> g, int n) {
    zetaSub(f, n), zetaSub(g, n);
    for (int m = 0; m < (1 << n); m++) f[m] *= g[m];
    return mobiusSub(f, n), f;
}
vector<Mint> andConv(vector<Mint> f, vector<Mint> g, int n) {
    zetaSuper(f, n), zetaSuper(g, n);
    for (int m = 0; m < (1 << n); m++) f[m] *= g[m];
    return mobiusSuper(f, n), f;
}

// SUBSET (disjoint) CONVOLUTION:  $h[m] = \sum_{a|b=m, a\&b=0} f[a] g[b]$ .  $O(2^n n^2)$ .
// The ranked-zeta trick: index by popcount, convolve the ranks
// like polynomials.
vector<Mint> subsetConv(const vector<Mint>& f, const
    vector<Mint>& g, int n) {
    int N = 1 << n;
    vector<vector<Mint>> F(N + 1, vector<Mint>(N, 0)), G = F, H =
    F;
    for (int m = 0; m < N; m++) F[__builtin_popcount(m)][m] =
    f[m], G[__builtin_popcount(m)][m] = g[m];
    for (int i = 0; i <= n; i++) zetaSub(F[i], n), zetaSub(G[i],
    n);
    for (int i = 0; i <= n; i++)
        for (int j = 0; i + j <= n; j++)
            for (int m = 0; m < N; m++) H[i + j][m] += F[i][m] *
    G[j][m];
    for (int i = 0; i <= n; i++) mobiusSub(H[i], n);
    vector<Mint> h(N);
    for (int m = 0; m < N; m++) h[m] =
    H[__builtin_popcount(m)][m];
    return h;
}
// USES: exact set cover / "partition the ground set into k valid
// groups" (repeated subset
// convolution, or exponentiate the EGF-like series); counting
// perfect matchings by subsets;
// "for each mask, best way to split it into two disjoint valid
// halves".
// SUBMASK ENUMERATION over all masks is  $O(3^n)$ : for  $(s = m; s; s
// = (s-1) \& m)$ 

```

5.8 SpecialSequences

Named sequences that keep showing up, with what each one counts. Needs Mint + buildFact().

```

// MOTZKIN M(n): lattice paths (0,0)->(n,0) with steps U/D/FLAT
// staying >= 0; also the number of
// ways to draw non-crossing chords on n points on a circle. M:
// 1,1,2,4,9,21,51,127
vector<Mint> motzkin(int n) {
    vector<Mint> m(n + 1, 0);
    m[0] = 1;
    if (n >= 1) m[1] = 1;
    for (int i = 2; i <= n; i++) { // M(i)
        // = M(i-1) + sum M(k)M(i-2-k)
        m[i] = m[i - 1];
        for (int k = 0; k <= i - 2; k++) m[i] += m[k] * m[i - 2 -
        k];
    }
    return m;
}
// NARAYANA N(n,k): Dyck paths of length 2n with exactly k peaks.
// sum_k N(n,k) = Catalan(n).
Mint narayana(int n, int k) { return k < 1 || k > n ? Mint(0) :
    C(n, k) * C(n, k - 1) * inv_n[n]; }
// SCHRODER: large R(n) counts lattice paths (0,0)->(n,n) with
// steps E/N/NE staying below y=x.
// R: 1,2,6,22,90,394. R(n) = R(n-1) + sum_{k} R(k) R(n-1-k)
// (little schroeder s(n) = R(n)/2 for n >= 1 only (s(0) = 1,
// not 1/2))
vector<Mint> schroder(int n) {
    vector<Mint> r(n + 1, 0);
    r[0] = 1;
    for (int i = 1; i <= n; i++) { r[i] = r[i - 1]; for (int k =
    0; k < i; k++) r[i] += r[k] * r[i - 1 - k]; }
    return r;
}
// EULERIAN A(n,k): permutations of n with exactly k ascents.
// A(n,k) = (k+1) A(n-1,k) + (n-k) A(n-1,k-1). sum_k A(n,k) = n!
vector<vector<Mint>> eulerian(int n) {
    vector<vector<Mint>> a(n + 1, vector<Mint>(n + 1, 0));
    a[0][0] = 1;
    for (int i = 1; i <= n; i++)
        for (int k = 0; k < i; k++)
            a[i][k] = Mint(k + 1) * a[i - 1][k] + Mint(i - k) *
    (k ? a[i - 1][k - 1] : Mint(0));
    return a;
}
// q-BINOMIAL (Gaussian) coefficient: #k-dim subspaces of an n-dim
// space over F_q; also
// #partitions fitting in a k x (n-k) box, weighted by q^area.
// PRECONDITION: ord_p(q) > k. The denominator is prod (q^i - 1),
// which vanishes in F_p
// whenever ord_p(q) <= k - NOT only at q == 1. (q = -1 mod
// 998244353 breaks every k >= 2.)
// Root-of-unity-safe fallback, O(nk): Pascal's rule [n,k]_q =
// [n-1,k-1]_q + q^k [n-1,k]_q.
Mint qBinom(int n, int k, Mint q) { //
    // prod_{i=1..k} (q^{n-k+i}-1)/(q^i-1)
    if (k < 0 || k > n) return 0;
    if (q == Mint(1)) return C(n, k); // q=1
    // is the 0/0 limit
    Mint num = 1, den = 1;
    for (int i = 1; i <= k; i++) num *= q.pow(n - k + i) - 1, den
    *= q.pow(i) - 1;
    return num / den;
}
// HOOK LENGTH FORMULA: #standard Young tableaux of shape lambda =
// n! / prod(hook lengths).
// hook(i,j) = (arm to the right) + (leg below) + 1.
Mint youngTableaux(const vector<int>& lam) {
    int n = 0;
    for (int x : lam) n += x;
    vector<int> conj(lam.empty() ? 0 : lam[0], 0);
    for (int i = 0; i < (int)lam.size(); i++) for (int j = 0; j <
    lam[i]; j++) conj[j]++;
    Mint den = 1;
    for (int i = 0; i < (int)lam.size(); i++)
        for (int j = 0; j < lam[i]; j++) den *= Mint(lam[i] - j +
    conj[j] - i - 1);
    return fact[n] / den;
}
// LGV LEMMA: for sources a_1..a_k and sinks b_1..b_k in a DAG,
// the number of NON-INTERSECTING
// path systems equals det(M) with M[i][j] = #paths a_i -> b_j
// (when only the identity
// permutation can be non-intersecting). Counts plane partitions,
// rhombus tilings, etc.

```

5.9 EulerTransformAndSymbolic

```

SYMBOLIC METHOD: spec -> generating function =====
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
/* ===== SYMBOLIC METHOD: spec -> generating function
   ↳ =====
Decide labelled (EGF) or unlabelled (OGF), then translate the
↳ construction:
UNLABELLED (OGF), A built from atoms with GF B(z):
SEQ (ordered list) A = 1 / (1 - B)
MSET (unordered, rep) A = exp( sum_{k>=1} B(z^k) / k )
↳ <- Euler transform
PSET (unordered, no rep) A = exp( sum_{k>=1} (-1)^(k-1)
↳ B(z^k)/k )
CYC (up to rotation) A = sum_{k>=1} (phi(k)/k) * log(1/(1
↳ - B(z^k)))
LABELLED (EGF):
SEQ A = 1/(1-B) SET A = exp(B) CYC A =
↳ log(1/(1-B))
"SET of connected pieces = exp(connected)" is the exponential
↳ formula.
=====
↳ */

// EULER TRANSFORM: a[n] = #atoms of size n -> b[n] = #MULTISETS
↳ of atoms of total size n.
// prod_{n>=1} (1 - x^n)^(-a_n) = 1 + sum b_n x^n. O(n log n)
↳ via the harmonic sum.
vector<Mint> eulerTransform(const vector<Mint>& a) { // a[0]
↳ unused; returns b with b[0] = 1
    int n = a.size() - 1;
    vector<Mint> c(n + 1, 0), b(n + 1, 0);
    for (int d = 1; d <= n; d++) // c_n =
↳ sum_{d | n} d * a_d
        for (int m = d; m <= n; m += d) c[m] += Mint(d) * a[d];
    b[0] = 1;
    for (int i = 1; i <= n; i++) { // b_n =
↳ (c_n + sum_{k<n} c_k b_{n-k}) / n
        Mint s = c[i];
        for (int k = 1; k < i; k++) s += c[k] * b[i - k];
        b[i] = s * inv_[i];
    }
    return b;
}

int muSmall(int n) { // Mobius
↳ of one small n, by trial division
    int r = 1;
    for (int p = 2; (ll)p * p <= n; p++) if (n % p == 0) {
        n /= p;
        if (n % p == 0) return 0;
        r = -r;
    }
    return n > 1 ? -r : r;
}

// INVERSE Euler transform: b -> a (given multiset counts,
↳ recover the atom counts).
vector<Mint> eulerInverse(const vector<Mint>& b) {
    int n = b.size() - 1;
    vector<Mint> c(n + 1, 0), a(n + 1, 0);
    for (int i = 1; i <= n; i++) // c_n =
↳ n*b_n - sum_{d<n} c_d b_{n-d}
        Mint s = Mint(i) * b[i];
        for (int d = 1; d < i; d++) s -= c[d] * b[i - d];
        c[i] = s;
    }
    for (int i = 1; i <= n; i++) { // a_n =
↳ (1/n) sum_{d|n} mu(n/d) c_d
        Mint s = 0;
        for (int d = 1; d <= i; d++) if (i % d == 0) s +=
↳ Mint(muSmall(i / d)) * c[d];
        a[i] = s * inv_[i];
    }
    return a;
}

// UNLABELLED ROOTED TREES: r_1 = 1, and r_{n+1} = (Euler
↳ transform of r_1..r_n)[n]
// (a rooted tree = a root plus a MULTISET of rooted subtrees).
vector<Mint> rootedTrees(int n) {
    vector<Mint> r(n + 1, 0);
    if (n >= 1) r[1] = 1;
    for (int i = 1; i < n; i++) {
        vector<Mint> a(i + 1, 0);
        for (int j = 1; j <= i; j++) a[j] = r[j];
        auto b = eulerTransform(a);
        r[i + 1] = b[i];
    }
    return r;
}
↳ 0, 1, 1, 2, 4, 9, 20, 48, 115, 286, 719, ...

```

```

}
// PARTITIONS INTO PARTS FROM A SET S = eulerTransform with a[s] =
↳ 1 for s in S.
// COMPOSITIONS (ordered) use SEQ instead: 1/(1 - sum x^s).

```

5.10 PermanentAndSymmetric

```

RYSER'S FORMULA - permanent of an n x n matrix in O(2^n * n).
// perm(A) counts PERFECT MATCHINGS of a bipartite graph
↳ (weighted: sum over matchings of the
// product of weights). Use for n <= -20: "count ways to assign n
↳ tasks to n people",
// "count systems of distinct representatives", "count
↳ permutations avoiding forbidden cells".
Mint permanent(const vector<vector<Mint>>& a) {
    int n = a.size();
    vector<Mint> row(n, 0); // row[i]
↳ = sum_{j in S} a[i][j]
    Mint total = 0;
    int prev = 0;
    for (int g = 1; g < (1 << n); g++) { // Gray
↳ code: one column toggles per step
        int cur = g ^ (g >> 1), diff = cur ^ prev, j =
↳ __builtin_ctz(diff);
        if (cur & diff) for (int i = 0; i < n; i++) row[i] +=
↳ a[i][j]; // column j entered S
        else for (int i = 0; i < n; i++) row[i] -= a[i][j];
↳ // column j left S
        prev = cur;
        Mint p = 1;
        for (int i = 0; i < n; i++) p *= row[i];
        (__builtin_popcount(cur) & 1) ? total -= p : total += p;
    }
    return (n & 1) ? Mint(0) - total : total; // perm =
↳ (-1)^n * sum_S (-1)^|S| prod_i rowSum
}

// PERMANENT vs DETERMINANT: identical formula without the signs.
↳ det is polynomial-time,
// perm is #P-hard - never expect an O(n^3) permanent.

// NEWTON'S IDENTITIES - convert between power sums p_k = sum
↳ x_i^k and elementary symmetric e_k.
// e_k = sum over k-subsets of the product; this is exactly "for
↳ every k, the sum over all
// k-subsets of the product of the chosen values".
// e_0 = 1; k * e_k = sum_{i=1..k} (-1)^(i-1) * e_{k-i} * p_i
vector<Mint> powerToElementary(const vector<Mint>& p, int n) {
↳ // p[1..n] -> e[0..n]
    vector<Mint> e(n + 1, 0);
    e[0] = 1;
    for (int k = 1; k <= n; k++) {
        Mint s = 0;
        for (int i = 1; i <= k; i++) {
            Mint t = e[k - i] * p[i];
            (i & 1) ? s += t : s -= t;
        }
        e[k] = s * inv_[k];
    }
    return e;
}

// p_k = (-1)^(k-1) * k * e_k + sum_{i=1..k-1} (-1)^(k-1+i) *
↳ e_{k-i} * p_i
vector<Mint> elementaryToPower(const vector<Mint>& e, int n) {
↳ // e[0..n] -> p[1..n]
    vector<Mint> p(n + 1, 0);
    for (int k = 1; k <= n; k++) {
        Mint s = Mint(k) * e[k];
        if ((k - 1) & 1) s = Mint(0) - s; //
↳ (-1)^(k-1) * k * e_k
        for (int i = 1; i < k; i++) {
            Mint t = e[k - i] * p[i];
            ((k - 1 + i) & 1) ? s -= t : s += t;
        }
        p[k] = s;
    }
    return p;
}

// e_k are also the coefficients of prod (x + x_i): prod_{i} (x +
↳ x_i) = sum_k e_k * x^(n-k).
// So "sum over k-subsets of the product" = coefficient extraction
↳ from that product, which you
// can also get in O(n log^2 n) by divide-and-conquer
↳ multiplication of the linear factors.

```


5.11 BernoulliFaulhaber

```

BERNOULLI NUMBERS and the CLOSED-FORM FAULHABER polynomial.
// Use this (not Lagrange interpolation) when you need the
//   ↪ POLYNOMIAL COEFFICIENTS of
//   sum_{k=1..n} k^p, or when p is large and many n are queried.
// Convention here: B_1 = -1/2 (the "B~-" convention that the
//   ↪ formula below expects).
// B_0 = 1 ; sum_{k=0}^{m-1} C(m+1, k) * B_k = [m == 0]
// => B_m = -(1/(m+1)) * sum_{k=0}^{m-1} C(m+1, k) * B_k
vector<Mint> bernoulli(int m) {
    ↪ B[0..m], O(m^2)
    vector<Mint> B(m + 1, 0);
    B[0] = 1;
    for (int i = 1; i <= m; i++) {
        Mint s = 0;
        for (int k = 0; k < i; k++) s += C(i + 1, k) * B[k];
        B[i] = Mint(0) - s * inv_[i + 1];
    }
    return B;
    ↪ -1/2, 1/6, 0, -1/30, 0, 1/42, ...
}

// O(m log m) alternative: B_m / m! = [t^m] inv( sum_{i>=0} t^i /
//   ↪ (i+1)! ) -- one Poly::inv.

// FAULHABER: sum_{k=1}^{n} k^p = (1/(p+1)) * sum_{r=0}^{p}
//   ↪ (-1)^r * C(p+1, r) * B_r * n^(p+1-r)
Mint powerSumClosed(ll n, int p, const vector<Mint>& B) {
    Mint s = 0, np = Mint(n).pow(p + 1);
    ↪ n^(p+1-r), decreasing r
    vector<Mint> pw(p + 2);
    pw[0] = 1;
    for (int i = 1; i <= p + 1; i++) pw[i] = pw[i - 1] * Mint(n);
    for (int r = 0; r <= p; r++) {
        Mint t = C(p + 1, r) * B[r] * pw[p + 1 - r];
        (r & 1) ? s -= t : s += t;
    }
    return s * inv_[p + 1];
}

// The COEFFICIENTS of the Faulhaber polynomial in n (degree p+1):
// [n^(p+1-r)] = (-1)^r * C(p+1, r) * B_r / (p+1)
vector<Mint> faulhaberPoly(int p, const vector<Mint>& B) {
    vector<Mint> c(p + 2, 0);
    for (int r = 0; r <= p; r++) {
        Mint t = C(p + 1, r) * B[r] * inv_[p + 1];
        c[p + 1 - r] = (r & 1) ? Mint(0) - t : t;
    }
    return c;
    ↪ coefficient of n^i
}

// RELATED SUMS
// sum_{k=0}^{n-1} C(k, r) = C(n+1, r+1)
//   ↪ (hockey stick)
// sum_{k=1}^{n} k^p * r^k : degree-p polynomial times r^n;
//   ↪ solve by ansatz or by
//   ↪ differentiating the geometric
//   ↪ series p times.
// Bernoulli also gives Euler-Maclaurin and the von
//   ↪ Staudt-Clausen denominator facts.

```

5.12 CycleLemmaFussCatalan

CYCLE LEMMA (Dvoretzky-Motzkin) and the FUSS-CATALAN / rational-Catalan family.

```

// The general-k version of the ballot problem: it handles every
//   ↪ "+1 / -k step" path problem.
//
// CYCLE LEMMA: a word with m a's and n b's, m >= k*n, has EXACTLY
//   ↪ m - k*n cyclic shifts that
//   ↪ are k-dominating (every non-empty prefix has #a > k * #b).
//   ↪ k = 1 ballot: p A's and q B's with p > q -> exactly p - q
//   ↪ dominating shifts,
//   ↪ so the probability that A leads the whole count
//   ↪ is (p - q) / (p + q).

// #paths with p up-steps and q down-steps that stay STRICTLY
//   ↪ positive after the first step.
Mint ballotStrict(ll p, ll q) {
    ↪ >= 0
    if (p <= q) return 0;
    return C(p + q, q) * Mint(p - q) * inv_[(int)(p + q)];
}

// FUSS-CATALAN: A_m(p, r) = (r / (m*p + r)) * C(m*p + r, m)
Mint fussCatalan(ll m, ll p, ll r) {
    if (m < 0) return 0;
    return C(m * p + r, m) * Mint(r) * Mint(m * p + r).inv();
}

// #p-ary trees with m internal nodes = A_m(p, 1) = C(m*p + 1, m)
//   ↪ / (m*p + 1)
Mint pAryTrees(ll m, ll p) { return fussCatalan(m, p, 1); }
// Catalan is the case p = 2, r = 1.
// #sequences of m opens and (p-1)*m closes where every prefix
//   ↪ keeps the stack non-negative
// is also A_m(p, 1).

// RATIONAL CATALAN: for gcd(a, b) = 1, the number of monotone
//   ↪ lattice paths from (0,0) to
//   ↪ (b, a) that never rise above the line a*x = b*y is C(a + b, a)
//   ↪ / (a + b).
Mint rationalCatalan(ll a, ll b) { return C(a + b, a) * Mint(a +
    ↪ b).inv(); }

// CHUNG-FELLER: among the C(2n,n) balanced +-1 paths of length
//   ↪ 2n, the number having exactly
//   ↪ 2r steps above the axis is Catalan(n) for EVERY r in 0..n.
//   ↪ Hence C(2n,n) = (n+1)*Cat(n).
// Use it for "count sequences with exactly k flaws" - the answer
//   ↪ does not depend on k.

// REFLECTION PRINCIPLE (the tool behind all of the above):
// #paths (0,0)->(x,y) touching the line y = m = #paths
//   ↪ (0,0)->(x, 2m - y)
// so "#paths avoiding the line" = C(total, up) - C(total, up
//   ↪ shifted by the reflection).
// PRECONDITION: the barrier must SEPARATE the start (height 0)
//   ↪ from the end y, i.e. m != 0
// and 0, y strictly on the same side of m. Without it the
//   ↪ reflection subtracts too much:
// (x,y,m) = (2,2,1) returns -1 where the true count is 0.
Mint pathsAvoidingLine(ll x, ll y, ll m) {
    if (m == 0 || (m > 0 && y >= m) || (m < 0 && y <= m)) return
    ↪ 0;
    // steps +1/-1, must never reach y = m
    if (((x + y) & 1) || y > x) return 0;
    ll up = (x + y) / 2, up2 = (x + 2 * m - y) / 2;
    return C(x, up) - ((x + 2 * m - y) % 2 || up2 < 0 || up2 > x
    ↪ ? Mint(0) : C(x, up2));
}

```

5.13 LGVandPolya

LGV LEMMA and the POLYA CYCLE INDEX - the two theorems that were stated but never coded.

```
// Needs: Mint, determinant (06 - Math/04/02), phi (04 -
    ↳ NT/02/06), muSmall (09 - EulerTransform...).

// LINDSTROM-GESSEL-VIENNOT: with sources a_1..a_k and sinks
    ↳ b_1..b_k in a DAG,
// det(M), M[i][j] = #paths a_i -> b_j
// counts NON-INTERSECTING path systems, PROVIDED the only
    ↳ permutation that admits a
// non-intersecting system is the identity (true for "sorted"
    ↳ sources/sinks in a grid).
// Grid instance: monotone lattice paths, so M[i][j] = C(dx + dy,
    ↳ dx).
Mint lgvGrid(const vector<pair<ll, ll>>& src, const
    ↳ vector<pair<ll, ll>>& dst) {
    int k = src.size();
    vector<vector<Mint>> M(k, vector<Mint>(k));
    for (int i = 0; i < k; i++)
        for (int j = 0; j < k; j++) {
            ll dx = dst[j].first - src[i].first, dy =
            ↳ dst[j].second - src[i].second;
            M[i][j] = (dx < 0 || dy < 0) ? Mint(0) : C(dx + dy,
            ↳ dx);
        }
    return determinant(M); // needs
    ↳ GaussianElimination.cpp
}

// USES: k non-crossing monotone paths, rhombus/domino tilings of
    ↳ a region, plane partitions,
// "k tokens walk on a grid and must never share a cell".

// POLYA CYCLE INDEX. Z(G)(a_1..a_n) = (1/|G|) sum_{g in G} prod_k
    ↳ a_k^{(#k-cycles of g)}.
// The UNWEIGHTED orbit count is Z(G)(m, m, ..., m) for m colours
    ↳ - that is what necklaces()
// and bracelets() already give. The reason to carry the cycle
    ↳ index is the WEIGHTED case:
// "necklaces with exactly c_1 of colour 1, c_2 of colour 2, ..."
    ↳ which the numeric form cannot do.
// Z(C_n) = (1/n) * sum_{d|n} phi(d) * a_d^{(n/d)}
// Z(D_n) = (1/2) Z(C_n) + (1/2) a_1 a_2^{((n-1)/2)}
    ↳ [n odd]
// Z(D_n) = (1/2) Z(C_n) + (1/4) (a_1^2 a_2^{((n-2)/2)} +
    ↳ a_2^{(n/2)}) [n even]
// Z(S_n) = (1/n) * sum_{l=1..n} a_l * Z(S_{n-l}), Z(S_0) = 1
// Weighted PET: substitute a_k -> (t_1^k + t_2^k + ... + t_m^k)
    ↳ and read off the coefficient
// of t_1^{c_1} * ... * t_m^{c_m}.
// Necklaces with a PRESCRIBED colour multiset, n beads, counts
    ↳ c[] summing to n (rotations only):
Mint necklacesWithCounts(int n, const vector<int>& c) {
    Mint total = 0;
    for (int d = 1; d <= n; d++) {
        if (n % d) continue; //
        ↳ rotations of order n/d have d cycles
        bool ok = true;
        for (int x : c) if (x % (n / d)) ok = false; // each
        ↳ colour must split evenly
        if (!ok) continue;
        Mint ways = fact[d]; //
        ↳ multinomial over the d cycle slots
        for (int x : c) ways *= invFact[x / (n / d)];
        total += Mint(phi(n / d)) * ways;
    }
    return total * Mint(n).inv();
}

// APERIODIC necklaces (Lyndon words) with k colours: M(n,k) =
    ↳ (1/n) sum_{d|n} mu(d) k^{(n/d)}.
Mint lyndonCount(ll n, ll k) {
    Mint s = 0;
    for (ll d = 1; d <= n; d++) if (n % d == 0) s +=
    ↳ Mint(muSmall(d)) * Mint(k).pow(n / d);
    return s * Mint(n).inv();
}

// TRANSFER MATRIX: count length-L walks in a finite automaton.
    ↳ Build A[s][t] = #transitions,
// then (A^L)[s][t]. The generating function sum_L (A^L)[i][j] x^L
    ↳ = ((I - xA)^{-1})[i][j] is
// RATIONAL with denominator det(I - xA) of degree <= #states, so
    ↳ the sequence satisfies a linear
// recurrence of that order - feed a prefix to Berlekamp-Massey
    ↳ and finish in O(k^2 log L)
// instead of O(states^3 log L).
```

6 Math

6.1 Modular Arithmetic

6.1.1 Mint

Modular integer (MOD must be PRIME for inv/division) + O(N) factorial tables.

```
struct Mint {
    int v;
    Mint(ll x = 0) { v = int(x % MOD); if (v < 0) v += MOD; }
    Mint& operator+=(Mint o) { if ((v += o.v) >= MOD) v -= MOD;
    ↳ return *this; }
    Mint& operator-=(Mint o) { if ((v -= o.v) < 0) v += MOD;
    ↳ return *this; }
    Mint& operator*=(Mint o) { v = int(1LL * v * o.v % MOD);
    ↳ return *this; }
    Mint& operator/=(Mint o) { return *this *= o.inv(); }
    friend Mint operator+(Mint a, Mint b) { return a += b; }
    friend Mint operator-(Mint a, Mint b) { return a -= b; }
    friend Mint operator*(Mint a, Mint b) { return a *= b; }
    friend Mint operator/(Mint a, Mint b) { return a /= b; }
    friend bool operator==(Mint a, Mint b) { return a.v == b.v; }
    Mint pow(ll e) const { Mint r = 1, b = *this; for (; e > 0; e
    ↳ >>= 1, b *= b) if (e & 1) r *= b; return r; }
    Mint inv() const { return pow(MOD - 2); } //
    ↳ Fermat: a^{p-2} = a^{-1} (mod p)
    friend ostream& operator<<(ostream& o, Mint m) { return o <<
    ↳ m.v; }
    friend istream& operator>>(istream& i, Mint& m) { ll x; i >>
    ↳ x; m = Mint(x); return i; }
};

const int N = 200005; // ALSO
    ↳ in 01 - Template.cpp: keep one
Mint fact[N], invFact[N], inv_[N];
void buildFact() { // O(N)
    ↳ total, not O(N log MOD)
    fact[0] = invFact[0] = inv_[1] = 1;
    for (int i = 1; i < N; i++) fact[i] = fact[i - 1] * i;
    invFact[N - 1] = fact[N - 1].inv();
    for (int i = N - 1; i > 0; i--) invFact[i - 1] = invFact[i] *
    ↳ i;
    for (int i = 2; i < N; i++) inv_[i] = Mint(MOD - MOD / i) *
    ↳ inv_[MOD % i];
}

Mint C(ll n, ll r) { if (r < 0 || r > n) return 0; return fact[n]
    ↳ * invFact[r] * invFact[n - r]; }
Mint Perm(ll n, ll r) { if (r < 0 || r > n) return 0; return
    ↳ fact[n] * invFact[n - r]; }
// C(n, r) for HUGE n and small r (no tables): prod_{i<r} (n-i) / r!
Mint Cbig(ll n, ll r) {
    if (r < 0 || n < 0 || r > n) return 0;
    Mint num = 1;
    for (ll i = 0; i < r; i++) num *= Mint(n - i);
    return num * invFact[r];
}
```

6.2 Multiplicative Functions

6.2.1 EulerTotientPhi

EULER TOTIENT SIEVE - phiArr[i] = #{1 <= k <= i : gcd(k,i)=1} for all i < N in O(N log log N).

```
// Needed for: a^b mod m with gcd(a,m)=1 (Euler's theorem),
    ↳ counting coprime pairs, Mobius identities
// (sum_{d|n} phi(d) = n), and the order of the multiplicative
    ↳ group mod n.
// NAME: phiArr, not phi - a global 'phi' collides with the phi()
    ↳ FUNCTION in
// 04 - Number Theory/02 - Primes and Divisors/06 -
    ↳ MillerRabinPollard.cpp.
int phiArr[N];

void calculatePhi() {
    for (int i = 0; i < N; ++i) phiArr[i] = i & 1 ? i : i / 2;
    for (int i = 3; i < N; i += 2)
        if (phiArr[i] == i)
            for (int j = i; j < N; j += i) phiArr[j] -= phiArr[j]
            ↳ / i;
}
```

6.2.2 MobiusFunction

Mobius Function $\mu(n)$ Sieve - Time: $O(N)$, Space: $O(N)$

```
// mu(n) = 1 if n is square-free with an even number of prime
  ↳ factors
// mu(n) = -1 if n is square-free with an odd number of prime
  ↳ factors
// mu(n) = 0 if n has a squared prime factor
struct Mobius {
    int n;
    vector<int> mu, primes;
    vector<bool> is_prime;

    Mobius(int n = 1e7) : n(n), mu(n + 1, 0), is_prime(n + 1,
  ↳ true) {
        is_prime[0] = is_prime[1] = false;
        mu[1] = 1;

        for (int i = 2; i <= n; i++) {
            if (is_prime[i]) {
                primes.pb(i);
                mu[i] = -1;
            }
            for (int p : primes) {
                if ((ll)i * p > n) break;
                is_prime[i * p] = false;
                if (i % p == 0) {
                    mu[i * p] = 0;
                    break;
                } else {
                    mu[i * p] = -mu[i];
                }
            }
        }
    }
};
```

6.3 Matrices

6.3.1 MatrixExponentiation

Matrix over Mint. mul $O(n^3)$ (cache-friendly i-k-j order), pow $O(n^3 \log e)$.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).
// Fix the size at compile time (array<array<Mint,K>,K>) if you
  ↳ need the last 2x speedup.
typedef vector<vector<Mint>> Mat;

Mat operator*(const Mat& a, const Mat& b) {
    int n = a.size(), k = b.size(), m = b[0].size();
    Mat c(n, vector<Mint>(m));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < k; t++) if (a[i][t].v)
            for (int j = 0; j < m; j++) c[i][j] += a[i][t] *
  ↳ b[t][j];
    return c;
}

Mat eye(int n) { Mat r(n, vector<Mint>(n)); for (int i = 0; i <
  ↳ n; i++) r[i][i] = 1; return r; }

Mat mpow(Mat a, ll e) {
    Mat r = eye(a.size());
    for (; e > 0; e >>= 1, a = a * a) if (e & 1) r = r * a;
    return r;
}

// RECIPES
// Linear recurrence  $f(n) = c_1 f(n-1) + \dots + c_k f(n-k)$ :
// companion matrix, first row =  $(c_1 \dots c_k)$ , subdiagonal = 1.
  ↳  $f(n) = (M^{n-k} * [f(k-1) \dots f(0)]^T)[0]$ 
// Add a "+d" constant term: enlarge by one row/col with a 1 on
  ↳ the diagonal, put d in the first row.
// Count walks of length L between u and v:  $(A^L)[u][v]$  where A
  ↳ is the adjacency matrix.
// Shortest walk with exactly L edges: same, but replace (+,*)
  ↳ with (min,+) - "min-plus product".
```

6.4 Linear Algebra

6.4.1 LinearXorBasis

XOR basis (vector space over $GF(2)$). insert / query $O(B)$.

```
// span size =  $2^{sz}$ . Use MERGE to keep a basis in each
  ↳ segment-tree node -> max-xor over a RANGE.
struct Basis {
    static const int B = 62; // 60 silently drops bits 61-62
    ↳ of an ll up to  $9e18$ 
    ll b[B + 1] = {}; // b[i]
    ↳ has leading bit i
    int sz = 0;

    bool insert(ll x) { // false
    ↳ if x already in the span
        for (int i = B; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { b[i] = x, sz++; return true; }
            x ^= b[i];
        }
        return false;
    }
    bool contains(ll x) const {
        for (int i = B; i >= 0; i--) if (x >> i & 1) { if (!b[i])
  ↳ return false; x ^= b[i]; }
        return true;
    }
    ll maxXor(ll r = 0) const { for (int i = B; i >= 0; i--) r =
  ↳ max(r, r ^ b[i]); return r; }
    ll minXor(ll r = 0) const { for (int i = B; i >= 0; i--) if
  ↳ (r >> i & 1) r ^= b[i]; return r; }
    void reduce() { //
    ↳ reduced row echelon form
        for (int i = B; i >= 0; i--) if (b[i])
            for (int j = i - 1; j >= 0; j--) if (b[i] >> j & 1)
  ↳ b[i] ^= b[j];
    }
    ll kth(ll k) { // k-th
    ↳ SMALLEST value in the span, 0-indexed
        reduce(); // kth(0)
    ↳ == 0; -1 if k >=  $2^{sz}$ 
        ll r = 0;
        for (int i = 0; i <= B; i++) if (b[i]) { if (k & 1) r ^=
  ↳ b[i]; k >>= 1; }
        return k ? -1 : r;
    }
    ll spanSize() const { return sz >= 63 ? -1 : 1LL << sz; }
    friend Basis merge(Basis a, const Basis& c) {
        for (int i = 0; i <= B; i++) if (c.b[i]) a.insert(c.b[i]);
        return a;
    }
};
```

6.4.2 GaussianElimination

Gaussian elimination mod a PRIME. Returns rank; solves $Ax = b$; also det and inverse.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp) for the
// modular routines.
// O(n^2 m). Pass the augmented matrix (n rows, m+1 cols) to
// solve().
int gaussMod(vector<vector<Mint>>& a, int m) {
    // reduce first m columns to RREF
    int n = a.size(), rank = 0;
    for (int col = 0; col < m && rank < n; col++) {
        int piv = -1;
        for (int i = rank; i < n; i++) if (a[i][col].v) { piv = i; break; }
        if (piv < 0) continue;
        swap(a[rank], a[piv]);
        Mint iv = a[rank][col].inv();
        for (auto& x : a[rank]) x *= iv;
        for (int i = 0; i < n; i++) if (i != rank && a[i][col].v)
            Mint f = a[i][col];
            for (size_t j = 0; j < a[i].size(); j++) a[i][j] -= f * a[rank][j];
            rank++;
    }
    return rank;
}

// Solve  $Ax = b$ . Returns {} if inconsistent, else one solution
// (free vars = 0).
// 'free' receives the indices of the free variables (solution
// space has MOD^|free| points).
vector<Mint> solveLinear(vector<vector<Mint>> a, vector<Mint> b,
    vector<int>* freeIdx = nullptr) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) a[i].push_back(b[i]);
    int rank = gaussMod(a, m);
    for (int i = rank; i < n; i++) if (a[i][m].v) return {};
    // 0 = nonzero
    vector<Mint> x(m, 0);
    int r = 0;
    for (int c = 0; c < m; c++) {
        // must run
        over EVERY column: stopping at
        if (r < rank && a[r][c].v) x[c] = a[r][m], r++;
        // rank drops the free columns
        else if (freeIdx) freeIdx->push_back(c);
        // that
    }
    return x;
}

Mint determinant(vector<vector<Mint>> a) {
    // O(n^3), destroys a
    int n = a.size();
    Mint det = 1;
    for (int col = 0; col < n; col++) {
        int piv = -1;
        for (int i = col; i < n; i++) if (a[i][col].v) { piv = i; break; }
        if (piv < 0) return 0;
        if (piv != col) swap(a[col], a[piv]), det = Mint(0) - det;
        det *= a[col][col];
        Mint iv = a[col][col].inv();
        for (int i = col + 1; i < n; i++) if (a[i][col].v) {
            Mint f = a[i][col] * iv;
            for (int j = col; j < n; j++) a[i][j] -= f * a[col][j];
        }
    }
    return det;
}

vector<vector<Mint>> inverse(vector<vector<Mint>> a) {
    // {} if singular
    int n = a.size();
    for (int i = 0; i < n; i++) { a[i].resize(2 * n, 0); a[i][n + i] = 1; }
    if (gaussMod(a, n) < n) return {};
    vector<vector<Mint>> r(n, vector<Mint>(n));
    for (int i = 0; i < n; i++) for (int j = 0; j < n; j++)
        r[i][j] = a[i][n + j];
    return r;
}

// MATRIX-TREE: #spanning trees = determinant of the Laplacian
// (deg on diagonal, -1 per edge)
// with any one row AND the matching column deleted.
```

6.4.3 GaussXorAndReal

XOR-equation systems over GF(2) with bitset - $O(n*m/64)$. n equations, m variables,

```
// row[i][m] holds the RHS. Returns rank, or -1 if inconsistent.
template<int M>
int gaussXor(vector<bitset<M + 1>>& a, int m, bitset<M>* sol =
    nullptr) {
    int n = a.size(), rank = 0;
    vector<int> where(m, -1);
    for (int col = 0; col < m && rank < n; col++) {
        int piv = -1;
        for (int i = rank; i < n; i++) if (a[i][col]) { piv = i; break; }
        if (piv < 0) continue;
        swap(a[rank], a[piv]);
        for (int i = 0; i < n; i++) if (i != rank && a[i][col])
            a[i] ^= a[rank];
        where[col] = rank++;
    }
    for (int i = rank; i < n; i++) if (a[i][m]) return -1;
    if (sol) { sol->reset(); for (int c = 0; c < m; c++) if
        (where[c] >= 0) (*sol)[c] = a[where[c]][m]; }
    return rank;
}

// USES: "light switch" puzzles, xor-equation counting, rank of a
// set of vectors over GF(2),
// linear independence of xor values (dense sibling of the
// XOR basis).

// ---- Gaussian elimination over the reals, with partial pivoting
// ----
// Returns rank; a is n x (m+1) augmented. Solution in x (free
// vars = 0). eps-based.
int gaussReal(vector<vector<ld>> a, int m, vector<ld>& x) {
    int n = a.size(), rank = 0;
    vector<int> where(m, -1);
    for (int col = 0; col < m && rank < n; col++) {
        int piv = rank;
        for (int i = rank; i < n; i++) if (fabsl(a[i][col]) >
            fabsl(a[piv][col])) piv = i;
        if (fabsl(a[piv][col]) < 1e-12) continue;
        swap(a[rank], a[piv]);
        for (int i = 0; i < n; i++) if (i != rank) {
            ld f = a[i][col] / a[rank][col];
            for (int j = col; j <= m; j++) a[i][j] -= f * a[rank][j];
        }
        where[col] = rank++;
    }
    x.assign(m, 0);
    for (int c = 0; c < m; c++) if (where[c] >= 0) x[c] =
        a[where[c]][m] / a[where[c]][c];
    return rank;
}

// The classic use is EXPECTED-VALUE DP with cyclic state
// dependencies: write one equation per
// state,  $E[s] = 1 + \sum p(s \rightarrow t) E[t]$ , and solve. n <= ~500 fits
// O(n^3).
```

6.5 Convolution

6.5.1 FFT

FFT over complex doubles. `conv()` is exact while `|result coefficient| < ~1e15`.

```
// Prefer NTT when the answer is taken modulo an NTT prime; use
//   ↪ convMod for arbitrary moduli.
typedef complex<double> C;
void fft(vector<C>& a) {
    int n = a.size(), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1);
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n), rt.resize(n);
        auto x = polar(1.0L, acos(-1.0L) / k);
        for (int i = k; i < 2 * k; i++) rt[i] = R[i] = i & 1 ?
        ↪ R[i / 2] * x : R[i / 2];
    }
    vector<int> rev(n);
    for (int i = 0; i < n; i++) rev[i] = (rev[i / 2] | (i & 1) <<
    ↪ L) / 2;
    for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i],
    ↪ a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k)
            for (int j = 0; j < k; j++) {
                C z = rt[j + k] * a[i + j + k];
                a[i + j + k] = a[i + j] - z;
                a[i + j] += z;
            }
}
// Real convolution. Result is rounded to the nearest integer.
vector<ll> conv(const vector<ll>& a, const vector<ll>& b) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <= 1;
    vector<C> in(n), out(n);
    for (size_t i = 0; i < a.size(); i++)
        ↪ in[i].real((double)a[i]);
    for (size_t i = 0; i < b.size(); i++)
        ↪ in[i].imag((double)b[i]);
    fft(in);
    for (C& x : in) x *= x; //
    ↪ (a+bi)^2 = a^2-b^2+2abi
    for (int i = 0; i < n; i++) out[i] = in[-i & (n - 1)] -
    ↪ conj(in[i]);
    fft(out);
    vector<ll> res(rs);
    for (int i = 0; i < rs; i++) res[i] = llround(imag(out[i]) /
    ↪ (4 * n));
    return res;
}
// Convolution modulo an ARBITRARY M, splitting coefficients at
//   ↪ sqrt(M) to keep precision.
vector<ll> convMod(const vector<ll>& a, const vector<ll>& b, ll
    ↪ M) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, B = 32 - __builtin_clz(rs),
    ↪ n = 1 << B;
    ll cut = (ll)sqrtl((ld)M);
    vector<C> L(n), R(n), os(n), ol(n);
    for (size_t i = 0; i < a.size(); i++) L[i] = C((double)a[i]
    ↪ / cut), (double)(a[i] % cut));
    for (size_t i = 0; i < b.size(); i++) R[i] = C((double)(b[i]
    ↪ / cut), (double)(b[i] % cut));
    fft(L), fft(R);
    for (int i = 0; i < n; i++) {
        int j = -i & (n - 1);
        ol[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        os[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / C(0, 1);
    }
    fft(ol), fft(os);
    vector<ll> res(rs);
    for (int i = 0; i < rs; i++) {
        ll av = llround(real(ol[i])), cv = llround(imag(os[i]));
        ll bv = llround(imag(ol[i])) + llround(real(os[i]));
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}
```

6.5.2 NTT

NTT modulo 998244353 (= $119 \cdot 2^{23} + 1$, primitive root 3). $O(n \log n)$.

```
// multiply() is the entry point. MAX LENGTH  $2^{23}$  for 998244353
//   ↪ (see the assert in ntt()).
// For an ARBITRARY modulus: run this under three of the primes
//   ↪ below and Garner them together
// (06 - Garner.cpp is written for exactly that), or split the
//   ↪ coefficients at sqrt(M) and use
// convMod in 01 - FFT.cpp.
const ll NMOD = 998244353, NROOT = 3;
ll npow(ll a, ll e, ll m = NMOD) { ll r = 1; a %= m; for (; e; e
    ↪ >>= 1, a = a * a % m) if (e & 1) r = r * a % m; return r; }

void ntt(vector<ll>& a, bool inv) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        assert((NMOD - 1) % len == 0); // 998244352 =  $119 \cdot 2^{23}$ :
        ↪ len >  $2^{23}$  truncates the division
        ll w = npow(NROOT, (NMOD - 1) / len); // and returns
        ↪ GARBAGE with no other symptom
        if (inv) w = npow(w, NMOD - 2);
        for (int i = 0; i < n; i += len) {
            ll cur = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] * cur %
                ↪ NMOD;
                a[i + j] = (u + v) % NMOD;
                a[i + j + len / 2] = (u - v + NMOD) % NMOD;
                cur = cur * w % NMOD;
            }
        }
        if (inv) { ll ninv = npow(n, NMOD - 2); for (ll& x : a) x = x
        ↪ * ninv % NMOD; }
    }
}
vector<ll> multiply(vector<ll> a, vector<ll> b) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <= 1;
    a.resize(n), b.resize(n);
    ntt(a, false), ntt(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % NMOD;
    ntt(a, true);
    a.resize(rs);
    return a;
}
// ARBITRARY MODULUS: split each coefficient as  $a = hi \cdot 2^{15} + lo$ 
//   ↪ and combine three products
// (or run NTT under three NTT-friendly primes and CRT). Values
//   ↪ must stay below  $\sim 1e18$ .
// COMMON NTT PRIMES: 998244353 (g=3), 167772161 (g=3), 469762049
//   ↪ (g=3), 1004535809 (g=3).
// USES: polynomial multiplication; "count pairs/triples with a
//   ↪ given sum"; convolving
// distributions; string matching with wildcards; sum of  $C(a_i, k)$ 
//   ↪ over all i for every k.
```


6.5.3 FWHT

Fast Walsh-Hadamard Transform (FWHT) - Time: $O(N \log N)$

```
// Computes Bitwise Convolutions: XOR, AND, OR for array of size N
//  $\hookrightarrow = 2^k$ 
struct FWHT {
    enum Type { XOR, AND, OR };

    static void transform(vector<ll>& a, bool invert, Type type =
     $\hookrightarrow$  XOR) {
        int n = a.size();
        for (int len = 1; 2 * len <= n; len <= 1) {
            for (int i = 0; i < n; i += 2 * len) {
                for (int j = 0; j < len; j++) {
                    ll u = a[i + j], v = a[i + len + j];
                    if (type == XOR) {
                        a[i + j] = u + v;
                        a[i + len + j] = u - v;
                    } else if (type == AND) {
                        a[i + j] = invert ? u - v : u + v;
                    } else if (type == OR) {
                        a[i + len + j] = invert ? v - u : u + v;
                    }
                }
            }
        }
        if (type == XOR && invert) {
            for (ll &x : a) x /= n;
        }
    }

    // Computes  $C[k] = \sum_{i \oplus j = k} (A[i] * B[j])$  where  $\oplus$  in
     $\hookrightarrow \{XOR, AND, OR\}$ 
    static vector<ll> multiply(vector<ll> a, vector<ll> b, Type
     $\hookrightarrow$  type = XOR) {
        int n = 1;
        while (n < max(a.size(), b.size())) n <= 1;
        a.resize(n); b.resize(n);

        transform(a, false, type);
        transform(b, false, type);
        for (int i = 0; i < n; i++) a[i] *= b[i];
        transform(a, true, type);
        return a;
    }
};
```

6.5.4 Polynomial

FORMAL POWER SERIES over $\text{NMOD} = 998244353$, on top of multiply() from 02 - NTT.cpp.

```
// Every routine works "mod  $x^k$ ". inv/log/exp/sqrt/pow are all
//  $\hookrightarrow$  Newton iteration: each step
// doubles the number of correct coefficients, so the cost is the
//  $\hookrightarrow$  cost of the last multiply.
typedef vector<ll> Poly;

Poly cut(Poly a, int k) { a.resize(k, 0); return a; }
Poly operator+(Poly a, const Poly& b) {
    a.resize(max(a.size(), b.size()), 0);
    for (size_t i = 0; i < b.size(); i++) a[i] = (a[i] + b[i]) %
     $\hookrightarrow$  NMOD;
    return a;
}
Poly operator-(Poly a, const Poly& b) {
    a.resize(max(a.size(), b.size()), 0);
    for (size_t i = 0; i < b.size(); i++) a[i] = (a[i] - b[i] +
     $\hookrightarrow$  NMOD) % NMOD;
    return a;
}
Poly operator*(const Poly& a, ll c) { Poly r = a; for (ll& x : r)
     $\hookrightarrow$  x = x * (c % NMOD) % NMOD; return r; }

Poly deriv(const Poly& a) {
    if (a.size() <= 1) return {0};
    Poly r(a.size() - 1);
    for (size_t i = 1; i < a.size(); i++) r[i - 1] = a[i] * i %
     $\hookrightarrow$  NMOD;
    return r;
}
Poly integ(const Poly& a) {
    Poly r(a.size() + 1, 0);
    for (size_t i = 0; i < a.size(); i++) r[i + 1] = a[i] *
     $\hookrightarrow$  npow(i + 1, NMOD - 2) % NMOD;
    return r;
}
//  $1/a \bmod x^k$ . Requires  $a[0] \neq 0$ .  $g_{2m} = g_m * (2 - a * g_m)$ 
Poly inv(const Poly& a, int k) {
    Poly g = {npow(a[0], NMOD - 2)};
    for (int m = 1; m < k; m <= 1) {
        Poly t = multiply(cut(a, 2 * m), g);
        t = cut(t, 2 * m);
        for (ll& x : t) x = (NMOD - x) % NMOD;
        t[0] = (t[0] + 2) % NMOD;
        g = cut(multiply(g, t), 2 * m);
    }
    return cut(g, k);
}
//  $\log(a) \bmod x^k$ . Requires  $a[0] = 1$ .  $\log a = \text{integral}(a' / a)$ 
Poly log_(const Poly& a, int k) { return
     $\hookrightarrow$  cut(integ(cut(multiply(deriv(a), inv(a, k)), k - 1)), k); }
//  $\exp(a) \bmod x^k$ . Requires  $a[0] = 0$ .  $g_{2m} = g_m * (1 + a -$ 
 $\hookrightarrow \log g_m)$ 
Poly exp_(const Poly& a, int k) {
    Poly g = {1};
    for (int m = 1; m < k; m <= 1) {
        Poly t = cut(a, 2 * m) - log_(cut(g, 2 * m), 2 * m);
        t[0] = (t[0] + 1) % NMOD;
        g = cut(multiply(g, t), 2 * m);
    }
    return cut(g, k);
}
//  $\sqrt{a} \bmod x^k$ , for  $a[0] = 1$  (general  $a[0]$  needs a modular
 $\hookrightarrow$  square root of  $a[0]$ ).
Poly sqrt_(const Poly& a, int k) {
    Poly g = {1};
    ll i2 = npow(2, NMOD - 2);
    for (int m = 1; m < k; m <= 1) {
        Poly t = cut(a, 2 * m) + multiply(g, g);
        t = cut(t, 2 * m);
        g = cut(multiply(t, inv(cut(g, 2 * m) * 2, 2 * m)), 2 *
     $\hookrightarrow$  m);
    }
    return cut(g, k);
}
//  $a^e \bmod x^k$ , for  $a[0] = 1$ :  $\exp(e * \log a)$ . For  $a[0] \neq 1$ 
 $\hookrightarrow$  factor out  $c * x^d$  first.
Poly pow_(const Poly& a, ll e, int k) { return exp_(log_(a, k) *
     $\hookrightarrow$  (e % NMOD), k); }

// Polynomial DIVISION with remainder:  $a = q * b + r$ ,  $\deg r < \deg b$ .
 $\hookrightarrow O(n \log n)$ 
pair<Poly, Poly> divmod(Poly a, Poly b) {
    while (a.size() > 1 && !a.back()) a.pop_back(); // trim
     $\hookrightarrow$  BOTH, or the reversal is wrong
```



```

while (b.size() > 1 && !b.back()) b.pop_back();
int n = a.size(), m = b.size();
if (n < m) return {{0}, a};
Poly ra(a.rbegin(), a.rend()), rb(b.rbegin(), b.rend());
Poly q = cut(multiply(cut(ra, n - m + 1), inv(cut(rb, n - m +
→ 1), n - m + 1)), n - m + 1);
reverse(all(q));
Poly r = cut(a - multiply(q, b), max(1, m - 1)); // cut
→ AFTER subtracting, not before
return {q, r};
}
// USES: counting with generating functions (exp of a "connected"
→ series = all structures),
// partition numbers via prod 1/(1-x^i) = exp(sum ...),
→ Catalan-style algebraic equations via
// sqrt, k-th power of a GF, linear recurrence coefficients via
→ inv, and Lagrange inversion.

```

6.6 Interpolation

6.6.1 LagrangeAndFaulhaber

Lagrange interpolation + power sums. Needs Mint + buildFact().

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).

// Given y[0..n] = f(0), f(1), ..., f(n) for a polynomial f of
→ degree ≤ n, evaluate f(x). O(n).
// The consecutive-points case is the one you actually want; it
→ kills a whole class of problems
// whose answer is "a polynomial in n of degree d" (guess d,
→ sample d+1 values, interpolate).
Mint lagrangeConsecutive(const vector<Mint>& y, ll x) {
    int n = y.size() - 1;
    if (0 ≤ x && x ≤ n) return y[x]; // x < 0 would
→ index y[negative]
    vector<Mint> pre(n + 2, 1), suf(n + 2, 1);
    for (int i = 0; i ≤ n; i++) pre[i + 1] = pre[i] * Mint(x -
→ i);
    for (int i = n; i ≥ 0; i--) suf[i] = suf[i + 1] * Mint(x -
→ i);
    Mint r = 0;
    for (int i = 0; i ≤ n; i++) {
        Mint t = y[i] * pre[i] * suf[i + 1] * invFact[i] *
→ invFact[n - i];
        ((n - i) & 1) ? r -= t : r += t;
    }
    return r;
}
// Arbitrary sample points, O(n^2).
Mint lagrange(const vector<Mint>& xs, const vector<Mint>& ys,
→ Mint x) {
    int n = xs.size();
    Mint r = 0;
    for (int i = 0; i < n; i++) {
        Mint num = ys[i], den = 1;
        for (int j = 0; j < n; j++) if (j != i) num *= x - xs[j],
→ den *= xs[i] - xs[j];
        r += num / den;
    }
    return r;
}
// FAULHABER: sum_{i=1..n} i^k is a polynomial in n of degree k+1
→ -> sample k+2 points. O(k log k).
Mint powerSum(ll n, int k) {
    vector<Mint> y(k + 2, 0);
    for (int i = 1; i ≤ k + 1; i++) y[i] = y[i - 1] +
→ Mint(i).pow(k);
    return lagrangeConsecutive(y, n);
}
// Same trick works for: sum of C(i, k), sum of i^k * r^i (degree k
→ in n after dividing by r^n),
// and any DP whose answer is eventually polynomial.

```

6.6.2 BerlekampMassey

Berlekamp-Massey: find the shortest linear recurrence fitting a sequence, $O(n^2)$.

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).
// Then jump to the k-th term in O(len^2 log k). THE weapon for
→ "brute force small n, find the
// pattern, extrapolate" - works on any sequence that satisfies a
→ constant-coefficient recurrence.
// Feed it ~2*len terms (60-100 is usually plenty).
vector<Mint> berlekampMassey(vector<Mint> s) {
    int n = s.size(), L = 0, m = 0;
    vector<Mint> C(n), B(n), T;
    C[0] = B[0] = 1;
    Mint b = 1;
    for (int i = 0; i < n; i++) {
        m++;
        Mint d = s[i];
        for (int j = 1; j ≤ L; j++) d += C[j] * s[i - j];
        if (d.v == 0) continue;
        T = C;
        Mint coef = d / b;
        for (int j = m; j < n; j++) C[j] -= coef * B[j - m];
        if (2 * L > i) continue;
        L = i + 1 - L, B = T, b = d, m = 0;
    }
    C.resize(L + 1), C.erase(C.begin());
    for (auto& x : C) x = Mint(0) - x;
    return C; // s[i] =
→ sum_j C[j] * s[i-1-j]
}
// k-th term (0-indexed) of the recurrence tr with initial values
→ S. O(|tr|^2 log k).
Mint linearRec(const vector<Mint>& S, const vector<Mint>& tr, ll
→ k) {
    if (tr.empty()) return 0; // e[1] would be
→ out of bounds
    int n = tr.size();
    auto comb = [&](vector<Mint> a, vector<Mint> b) {
        vector<Mint> r(2 * n + 1, 0);
        for (int i = 0; i ≤ n; i++) if (a[i].v)
            for (int j = 0; j ≤ n; j++) r[i + j] += a[i] * b[j];
        for (int i = 2 * n; i > n; i--)
            for (int j = 0; j < n; j++) r[i - 1 - j] += r[i] *
→ tr[j];
        r.resize(n + 1);
        return r;
    };
    vector<Mint> pol(n + 1, 0), e(pol);
    pol[0] = e[1] = 1;
    for (++k; k; k /= 2) {
        if (k % 2) pol = comb(pol, e);
        e = comb(e, e);
    }
    Mint r = 0;
    for (int i = 0; i < n; i++) r += pol[i + 1] * S[i];
    return r;
}

```

6.7 Numerical

6.7.1 Integration

ADAPTIVE SIMPSON - numeric integral of a black-box f on $[a, b]$ to a requested absolute error.

```
// Simpson approximates f by a parabola through the two ends and
//   ↳ the midpoint; "adaptive" means it
// splits an interval only where the parabola does not yet agree
//   ↳ with the two half-parabolas.
// Reach for it when the answer is an integral you cannot do in
//   ↳ closed form: area of a union of
// circles/ellipses cut by a line, area swept by a moving shape,
//   ↳ probability over a continuous
// parameter, arc lengths. Cost is proportional to how wiggly f
//   ↳ is, not to (b-a).
template <class F> ld simpson(F f, ld a, ld b) {
    return (b - a) * (f(a) + 4 * f((a + b) / 2) + f(b)) / 6;
}
template <class F> ld asr(F f, ld a, ld b, ld eps, ld S, int dep)
    ↳ {
    ld c = (a + b) / 2, L = simpson(f, a, c), R = simpson(f, c,
    ↳ b);
    if (dep <= 0 || fabs1(L + R - S) <= 15 * eps) return L + R +
    ↳ (L + R - S) / 15;
    return asr(f, a, c, eps / 2, L, dep - 1) + asr(f, c, b, eps /
    ↳ 2, R, dep - 1);
}
template <class F> ld integrate(F f, ld a, ld b, ld eps = 1e-9) {
    return asr(f, a, b, eps, simpson(f, a, b), 50);
}
// TRAPS. (1) If f has a KINK or a jump, split the interval at
//   ↳ that point by hand and integrate the
// pieces - adaptive Simpson will burn its depth budget trying to
//   ↳ resolve it. (2) The (L+R-S)/15
// term is Richardson extrapolation; keeping it makes the result
//   ↳ much more accurate for free.
// (3) A too-loose eps on a nearly flat region can stop early and
//   ↳ miss a spike: if f can be zero on
// most of the range and huge somewhere, integrate the pieces
//   ↳ separately.
//
// OTHER NUMERICAL TOOLS WORTH THE PAGE:
// TERNARY SEARCH on a strictly unimodal f over reals: 100
//   ↳ iterations of
// m1 = l + (r-l)/3, m2 = r - (r-l)/3; f(m1) < f(m2) ? r = m2 :
//   ↳ l = m1; (minimising)
// fixes the answer to -1e-30 relative. On INTEGERS use binary
//   ↳ search on the difference
// f(m+1) - f(m) instead - the three-way split has off-by-one
//   ↳ traps at the last two points.
// NEWTON'S METHOD for a root of g: x -= g(x) / g'(x). Doubles the
//   ↳ correct digits per step but
// needs a good start; bisection first if the sign changes, then
//   ↳ Newton.
// BISECTION on a monotone predicate: 100 iterations of (l+r)/2 is
//   ↳ always safe and never diverges.
// GOLDEN-SECTION SEARCH: like ternary search but one function
//   ↳ evaluation per step instead of two -
// worth it only when f is expensive.
// SOLVING f(x) = 0 WHERE f IS A POLYNOMIAL: find all real roots
//   ↳ by recursing on the derivative
// (roots of f' split the line into monotone pieces; bisection
//   ↳ inside each).
// SIMULATED ANNEALING / HILL CLIMBING when the objective is not
//   ↳ unimodal: start from several
// random points, take a step of size T, accept worse solutions
//   ↳ with probability exp(-dE/T),
// cool T geometrically (T *= 0.995). Good for geometric-median
//   ↳ / facility-location style
// problems where an exact method is out of reach; always seed
//   ↳ the RNG from the clock.
// WEISZFELD for the geometric median (minimise sum of Euclidean
//   ↳ distances): iterate
// x <- (sum p_i / |x - p_i|) / (sum 1 / |x - p_i|), guarding
//   ↳ the case x == p_i. Converges
// linearly; nested ternary search is the safer contest answer
//   ↳ (the objective is convex).
```

6.7.2 LinearSolvers

SPECIAL-SHAPE LINEAR SOLVERS. Gaussian elimination is $O(n^3)$; these are the shapes where the

```
// structure buys you a factor of n or removes a precondition.

// --- THOMAS ALGORITHM: TRIDIAGONAL system in  $O(n)$ . ---
// a[i] x[i-1] + b[i] x[i] + c[i] x[i+1] = d[i], with a[0] =
//   ↳ c[n-1] = 0.
// This is THE closer for random walks on a path ("expected steps
//   ↳ to escape [0, n]"), 1-D heat /
// diffusion DPs, and any DP whose state depends on its two
//   ↳ neighbours. Gaussian elimination on
// the same system is  $O(n^3)$  and will TLE at n = 1e5; this is 8
//   ↳ lines.
// STABLE without pivoting when the system is diagonally dominant
//   ↳ (|b| > |a| + |c|), which every
// hitting-time system is.
vector<ld> thomas(vector<ld> a, vector<ld> b, vector<ld> c,
    ↳ vector<ld> d) {
    int n = b.size();
    for (int i = 1; i < n; i++) {
        ld m = a[i] / b[i - 1];
        b[i] -= m * c[i - 1], d[i] -= m * d[i - 1];
    }
    vector<ld> x(n);
    x[n - 1] = d[n - 1] / b[n - 1];
    for (int i = n - 2; i >= 0; i--) x[i] = (d[i] - c[i] * x[i +
    ↳ 1]) / b[i];
    return x;
}
// BANDED systems of half-width w (state depends on w neighbours)
//   ↳ generalise this at  $O(n w^2)$  -
// the shape of "robot on a grid with m columns", where w = m.
// CYCLIC tridiagonal (x[0] and x[n-1] also coupled):
//   ↳ Sherman-Morrison - solve two tridiagonal
// systems and combine, still  $O(n)$ .

// --- DETERMINANT MODULO AN ARBITRARY m (composite allowed),
//   ↳  $O(n^3 \log m)$ . ---
// Gaussian elimination inverts the pivot, so it needs a PRIME
//   ↳ modulus. This version never
// divides: it runs the Euclidean algorithm between two rows,
//   ↳ swapping them, until the lower
// entry is 0. Use it when the modulus is  $2^k$ ,  $1e9$ , or anything
//   ↳ you cannot invert in.
ll detMod(vector<vector<ll>> a, ll m) {
    int n = a.size();
    ll det = 1 % m;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++)
            while (a[j][i]) { //
                // Euclid on the two rows
                ll t = a[i][i] / a[j][i];
                for (int k = i; k < n; k++) a[i][k] = (a[i][k] -
                ↳ t * a[j][k]) % m;
                swap(a[i], a[j]), det = -det;
            }
        det = det * a[i][i] % m;
        if (!a[i][i]) return 0;
    }
    return (det % m + m) % m;
}
// MATRIX RANK over a field: the usual elimination, counting
//   ↳ non-zero pivot rows.
// RANK over GF(2): use bitsets -  $O(n^3 / 64)$ , see 04 - Linear
//   ↳ Algebra/03.
// INTEGER determinant with no modulus: compute it modulo several
//   ↳ primes and CRT. Size it with
// HADAMARD'S BOUND |det| <= prod of the row 2-norms <=  $n^{(n/2)}$  *
//   ↳ max|a_ij|^n, and take enough
// primes that the product exceeds  $2x$  that.
// SPARSE determinant (n = 1e5 with few non-zeros): Wiedemann -
//   ↳ pick random u, v, feed the Krylov
// sequence  $u^T A^k v$  to Berlekamp-Massey to get the minimal
//   ↳ polynomial, whose constant term is
// +-det after a random diagonal preconditioner.  $O(n * nnz)$ .
// CHARACTERISTIC POLYNOMIAL in  $O(n^3)$ : reduce to upper HESSENBERG
//   ↳ form by similarity transforms
// (so det(xI - H) satisfies a short recurrence in the leading
//   ↳ principal minors), then read the
// coefficients off. With Cayley-Hamilton that gives  $A^k$  in  $O(n^3$ 
//   ↳ +  $n^2 \log k)$  instead of
//  $O(n^3 \log k)$  - worth it only for large k and small n.
```

6.8 Linear Programming

6.8.1 Simplex

```

TWO-PHASE SIMPLEX. maximise  $c \cdot x$  subject to  $A \cdot x \leq b, x \geq 0$ .
// Returns the optimum, or +inf if unbounded, or -inf if
  ↳ infeasible; x is filled with a solution.
// Exponential in theory,  $O(n \cdot m)$  per pivot and instant at contest
  ↳ sizes (n, m up to a few hundred).
// REACH FOR IT when the problem is "optimise a linear objective
  ↳ under linear constraints" and no
// combinatorial structure presents itself - resource allocation,
  ↳ blending, fractional covering,
// zero-sum matrix games, and any
  ↳ min-cost-flow-with-side-constraints that flow cannot express.
typedef vector<ld> vd;
struct Simplex {
    int m, n;
    vector<int> B, N;
    vector<vd> D;
    Simplex(const vector<vd>& A, const vd& b, const vd& c)
        : m(b.size()), n(c.size()), B(m), N(n + 1), D(m + 2, vd(n
  ↳ + 2, 0)) {
        for (int i = 0; i < m; i++) for (int j = 0; j < n; j++)
  ↳ D[i][j] = A[i][j];
        for (int i = 0; i < m; i++) B[i] = n + i, D[i][n] = -1,
  ↳ D[i][n + 1] = b[i];
        for (int j = 0; j < n; j++) N[j] = j, D[m][j] = -c[j];
        N[n] = -1, D[m + 1][n] = 1;
    }
    void pivot(int r, int s) {
        ld inv = 1 / D[r][s];
        for (int i = 0; i < m + 2; i++) if (i != r &&
  ↳ fabs1(D[i][s]) > 1e-9) {
            ld f = D[i][s] * inv;
  ↳ the pivot COLUMN is fixed up by
            for (int j = 0; j < n + 2; j++) if (j != s) D[i][j]
  ↳ -= f * D[r][j]; // the loop below
        }
        for (int j = 0; j < n + 2; j++) if (j != s) D[r][j] *=
  ↳ inv;
        for (int i = 0; i < m + 2; i++) if (i != r) D[i][s] *=
  ↳ -inv;
        D[r][s] = inv;
        swap(B[r], N[s]);
    }
    bool simplex(int phase) {
        int x = m + phase - 1;
        for (;;) {
            int s = -1;
            for (int j = 0; j < n; j++)
                if (N[j] != -phase && (s < 0 ||
  ↳ make_pair(D[x][j], N[j]) < make_pair(D[x][s], N[s]))) s = j;
            if (D[x][s] >= -1e-9) return true;
            int r = -1;
            for (int i = 0; i < m; i++) {
                if (D[i][s] <= 1e-9) continue;
                if (r < 0 || make_pair(D[i][n + 1] / D[i][s],
  ↳ B[i]) < make_pair(D[r][n + 1] / D[r][s], B[r])) r = i;
            }
            if (r < 0) return false;
  ↳ unbounded
            pivot(r, s);
        }
        ld solve(vd& x) {
            int r = 0;
            for (int i = 1; i < m; i++) if (D[i][n + 1] < D[r][n +
  ↳ 1]) r = i;
            if (D[r][n + 1] < -1e-9) {
  ↳ phase 1: find any feasible point
                pivot(r, n);
                if (!simplex(2) || D[m + 1][n + 1] < -1e-9) return
  ↳ -1.0L / 0.0L; // infeasible
                for (int i = 0; i < m; i++) if (B[i] == -1) {
                    int s = 0;
                    for (int j = 1; j <= n; j++)
                        if (make_pair(D[i][j], N[j]) <
  ↳ make_pair(D[i][s], N[s])) s = j;
                    pivot(i, s);
                }
            }
            bool ok = simplex(1);
            x.assign(m, 0);
            for (int i = 0; i < m; i++) if (B[i] < n) x[B[i]] =
  ↳ D[i][n + 1];
            return ok ? D[m][n + 1] : 1.0L / 0.0L;
  ↳ +inf = unbounded
        }
    }
};

```

```

/* MODELLING NOTES - the part that actually costs you time
MINIMISE instead: negate c and negate the answer.
>= CONSTRAINT: multiply the row by -1. EQUALITY: add it twice,
  ↳ as <= and as >=.
FREE VARIABLE (no x >= 0): substitute  $x = u - v$  with  $u, v \geq 0$ .
  ↳ Doubles that column.
UPPER BOUND  $x_i \leq U$ : just another row; simplex has no special
  ↳ handling and does not need any.
LINEAR-FRACTIONAL objective  $(c \cdot x + a) / (d \cdot x + b)$ : CHARNES-COOPER
  ↳ - substitute  $y = x / (d \cdot x + b)$ ,
         $t = 1 / (d \cdot x + b)$  and maximise  $c \cdot y + a \cdot t$  subject to  $A \cdot y \leq b \cdot t, d \cdot y
  ↳ + b \cdot t = 1, t \geq 0$ . Linear again.
INTEGER answers are NOT guaranteed. If the constraint matrix is
  ↳ TOTALLY UNIMODULAR (network
        matrices, interval matrices, bipartite incidence) the optimum
  ↳ IS integral - that is exactly why
        flow problems have integral optima. Otherwise you need ILP,
  ↳ which is NP-hard.
ZERO-SUM MATRIX GAME: shift A so every entry is > 0, then min
  ↳ sum(y) s.t.  $A \cdot y \geq 1, y \geq 0$  gives
        value  $v = 1 / \text{sum}(y)$  and the column player's mixed strategy  $y \cdot v$ ;
  ↳ the DUAL gives the row player's.
Von Neumann:  $\max_x \min_y x^T A y = \min_y \max_x x^T A y$ .
DUALITY, which is often the actual solution:  $\max c \cdot x, Ax \leq b, x
  ↳ \geq 0$  is dual to
        min  $b \cdot y, A^T y \geq c, y \geq 0$ ; the two optima are equal.
  ↳ COMPLEMENTARY SLACKNESS at the optimum:
         $x_j > 0 \Rightarrow$  the  $j$ -th dual constraint is tight, and  $y_i > 0 \Rightarrow$ 
  ↳ the  $i$ -th primal row is tight.
        The combinatorial specialisations you already know are
  ↳ max-flow/min-cut and Konig.
PRECISION: this is floating point. If the data is integral and
  ↳ small, and you need an exact
        answer, prefer a combinatorial model (flow, matching) over
  ↳ simplex. */

```

7 Strings

7.1 String Matching

7.1.1 KMP

KMP. $\text{pi}[i]$ = length of the longest proper border of $s[0..i]$ (prefix that is also a suffix).

```

// Gives  $O(n \cdot m)$  substring search, the period of a prefix ( $i+1 -
  ↳ \text{pi}[i]$ ), and the automaton for DP.
vector<int> compute_pi(const string &s) {
    int n = s.size();
    vector<int> pi(n);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j])
            j = pi[j - 1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}

vector<int> find_matches(const string &text, const string &pat) {
    int n = pat.length(), m = text.length();
    string s = pat + "#" + text;
    vector<int> pi = compute_pi(s), ans;
    for (int i = n + 1; i <= n + m; i++) { /* n + 1 is where the
  ↳ text starts */
        if (pi[i] == n) {
            ans.push_back(i - 2 * n); /* i - (n - 1) - (n + 1) */
        }
    }
    return ans;
}

void compute_automaton(string s, vector<vector<int>> &aut) {
    s += '#';
    int n = s.size();
    vector<int> pi = compute_pi(s);
    aut.assign(n, vector<int>(26));
    for (int i = 0; i < n; i++) {
        for (int c = 0; c < 26; c++) {
            if (i > 0 && 'a' + c != s[i])
                aut[i][c] = aut[pi[i - 1]][c];
            else
                aut[i][c] = i + ('a' + c == s[i]);
        }
    }
}

```

7.1.2 RabinKarp

RABIN-KARP - rolling-hash substring search in $O(n+m)$ expected. Prefer 05 - Hashing for anything

```
// serious (this uses a FIXED base, which is hackable); keep this
// for one-off matching.
vector<int> rabin_karp(string const &s, string const &t) {
    const int p = 31;
    const int m = 1e9 + 9;
    int S = s.size(), T = t.size();

    vector<ll> p_pow(max({S, T, 1})); // max(0,0)
    // would leave it empty
    p_pow[0] = 1;
    for (int i = 1; i < (int) p_pow.size(); i++)
        p_pow[i] = (p_pow[i - 1] * p) % m;

    vector<ll> h(T + 1, 0);
    for (int i = 0; i < T; i++)
        h[i + 1] = (h[i] + (t[i] - 'a' + 1) * p_pow[i]) % m;
    ll h_s = 0;
    for (int i = 0; i < S; i++)
        h_s = (h_s + (s[i] - 'a' + 1) * p_pow[i]) % m;

    vector<int> occurrences;
    for (int i = 0; i + S - 1 < T; i++) {
        ll cur_h = (h[i + S] + m - h[i]) % m;
        if (cur_h == h_s * p_pow[i] % m)
            occurrences.push_back(i);
    }
    return occurrences;
}
```

7.1.3 ZFunction

$z[i]$ is the length of the longest substring starting from $s[i]$ which is also a prefix of s

```
vector<int> z_function(string s) {
    int n = s.length();
    vector<int> z(n);
    int l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if (i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}
```

7.1.4 WildcardMatching

STRING MATCHING WITH WILDCARDS in $O(n \log n)$. A wildcard matches any character, on EITHER side.

```
// Needs: conv (06 - Math/05 - Convolution/01 - FFT.cpp).
// Map every real character to a positive value and every wildcard
// to 0. Then position i matches iff
// sum_j (p[j] - t[i+j])^2 * p[j] * t[i+j] == 0,
// because each term is >= 0 and vanishes exactly when the
// characters agree or one of them is 0.
// Expanding gives three correlations: sum p^3 t - 2 sum p^2 t^2 +
// sum p t^3, and a correlation is a
// convolution with the pattern reversed.
vector<int> wildcardMatch(const string& t, const string& p, char
    wild = '?') {
    int n = t.size(), m = p.size();
    if (m > n) return {};
    // The identity needs values >= 0 and wildcard == 0. Compress
    // the characters that actually
    // occur to 1..k - with a raw 'c - 'a' + 1' any character
    // below 'a' is NEGATIVE and the terms
    // can cancel, which reports matches that do not exist (e.g.
    // text "_b" against pattern "Z").
    map<char, ll> id;
    for (char c : t) if (c != wild) id.emplace(c, 0);
    for (char c : p) if (c != wild) id.emplace(c, 0);
    ll k = 0;
    for (auto& [c, v] : id) v = ++k;
    auto val = [&](char c) { return c == wild ? 0LL : id[c]; };
    vector<ll> a(m), b(n);
    for (int i = 0; i < m; i++) a[i] = val(p[m - 1 - i]);
    // reversed pattern
    for (int i = 0; i < n; i++) b[i] = val(t[i]);
    auto pw = [](vector<ll> v, int k) { for (ll& x : v) { ll y =
    1; for (int i = 0; i < k; i++) y *= x; x = y; } return v; };
    auto c1 = conv(pw(a, 3), b), c2 = conv(pw(a, 2), pw(b, 2)),
    c3 = conv(a, pw(b, 3));
    vector<int> res;
    for (int i = 0; i + m <= n; i++)
        if (c1[i + m - 1] - 2 * c2[i + m - 1] + c3[i + m - 1] ==
        0) res.push_back(i);
    return res;
    // 0-based start positions
}
// PRECISION: the largest intermediate is about  $k^4 * m$ , where  $k$ 
// is the number of DISTINCT real
// characters (which is why compressing is free). With  $k = 26$  and
//  $m = 1e5$  that is
//  $\sim 4.6e10$ , comfortable for double FFT. For a big alphabet,
// compress the characters that actually
// occur first, or run the three convolutions under two NTT primes
// and CRT.
// COUNT MISMATCHES at every shift (no wildcards, small alphabet):
// for each letter c convolve the
// 0/1 indicator of c in p (reversed) with the indicator of c in
// t; summing gives the number of
// MATCHES per shift, and mismatches =  $m - \text{matches}$ .  $O(|\Sigma| n \log n)$ .
// k-MISMATCH: same counts, keep shifts with matches  $\geq m - k$ .
// SUBSET / "at most one character class" matching: encode each
// position as a bitmask over the
// alphabet and convolve the complement indicators; a shift is
// valid iff the total is 0.
```

7.2 Hashing

7.2.1 StringHashing

Double-mod polynomial hashing. RANDOM bases => not anti-hackable.

```
// No modular inverses: h(l,r) is compared already scaled, so
//    get() is 2 mults.
ll HB1, HB2;
struct Hash {
    static const ll M1 = 1000000007, M2 = 998244353;
    int n;
    vector<ll> h1, h2, p1, p2;

    Hash(const string& s) : n(s.size()), h1(n + 1, 0), h2(n + 1,
    0), p1(n + 1, 1), p2(n + 1, 1) {
        if (!HB1) {
            mt19937_64
            rng(chrono::steady_clock::now().time_since_epoch().count());
            HB1 = 256 + rng() % (M1 - 300), HB2 = 256 + rng() %
            (M2 - 300); // FULL range: a base drawn
            // from only 1e6 values gives collision probability n/1e6
            // per modulus (0.1 at n = 1e5)
        }
        for (int i = 0; i < n; i++) {
            h1[i + 1] = (h1[i] * HB1 + s[i]) % M1, p1[i + 1] =
            p1[i] * HB1 % M1;
            h2[i + 1] = (h2[i] * HB2 + s[i]) % M2, p2[i + 1] =
            p2[i] * HB2 % M2;
        }
        pair<ll, ll> get(int l, int r) const { // [l, r]
            // inclusive, 0-based
            ll a = (h1[r + 1] - h1[l] * p1[r - l + 1]) % M1;
            ll b = (h2[r + 1] - h2[l] * p2[r - l + 1]) % M2;
            return {a < 0 ? a + M1 : a, b < 0 ? b + M2 : b};
        }
        bool eq(int l1, int r1, int l2, int r2) const {
            return r1 - l1 == r2 - l2 && get(l1, r1) == get(l2, r2);
        }
        // concat(A over lenA, B): h = hA * base^lenB + hB (works
        // per modulus)
    };
    // USES: substring equality O(1) * longest common prefix of two
    //    suffixes by binary search
    //    * dedup of substrings * 2D grid patterns (hash rows, then
    //    hash the row-hashes)
    //    * tree isomorphism (hash the sorted multiset of child
    //    hashes)
```

7.2.2 Hashing2D

2D HASHING - O(1) hash of any submatrix, for grid pattern matching.

```
// 2D prefix hash with random bases b1 (down) and b2 (right),
//    single 62-bit modulus via __int128.
ll HB2D1 = 0, HB2D2 = 0; // ONE base
//    pair per RUN, at file scope.
struct Hash2D { //
    // Per-object bases would make two grids
    static const ll M = (1ll)1e18 + 9; //
    // incomparable - and "find this pattern in
    int n, m; // that
    // grid" needs exactly two objects.
    ll b1, b2;
    vector<ll> P1, P2;
    vector<vector<ll>> h;

    static ll mul(ll a, ll b) { return (ll)((__int128)a * b % M);
    }
    Hash2D(const vector<string>& g) : n(g.size()), m(n ?
    g[0].size() : 0),
        P1(n + 1, 1), P2(m + 1, 1),
    h(n + 1, vector<ll>(m + 1, 0)) {
        if (!HB2D1) {
            mt19937_64
            rng(chrono::steady_clock::now().time_since_epoch().count());
            HB2D1 = 256 + rng() % (M - 300), HB2D2 = 256 + rng()
            % (M - 300);
        }
        b1 = HB2D1, b2 = HB2D2;
        for (int i = 1; i <= n; i++) P1[i] = mul(P1[i - 1], b1);
        for (int j = 1; j <= m; j++) P2[j] = mul(P2[j - 1], b2);
        for (int i = 1; i <= n; i++)
            for (int j = 1; j <= m; j++) {
                __int128 v = (__int128)mul(h[i - 1][j], b1) +
                mul(h[i][j - 1], b2)
                - mul(mul(h[i - 1][j - 1], b1), b2) +
                g[i - 1][j - 1];
                h[i][j] = (ll)((v % M + M) % M);
            }
        // hash of the submatrix with top-left (r, c) and size dr x
        // dc, 0-based
        ll get(int r, int c, int dr, int dc) const {
            int r2 = r + dr, c2 = c + dc;
            __int128 v = (__int128)h[r2][c2] - mul(h[r][c2], P1[dr])
            - mul(h[r2][c], P2[dc])
            + mul(mul(h[r][c], P1[dr]), P2[dc]);
            return (ll)((v % M + M) % M);
        }
    };
    // USES: count occurrences of a pattern grid * largest common
    //    square submatrix (binary search the
    //    side, hash all windows of both grids) * count distinct k x k
    //    submatrices.
    // CHEAPER when the pattern is fixed: 1D-hash each row of the
    //    pattern, then run KMP /
    // Aho-Corasick over the rows treated as single symbols.
```

7.3 Aho Corasick

7.3.1 AhoCorasick

AHO-CORASICK - a trie of all patterns plus suffix (fail) links, so one pass over the text finds

```
// every occurrence of every pattern in  $O(|text| + \text{total pattern length} + \text{\#matches})$ .
// The goto-automaton form (nxt filled in by BFS) is also the
// transition table for DP over strings.
struct Mapper {
    int operator()(char c) const { return c - 'a'; }
};

template <int ALPHABET, typename Mapper>
struct Aho {
    struct Node {
        array<int, ALPHABET> nxt;
        int link = 0;
        int terminal_idx = -1; // -1 if not a terminal, else idx
        // of string insertion
        int exit_link = 0; // Points to the nearest terminal suffix
        int depth = 0; // Size of the prefix the node represents
        Node() { nxt.fill(0); }
    };

    vector<Node> t;
    vector<int> bfs_order;
    Mapper get_idx; // Instance of the mapping function
    int word_count = 0; // Tracks the order of inserted strings

    // Constructor optionally accepts a custom mapper instance
    Aho(Mapper mapper = Mapper()) : get_idx(mapper) {
        t.emplace_back();
    }

    int insert(const string& p) {
        int u = 0;
        for (char c : p) {
            int v = get_idx(c);
            if (!t[u].nxt[v]) {
                t[u].nxt[v] = t.size();
                t.emplace_back();
                t.back().depth = t[u].depth + 1; // New node's
                // depth is parent's depth + 1
            }
            u = t[u].nxt[v];
        }

        // If this exact string hasn't been inserted before,
        // assign it the current index
        if (t[u].terminal_idx == -1) {
            t[u].terminal_idx = word_count;
        }
        word_count++; // Increment for the next insertion

        return u;
    }

    void build() {
        queue<int> q;
        for (int c = 0; c < ALPHABET; ++c) {
            if (t[0].nxt[c]) {
                q.push(t[0].nxt[c]);
            }
        }

        while (!q.empty()) {
            int u = q.front();
            q.pop();
            bfs_order.push_back(u);

            for (int c = 0; c < ALPHABET; ++c) {
                int v = t[u].nxt[c];
                if (v) {
                    t[v].link = t[t[u].link].nxt[c];

                    // Compute the exit link:
                    // If the immediate suffix link is a terminal,
                    // point to it.
                    // Otherwise, copy its exit link.
                    t[v].exit_link = (t[t[v].link].terminal_idx
                    != -1) ? t[v].link : t[t[v].link].exit_link;

                    q.push(v);
                }
                else {
                    t[u].nxt[c] = t[t[u].link].nxt[c];
                }
            }
        }
    }
};
```

```
    }
}

int advance(int u, char c) {
    return t[u].nxt[get_idx(c)];
}

};
```


7.3.2 AhoCorasickDP

DP OVER THE AHO-CORASICK AUTOMATON - the high-value use of Aho, more than matching itself.

```
// Needs: Mint (06 - Math/01) and AhoCorasick (01 -
// AhoCorasick.cpp).
// After build(), nxt[] is a total transition function, so the
// automaton is a DFA over states
// 0..S-1 and you can run any DP on (position, state, extra).
// "bad" marks states that contain a forbidden pattern as a
// suffix; propagate it along exit links.
struct AhoDP {
    static const int A = 26; // trie fan-out
    int alpha = 26; // alphabet
    // the DP counts over (set smaller!)
    vector<array<int, A>> nxt;
    vector<int> link, term;
    vector<char> bad;
    vector<int> order; // BFS
    // order of states

    AhoDP() { newNode(); }
    int newNode() { nxt.push_back({}); nxt.back().fill(0);
    // link.push_back(0); term.push_back(0); bad.push_back(0);
    // return nxt.size() - 1; }
    void insert(const string& p, int id = 1) {
        int u = 0;
        for (char c : p) {
            int v = c - 'a';
            if (!nxt[u][v]) nxt[u][v] = newNode();
            u = nxt[u][v];
        }
        term[u] = id, bad[u] = 1;
    }
    void build() {
        queue<int> q;
        for (int c = 0; c < A; c++) if (nxt[0][c])
        // q.push(nxt[0][c]);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            order.push_back(u);
            bad[u] = bad[u] || bad[link[u]]; // a state
            // is bad if any SUFFIX is a pattern
            for (int c = 0; c < A; c++) {
                int v = nxt[u][c];
                if (v) link[v] = link[u][c], q.push(v);
                else nxt[u][c] = link[u][c]; // total
            }
        }
        // #strings of length L over the first 'alpha' letters
        // containing NO pattern. O(L*S*alpha).
        Mint countAvoiding(int L) {
            int S = nxt.size();
            vector<Mint> dp(S, 0), nd;
            dp[0] = 1;
            for (int i = 0; i < L; i++) {
                nd.assign(S, 0);
                for (int u = 0; u < S; u++) if (!bad[u] && dp[u].v)
                for (int c = 0; c < alpha; c++) {
                    int v = nxt[u][c];
                    if (!bad[v]) nd[v] += dp[u];
                }
                dp = nd;
            }
            Mint r = 0;
            for (int u = 0; u < S; u++) if (!bad[u]) r += dp[u];
            return r;
        }
        // #occurrences of every pattern in a text: walk the text,
        // count visits per state, then push
        // the counts DOWN the suffix-link tree (reverse BFS order).
        vector<ll> occurrences(const string& s) {
            vector<ll> cnt(nxt.size(), 0);
            int u = 0;
            for (char c : s) u = nxt[u][c - 'a'], cnt[u]++;
            for (int i = order.size() - 1; i >= 0; i--)
            // cnt[link[order[i]]] += cnt[order[i]];
            // return cnt;
            // cnt[state of pattern p] = #occurrences
        }
};
// SHORTEST / LEXICOGRAPHICALLY SMALLEST string avoiding all
// patterns: BFS over (state) with
// the alphabet in order. If a cycle among good states is
// reachable, arbitrarily long strings
// exist - detect it with a cycle check on the good-state subgraph.
```

7.4 Suffix Structures

7.4.1 SuffixArray

SUFFIX ARRAY + LCP + sparse table for O(1) LCE. Build is O(n log n) (prefix doubling with a radix

```
// sort), Kasai gives the LCP array in O(n). Everything you can
// ask about substrings reduces to a
// range on this array - see 03 - SuffixArrayApplications.cpp.
struct SuffixArray {
    string S;
    // sa is the suffix array with the empty suffix being sa[0]
    // lcp[i] holds the lcp between sa[i], sa[i - 1]
    vector<int> logs, sa, lcp, rank;
    vector<vector<int>> table;

    SuffixArray() {}

    SuffixArray(string &s, int lim = 256) {
        S = s;
        int n = s.size() + 1, k = 0, a, b;
        vector<int> c(s.begin(), s.end() + 1), tmp(n), frq(max(n,
        // lim));
        c.back() = 0; // 0 is less than any character
        sa = lcp = rank = tmp, iota(sa.begin(), sa.end(), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p)
        // {
            p = j, iota(tmp.begin(), tmp.end(), n - j);
            for (int i = 0; i < n; i++) {
                if (sa[i] >= j)
                    tmp[p++] = sa[i] - j;
            }
            fill(frq.begin(), frq.end(), 0);

            for (int i = 0; i < n; i++) frq[c[i]]++;
            for (int i = 1; i < lim; i++) frq[i] += frq[i - 1];
            for (int i = n; i--;) sa[--frq[c[tmp[i]]]] = tmp[i];

            swap(c, tmp), p = 1, c[sa[0]] = 0;
            for (int i = 1; i < n; i++)
                a = sa[i - 1], b = sa[i], c[b] = (tmp[a] ==
                // tmp[b] && tmp[a + j] == tmp[b + j]) ? p - 1 : p++;
            }

            for (int i = 1; i < n; i++) rank[sa[i]] = i;
            for (int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
                for (k && k--, j = sa[rank[i] - 1];
                    s[i + k] == s[j + k];
                        k++);
        }

        void preLcp() {
            int n = S.size() + 1;
            logs = vector<int>(n + 5);
            for (int i = 2; i < n + 5; ++i) {
                logs[i] = logs[i / 2] + 1;
            }
            // level-major: sized by logs[n], not a hard-coded 20
            // (which is OUT OF BOUNDS at
            // |s| >= 2^20 - and Codeforces routinely gives |s| up to
            // 1e6), and no dead columns.
            table = vector<vector<int>>(logs[n] + 1, vector<int>(n));
            for (int i = 0; i < n; ++i) table[0][i] = lcp[i];
            for (int j = 1; j <= logs[n]; ++j)
                for (int i = 0; i + (1 <= j) <= n; ++i)
                    table[j][i] = min(table[j - 1][i], table[j - 1][i
                    // + (1 <= (j - 1))]);
        }

        // ARGUMENTS ARE SUFFIX-ARRAY INDICES, not string positions.
        // For two string positions
        // p and q use lce(p, q) below, which is what every
        // application actually wants.
        int queryLcp(int i, int j) {
            if (i == j) return (int)S.size() - sa[i];
            if (i > j) swap(i, j);
            i++;
            int len = logs[j - i + 1];
            return min(table[len][i], table[len][j - (1 <= len) + 1]);
        }

        int lce(int p, int q) { return queryLcp(rank[p], rank[q]); }
        // longest common extension
    };
};
```

7.4.2 SuffixAutomaton

SUFFIX AUTOMATON - $O(n)$ states/transitions, accepts exactly the substrings of s .

```
// st[v].len = longest string in v's endpos class    link =
// suffix link (v's parent)
// cnt[v] = |endpos(v)| = #occurrences of every string in v
// distinct substrings = sum over v != root of (len[v] -
// len[link[v]])
struct SAM {
    struct S { int len = 0, link = -1; array<int, 26> nxt; S() {
        ↪ nxt.fill(-1); } };
    vector<S> st;
    vector<ll> cnt; // occurrence counts (after countOcc)
    vector<int> firstPos; // position of the first occurrence
    int last = 0;
    bool counted = false; // countOcc() is idempotent

    SAM(int reserve = 0) { st.reserve(2 * reserve);
        ↪ st.push_back(S()), cnt.push_back(0), firstPos.push_back(0); }

    void extend(char ch) {
        int c = ch - 'a', cur = st.size();
        st.push_back(S()), cnt.push_back(1);
        ↪ firstPos.push_back(0);
        st[cur].len = st[last].len + 1, firstPos[cur] =
        ↪ st[cur].len - 1;
        int p = last;
        while (p != -1 && st[p].nxt[c] == -1) st[p].nxt[c] = cur,
        ↪ p = st[p].link;
        if (p == -1) st[cur].link = 0;
        else {
            int q = st[p].nxt[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;
            else {
                int cl = st.size();
                st.push_back(st[q]), cnt.push_back(0);
                ↪ firstPos.push_back(firstPos[q]);
                st[cl].len = st[p].len + 1;
                while (p != -1 && st[p].nxt[c] == q) st[p].nxt[c]
                ↪ = cl, p = st[p].link;
                st[q].link = st[cur].link = cl;
            }
        }
        last = cur;
    }

    void build(const string& s) { for (char c : s) extend(c); }

    vector<int> byLen() const { // states
        ↪ sorted by len (counting sort)
        int n = st.size(), mx = st[0].len;
        for (auto& x : st) mx = max(mx, x.len);
        vector<int> c(mx + 2, 0), ord(n);
        for (auto& x : st) c[x.len]++;
        for (int i = 1; i <= mx; i++) c[i] += c[i - 1];
        for (int v = n - 1; v >= 0; v--) ord[--c[st[v].len]] = v;
        ↪ return ord; // increasing len; reverse for link-tree order
    }

    void countOcc() { // push
        ↪ counts down the suffix-link tree
        if (counted) return; // NOT
        ↪ idempotent without this guard
        counted = true;
        auto ord = byLen();
        for (int i = (int)ord.size() - 1; i > 0; i--) {
            int v = ord[i];
            if (st[v].link >= 0) cnt[st[v].link] += cnt[v];
        }
        cnt[0] = 0;
    }

    ll distinctSubstrings() const { // number of
        ↪ DISTINCT non-empty substrings
        ll r = 0;
        for (int v = 1; v < (int)st.size(); v++) r += st[v].len -
        ↪ st[st[v].link].len;
        return r;
    }

    ll totalOccurrences() { // sum over
        ↪ distinct substrings of #occurrences
        countOcc();
        ll r = 0;
        for (int v = 1; v < (int)st.size(); v++) r += cnt[v] *
        ↪ (st[v].len - st[st[v].link].len);
        return r;
    }
}
```

```
int countOf(const string& p) { //
    ↪ #occurrences of p in s (call countOcc first)
    int v = 0;
    for (char ch : p) { v = st[v].nxt[ch - 'a']; if (v < 0)
    ↪ return 0; }
    return cnt[v];
}

// K-th SMALLEST distinct substring (k is 1-indexed). Needs
    ↪ 'paths' = #substrings from v.
vector<ll> paths;
void buildPaths() {
    paths.assign(st.size(), 1); // 1 counts
    ↪ the empty continuation
    auto ord = byLen();
    for (int i = (int)ord.size() - 1; i >= 0; i--) {
        int v = ord[i];
        paths[v] = 1;
        for (int c = 0; c < 26; c++) if (st[v].nxt[c] >= 0)
        ↪ paths[v] += paths[st[v].nxt[c]];
    }

    string kthSubstring(ll k) { // "" if
        ↪ fewer than k distinct substrings
        string r;
        int v = 0;
        while (k > 0) {
            int c = 0;
            for (; c < 26; c++) {
                int u = st[v].nxt[c];
                if (u < 0) continue;
                if (k <= paths[u]) { r += char('a' + c), v = u,
                ↪ k--; break; }
                k -= paths[u];
            }
            if (c == 26) break;
        }
        return r;
    }
};

// LONGEST COMMON SUBSTRING of s and t: build the SAM of s, then
    ↪ walk t keeping the current
// length; on a missing transition follow suffix links and clamp
    ↪ the length to st[v].len.
int longestCommonSubstring(SAM& A, const string& t) {
    int v = 0, l = 0, best = 0;
    for (char ch : t) {
        int c = ch - 'a';
        while (v && A.st[v].nxt[c] < 0) v = A.st[v].link, l =
        ↪ A.st[v].len;
        if (A.st[v].nxt[c] >= 0) v = A.st[v].nxt[c], l++;
        best = max(best, l);
    }
    return best;
}

// GENERALISED SAM (several strings): reset 'last = 0' before each
    ↪ string and, in extend(),
// if a transition already exists reuse/clone it instead of
    ↪ creating a new state.
```

7.4.3 SuffixArrayApplications

WHAT YOU ACTUALLY EXTRACT FROM A SUFFIX ARRAY.

```
// Convention of 01 - SuffixArray.cpp: sa has n+1 entries (sa[0] =
//   ↳ the empty suffix),
// and lcp[i] = LCP(sa[i], sa[i-1]). Everything below assumes
//   ↳ that.

// 1) NUMBER OF DISTINCT SUBSTRINGS = sum over i of (len of suffix
//   ↳ i) - lcp with the previous.
ll distinctSubstrings(const SuffixArray& S) {
    ll n = S.S.size(), r = 0;
    for (ll i = 1; i <= n; i++) r += (n - S.sa[i]) - S.lcp[i];
    return r;
}

// 2) LONGEST REPEATED SUBSTRING = max lcp; the substring starts
//   ↳ at sa[argmax].
pair<int, int> longestRepeated(const SuffixArray& S) {
    ↳ {length, start position}
    int n = S.S.size(), bi = 0, best = 0;
    for (int i = 1; i <= n; i++) if (S.lcp[i] > best) best =
    ↳ S.lcp[i], bi = S.sa[i];
    return {best, bi};
}

// 3) OCCURRENCES of a pattern = the contiguous SA block whose
//   ↳ suffixes start with it.
// Two binary searches, O(|P| log n).
pair<int, int> patternRange(const SuffixArray& S, const string&
    ↳ p) { // [lo, hi], empty if lo > hi
    int n = S.S.size();
    auto cmp = [&](int idx, const string& q) { return
    ↳ S.S.compare(idx, q.size(), q) < 0; };
    int lo = 1, hi = n, L = n + 1, R = n;
    while (lo <= hi) { int m = (lo + hi) / 2; cmp(S.sa[m], p) ?
    ↳ lo = m + 1 : (L = m, hi = m - 1); }
    lo = L, hi = n, R = L - 1;
    ↳ R MUST be reset here:
    ↳ otherwise an absent pattern
    ↳ returns the range [L, n]
    while (lo <= hi) {
        int m = (lo + hi) / 2;
        S.S.compare(S.sa[m], p.size(), p) == 0 ? (R = m, lo = m +
        ↳ 1) : hi = m - 1;
    }
    return {L, R};
    ↳ count = R - L + 1
}

// 4) LONGEST COMMON SUBSTRING of two strings (named
//   ↳ lcsViaSuffixArray to avoid clashing
// with the SAM version in 02 - SuffixAutomaton.cpp): build SA
//   ↳ of a + sep + b (sep smaller than every
// real character), then take the max lcp over ADJACENT pairs
//   ↳ coming from different sides.
int lcsViaSuffixArray(const string& a, const string& b) {
    string t = a + char(1) + b;
    SuffixArray S(t);
    int n = t.size(), best = 0, cut = a.size();
    auto side = [&](int p) { return p < cut ? 0 : (p == cut ? -1
    ↳ : 1); };
    for (int i = 2; i <= n; i++) {
        int s1 = side(S.sa[i]), s2 = side(S.sa[i - 1]);
        if (s1 >= 0 && s2 >= 0 && s1 != s2) best = max(best,
        ↳ S.lcp[i]);
    }
    return best;
}

// 5) K-TH SMALLEST DISTINCT SUBSTRING: SA entry i contributes (n
//   ↳ - sa[i]) - lcp[i] new
// substrings, all sharing the prefix of length lcp[i].
//   ↳ Prefix-sum those counts and binary
// search for k; inside the block the answer is a prefix of
//   ↳ suffix sa[i].
// 6) LCS OF K STRINGS: concatenate with distinct separators, then
//   ↳ slide a window over the SA
// that contains at least one suffix of every string; the
//   ↳ answer is the max over windows of
// the minimum lcp inside (monotonic deque), O(n) after the SA.
// 7) COMPARE TWO SUBSTRINGS in O(1): let l = S.lce(i, j) (STRING
//   ↳ positions - queryLcp takes
// SUFFIX-ARRAY indices, do not mix them up); if l >= both
// lengths they are equal, otherwise compare s[i+l] with s[j+l].
```

7.4.4 GeneralizedSAM

GENERALIZED SUFFIX AUTOMATON - one automaton accepting every substring of a SET of strings.

```
// Same size bound O(total length). The only change vs a plain
//   ↳ SAM: when extending, the transition
// may ALREADY exist (a previous string walked here), so handle
//   ↳ that case instead of blindly adding.
// Feeding strings one after another into a plain SAM is WRONG -
//   ↳ it creates junk states.
struct GenSAM {
    struct S { int len = 0, link = -1; array<int, 26> nxt; S() {
        ↳ nxt.fill(-1); } };
    vector<S> st;
    GenSAM() { st.push_back(S()); }
    int clone(int q, int len) { int c = st.size();
        ↳ st.push_back(st[q]), st[c].len = len; return c; }
    int extend(int last, int c) {
        ↳ returns the new "last"
        if (st[last].nxt[c] != -1) {
            ↳ the edge is already there
            int q = st[last].nxt[c];
            if (st[q].len == st[last].len + 1) return q;
            int cl = clone(q, st[last].len + 1);
            for (int p = last; p != -1 && st[p].nxt[c] == q; p =
            ↳ st[p].link) st[p].nxt[c] = cl;
            st[q].link = cl;
            return cl;
        }
        int cur = st.size();
        st.push_back(S()); st[cur].len = st[last].len + 1;
        int p = last;
        while (p != -1 && st[p].nxt[c] == -1) st[p].nxt[c] = cur,
        ↳ p = st[p].link;
        if (p == -1) st[cur].link = 0;
        else {
            int q = st[p].nxt[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;
            else {
                int cl = clone(q, st[p].len + 1);
                for (; p != -1 && st[p].nxt[c] == q; p =
                ↳ st[p].link) st[p].nxt[c] = cl;
                st[q].link = st[cur].link = cl;
            }
        }
        return cur;
    }
    void add(const string& s) { int last = 0; for (char ch : s)
        ↳ last = extend(last, ch - 'a'); }
    vector<int> byLen() const {
        ↳ states in increasing len
        int n = st.size(), mx = 0;
        for (auto& x : st) mx = max(mx, x.len);
        vector<int> c(mx + 2, 0), ord(n);
        for (auto& x : st) c[x.len]++;
        for (int i = 1; i <= mx; i++) c[i] += c[i - 1];
        for (int v = n - 1; v >= 0; v--) ord[--c[st[v].len]] = v;
        return ord;
    }
    // HOW MANY OF THE k STRINGS CONTAIN each substring: mark the
    ↳ states each string walks through,
    // then push the marks up the link tree. k <= 64 here; for
    ↳ larger k use small-to-large sets, or
    // a bitset<K>, or process the strings in batches of 64 and
    ↳ add up the popcounts.
    vector<int> countStringsContaining(const vector<string>& v) {
        ↳ requires v.size() <= 64
        vector<unsigned long long> mask(st.size(), 0);
        for (int i = 0; i < (int)v.size(); i++)
            for (int p = 0, j = 0; j < (int)v[i].size(); j++)
                p = st[p].nxt[v[i][j] - 'a'], mask[p] |= 1ULL <<
        ↳ i;
        auto ord = byLen();
        for (int i = ord.size() - 1; i > 0; i--)
            mask[st[ord[i]].link] |= mask[ord[i]];
        vector<int> r(st.size());
        for (size_t i = 0; i < st.size(); i++) r[i] =
        ↳ __builtin_popcountll(mask[i]);
        return r;
        ↳ r[v] for every state v
    }
};

// LONGEST COMMON SUBSTRING OF k STRINGS: build the generalized
//   ↳ SAM, take the deepest state with
// count == k; its longest string is len[v] (walk suffix links for
//   ↳ the actual string).
// DISTINCT SUBSTRINGS OF THE UNION = sum over v != 0 of (len[v] -
//   ↳ len[link[v]]), same formula.
```

```
// SUBSTRING PRESENT IN EXACTLY ONE STRING / IN AT LEAST k: filter
    ↪ by the same counts.
// ALTERNATIVE when the strings are few: concatenate with distinct
    ↪ separators and use one SUFFIX
// ARRAY + a sliding window over lcp. Shorter to write,  $O(n \log n)$ , but no online extension.
```

7.5 Palindromes

7.5.1 Manacher

MANACHER - the radius of the longest palindrome centred at every position, in $O(n)$.

```
// Answers "is s[l..r] a palindrome" in  $O(1)$  and counts all
    ↪ palindromic substrings in one pass.
struct Manacher {
    vector<int> p;

    // Returns a vector where p[i] is the radius of the longest
    ↪ palindrome around center i
    // Centers are interleaved: 0 is before string, 1 is first
    ↪ char, 2 is between 1st & 2nd, etc.
    Manacher(string s) {
        string t = "#";
        for (char c : s) {
            t += c;
            t += "#";
        }
        int n = t.size();
        p.assign(n, 0);
        int c = 0, r = 0;
        for (int i = 0; i < n; i++) {
            int mirror = 2 * c - i;
            if (i < r) p[i] = min(r - i, p[mirror]);
            while (i - 1 - p[i] >= 0 && i + 1 + p[i] < n && t[i -
    ↪ 1 - p[i]] == t[i + 1 + p[i]]) {
                p[i]++;
            }
            if (i + p[i] > r) {
                c = i;
                r = i + p[i];
            }
        }

        // Check if substring [l, r] (0-based) is a palindrome in  $O(1)$ 
        bool is_palindrome(int l, int r) {
            int len = r - l + 1;
            int center = l + r + 1;
            return p[center] >= len;
        }
    };
};
```

7.5.2 PalindromicTree

PALINDROMIC TREE (eertree) - $O(n)$ nodes, one per DISTINCT palindromic substring, built online in

```
//  $O(n)$ . Gives the count of distinct palindromes, occurrence
    ↪ counts, and the longest palindromic
// suffix at every prefix - which is what
    ↪ palindromic-factorization DP needs.
struct PalindromeTree {
    int n, id, cur, tot;
    vector<array<int, 26>> go;

    // Core structural links
    vector<int> suflink, len, cnt;

    // Extensions for Hard DP (Palindrome Partitioning)
    vector<int> diff; // Difference in length to the suffix link
    vector<int> slink; // Series link (skips arithmetic
    ↪ progressions of suffix links)

    // Extensions for counting
    vector<int> num; // Number of palindromic suffixes ending
    ↪ at this node
    long long total_overlapping; // Running sum of total
    ↪ palindromes

    PalindromeTree() {}

    PalindromeTree(const string &s) {
        build(s);
    }

    void init(int max_len) {
        n = max_len;
        go.assign(n + 2, {});
        suflink.assign(n + 2, 0);
        len.assign(n + 2, 0);
        cnt.assign(n + 2, 0);
        diff.assign(n + 2, 0);
        slink.assign(n + 2, 0);
        num.assign(n + 2, 0);

        suflink[0] = suflink[1] = 1;
        len[1] = -1;
        id = 2;
        cur = 0;
        tot = 0;
        total_overlapping = 0;
    }

    int get(const string &s, int i, int v) {
        while (i - len[v] - 1 < 0 || s[i - len[v] - 1] != s[i]) {
            v = suflink[v];
        }
        return v;
    }

    void add(const string &s, int i) {
        int ch = s[i] - 'a';
        cur = get(s, i, cur);

        if (go[cur][ch] == 0) {
            len[id] = 2 + len[cur];
            suflink[id] = go[get(s, i, suflink[cur])][ch];

            // 1. Number of palindromes ending exactly here
            num[id] = num[suflink[id]] + 1;

            // 2. Arithmetic Progression / Series Links for DP
            diff[id] = len[id] - len[suflink[id]];
            if (diff[id] == diff[suflink[id]]) {
                slink[id] = slink[suflink[id]];
            } else {
                slink[id] = suflink[id];
            }

            tot++;
            go[cur][ch] = id++;
        }
        cur = go[cur][ch];
        cnt[cur]++;
        total_overlapping += num[cur];
    }

    // Standard build
    void build(const string &s) {
        init(s.length());
        for (int i = 0; i < n; i++) {
```

```

        add(s, i);
    }
    countAll();
}

// Call this to push frequencies down the suffix links
void countAll() {
    for (int i = id - 1; i >= 2; --i) {
        cnt[suflink[i]] += cnt[i];
    }
}

// --- Black-Box Solvers ---

// 1. Returns number of unique palindromes
int cntDistinct() {
    return tot;
}

// 2. Returns total number of palindromes (including
    ↪ overlaps/duplicates)
long long cntTotal() {
    return total_overlapping;
}

// 3. Solves Minimum Palindrome Partitioning in  $O(N \log N)$ 
// Rebuilds the tree internally to compute the DP concurrently.
int min_partitions(const string &s) {
    init(s.length());
    const int INF = 1e9;
    vector<int> dp(s.length() + 1, INF);
    vector<int> series_ans(s.length() + 2, INF);

    dp[0] = 0;

    for (int i = 0; i < s.length(); ++i) {
        add(s, i);

        for (int v = cur; v > 1; v = slink[v]) {
            series_ans[v] = dp[i + 1 - (len[slink[v]] +
    ↪ diff[v])];

            if (diff[v] == diff[suflink[v]]) {
                series_ans[v] = min(series_ans[v],
    ↪ series_ans[suflink[v]]);
            }

            dp[i + 1] = min(dp[i + 1], series_ans[v] + 1);
        }
    }
    return dp[s.length()];
}

// 4. Returns an array of frequencies for string B's
    ↪ palindromes inside this tree (String A)
// Note: You must 'build()' the tree with string A before
    ↪ calling this with B.
vector<int> count_string(const string& B) {
    vector<int> out_cnt(id, 0);
    int curr = 0;

    for (int i = 0; i < B.length(); ++i) {
        int ch = B[i] - 'a';

        while (true) {
            if (i - len[curr] - 1 >= 0 && B[i - len[curr] -
    ↪ 1] == B[i]) {
                if (go[curr][ch]) {
                    curr = go[curr][ch];
                    break;
                }
            }
            if (curr == 1) {
                curr = 0;
                break;
            }
            curr = suflink[curr];
        }
        out_cnt[curr]++;
    }

    for (int i = id - 1; i >= 2; --i) {
        out_cnt[suflink[i]] += out_cnt[i];
    }

    return out_cnt;
}

```

};

7.5.3 PalindromicTreeRollback

PALINDROMIC TREE WITH ROLLBACK - same structure, but every extend() can be undone, so you can DFS

```
// NOTE: 02 - PalindromicTree.cpp declares the same 'struct
↳ PalindromeTree'. Paste one.
// a trie/tree of characters and maintain the palindromic
↳ structure of the current root-to-node string.
struct PalindromeTree {
    int n, id, cur, tot;
    vector<array<int, 26>> go;

    // Core structural links
    vector<int> suflink, len, cnt;

    // Extensions for Hard DP (Palindrome Partitioning)
    vector<int> diff; // Length difference to suffix link
    vector<int> slink; // Series link

    // Extensions for counting
    vector<int> num; // Number of palindromic suffixes ending
↳ at this node
    long long total_overlapping; // Running total of all
↳ palindromic substrings

    // --- ROLLBACK EXTENSIONS ---
    string S; // Maintains current active string
    struct History {
        int old_cur;
        int added_node; // ID of created node, or 0 if edge
↳ already existed
        int parent_node; // Node where go[parent_node][ch] was
↳ updated
        int ch; // Character index (0..25)
    };
    vector<History> history;

    PalindromeTree() {}

    PalindromeTree(int max_len) {
        init(max_len);
    }

    void init(int max_len) {
        n = max_len;
        go.assign(n + 2, {});
        suflink.assign(n + 2, 0);
        len.assign(n + 2, 0);
        cnt.assign(n + 2, 0);
        diff.assign(n + 2, 0);
        slink.assign(n + 2, 0);
        num.assign(n + 2, 0);

        suflink[0] = suflink[1] = 1;
        len[1] = -1;
        id = 2;
        cur = 0;
        tot = 0;
        total_overlapping = 0;

        S.clear();
        history.clear();
    }

    int get(int i, int v) {
        while (i - len[v] - 1 < 0 || S[i - len[v] - 1] != S[i]) {
            v = suflink[v];
        }
        return v;
    }

    // 1. PUSH / ADD CHARACTER (WITH ROLLBACK HISTORY)
    void push_back(char c) {
        S.push_back(c);
        int i = (int)S.length() - 1;
        int ch = c - 'a';

        int old_cur = cur;
        cur = get(i, cur);

        int added_node = 0;
        int parent_node = 0;

        if (go[cur][ch] == 0) {
            added_node = id;
            parent_node = cur;

            len[id] = 2 + len[cur];
```

```
            suflink[id] = go[get(i, suflink[cur])][ch];

            num[id] = num[suflink[id]] + 1;

            diff[id] = len[id] - len[suflink[id]];
            if (diff[id] == diff[suflink[id]]) {
                slink[id] = slink[suflink[id]];
            } else {
                slink[id] = suflink[id];
            }

            tot++;
            go[cur][ch] = id++;
        }

        cur = go[cur][ch];
        cnt[cur]++;
        total_overlapping += num[cur];

        // Save state transition into history stack
        history.push_back({old_cur, added_node, parent_node, ch});
    }

    // 2. POP / ROLLBACK LAST CHARACTER
    void pop_back() {
        if (history.empty()) return;

        History h = history.back();
        history.pop_back();

        // Revert occurrence count & running palindrome count
        total_overlapping -= num[cur];
        cnt[cur]--;

        // If a new node was created during this push_back, delete
↳ it
        if (h.added_node != 0) {
            int u = h.added_node;
            go[h.parent_node][h.ch] = 0; // Disconnect child edge

            // Reset node u's properties
            go[u].fill(0);
            suflink[u] = len[u] = cnt[u] = diff[u] = slink[u] =
↳ num[u] = 0;

            id--;
            tot--;
        }

        // Restore previous active node and string length
        cur = h.old_cur;
        S.pop_back();
    }

    // --- Black-Box Solvers ---

    int cntDistinct() {
        return tot;
    }

    long long cntTotal() {
        return total_overlapping;
    }

    // Call this if you need to push end-occurrence counts down
↳ suffix links
    void countAll() {
        for (int i = id - 1; i >= 2; --i) {
            cnt[suflink[i]] += cnt[i];
        }
    }
};
```

7.6 Rotations

7.6.1 LyndonAndRotations

LYNDON FACTORISATION (Duval) and MINIMAL CYCLIC ROTATION (Duval), both $O(n)$, $O(1)$ memory.

```
// A Lyndon word is strictly smaller than all of its proper
//   suffixes. Every string factors
//   UNIQUELY as  $w_1 \geq w_2 \geq \dots \geq w_k$  with each  $w_i$  Lyndon.
vector<string> duval(const string& s) {
    int n = s.size(), i = 0;
    vector<string> f;
    while (i < n) {
        int j = i + 1, k = i;
        while (j < n && s[k] <= s[j]) s[k] < s[j] ? k = i : k++,
        j++;
        while (i <= k) f.push_back(s.substr(i, j - k)), i += j -
        k;
        return f;
    }
}
// Smallest cyclic rotation: run Duval on  $s+s$  and take the factor
//   starting before  $n$ .
int minRotation(string s) {
    int n = s.size();
    s += s;
    int i = 0, ans = 0;
    while (i < n) {
        ans = i;
        int j = i + 1, k = i;
        while (j < (int)s.size() && s[k] <= s[j]) s[k] < s[j] ? k
        = i : k++, j++;
        while (i <= k) i += j - k;
    }
    return ans; // index in the
    // ORIGINAL string
}
// USES: canonical form of a cyclic string (necklace dedup ->
//   pairs with Burnside);
//   "is A a rotation of B" (compare minRotation forms, or
//   search A in B+B);
//   lexicographically smallest result of rotating; runs /
//   repetitions via Lyndon roots.
```

7.7 Sequences

7.7.1 DeBruijnAndBitap

DE BRUIJN SEQUENCES and BIT-PARALLEL MATCHING.

```
// --- DE BRUIJN SEQUENCE  $B(k, n)$ : a CYCLIC string over an
//   alphabet of size  $k$ , of length  $k^n$ , in
//   which every one of the  $k^n$  words of length  $n$  appears exactly
//   once as a (cyclic) substring.
// USES: "open the safe with the fewest keypresses" (the classic),
//   minimum-length probe sequences,
//   and hashing tricks (a de Bruijn constant is how you find the
//   index of the lowest set bit).
// CONSTRUCTION (FKM / Lyndon): concatenate, in lexicographic
//   order, every Lyndon word whose
//   length DIVIDES  $n$ . That is exactly what this  $O(k^n)$  recursion
//   emits.
string deBruijn(int k, int n) {
    string s;
    vector<int> a(k * n, 0);
    function<void(int, int)> db = [&](int t, int p) {
        if (t > n) {
            if (n % p == 0) for (int i = 1; i <= p; i++) s +=
            char('0' + a[i]);
        } else {
            a[t] = a[t - p];
            db(t + 1, p);
            for (int j = a[t - p] + 1; j < k; j++) a[t] = j, db(t
            + 1, t);
        }
    };
    db(1, 1);
    return s; //
    // length  $k^n$ , read cyclically
}
// The same object is an EULERIAN CIRCUIT in the de Bruijn graph
//   (vertices = words of length  $n-1$ ,
//   edges = words of length  $n$ ), so Hierholzer on that graph is the
//   alternative construction - and
//   the BEST theorem then counts how many distinct de Bruijn
//   sequences exist:  $(k!)^{(k^{n-1})} / k^n$ .

// --- BITAP / SHIFT-AND: exact matching in  $O(n*m/64)$  with
//   bitsets, and the natural home for
//   approximate matching. Build one bitmask per character marking
//   where it occurs in the pattern;
//   then one pass over the text ANDs shifted copies together.
//   Beats KMP only when you need the extra flexibility (k
//   mismatches, wildcards, "any of these
//   patterns"), or when you want ALL occurrences as a bitset to
//   intersect with a range.
vector<int> bitap(const string& t, const string& p) {
    int n = t.size(), m = p.size();
    if (!m || m > n) return {};
    vector<unsigned long long> mask(256, 0);
    // Works for  $m \leq 64$  with a single word; for longer patterns
    // use bitset<M> and shift it.
    if (m > 64) return {};
    for (int i = 0; i < m; i++) mask[(unsigned char)p[i]] |= 1ULL
    << i;
    unsigned long long st = 0, hit = 1ULL << (m - 1);
    vector<int> res;
    for (int i = 0; i < n; i++) {
        st = ((st << 1) | 1) & mask[(unsigned char)t[i]];
        if (st & hit) res.push_back(i - m + 1);
    }
    return res;
}
/* WHAT BITAP GENERALISES TO, which is the reason to carry it
k MISMATCHES: keep  $k+1$  state words  $RO..Rk$ , where  $Rj$  = "matched a
prefix with at most  $j$  errors".
 $Rj\_new = ((Rj << 1) \& mask[c]) | Rj-1 | (Rj-1 << 1) | (Rj-1\_old$ 
<< 1) | 1
covering substitution, insertion and deletion respectively -
that is Wu-Manber,  $O(k n m / 64)$ .
WILDCARDS in the PATTERN: set mask[c] bits for every character
the class accepts. A '?' just
means "every mask has that bit". This is free - no FFT needed -
and is the right tool when the
alphabet is small and the pattern is short. Use the FFT method
(O1 - String Matching/O4) when
the pattern is long or wildcards appear in the TEXT too.
MULTIPLE PATTERNS of the same length: OR their masks and check
which one actually matched.
OCCURRENCES INSIDE  $[l, r]$ : keep the whole occurrence set as a
bitset and popcount a range - that
is what makes "how many times does  $p$  occur in  $t[l..r]$ "  $O(n/64)$ 
per query with no suffix
structure at all.
```

Dynamic Programming (DP)

MYERS' BIT-VECTOR ALGORITHM computes full edit distance in $O(nm)$ ($\hookrightarrow / 64$) with the same idea (carry-save arithmetic on the DP's difference vectors); it is $\hookrightarrow$ the fastest practical exact edit distance and the escape hatch when the $O(nm)$ DP is 1e10. $\ast/$

8 Dynamic Programming (DP)

8.1 DP Optimizations

8.1.1 DivideAndConquerDP

Divide & Conquer DP Optimization - Time: $O(K * N \log N)$, Space: $O(N)$

```
// Applies when DP transition is  $dp[k][i] = \min_{j < i} \{ \dots \}$ 
 $\hookrightarrow (dp[k-1][j] + cost(j+1, i))$ 
// Condition:  $opt(i, j) \leq opt(i, j+1)$  (Monotonicity of optimal
 $\hookrightarrow$  split points)
const int N = 200005; // ALSO in 01 -
 $\hookrightarrow$  Template.cpp: keep one copy

int a[N];
ll curCost;
int curL = 0, curR = -1, freq[N];
pair<int, ll> pre[N], cur[N];
// Modify at will :
int lo(int ind) { return 1; }
int hi(int ind) { return ind; }

void add(int val) {
    curCost += freq[val]++;
}

void remove(int val) {
    curCost -= --freq[val];
}

ll Cost(int l, int r) {
    while (curR < r)
        add(a[++curR]);
    while (curL > l)
        add(a[--curL]);
    while (curR > r)
        remove(a[curR--]);
    while (curL < l)
        remove(a[curL++]);
    return curCost;
}

ll f(int ind, int k) { return pre[k-1].second + Cost(k, ind); }
void store(int ind, int k, ll v) { cur[ind] = pair(k, v); }

void rec(int L, int R, int LO, int HI) {
    if (L > R) return;
    int mid = (L + R) >> 1;
    pair best(llinf, LO);
    for (int k = max(LO, lo(mid)); k <= min(HI, hi(mid)); ++k)
        best = min(best, pair(f(mid, k), k));
    store(mid, best.second, best.first);
    rec(L, mid-1, LO, best.second);
    rec(mid+1, R, best.second, HI);
}

void solve(int L, int R) { rec(L, R, -inf, inf); }
```

8.1.2 CHT

DYNAMIC CONVEX HULL TRICK (UPPER hull $\rightarrow$ MAXIMUM of $m \cdot x + b$). Lines may be inserted in ANY order

```
// and queried at any x,  $O(\log n)$  each. If your slopes AND queries
 $\hookrightarrow$  are both monotone, the deque
// version at the bottom of this file is  $O(1)$  amortised and much
 $\hookrightarrow$  shorter - prefer it.
// For a MINIMUM, insert  $(-m, -b)$  and negate the answer.
// To maximise over all x in  $[l, r]$ , query only l and r: the upper
 $\hookrightarrow$  envelope is convex.

struct Line {
    ll m, b;
    mutable function<const Line *(>> succ;

    bool operator<(const Line &other) const {
        return m < other.m;
    }

    bool operator<(const ll &x) const {
        const Line *s = succ();
        if (!s)
            return 0;
        return (lll)(b - s->b) < (lll)(s->m - m) * x; //
 $\hookrightarrow$  __int128: |dm|*|x| overflows ll
    }
};

// will maintain upper hull for maximum
struct HullDynamic : public multiset<Line, less<>> {
    bool bad(iterator y) {
        auto z = next(y);
        if (y == begin()) {
            if (z == end())
                return 0;
            return y->m == z->m && y->b <= z->b;
        }
        auto x = prev(y);
        if (z == end())
            return y->m == x->m && y->b <= x->b;
        // __int128, NOT long double: each factor reaches  $\sim 2e18$ ,
 $\hookrightarrow$  the product needs  $\sim 123$  bits,
        // and this test is consulted exactly when three lines are
 $\hookrightarrow$  nearly collinear.
        return (lll)(x->b - y->b) * (z->m - y->m) >= (lll)(y->b -
 $\hookrightarrow$  z->b) * (y->m - x->m);
    }

    void insert_line(ll m, ll b) {
        // for minimum
        // m *= -1;
        // b *= -1;
        auto y = insert({m, b});
        y->succ = [=] { return next(y) == end() ? 0 : &*next(y);
 $\hookrightarrow$  };
        if (bad(y)) {
            erase(y);
            return;
        }
        while (next(y) != end() && bad(next(y)))
            erase(next(y));
        while (y != begin() && bad(prev(y)))
            erase(prev(y));
    }

    ll query(ll x) { // UB if
 $\hookrightarrow$  empty - insert at least one line first
        auto l = *lower_bound(x);
        // for minimum
        // return -(l.m * x + l.b);
        return l.m * x + l.b;
    }
};

// -----
// MONOTONE CHT (deque) - the version you actually want when the
 $\hookrightarrow$  DP feeds lines in slope order.
// MINIMISES  $m \cdot x + b$ . Requires slopes added in DECREASING order;
 $\hookrightarrow$  queries in increasing x are  $O(1)$ 
// amortised (move the head), arbitrary x is  $O(\log n)$  by binary
 $\hookrightarrow$  search on the hull.
// For a MAXIMUM, negate m and b on the way in and negate the
 $\hookrightarrow$  answer on the way out.

struct MonoCHT {
    vector<ll> M, B;
    int head = 0;
    bool bad(int a, int b, int c) { // is line b
 $\hookrightarrow$  unnecessary between a and c?
```

```

    return (lll)(B[c] - B[a]) * (M[a] - M[b]) <= (lll)(B[b] -
    ↪ B[a]) * (M[a] - M[c]);
}
void add(ll m, ll b) {                               // m must be
    ↪ <= the last slope added
    M.push_back(m), B.push_back(b);
    while (M.size() >= 3 && bad(M.size() - 3, M.size() - 2,
    ↪ M.size() - 1))
        M.erase(M.end() - 2), B.erase(B.end() - 2);
    head = min(head, (int)M.size() - 1);
}
ll f(int i, ll x) { return M[i] * x + B[i]; }
ll queryInc(ll x) {                                   // x
    ↪ non-decreasing across calls
    while (head + 1 < (int)M.size() && f(head + 1, x) <=
    ↪ f(head, x)) head++;
    return f(head, x);
}
ll query(ll x) {                                     // any x,
    ↪ O(log n)
    int lo = 0, hi = M.size() - 1;
    while (lo < hi) { int m = (lo + hi) / 2; f(m + 1, x) <=
    ↪ f(m, x) ? lo = m + 1 : hi = m; }
    return f(lo, x);
}
};
// WHEN EACH ONE: monotone slopes + monotone queries ->
    ↪ MonoCHT::queryInc, O(n).
//      monotone slopes, arbitrary queries ->
    ↪ MonoCHT::query, O(n log n).
//      arbitrary slopes ->
    ↪ HullDynamic above, or Li Chao (04),
//                                     which
    ↪ needs no ordering at all.
//      lines inserted and removed -> CHT with
    ↪ rollback (03) or Li Chao merge.

```

8.1.3 CHTRollback

Rollback Convex Hull Trick - Space: $O(N)$, Add: $O(\log N)$, Query: $O(\log N)$, Rollback: $O(1)$

```

// Supports inserting monotonic lines with history stack rollback
    ↪ for tree/dsu DP
// Template parameters:
// IS_MAX: Set to true for Maximum queries, false for Minimum
    ↪ queries.
// IS_INC: Set to true if slopes added are strictly Increasing,
    ↪ false for Decreasing.
template<bool IS_MAX = false, bool IS_INC = false>
struct RollbackCHT {
    struct Line {
        ll m, c;
        Line() : m(0), c(1llinf) {}
        Line(ll m, ll c) : m(m), c(c) {}
        ll eval(ll x) const { return m * x + c; }
    };

    struct History {
        int pos;
        int sz;
        Line old_line;
    };

    vector<Line> st;
    vector<History> hist;
    int sz;

    RollbackCHT(int max_nodes) {
        st.resize(max_nodes);
        sz = 0;
    }

    bool redundant(Line l1, Line l2, Line l3) {
        __int128 dx1 = l1.m - l2.m;
        __int128 dc1 = l2.c - l1.c;
        __int128 dx2 = l2.m - l3.m;
        __int128 dc2 = l3.c - l2.c;
        return dc1 * dx2 >= dc2 * dx1;
    }

    void add(ll m, ll c) {
        // The magic happens here: automatically formats internal
    ↪ state
        ll m_int = IS_INC ? -m : m;
        ll c_int = IS_MAX ? -c : c;

        Line l_new(m_int, c_int);
        int pos = sz;
        int low = 1, high = sz - 1;

        while (low <= high) {
            int mid = low + (high - low) / 2;
            if (redundant(st[mid - 1], st[mid], l_new)) {
                pos = mid;
                high = mid - 1;
            } else {
                low = mid + 1;
            }
        }

        hist.push_back({pos, sz, st[pos]});
        st[pos] = l_new;
        sz = pos + 1;
    }

    void rollback() {
        if (hist.empty()) return;
        History h = hist.back();
        hist.pop_back();
        sz = h.sz;
        st[h.pos] = h.old_line;
    }

    ll query(ll x) {
        if (sz == 0) return IS_MAX ? -1llinf : 1llinf;

        // The magic happens here: matches x to the internal
    ↪ geometry
        ll x_int = (IS_INC != IS_MAX) ? -x : x;

        int low = 0, high = sz - 2;
        int best = sz - 1;

        while (low <= high) {
            int mid = low + (high - low) / 2;

```

```

    if (st[mid].eval(x_int) <= st[mid + 1].eval(x_int)) {
        best = mid;
        high = mid - 1;
    } else {
        low = mid + 1;
    }
}

ll ans = st[best].eval(x_int);
return IS_MAX ? -ans : ans;
};

```

8.1.4 LiChaoTree

LI CHAO TREE - MAXIMUM of a set of lines $y = m \cdot x + b$ at any query x , with NO order requirement

```

// on the slopes or on the queries (that is the whole point; CHT
// needs monotone slopes).
// O(log X) per line insert and per query, O(log^2 X) to insert a
// line only on a SEGMENT [lx, rx].
// Space O(n log X). Works for any family of functions where two
// members cross AT MOST ONCE -
// lines, and also  $a \cdot x^2 + b \cdot x + c$  with a FIXED  $a$ .
// EXACT INTEGER ARITHMETIC: never compute the crossing abscissa.
// Compare the two candidates at
// ns, mid and ne only - a division in double is off by  $\sim 1e3$  in  $x$ 
// when the coefficients reach  $1e18$ ,
// which silently sends the loser down the wrong half.
// SET the domain [X0, X1] before use; it must cover every  $x$  you
// will ever query.
struct LiChao {
    struct Line { ll m = 0, b = LLONG_MIN / 4; }; //
    // neutral = -infinity
    ll X0, X1;
    vector<Line> t;
    vector<int> lf, rt;
    LiChao(ll X0, ll X1) : X0(X0), X1(X1) { newNode(); }
    int newNode() { t.push_back({}); lf.push_back(-1),
    // rt.push_back(-1); return t.size() - 1; }
    static ll f(const Line& L, ll x) { return L.m == 0 && L.b ==
    // LLONG_MIN / 4 ? L.b : L.m * x + L.b; }
    void add(Line nw, int v, ll l, ll r) { //
    // insert on the whole node range
        while (true) {
            ll m = l + (r - l) / 2; //
            // NOT (l+r)/2: negative l+r truncates up
            bool lef = f(nw, l) > f(t[v], l), mid = f(nw, m) >
            // f(t[v], m);
            if (mid) swap(t[v], nw);
            if (l == r) return;
            if (lef != mid) {
                if (lf[v] < 0) lf[v] = newNode();
                v = lf[v], r = m;
            } else {
                if (f(nw, r) > f(t[v], r)) {
                    if (rt[v] < 0) rt[v] = newNode();
                    v = rt[v], l = m + 1;
                } else return;
            }
        }
    }
    void addLine(ll m, ll b) { add({m, b}, 0, X0, X1); }
    void addSegment(ll m, ll b, ll lx, ll rx, int v = 0, ll l =
    // LLONG_MIN, ll r = LLONG_MIN) {
        if (l == LLONG_MIN) l = X0, r = X1;
        if (rx < l || r < lx) return;
        if (lx <= l && r <= rx) { add({m, b}, v, l, r); return; }
        ll md = l + (r - l) / 2;
        if (lf[v] < 0) lf[v] = newNode();
        if (rt[v] < 0) rt[v] = newNode();
        addSegment(m, b, lx, rx, lf[v], l, md), addSegment(m, b,
        // lx, rx, rt[v], md + 1, r);
    }
    ll query(ll x) {
        ll r = LLONG_MIN, l = X0, rr = X1;
        for (int v = 0; v >= 0;) {
            r = max(r, f(t[v], x));
            if (l == rr) break;
            ll m = l + (rr - l) / 2;
            if (x <= m) v = lf[v], rr = m;
            else v = rt[v], l = m + 1;
        }
        return r; //
    }
    // LLONG_MIN/4 if no line covers x
};

// FOR A MINIMUM: insert (-m, -b) and negate the answer.
// PERSISTENT / MERGEABLE: see 05 - PersistentLiChaoTree.cpp.
// MERGING two Li Chao trees costs
// O(n log X) amortised in total (push the loser line down into
// the other subtree, exactly like
// segment-tree merging) - that is how you do convex-hull DP on a
// tree without small-to-large.

```

8.1.5 PersistentLiChaoTree

Persistent Li Chao Tree - Space: $O(N \log X)$, Insert Line: $O(\log X)$, Query: $O(\log X)$

```
// Version-persistent Li Chao Segment Tree for tree / history DP
↳ queries
const ll maxN = 1e7 + 5;

struct Line {
    ll m, c;

    Line() : m(0), c(1llinf) {}

    Line(ll m, ll c) : m(m), c(c) {}
};

ll sub(ll x, Line l) {
    return x * l.m + l.c;
}

// Persistent Li Chao
struct Node {
    // range I am responsible for
    Line line;
    Node *left, *right;

    Node() {
        left = right = NULL;
    }

    Node(ll m, ll c) {
        line = Line(m, c);
        left = right = NULL;
    }

    void extend(int l, int r) {
        if (left == NULL && l != r) {
            left = new Node();
            right = new Node();
        }

        Node *copy(Node *node) {
            Node *newNode = new Node;
            newNode->left = node->left;
            newNode->right = node->right;
            newNode->line = node->line;
            return newNode;
        }

        Node *add(Line toAdd, int l, int r) {
            int mid = 1 + (r - l) / 2; // NOT (l+r)/2: that
            ↳ truncates toward ZERO, so on a negative
            // interval the left child
            ↳ equals the parent -> infinite recursion
            Node *cur = copy(this);
            if (l == r) {
                if (sub(l, toAdd) < sub(l, cur->line))
                    swap(toAdd, cur->line);
                return cur;
            }
            bool lef = sub(l, toAdd) < sub(l, cur->line);
            bool midE = sub(mid + 1, toAdd) < sub(mid + 1, cur->line);
            if (midE)
                swap(cur->line, toAdd);
            cur->extend(l, r);
            if (lef != midE) {
                cur->left = cur->left->add(toAdd, l, mid);
            } else {
                if (mid + 1 > r)
                    return cur;
                cur->right = cur->right->add(toAdd, mid + 1, r);
            }
            return cur;
        }

        Node *add(Line toAdd) {
            return add(toAdd, -maxN + 1, maxN - 1);
        }
    }

    ll query(ll x, int l, int r) {
        int mid = 1 + (r - l) / 2; // NOT (l+r)/2: that
        ↳ truncates toward ZERO, so on a negative
        // interval the left child
        ↳ equals the parent -> infinite recursion
        if (l == r || left == NULL)
            return sub(x, line);
        extend(l, r);
        if (x <= mid)
```

```
        return min(sub(x, line), left->query(x, l, mid));
    } else
        return min(sub(x, line), right->query(x, mid + 1, r));
}

ll query(ll x) {
    return query(x, -maxN + 1, maxN - 1);
}

void clear() {
    if (left != NULL) {
        left->clear();
        right->clear();
    }
    delete this;
}

};

const int N = 200005; // #versions;
↳ also in 01 - Template.cpp
Node *tree[N];
```

8.1.6 KnuthDP

KNUTH OPTIMIZATION - interval DP $dp[i][j] = \min_{i \leq k < j} \{dp[i][k] + dp[k+1][j]\} + cost(i, j)$

```
// from  $O(n^3)$  to  $O(n^2)$ , by restricting  $k$  to  $[opt[i][j-1],$ 
↳  $opt[i+1][j]]$ .
// THE HYPOTHESIS IS ON cost, NOT ON opt. 'opt[i][j-1] <=
↳  $opt[i][j] <= opt[i+1][j]$ ' is the
// CONCLUSION - you cannot test it before running. What you must
↳ check is BOTH of:
// (1) QUADRANGLE INEQUALITY (Monge):  $cost(a, c) + cost(b, d) \leq$ 
↳  $cost(a, d) + cost(b, c)$ 
// for all  $a \leq b \leq c \leq d$ . Equivalent local form, and the
↳ only one checkable by hand:
//  $cost(a, c) + cost(a+1, c+1) \leq cost(a+1, c) + cost(a, c+1)$ .
// (2) MONOTONE ON INTERVALS:  $a \leq b \leq c \leq d \Rightarrow cost(b, c)$ 
↳  $\leq cost(a, d)$ .
// Both hold for: sum of a non-negative array over  $[i, j]$ ;  $(S_j -$ 
↳  $S_i)^2$ ;  $f(S_j - S_i)$  with  $f$  convex
// and  $a[] \geq 0$ ; "number of equal pairs inside  $[i, j]$ ". They FAIL
↳ for anything built from max, and
// monotonicity fails as soon as  $a[]$  can be negative (QI can still
↳ hold - DEC needs only QI,
// Knuth needs both). Applied without them there is no crash and
↳ no TLE, just a wrong answer.
// Classic uses: optimal BST, merging  $n$  piles of stones, CF
↳ 1101F-style interval costs.
```

```
struct KnuthDP {
    int n;
    vector<vector<ll>> dp;
    vector<vector<int>> opt;

    KnuthDP(int n = 0) : n(n), dp(n + 2, vector<ll>(n + 2, 0)),
    ↳ opt(n + 2, vector<int>(n + 2, 0)) {}

    // User-defined cost function cost(i, j)
    ll solve(function<ll(int, int)> cost) {
        for (int i = 1; i <= n; i++) {
            opt[i][i] = i;

            for (int len = 2; len <= n; len++) {
                for (int i = 1; i + len - 1 <= n; i++) {
                    int j = i + len - 1;
                    dp[i][j] = 1llinf;
                    int L = opt[i][j - 1], R = opt[i + 1][j];
                    for (int k = L; k <= min(R, j - 1); k++) {
                        ll cur = dp[i][k] + dp[k + 1][j] + cost(i, j);
                        if (cur < dp[i][j]) {
                            dp[i][j] = cur;
                            opt[i][j] = k;
                        }
                    }
                }
            }
            return dp[1][n];
        }
    }
};
```

8.1.7 AliensTrick

ALIENS TRICK (Lagrangian relaxation / wqs binary search) - $O(n \log(\text{range}))$.

```
// Drops an "exactly k pieces" constraint: charge lambda per
//   piece, solve the UNCONSTRAINED DP,
// and binary search lambda until the optimum uses k pieces.
//   Answer = cost(lambda) - k*lambda.
// PRECONDITION: f(k), the optimum with exactly k pieces, must be
//   CONVEX in k (for a minimum;
//   concave for a maximum). NOT "strictly" - strictness is never
//   needed and rarely true.
// THREE THINGS THAT MAKE IT CORRECT, all easy to get wrong:
// (1) dp_solver MUST return the LARGEST optimal count for that
//   lambda. With f convex, the
//   difference d(k) = f(k) - f(k-1) is non-decreasing; the
//   search below finds the largest
//   lambda with count >= k, and then d(k) <= -lambda <=
//   d(k+1), so k really is optimal at
//   that lambda and cost - k*lambda = f(k). Any other
//   tie-break makes the last line wrong.
// (2) An integer lambda suffices ONLY if all costs are integers.
// (3) [min_penalty, max_penalty] must straddle every slope of f;
//   assert it.
// CONVEX families: "k pairwise non-adjacent items", "exactly k
//   segments with a Monge cost",
// "k disjoint intervals", "min-cost flow of value k" (flow cost
//   is always convex in the value),
// matroid-rank-constrained optima. TRAPS: two coupled
//   constraints, or a cost rewarding a parity
// of k. CHECK IT: brute-force f(k) for n <= 60 and assert
//   f(k+1)-f(k) >= f(k)-f(k-1).
struct AliensTrick {
    // Binary searches optimal penalty lambda to choose exactly K
    // items
    // Returns minimum cost to choose exactly K items
    static ll solve(int n, int k, ll min_penalty, ll max_penalty,
        function<pair<ll, int>(ll)> dp_solver) {
        ll l = min_penalty, r = max_penalty, best_lambda = 0;

        while (l <= r) {
            ll mid = l + (r - l) / 2;
            auto [cost, count] = dp_solver(mid);
            if (count >= k) {
                best_lambda = mid;
                l = mid + 1; // Increase penalty if we used too
                // many items
            } else {
                r = mid - 1; // Decrease penalty if we used too
                // few items
            }
        }

        // True cost = cost_with_penalty - (k * best_lambda)
        auto [cost, count] = dp_solver(best_lambda);
        return cost - (ll)k * best_lambda;
    }
};
```

8.1.8 AliensTrickExample

ALIENS TRICK (Lagrangian / wqs binary search) - drops the "exactly k pieces" constraint by charging

```
// a penalty lambda per piece, solving the unconstrained DP, and
//   binary searching lambda until the
// optimal solution uses k. REQUIRES the optimum as a function of
//   k to be CONVEX; check that first.
// 1. DP State Structure
struct DPState {
    long long cost;
    long long count; // Number of items/segments picked

    // TIE-BREAKER LOGIC:
    // If costs are equal, we maximize the count to handle
    // collinear points on the convex hull.
    // If you are writing a MINIMIZATION problem with min(),
    // change this to:
    // if (cost != other.cost) return cost > other.cost; (since
    // priority queue / max naturally takes the "biggest")
    // Actually, when using standard >, < operators for min/max DP:
    bool operator<(const DPState& other) const {
        if (cost != other.cost) return cost < other.cost;
        return count < other.count; // Always prefer taking MORE
        // items if costs are tied
    }
};

void solve_alien() {
    int n, k;
    cin >> n >> k;
    // Read ABC 218 H input: A array is of size N-1
    vector<long long> A(n);
    for(int i = 1; i < n; i++) cin >> A[i];
    // If K > N / 2, swap colors so K <= N / 2 (the score is
    // symmetric)
    if (k > n / 2) k = n - k;
    // Construct 1-indexed values array 'a' where a[i] = A[i-1] +
    // A[i]
    vector<long long> a(n + 1);
    for (int i = 1; i <= n; i++) {
        a[i] = A[i - 1] + (i < n ? A[i] : 0);
    }

    // 2. The Unconstrained DP
    // Returns the optimal {cost, count} for a given penalty.
    auto check_dp = [&](long long penalty) -> DPState {
        // dp[i][0]: Max score in prefix i, ending completely
        // outside a segment
        // dp[i][1]: Max score in prefix i, ending inside an
        // active segment
        vector<vector<DPState>> dp(n + 1, vector<DPState>(2, {0,
        // 0}));

        for (int i = 1; i <= n; i++) {
            // Transition 0: We don't pick item i
            dp[i][0] = max(dp[i-1][0], dp[i-1][1]);

            // Transition 1: We pick item i
            // Rule: ONLY apply penalty when STARTING a new
            // pick/segment!
            // Type B logic: To ensure non-adjacency, we must
            // transition from dp[i-1][0].

            // --- IF MAXIMIZING PROFIT ---
            DPState start_new = {dp[i-1][0].cost + a[i] -
            // penalty, dp[i-1][0].count + 1};

            // Replaced max(start_new, continue_old) with just
            // start_new
            dp[i][1] = start_new;
        }
        return max(dp[n][0], dp[n][1]);
    };

    // 3. WQS Binary Search
    // The bounds must be large enough to cover the maximum
    // possible penalty.
    // FIXED: Dropped bounds to 2e9 to prevent k * mid_penalty
    // from overflowing long long
    long long low = -2e9 - 7, high = 2e9 + 7;
    long long best_ans = 0;

    while (low <= high) {
        long long mid_penalty = low + (high - low) / 2;

        DPState res = check_dp(mid_penalty);

        // We want EXACTLY k items.
    }
}
```



```

// If we picked >= k items, it means the penalty wasn't
↳ harsh enough to stop us.
// Even if res.count > k, we save the answer because of
↳ collinear points on the hull.
if (res.count >= k) {

    // --- IF MAXIMIZING PROFIT ---
    // We subtracted the penalty K times, so we add it
↳ back.

    best_ans = res.cost + (1LL * k * mid_penalty);

    // --- IF MINIMIZING COST ---
    // best_ans = res.cost - (1LL * k * mid_penalty);

    // Penalty was too weak, increase it to force picking
↳ fewer items next time
    low = mid_penalty + 1;
} else {
    // We picked too few items. The penalty was too harsh.
    high = mid_penalty - 1;
}

// Output the final answer
cout << best_ans << "\n";
}

```

8.1.9 MaxAverageSubarray

UPPER hull (maximum) and this one is the LOWER hull (minimum) - they are NOT

```

// NOTE: 02 - CHT.cpp also declares 'struct Line' / 'struct
↳ HullDynamic', but that one is the
// UPPER hull (maximum) and this one is the LOWER hull
↳ (minimum) - they are NOT
// interchangeable. If you need both, rename one pair to
↳ UpperHull / LowerHull.
/*
↳ =====

```

* DYNAMIC CONVEX HULL TRICK (CHT) - AVERAGE SUBARRAY TEMPLATE

*
↳ =====

* HOW TO USE THE 4 VARIATIONS:

* 1. MAX Average, LONGEST length (Default)

* - Code: Use 'lhs < rhs' in Query::operator<

* - Call: hull.insert_line(i, S[i]);

↳ hull.query_tangent_index(i, S[i]);

* 2. MAX Average, SHORTEST length

* - Code: Use 'lhs <= rhs' in Query::operator<

* - Call: hull.insert_line(i, S[i]);

↳ hull.query_tangent_index(i, S[i]);

* 3. MIN Average, LONGEST length

* - Code: Use 'lhs < rhs' in Query::operator<

* - Call: hull.insert_line(i, -S[i]);

↳ hull.query_tangent_index(i, -S[i]); // Negated

* 4. MIN Average, SHORTEST length

* - Code: Use 'lhs <= rhs' in Query::operator<

* - Call: hull.insert_line(i, -S[i]);

↳ hull.query_tangent_index(i, -S[i]); // Negated

*
↳ =====

↳ */

```

struct Query {
    ll i, Si;
};

```

```

struct Line {
    ll m, b;
    mutable function<const Line *(> succ;

    bool operator<(const Line &other) const {
        return m < other.m;
    }

    bool operator<(const Query &q) const {
        const Line *s = succ();
        if (!s) return false;

        // CORRECTED ALGEBRA: Direct cross-multiplication of slopes
        __int128_t lhs = (__int128_t)(q.Si - b) * (q.i - s->m);
        __int128_t rhs = (__int128_t)(q.Si - s->b) * (q.i - m);

        // --- TIE-BREAKING CONTROL ---
        // Change this return statement based on length
↳ requirements:
        // return lhs < rhs; // <-- Use for LONGEST subarray
        // return lhs <= rhs; // <-- Use for SHORTEST subarray
        return lhs < rhs;
    }
};

```

```

struct HullDynamic : public multiset<Line, less<>> {
    bool bad(iterator y) {
        auto z = next(y);
        if (y == begin()) {
            if (z == end()) return false;
            // CORRECTED: Lower hull keeps smallest y
            return y->m == z->m && y->b >= z->b;
        }
        auto x = prev(y);
        if (z == end()) return y->m == x->m && y->b >= x->b;

        return (__int128_t)(x->b - y->b) * (z->m - y->m) <=
↳ (__int128_t)(y->b - z->b) * (y->m - x->m);
    }

    void insert_line(ll m, ll b) {
        auto y = insert({m, b});
        y->succ = [=] { return next(y) == end() ? 0 : &*next(y);
↳ };
    }
};

```

```

    if (bad(y)) {
        erase(y);
        return;
    }
    while (next(y) != end() && bad(next(y))) erase(next(y));
    while (y != begin() && bad(prev(y))) erase(prev(y));
}

int query_tangent_index(ll i, ll Si) {
    auto l = *lower_bound(Query[i], Si);
    return l.m;
}

// USAGE (longest subarray with average >= 0, after subtracting
//   the target from every element):
// S[0..n] = prefix sums; the answer for each i is i - (tangent
//   index from the hull of (j, S[j])).
// HullDynamic hull; hull.insert_line(0, 0);
// for (int i = 1; i <= n; i++) {
//     int k = hull.query_tangent_index(i, S[i]);
//     best j < i
//     ans[i] = i - k;
//     hull.insert_line(i, S[i]);
// }
// WHY IT WORKS: avg(j+1..i) >= 0 <=> S[i] - S[j] >= 0, and
//   among all valid j you want the
// SMALLEST, which is the point of tangency from (i, S[i]) to the
//   lower hull of the points (j, S[j]).
// Binary search on the answer + prefix sums is the shorter O(n
//   log n) alternative; this is O(n).

```

8.1.10 SlopeTrick

SLOPE TRICK - maintain a CONVEX piecewise-linear $f(x)$ as two heaps of its break-points.

```

// L = left slopes (max-heap), R = right slopes (min-heap), minf =
//   current minimum value.
// Every operation below is O(log n). Classic use: "minimum total
//   |change| to make an array
// non-decreasing", scheduling with earliness/tardiness penalties,
//   and any DP where
// dp_i(x) = min over y<=x of dp_{i-1}(y) + |x - a_i|.
struct SlopeTrick {
    priority_queue<ll> L;
    // left breakpoints (max-heap)
    priority_queue<ll, vector<ll>, greater<ll>> R;
    // right breakpoints (min-heap)
    ll minf = 0, addL = 0, addR = 0;
    // lazy shifts of each side

    ll topL() { return L.top() + addL; }
    ll topR() { return R.top() + addR; }
    void pushL(ll x) { L.push(x - addL); }
    void pushR(ll x) { R.push(x - addR); }

    void addConst(ll c) { minf += c; }
    void addRampL(ll a) {
        // f(x) += max(0, x - a)
        if (!L.empty()) minf += max(0LL, topL() - a);
        pushL(a);
        pushR(topL()); L.pop();
    }
    void addRampR(ll a) {
        // f(x) += max(0, a - x)
        if (!R.empty()) minf += max(0LL, a - topR());
        pushR(a);
        pushL(topR()); R.pop();
    }
    void addAbs(ll a) { addRampL(a), addRampR(a); }
    // f(x) += |x - a|
    void prefixMin() { while (!R.empty()) R.pop(); }
    // f(x) = min_{y<=x} f(y)
    void suffixMin() { while (!L.empty()) L.pop(); }
    // f(x) = min_{y>=x} f(y)
    void shift(ll dl, ll dr) { addL += dl, addR += dr; }
    // f(x) -> min over [x-dr, x+dl]
    ll minVal() const { return minf; }
};

// RECIPE for "make a[] non-decreasing with min sum |b_i - a_i|":
// for each a_i: st.prefixMin(); st.addAbs(a_i); answer =
//   st.minVal()
// For "strictly increasing", first replace a_i by a_i - i.

```

8.1.11 MinPlusConvolution

MIN-PLUS / MAX-PLUS CONVOLUTION: $c[k] = \min_i (a[i] + b[k-i])$.

```

// In general this is quadratic and CONJECTURED to stay that way
//   (the MinConv hypothesis:
// MinConv, MaxConv, unbounded knapsack and 0/1 knapsack are all
//   equivalent under subquadratic
// reductions). So the ONLY fast versions rely on convexity.

// BOTH CONVEX -> O(n + m). The difference sequence of the answer
//   is the sorted MERGE of the two
// difference sequences: greedily always take the smaller next
//   slope.
vector<ll> minPlusConvexConvex(const vector<ll>& a, const
//   vector<ll>& b) {
    int n = a.size(), m = b.size();
    vector<ll> c(n + m - 1);
    c[0] = a[0] + b[0];
    int i = 0, j = 0;
    for (int k = 1; k < n + m - 1; k++) {
        ll da = (i + 1 < n) ? a[i + 1] - a[i] : LLONG_MAX;
        ll db = (j + 1 < m) ? b[j + 1] - b[j] : LLONG_MAX;
        if (da <= db) c[k] = c[k - 1] + da, i++;
        else c[k] = c[k - 1] + db, j++;
    }
    return c;
}

// MAX-PLUS with both CONCAVE: same code, take the LARGER slope
//   first (or negate and reuse).
vector<ll> maxPlusConcaveConcave(vector<ll> a, vector<ll> b) {
    for (ll& x : a) x = -x;
    for (ll& x : b) x = -x;
    auto c = minPlusConvexConvex(a, b);
    for (ll& x : c) x = -x;
    return c;
}

// ONLY ONE CONVEX -> the matrix X[k][i] = a[i] + b[k-i] is
//   totally monotone (Monge), so the row
// minima can be found with SMAWK in O(n + m), or with divide &
//   conquer in O(n log n):
// solve(kl, kr, il, ir): take k = mid, scan i in [il, ir] for
//   the best, recurse with the
// split at the optimum. Same skeleton as DivideAndConquerDP.
// 'valid(k)' must return the legal range of i for output index k
//   - for the convolution above
// that is [max(0, k-m+1), min(n-1, k)]. Intersecting with it is
//   what keeps 'best' well defined;
// without it an empty candidate set leaves best = -1 and poisons
//   both recursive calls.
template <class F, class V>
void dcOpt(int kl, int kr, int il, int ir, vector<ll>& c, F cost,
//   V valid) {
    if (kl > kr) return;
    int km = (kl + kr) / 2, best = -1;
    ll bv = LLONG_MAX;
    auto [lo, hi] = valid(km);
    for (int i = max(il, lo); i <= min(ir, hi); i++) {
        ll v = cost(km, i);
        if (v < bv) bv = v, best = i;
    }
    c[km] = bv;
    if (best < 0) best = max(il, lo);
    // candidate: keep the bounds sane
    dcOpt(kl, km - 1, il, best, c, cost, valid);
    dcOpt(km + 1, kr, best, ir, c, cost, valid);
}

// NEITHER CONVEX -> O(n*m) is the best you should plan for, and
//   that is not laziness: min-plus
// convolution, 0/1 knapsack, unbounded knapsack and tree knapsack
//   are subquadratic-EQUIVALENT,
// so a general fast merge would break all four. Look for
//   convexity, a small value range, or a
// bitset instead of a cleverer convolution.
// WHERE THIS APPEARS: merging two "cost as a function of how many
//   you take" curves - group
// knapsack, tree DP where each child contributes a convex cost
//   curve, and combining the
// per-part curves produced by Alien's trick.

```

8.1.12 OneDOneD

1D/1D DP OPTIMISATION - the self-referential case that divide & conquer structurally cannot do.

```
// dp[i] = min over j < i of ( dp[j] + w(j, i) )
// D&C optimisation needs the LAYERED form dp[k][i] =
//   ↪ min(dp[k-1][j] + w) so that the whole
// previous layer is already known. Here dp[j] is produced by the
//   ↪ very loop that consumes it, so
// you must decide dp[i] before you may use it. That is what this
//   ↪ file solves.
// PRECONDITION: w satisfies the QUADRANGLE INEQUALITY (Monge)
//   w(a,c) + w(b,d) <= w(a,d) + w(b,c)   for all a <= b <= c <= d
// checkable in the local form w(a,c) + w(a+1,c+1) <= w(a+1,c) +
//   ↪ w(a,c+1).
// Then the optimal j for i is NON-DECREASING in i, so the array
//   ↪ of "who is currently best for
// each future i" is a sequence of INTERVALS, each owned by one j
//   ↪ - and finalising dp[i] can only
// append one new interval and overwrite a suffix of the old ones.
//   ↪ Keep them on a stack.
// O(n log n) (the binary search for the crossover); O(n) if the
//   ↪ crossover is O(1) to find.
template <class F>
vector<ll> onedOned(int n, F w) {
    ↪ w(j, i) for j < i
    vector<ll> dp(n + 1, llinf);
    dp[0] = 0;
    vector<array<int, 3>> st;
    ↪ {owner j, from, to} intervals
    st.push_back({0, 1, n});
    int head = 0;
    for (int i = 1; i <= n; i++) {
        while (head < (int)st.size() && st[head][2] < i) head++;
        dp[i] = dp[st[head][0]] + w(st[head][0], i);
        ↪ // the owner of position i
        st[head][1] = i;
        // now insert i as a candidate owner for the suffix it wins
        auto better = [&](int a, int b, int x) {
            ↪ // is a better than b at x?
            return dp[a] + w(a, x) <= dp[b] + w(b, x);
        };
        while (!st.empty() && st.back()[1] > i && better(i,
            ↪ st.back()[0], st.back()[1])) st.pop_back();
        if (st.empty() || head >= (int)st.size())
            ↪ st.push_back({i, i + 1, n});
        else {
            auto [j, l, r] = st.back();
            if (better(i, j, r)) {
                ↪ // i wins part of this interval
                int lo = max(l, i + 1), hi = r, pos = r + 1;
                while (lo <= hi) {
                    ↪ // first x where i beats j
                    int m = (lo + hi) / 2;
                    better(i, j, m) ? (pos = m, hi = m - 1) : lo
                    ↪ = m + 1;
                }
                st.back()[2] = pos - 1;
                if (pos <= n) st.push_back({i, pos, n});
                ↪ else if (r < n) st.push_back({i, r + 1, n});
            }
        }
    }
    return dp;
}

/* WHEN YOU NEED WHICH OPTIMISATION - the whole family on one card
dp[i] = min_j (dp[j] + w(j,i)),   w Monge, SELF-REFERENTIAL
↪ -> THIS FILE, O(n log n)
dp[k][i] = min_j (dp[k-1][j] + w(j,i)),   w Monge, LAYERED
↪ -> D&C opt (01), O(n k log n)
dp[i][j] = min_k (dp[i][k] + dp[k+1][j]) + w(i,j),   w Monge AND
    monotone on intervals
↪ -> Knuth (06), O(n^2)
dp[i] = min_j (m_j * x_i + b_j)
↪ -> CHT (02) / Li Chao (04)
"exactly k pieces", optimum convex in k
↪ -> Aliens (07)
min-plus convolution of a CONVEX sequence with anything
↪ -> SMAWK / D&C (11)
the query point moves monotonically and lines are range-updated
↪ -> kinetic segment tree
VERIFY the quadrangle inequality numerically on random
↪ quadruples before trusting any of them:
a DP optimisation applied without its precondition is a silent
↪ wrong answer, never a crash.
COMMON MONGE COSTS: (S_j - S_i)^2; f(S_j - S_i) with f convex
↪ and a[j] >= 0; c * (j - i);
"number of equal pairs inside [i,j]"; a_i * b_j with a
```

↪ non-increasing and b non-decreasing.
 NOT MONGE: anything built from max, and anything with negative
 ↪ a[] where you also need the
 interval-monotonicity half (D&C only needs QI; Knuth needs
 ↪ both). */

8.2 Bitmask DP

8.2.1 SOS_DP

Computes the sum of all subsets for every bitmask in $O(N * 2^N)$

```
vector<long long> SOS_DP(int n_bits, const vector<long long>& a) {
    int max_mask = 1 << n_bits;
    vector<long long> dp = a; // dp[mask] initially holds exactly
    ↪ the value for mask

    for (int i = 0; i < n_bits; ++i) {
        for (int mask = 0; mask < max_mask; ++mask) {
            if (mask & (1 << i)) {
                dp[mask] += dp[mask ^ (1 << i)];
            }
        }
    }
    return dp; // dp[mask] now contains sum of all submasks of mask
}
```

8.2.2 BitmaskDP

BITMASK DP - the $n \leq 20$ toolkit.

```
// MEMORY: vector<vector<ll>> dp(1<<20, vector<ll>>(20)) is ~193 MB
//   ↳ (24 B of vector header per row).
// Flatten to ONE vector<ll> of size (1<<n)*n (168 MB) or
//   ↳ vector<int> (84 MB), or iterate masks in
// increasing popcount and keep only two layers. At n = 20, 2^n *
//   ↳ n^2 = 4e8 - usually not intended.

// TSP: shortest Hamiltonian cycle from 0 back to 0. O(2^n * n^2).
ll tsp(const vector<vector<ll>>& d) {
    int n = d.size(), N = 1 << n;
    vector<vector<ll>> dp(N, vector<ll>(n, llinf));
    dp[1][0] = 0;
    for (int m = 1; m < N; m++)
        for (int u = 0; u < n; u++) if ((m >> u & 1) && dp[m][u]
            ↳ < llinf)
            for (int v = 0; v < n; v++) if (!(m >> v & 1) &&
            ↳ d[u][v] < llinf)
                dp[m | 1 << v][v] = min(dp[m | 1 << v][v],
            ↳ dp[m][u] + d[u][v]);
    ll best = llinf;
    for (int u = 1; u < n; u++) if (dp[N - 1][u] < llinf &&
            ↳ d[u][0] < llinf)
        best = min(best, dp[N - 1][u] + d[u][0]);
    return n == 1 ? 0 : best;
    ↳ +d[u][0] for a Hamiltonian PATH
}

// MINIMUM SET COVER / partition into the fewest valid groups.
//   ↳ O(3^n) via submask enumeration.
// ok[m] = true if the set m is a legal single group.
int minPartition(int n, const vector<char>& ok) {
    int N = 1 << n;
    vector<int> dp(N, inf);
    dp[0] = 0;
    for (int m = 1; m < N; m++) {
        int low = m & -m;
        ↳ lowest element: kills the n! factor
        for (int s = m; s; s = (s - 1) & m) if ((s & low) &&
            ↳ ok[s])
            dp[m] = min(dp[m], dp[m ^ s] + 1);
    }
    return dp[N - 1];
}

// COUNTING the partitions instead: same loop with dp[m] += dp[m ^
//   ↳ s].
// If groups are UNORDERED, fixing the lowest element (above)
//   ↳ already prevents double counting.

// ASSIGNMENT (n tasks to n workers, minimum cost), O(2^n * n).
//   ↳ dp[m] = best cost using the
// first popcount(m) workers on exactly the task set m.
ll assignment(const vector<vector<ll>>& c) {
    int n = c.size(), N = 1 << n;
    vector<ll> dp(N, llinf);
    dp[0] = 0;
    for (int m = 0; m < N - 1; m++) if (dp[m] < llinf) {
        int i = __builtin_popcount(m);
        ↳ worker
        for (int j = 0; j < n; j++) if (!(m >> j & 1))
            dp[m | 1 << j] = min(dp[m | 1 << j], dp[m] + c[i][j]);
    }
    return dp[N - 1];
    ↳ use the Hungarian algorithm
}

// COUNTING HAMILTONIAN PATHS / permutations with adjacency
//   ↳ constraints: same shape as tsp but
// dp[m][u] += dp[m ^ (1<<u)][u] over allowed v.
// SUBSET-SUM STYLE: when the state is "which items used" AND the
//   ↳ order does not matter, prefer
// dp over masks; when only a total matters, use a knapsack - 2^n
//   ↳ is not always necessary.
// MEET IN THE MIDDLE: n <= 40 -> split in half, enumerate 2^20
//   ↳ each side, sort + two pointers.
```

8.3 Knapsack

8.3.1 BoundedKnapsackBitset

BOUNDED KNAPSACK, FEASIBILITY ONLY, with a bitset: $O(n W / 64)$. Binary-group the counts

```
// (1,2,4,...) so "cnt copies of w" becomes O(log cnt) shift-ors.
//   ↳ Use when you only need which sums
// are reachable, not the best value.
const int MAX_W = 100005;
bitset<MAX_W> dp;

void multiset_bitset_dp() {
    // Example: We have 13 items, each with weight 5
    int count = 13;
    int weight = 5;

    vector<int> bundled_weights;

    // Binary grouping logic
    int current_power = 1;
    while (count >= current_power) {
        bundled_weights.push_back(current_power * weight);
        count -= current_power;
        current_power *= 2;
    }
    // Add whatever is left over
    if (count > 0) {
        bundled_weights.push_back(count * weight);
    }

    // Now run the ultra-fast Bitset DP on the bundles
    dp[0] = 1;
    for (int bundle_w : bundled_weights) {
        dp |= (dp << bundle_w);
    }
}
```

8.3.2 KnapsackFamily

KNAPSACK FAMILY. W = capacity. All $O(n^*W)$ unless noted.

```
// 0/1: iterate capacity DOWNWARD so each item is used once.
ll knap01(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    for (size_t i = 0; i < w.size(); i++)
        for (int c = W; c >= w[i]; c--) dp[c] = max(dp[c], dp[c -
            ↳ w[i]] + v[i]);
    return dp[W];
}

// UNBOUNDED: iterate capacity UPWARD.
ll knapInf(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    for (size_t i = 0; i < w.size(); i++)
        for (int c = w[i]; c <= W; c++) dp[c] = max(dp[c], dp[c -
            ↳ w[i]] + v[i]);
    return dp[W];
}

// BOUNDED (item i available cnt[i] times) via BINARY GROUPING:
//   ↳ split cnt into 1,2,4,...,rest
// and run 0/1 on the groups. O(W * sum log cnt).
ll knapBounded(const vector<int>& w, const vector<ll>& v, const
    ↳ vector<int>& cnt, int W) {
    vector<int> W2; vector<ll> V2;
    for (size_t i = 0; i < w.size(); i++) {
        int c = cnt[i];
        for (int k = 1; c > 0; k <= 1) {
            int t = min(k, c); c -= t;
            W2.push_back(w[i] * t), V2.push_back(v[i] * t);
        }
    }
    return knap01(W2, V2, W);
}

// FEASIBILITY ONLY (which sums are reachable) -> bitset,
//   ↳ O(n*W/64). Massively faster.
bitset<100001> reachableSums(const vector<int>& w) {
    bitset<100001> dp;
    dp[0] = 1;
    for (int x : w) dp |= dp << x;
    return dp;
}

// COUNTING ways (mod p): same loops, dp[c] += dp[c - w[i]].
// Order matters: item-outer/capacity-inner counts COMBINATIONS,
// capacity-outer/item-inner counts PERMUTATIONS
//   ↳ (ordered sequences).
```

8.4 Classics

8.4.1 DigitDP

DIGIT DP - count numbers in $[L, R]$ with a digit property. $f(R) - f(L-1)$.

```
// State: (position, tight, started, problem state). 'started'
// handles leading zeros.
// THIS CODE IS BASE 10 (to_string, digit cap 9, memo[20]
// positions). For another base, write the
// digits yourself, change the 9, and size memo by the digit count
// (base 2 needs 64 positions).
// L - 1 IS NOT ALWAYS AVAILABLE: if the bounds are given as huge
// decimal STRINGS you cannot
// subtract one, so carry two tight flags (tightLo, tightHi) and
// count [L, R] in a single pass.
string DIG;
ll memo[20][2][2][200];
bool vis[20][2][2][200];

ll go(int p, int tight, int started, int st) {
    if (p == (int)DIG.size()) return started ? /* ACCEPT? */ (st
    == 0) : 0; // <- edit
    if (vis[p][tight][started][st]) return
    memo[p][tight][started][st];
    vis[p][tight][started][st] = true;
    ll res = 0;
    int hi = tight ? DIG[p] - '0' : 9;
    for (int d = 0; d <= hi; d++) {
        int nStarted = started || d > 0;
        int nSt = nStarted ? (st + d) % 10 : 0;
        // <- edit
        res += go(p + 1, tight && d == hi, nStarted, nSt);
    }
    return memo[p][tight][started][st] = res;
}

bool qualifiesZero = 1; // does the
// number 0 satisfy the predicate?
ll countUpTo(ll x) {
    if (x < 0) return 0;
    DIG = to_string(x);
    memset(vis, 0, sizeof vis);
    return go(0, 1, 0, 0) + qualifiesZero; // set
    // qualifiesZero = 1 only if 0 itself counts
}

ll countRange(ll L, ll R) { return countUpTo(R) - countUpTo(L -
    1); }

// VARIANTS: track (sum of digits), (value mod k), (last digit),
// (a bitmask of used digits),
// (whether we already saw "13"), (count so far) - anything with a
// small state.
// For "k-th number with the property", binary search on countUpTo.
```

8.4.2 LIS

LIS in $O(n \log n)$, with reconstruction. STRICT increasing by default.

```
// Non-decreasing: change lower_bound -> upper_bound.
// Longest DEcreasing: negate the array. Longest non-increasing:
// negate + upper_bound.
vector<int> lis(const vector<ll>& a) {
    int n = a.size();
    vector<ll> tail; // tail[k] =
    // smallest possible tail of an LIS of length k+1
    vector<int> pos(n, -1), idx;
    for (int i = 0; i < n; i++) {
        int k = lower_bound(all(tail), a[i]) - tail.begin();
        if (k == (int)tail.size()) tail.push_back(a[i]),
        idx.push_back(i);
        else tail[k] = a[i], idx[k] = i;
        pos[i] = k;
        prv[i] = k ? idx[k - 1] : -1;
    }
    vector<int> res;
    for (int i = idx.empty() ? -1 : idx.back(); i >= 0; i =
    prv[i]) res.push_back(i);
    reverse(all(res));
    return res; // indices of
    // one LIS; length = tail.size()
}

// SAME SHAPE SOLVES: longest chain of pairs (sort by x, LIS on
// y); maximum non-overlapping
// intervals; "minimum number of non-increasing subsequences
// covering the array" = LIS length
// (Dilworth). For LIS with an extra weight, replace 'tail' with a
// max-BIT over compressed values.
```

8.4.3 IntervalDP

INTERVAL DP - dp over $[l, r]$, processed by increasing length. $O(n^3)$ ($O(n^2)$ with Knuth).

```
// This is the base pattern that Knuth's optimisation speeds up;
// write this first, optimise later.

// 1) SPLIT form: dp[l][r] = min over k of dp[l][k] + dp[k+1][r] +
// cost(l, r)
// matrix chain multiplication, merging stones, optimal BST,
// "cost to combine a range"
template <class Cost> ll intervalSplit(int n, Cost cost) {
    vector<vector<ll>> dp(n, vector<ll>(n, 0));
    for (int len = 2; len <= n; len++)
        for (int l = 0, r = len - 1; r < n; l++, r++) {
            dp[l][r] = llinf;
            for (int k = l; k < r; k++) dp[l][r] = min(dp[l][r],
            dp[l][k] + dp[k + 1][r]);
            dp[l][r] += cost(l, r);
        }
    return dp[0][n - 1];
}

// 2) ENDPOINT form: decide about a[l] or a[r] and shrink.
// Palindromes, "remove from either end".
// Minimum insertions to make s a palindrome:
int minInsertPalindrome(const string& s) {
    int n = s.size();
    vector<vector<int>> dp(n, vector<int>(n, 0));
    for (int len = 2; len <= n; len++)
        for (int l = 0, r = len - 1; r < n; l++, r++)
            dp[l][r] = s[l] == s[r] ? dp[l + 1][r - 1] : 1 +
            min(dp[l + 1][r], dp[l][r - 1]);
    return n ? dp[0][n - 1] : 0;
}

// 3) "BURST BALLOONS" form: choose the LAST element removed in
// the range, so the neighbours are
// the (still intact) boundaries l-1 and r+1.
ll burst(vector<ll> a) {
    int n = a.size();
    a.insert(a.begin(), 1), a.push_back(1);
    vector<vector<ll>> dp(n + 2, vector<ll>(n + 2, 0));
    for (int len = 1; len <= n; len++)
        for (int l = 1, r = len; r <= n; l++, r++)
            dp[l][r] = max(dp[l][r], dp[l][k - 1] + a[l - 1]
            * a[k] * a[r + 1] + dp[k + 1][r]);
    return dp[1][n];
}

// WHEN KNUTH APPLIES (form 1): cost is monotone on intervals and
// satisfies the quadrangle
// inequality; then opt[l][r-1] <= opt[l][r] <= opt[l+1][r] and
// the k-loop becomes amortised  $O(1)$ .
// COUNTING variants (e.g. #ways to parenthesise) replace min with
// + and cost with 1.
```


8.4.4 BrokenProfileDP

BROKEN PROFILE DP - sweep the grid cell by cell carrying a bitmask of the "broken" frontier.

```
// O(n * m * 2^m); always make m the SMALLER dimension.
// mask bit j = "cell (i, j) is already covered by something"
//   placed earlier".

// Count tilings of an n x m grid by 1x2 dominoes.
ll dominoTilings(int n, int m) {
    if (m > n) swap(n, m);
    if ((ll)n * m & 1) return 0;
    int M = 1 << m;
    vector<ll> cur(M, 0), nxt;
    cur[0] = 1;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) {
            nxt.assign(M, 0);
            for (int mask = 0; mask < M; mask++) if (cur[mask]) {
                if (mask >> j & 1) nxt[mask ^ (1 << j)] +=
                    cur[mask]; // already covered
            } else {
                if (i + 1 < n) nxt[mask | (1 << j)] +=
                    cur[mask]; // vertical domino
                if (j + 1 < m && !(mask >> (j + 1) & 1))
                    nxt[mask | (1 << (j + 1))] += cur[mask];
            } // horizontal domino
        }
    cur = nxt;
    return cur[0];
}

// BLOCKED CELLS: keep a grid 'bad[i][j]'; a blocked cell must
//   arrive already "covered"
// (skip it by clearing bit j and forbidding any placement on it).
// OTHER PIECES: add one transition per shape; the mask must
//   record every cell the shape
// occupies in a LATER position of the sweep.
// COUNTING INDEPENDENT SETS / no-two-adjacent colourings: mask =
//   which cells of the previous
// row are used; transition over masks with (a & b) == 0 and no
//   two adjacent bits in b.
// AS WRITTEN this is O(n * 4^m) (nested mask loops), not O(n * m
//   * 2^m): at m = 14 that is 2.7e8
// per row. To get O(n * 2^m), note sum over a with (a & b) == 0
//   of cur[a] = zetaSub(cur)[-b],
// so one subset-zeta pass per row replaces the inner loop (see 05
//   - Combinatorics/07).
ll independentSets(int n, int m) { // #ways
    to pick cells, none orthogonally adjacent
    if (m > n) swap(n, m);
    int M = 1 << m;
    vector<ll> cur(M, 0), nxt;
    for (int b = 0; b < M; b++) if (!(b & (b << 1))) cur[b] = 1;
    for (int i = 1; i < n; i++) {
        nxt.assign(M, 0);
        for (int a = 0; a < M; a++) if (cur[a])
            for (int b = 0; b < M; b++) if (!(b & (b << 1)) &&
                !(a & b)) nxt[b] += cur[a];
        cur = nxt;
    }
    ll r = 0;
    for (ll x : cur) r += x;
    return r;
}

// CONNECTED-COMPONENT DP ("plug DP") is the heavy version: the
//   state stores, for each frontier
// cell, WHICH connected component its plug belongs to (a
//   bracket-like labelling). Use it for
// "count Hamiltonian cycles / single-closed-loop tilings"; it is
//   long - only write it if the
// problem clearly needs connectivity, not just coverage.
```

8.4.5 MaximalRectangle

LARGEST ALL-ZERO (or all-one) RECTANGLE in a binary grid, O(n*m).

```
// Row by row, keep the histogram of consecutive zeros ending at
//   this row, then run the
// largest-rectangle-in-histogram monotonic stack.
ll maximalRectangle(const vector<vector<int>>& g, int target = 0)
{
    int n = g.size(), m = n ? g[0].size() : 0;
    vector<int> h(m, 0);
    ll best = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) h[j] = (g[i][j] == target) ?
            h[j] + 1 : 0;
        vector<int> st; //
        // indices, increasing height
        for (int j = 0; j <= m; j++) {
            int cur = (j == m) ? -1 : h[j];
            while (!st.empty() && h[st.back()] >= cur) {
                int ht = h[st.back()]; st.pop_back();
                int left = st.empty() ? -1 : st.back();
                best = max(best, (ll)ht * (j - left - 1));
            }
            st.push_back(j);
        }
        st.clear();
    }
    return best;
}

// LARGEST SQUARE instead: dp[i][j] = min(dp[i-1][j], dp[i][j-1],
//   dp[i-1][j-1]) + 1 when free.
// COUNTING all all-zero submatrices: for each row, for each
//   column j keep 'run[j]' = the number
// of rectangles ending at (i, j); with a monotonic stack, run[j]
//   = run[prevSmaller] + h[j] *
// (j - prevSmaller), and the total is the sum of run[j] over all
//   cells. O(n*m).
ll countAllZeroSubmatrices(const vector<vector<int>>& g) {
    int n = g.size(), m = n ? g[0].size() : 0;
    vector<ll> h(m, 0), run(m, 0);
    ll total = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) h[j] = g[i][j] == 0 ? h[j] +
            1 : 0;
        vector<int> st;
        for (int j = 0; j < m; j++) {
            while (!st.empty() && h[st.back()] >= h[j])
                st.pop_back();
            int p = st.empty() ? -1 : st.back();
            run[j] = (p < 0 ? 0 : run[p]) + h[j] * (j - p);
            total += run[j];
            st.push_back(j);
        }
    }
    return total;
}
```

8.5 Probability

8.5.1 ExpectedValueDP

EXPECTED VALUE / PROBABILITY DP.

```
// Needs: Mint (06 - Math/01) and solveLinear (06 - Math/04 -
// ↳ Linear Algebra/02 - GaussianElimination.cpp).
// LINEARITY OF EXPECTATION FIRST:  $E[\text{sum of indicators}] = \text{sum of}$ 
// ↳  $P(\text{indicator})$ . Most "expected
// number of  $X$ " problems collapse to summing per-element
// ↳ probabilities with no DP at all.
// Only when states depend on each other CYCLICALLY do you need a
// ↳ linear system.

//  $E[i] = 1 + \sum_j P[i][j] * E[j]$ , with  $E[\text{absorbing}] = 0$ .
// Move the unknowns left:  $E[i] - \sum_j P[i][j] E[j] = 1 \rightarrow (I$ 
// ↳  $- P) E = 1$ .  $O(n^3)$ .
// Needs solveLinear from GaussianElimination.cpp (values are
// ↳ Mint, so probabilities must be
// given as modular fractions  $p * \text{inv}(q)$ ).
vector<Mint> hittingTimes(const vector<vector<Mint>>& P, const
↳ vector<char>& absorbing) {
    int n = P.size();
    vector<vector<Mint>> A(n, vector<Mint>(n, 0));
    vector<Mint> b(n, 0);
    for (int i = 0; i < n; i++) {
        if (absorbing[i]) { A[i][i] = 1, b[i] = 0; continue; }
        A[i][i] = 1;
        for (int j = 0; j < n; j++) A[i][j] -= P[i][j];
        b[i] = 1;
    }
    return solveLinear(A, b);
}

/* ABSORBING MARKOV CHAIN, canonical form  $P = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix}$ :
fundamental matrix  $N = (I - Q)^{-1}$ ; expected steps before
↳ absorption =  $N * 1$ 
absorption probabilities  $B = N * R$  ( $B[i][j] = P(\text{absorbed in}$ 
↳  $j \mid \text{started in } i)$ )

TRIDIAGONAL SYSTEMS (grid random walks that only move
↳ left/right within a row) -  $O(n)$  with
the Thomas algorithm instead of  $O(n^3)$ :
 $a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i$ ,  $a_1 = c_n = 0$ 
forward:  $c'_i = c_i / (b_i - a_i c'_{i-1})$ ,  $d'_i = (d_i -$ 
↳  $a_i d'_{i-1}) / (b_i - a_i c'_{i-1})$ 
back:  $x_n = d'_n$ ,  $x_i = d'_i - c'_i x_{i+1}$ 
Diagonally dominant (always true for hitting times), so it is
↳ stable.

TREES: substitute  $E_v = A_v * E_{\text{parent}} + B_v$ , push A and B up
↳ in one DFS, then resolve at the
root and push back down.  $O(n)$ . (Folklore - no canonical
↳ reference; derive it, do not trust it.)

COMMON TRAPS
- "expected value of a maximum" is NOT the maximum of
↳ expected values; sum over thresholds:
 $E[\max] = \sum_{t \geq 1} P(\max \geq t)$ .
- Probabilities mod  $p$ : store  $p \cdot \text{inv}(q)$ ; never mix doubles and
↳ Mint.
- Infinite processes:  $E$  exists only if absorption has
↳ probability 1; check reachability first.
- Conditional expectation:  $E[X] = \sum_y P(Y=y) * E[X \mid Y=y]$ 
↳ - condition on the FIRST step. */
```

9.1.1 LCA

LCA by Euler tour + sparse table: $O(n \log n)$ build, $O(1)$ query. 0-based, rooted at 'root'.

```
// For k-th ancestor / "jump while predicate" use BinaryLifting
↳ instead (02 - BinaryLifting.cpp).
struct LCA {
    static const int B = 21;
    vector<array<int, 2>> tour;
    vector<array<array<int, 2>, B>> T;
    vector<int> d, in, lg;

    template <class G>
    LCA(int n, const G& adj, int root = 0) : d(n), in(n), lg(2 *
↳ n + 1) {
        tour.reserve(2 * n);
        dfs(root, -1, adj);
        int m = tour.size(); // == 2n - 1
        T.resize(m);
        for (int i = 2; i <= 2 * n; i++) lg[i] = lg[i / 2] + 1;
        for (int i = 0; i < m; i++) T[i][0] = tour[i];
        for (int j = 1; j < B; j++)
            for (int i = 0; i + (1 << j) - 1 < m; i++)
                T[i][j] = min(T[i][j - 1], T[i + (1 << (j -
↳ 1))][j - 1]);
    }
    template <class G>
    void dfs(int u, int p, const G& adj) {
        in[u] = tour.size();
        tour.push_back({d[u], u});
        for (int v : adj[u]) if (v != p) {
            d[v] = d[u] + 1;
            dfs(v, u, adj);
            tour.push_back({d[u], u});
        }
    }
    int get(int u, int v) const {
        int l = min(in[u], in[v]), r = max(in[u], in[v]), j =
↳ lg[r - l + 1];
        return min(T[l][j], T[r - (1 << j) + 1][j])[1];
    }
    int dist(int u, int v) const { return d[u] + d[v] - 2 *
↳ d[get(u, v)]; }
};
```

9 Trees

9.1 Lowest Common Ancestor (LCA)

9.1.2 BinaryLifting

BINARY LIFTING - k-th ancestor, LCA, and "jump while a predicate holds", all $O(\log n)$.

```
// This is NOT only a tree tool: the same table jumps  $2^k$  steps in
// ANY functional graph
// (i -> f(i)), which is how you do "apply the operation 1e18
// times", permutation powers,
// and "min lamps to cover [L,R]" (jump by farthest-reach).
struct Lift {
    int n, B;
    vector<vector<int>>> up; // up[k][v] =
    // 2^k-th ancestor (-1 above the root)
    vector<int> d;

    template <class G>
    Lift(int n, const G& adj, int root = 0) : n(n), B(1), d(n, 0)
    {
        while ((1 << B) < n) B++;
        up.assign(B, vector<int>(n, -1));
        dfs(root, -1, adj);
        for (int k = 1; k < B; k++)
            for (int v = 0; v < n; v++)
                up[k][v] = up[k-1][v] < 0 ? -1 : up[k-1][up[k-1][v]];
    }

    template <class G>
    void dfs(int u, int p, const G& adj) {
        up[0][u] = p;
        for (int v : adj[u]) if (v != p) d[v] = d[u] + 1, dfs(v, u, adj);
    }

    int kth(int v, ll k) const { // k-th ancestor,
    // -1 if it does not exist
        for (int i = 0; i < B && v >= 0; i++) if (k >> i & 1) v =
        up[i][v];
        return k >> B ? -1 : v;
    }

    int lca(int u, int v) const {
        if (d[u] < d[v]) swap(u, v);
        u = kth(u, d[u] - d[v]);
        if (u == v) return u;
        for (int i = B - 1; i >= 0; i--) if (up[i][u] !=
        up[i][v]) u = up[i][u], v = up[i][v];
        return up[0][u];
    }

    int dist(int u, int v) const { return d[u] + d[v] - 2 *
    d[lca(u, v)]; }

    int onPath(int u, int v, int k) const { // k-th vertex on
    // the path u -> v (k = 0 gives u)
        int a = lca(u, v), du = d[u] - d[a];
        return k <= du ? kth(u, k) : kth(v, d[u] + d[v] - 2 *
        d[a] - k);
    }

    // highest ancestor of v still satisfying pred (monotone: true
    // near v, false near the root)
    template <class F>
    int jumpWhile(int v, F pred) const {
        for (int i = B - 1; i >= 0; i--) if (up[i][v] >= 0 &&
        pred(up[i][v])) v = up[i][v];
        return v;
    }
};

// STANDALONE version for a functional graph f (no tree needed):
// up[0][v] = f(v); up[k][v] = up[k-1][up[k-1][v]]; then apply
// k steps by the bits of k.
```

9.1.3 LCAConstantTime

LCA IN $O(1)$ after $O(n \log n)$ preprocessing - Euler tour + sparse table on DEPTH.

```
// Binary lifting (O2) is  $O(\log n)$  per query and needs  $O(n \log n)$ 
// memory too, so reach for this
// when the query count dominates:  $O(1)$  LCA turns "distance
// between every pair in a list" and
// "is u an ancestor of v" from  $n \log n$  into  $n$ . It is also the
// primitive behind  $O(1)$  LCE on a
// suffix tree and behind virtual trees built in bulk.
// The tour has  $2n-1$  entries; the LCA of u and v is the SHALLOWEST
// vertex between their first
// occurrences, and since consecutive tour depths differ by
// exactly +1 a plain sparse table over
// (depth, vertex) pairs is enough - min over an overlapping range
// is idempotent.
struct LCA {
    int n, LOG;
    vector<int> first, depth, tour;
    vector<vector<pair<int, int>>> sp; //
    // (depth, vertex)
    LCA(int n, const vector<vector<int>>& adj, int root = 0)
    : n(n), first(n, -1), depth(n, 0) {
        tour.reserve(2 * n);
        vector<pair<int, int>> st{{root, -1}}; //
        // iterative: no stack overflow
        vector<int> it(n, 0);
        depth[root] = 0;
        while (!st.empty()) {
            auto& [u, p] = st.back();
            if (first[u] < 0) first[u] = tour.size();
            if (it[u] < (int)adj[u].size()) {
                int v = adj[u][it[u]++];
                if (v == p) continue;
                tour.push_back(u), depth[v] = depth[u] + 1;
                st.push_back({v, u});
            } else {
                tour.push_back(u);
                st.pop_back();
            }
        }
        int m = tour.size();
        LOG = 1;
        while ((1 << LOG) <= m) LOG++;
        sp.assign(LOG, vector<pair<int, int>>(m));
        for (int i = 0; i < m; i++) sp[0][i] = {depth[tour[i]],
        // tour[i]};
        for (int k = 1; k < LOG; k++)
            for (int i = 0; i + (1 << k) <= m; i++)
                sp[k][i] = min(sp[k-1][i], sp[k-1][i + (1 <<
        // (k-1)]];
        }
        int lca(int u, int v) const {
            int a = first[u], b = first[v];
            if (a > b) swap(a, b);
            int k = 31 - __builtin_clz(b - a + 1);
            return min(sp[k][a], sp[k][b - (1 << k) + 1]).second;
        }
        int dist(int u, int v) const { return depth[u] + depth[v] - 2
        // * depth[lca(u, v)]; }
        bool isAncestor(int u, int v) const { return lca(u, v) == u; }
    };

    // TRUE  $O(n)$  PREPROCESSING is possible (Farach-Colton-Bender):
    // -> split the +-1 depth array into
    // blocks of size  $\log(n)/2$ , precompute every possible block
    // -> pattern (there are only  $\sqrt{n}$  of
    // them), and put a sparse table over the block minima. It is ~60
    // -> lines and only worth it at
    //  $n \geq 5e5$ , where the  $O(n \log n)$  table's memory is the real
    // -> problem.

    // --- SUBTREE HASHING: are these two subtrees the same shape?  $O(n)$ 
    // ->  $\log n$ . ---
    // Canonical form: sort the children's ids and look the sorted
    // -> vector up in a map. Two subtrees
    // get the same id iff they are isomorphic as ROOTED trees.
    // Shorter than AHU (09 - Isomorphism) and merges naturally into
    // -> other tree DPs; AHU is still the
    // one to use when you need the UNROOTED comparison (root at the
    // -> centres and take the minimum).
    map<vector<int>, int> subtreeId;
    vector<int> subHash;
    int hashSubtree(int u, int p, const vector<vector<int>>& adj) {
        vector<int> ch;
        for (int v : adj[u]) if (v != p) ch.push_back(hashSubtree(v,
        // u, adj));
        sort(all(ch));
        auto it = subtreeId.find(ch);

```

```

    if (it == subtreeId.end()) it = subtreeId.emplace(ch,
    ↪ subtreeId.size()).first;
    return subHash[u] = it->second;
}

// RANDOMISED VARIANT, O(n) and mergeable: h(u) = XOR over
    ↪ children of f(h(child)) with f a fixed
// random 64-bit mixing function (splitmix64). No map, no sorting,
    ↪ and it composes under rerooting
// - which is what you want when you need the hash of every
    ↪ subtree AND of every "rest of tree".

// --- MERGING TWO TREE PATHS. A segment tree node stores the
    ↪ MINIMAL PATH CONTAINING all the
// vertices in its range, as that path's two endpoints ({-1,-1} =
    ↪ empty, {-2,-2} = they do not fit
// on any path). Merging is O(1) given an O(1) LCA: the covering
    ↪ path of two paths, if it exists,
// has its endpoints among the four given ones, so test each of
    ↪ the four candidate pairs (u, v)
// with dist(u,v) == dist(u,w) + dist(w,v) for every endpoint w.
// NOTE THE SEMANTIC: this is the smallest path CONTAINING both,
    ↪ not their union - two disjoint
// pieces of one long path merge into that long path, which is
    ↪ what a covering query wants.
// USES: "is this set of vertices contained in one path", "the
    ↪ smallest path covering a range of
// queries", and segment-tree-over-queries problems on trees.
struct PathNode { int x = -1, y = -1; };
PathNode mergePaths(PathNode a, PathNode b, const LCA& L) {
    if (a.x < 0) return b;
    if (b.x < 0) return a;
    int e[4] = {a.x, a.y, b.x, b.y};
    ↪ ALL SIX pairs, not just four:
    for (int i = 0; i < 4; i++) for (int j = i; j < 4; j++) {
    ↪ // (a.x,b.x) and (a.y,b.y) are
        int u = e[i], v = e[j];
    ↪ // spanning pairs too
        bool ok = true;
        for (int w : e) if (L.dist(u, v) != L.dist(u, w) +
    ↪ L.dist(w, v)) ok = false;
        if (ok) return {u, v};
    }
    return {-2, -2};
    ↪ not a path: mark it invalid
}

```

9.2 DSU on Tree

9.2.1 DSUOnTree

DSU ON TREE (small to large) - answers offline subtree queries in $O(n \log n)$ by keeping the heavy

*// child's data and re-adding only the light subtrees. Each vertex
 ↪ is added $O(\log n)$ times.
 // Use it for "most frequent colour in each subtree" style
 ↪ problems where merging maps would be $O(n^2)$.*

```

class DSUOnTree {
private:
    struct Query {
        int idx, x;
    };

    int n;
    vector<vector<int>> adj;
    vector<int> sz, depth;
    vector<bool> big;
    vector<vector<Query>> queries;
    vector<int> ans;

    // =====
    // PROBLEM SPECIFIC STATE VARIABLES
    // =====
    vector<char> c; // Example: Array of characters per node

    // Example state variables
    // int distinct_count = 0;
    // vector<int> freq;

    void add(int u, int p) {
        // --- ADD LOGIC HERE ---
        // Example: freq[c[u] - 'a']++;

        for (int v : adj[u]) {
            if (v == p || big[v]) continue; // Skip parent and
    ↪ heavy child
            add(v, u);
        }
    }

    void remove(int u, int p) {
        // --- REMOVE LOGIC HERE ---
        // Example: freq[c[u] - 'a']--;

        for (int v : adj[u]) {
            if (v == p || big[v]) continue; // Skip parent and
    ↪ heavy child
            remove(v, u);
        }
    }

    int get_answer(int u, int x = 0) {
        // --- QUERY ANSWER LOGIC HERE ---
        return 0;
    }

    // =====

    void pre(int u, int p = 0) {
        sz[u] = 1;
        depth[u] = depth[p] + 1;
        for (int v : adj[u]) {
            if (v == p) continue;
            pre(v, u);
            sz[u] += sz[v]; // Compute subtree size
        }
    }

    void dfs(int u, int p = 0, bool keep = false) {
        int mx = -1, bigChild = -1;

        // Find the heavy child
        for (int v : adj[u]) {
            if (v == p) continue;
            if (sz[v] > mx) {
                mx = sz[v];
                bigChild = v;
            }
        }

        // Process light children and clear them
        for (int v : adj[u]) {
            if (v == p || v == bigChild) continue;
            dfs(v, u, false);
        }

        // Process heavy child and keep it
        if (bigChild != -1) {

```

```

        dfs(bigChild, u, true);
        big[bigChild] = true;
    }

    // Add current node and its light children
    add(u, p);

    // Answer all queries for the current node
    for (const auto& q : queries[u]) {
        ans[q.idx] = get_answer(u, q.x);
    }

    // Reset the big child flag
    if (bigChild != -1) {
        big[bigChild] = false;
    }

    // Clear state if this node is not meant to be kept
    if (!keep) {
        remove(u, p);
    }
}

public:
    // Initialize with 1-based indexing in mind
    DSUOnTree(int nodes, const vector<char>& chars) : n(nodes),
    ↪ c(chars) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        depth.assign(n + 1, 0);
        big.assign(n + 1, false);
        queries.assign(n + 1, vector<Query>());
        // Initialize problem-specific arrays here (e.g.,
        ↪ freq.assign(26, 0));
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    void add_query(int node, int query_idx, int x = 0) {
        queries[node].push_back({query_idx, x});
    }

    vector<int> solve(int root, int num_queries) {
        ans.assign(num_queries, 0);
        pre(root, 0); // Precalculate sizes and depths
        dfs(root, 0, false); // Run the sack algorithm
        return ans;
    }
};

```

9.3 Heavy Light Decomposition (HLD)

9.3.1 HLD

HEAVY-LIGHT DECOMPOSITION - splits the tree into chains so any root-to-node path crosses $O(\log n)$

↪ of them; combined with a segment tree it gives path

↪ update/query in $O(\log^2 n)$.

↪ Subtrees are contiguous in `tin[]`, so subtree queries are a

↪ single range on the same tree.

```

class HLD {
public:
    vector<int> par, sz, head, tin, tout, who, depth;

    int dfs1(int u, vector<vector<int>> &adj) {
        for (int &v: adj[u]) {
            if (v == par[u]) continue;
            depth[v] = depth[u] + 1;
            par[v] = u;
            sz[u] += dfs1(v, adj);
            if (sz[v] > sz[adj[u][0]] || adj[u][0] == par[u])
        ↪ swap(v, adj[u][0]);
        }
        return sz[u];
    }

    void dfs2(int u, int &timer, const vector<vector<int>> &adj)
    ↪ {
        tin[u] = timer++;
        for (int v: adj[u]) {
            if (v == par[u]) continue;
            head[v] = (timer == tin[u] + 1 ? head[u] : v);
            dfs2(v, timer, adj);
        }
        tout[u] = timer - 1;
    }

    HLD(vector<vector<int>> adj, int r = 0)
        : par(adj.size(), -1), sz(adj.size(), 1),
    ↪ head(adj.size(), r), tin(adj.size()), tout(adj.size()),
        who(adj.size()),
        depth(adj.size()) {
        dfs1(r, adj);
        int x = 0;
        dfs2(r, x, adj);
        for (int i = 0; i < adj.size(); ++i) who[tin[i]] = i;
    }

    vector<pair<int, int>> path(int u, int v) {
        vector<pair<int, int>> res;
        for (; v = par[head[v]]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            if (head[u] != head[v]) {
                res.emplace_back(tin[head[v]], tin[v]);
            } else {
                if (depth[u] > depth[v]) swap(u, v);
                res.emplace_back(tin[u], tin[v]);
                return res;
            }
        }
    }

    pair<int, int> subtree(int u) {
        return {tin[u], tout[u]};
    }

    int dist(int u, int v) {
        return depth[u] + depth[v] - 2 * depth[lca(u, v)];
    }

    int lca(int u, int v) {
        for (; v = par[head[v]]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            if (head[u] == head[v]) {
                if (depth[u] > depth[v]) swap(u, v);
                return u;
            }
        }
    }

    bool isAncestor(int u, int v) {
        return tin[u] <= tin[v] && tout[u] >= tout[v];
    }
};

```

9.4 Centroid Decomposition

9.4.1 Centroid

CENTROID DECOMPOSITION - recursively remove the centroid (the vertex whose largest remaining

```
// component is  $\leq n/2$ ), giving a decomposition tree of depth
//  $\hookrightarrow O(\log n)$ . Every path in the original
// tree passes through the centroid of some level, which turns
//  $\hookrightarrow$  path-counting problems into
// "for each centroid, combine the distance lists of its
//  $\hookrightarrow$  components" in  $O(n \log n)$ .
class CentroidDecomposition {
private:
    int n;
    vector<vector<int>>> adj;
    vector<int> sz, par, nodes;
    vector<bool> vis;

    int dfsSz(int u, int p) {
        sz[u] = 1;
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            sz[u] += dfsSz(v, u);
        }
        return sz[u];
    }

    int getCentroid(int u, int p, int total_nodes) {
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            if (sz[v] * 2 > total_nodes) {
                return getCentroid(v, u, total_nodes);
            }
        }
        return u;
    }

    void collect(int u, int p) {
        nodes.push_back(u);
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            collect(v, u);
        }
    }

    void build(int u, int p) {
        int total_nodes = dfsSz(u, p);
        int centroid = getCentroid(u, p, total_nodes);

        if (p != -1) {
            par[centroid] = p;
        } else {
            par[centroid] = centroid;
        }

        vis[centroid] = true;

        // =====
        // PROBLEM SPECIFIC PROCESSING
        // =====
        // Process paths going through the 'centroid' here
        for (int v : adj[centroid]) {
            if (vis[v]) continue;
            nodes.clear();
            collect(v, centroid);

            // e.g., Update answers using the nodes collected from
            //  $\hookrightarrow$  this subtree branch
        }
        // =====

        // Recursively decompose the remaining components
        for (int v : adj[centroid]) {
            if (vis[v]) continue;
            build(v, centroid);
        }
    }

public:
    // Initialize with 1-based indexing
    CentroidDecomposition(int nodes_count) : n(nodes_count) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        par.assign(n + 1, 0);
        vis.assign(n + 1, false);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    // Start decomposition, usually from node 1
    void decompose(int root = 1) {
        build(root, -1);
    }

    // Returns the parent of a node in the centroid tree
    int get_centroid_parent(int u) const {
        return par[u];
    }
};
```

```
};

// Returns the parent of a node in the centroid tree
int get_centroid_parent(int u) const {
    return par[u];
}

};

// =====
// PROBLEM SPECIFIC PROCESSING
// =====
// Process paths going through the 'centroid' here
for (int v : adj[centroid]) {
    if (vis[v]) continue;
    nodes.clear();
    collect(v, centroid);

    // e.g., Update answers using the nodes collected from
    //  $\hookrightarrow$  this subtree branch
}
// =====

// Recursively decompose the remaining components
for (int v : adj[centroid]) {
    if (vis[v]) continue;
    build(v, centroid);
}

public:
    // Initialize with 1-based indexing
    CentroidDecomposition(int nodes_count) : n(nodes_count) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        par.assign(n + 1, 0);
        vis.assign(n + 1, false);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    // Start decomposition, usually from node 1
    void decompose(int root = 1) {
        build(root, -1);
    }

    // Returns the parent of a node in the centroid tree
    int get_centroid_parent(int u) const {
        return par[u];
    }
};
```

9.5 Tree Diameter

9.5.1 TreeDiameter

Tree Diameter via 2-BFS - Time: $O(V + E)$, Space: $O(V + E)$

// Computes diameter length, endpoints (u, v), and the full
 $\hookrightarrow$ diameter path

```
struct TreeDiameter {
    int n;
    vector<vector<int>>> adj;
    vector<int> dist, par;

    TreeDiameter(int n = 0) : n(n), adj(n + 1), dist(n + 1),
        par(n + 1) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        adj[v].pb(u);
    }

    int bfs(int src) {
        fill(all(dist), -1);
        queue<int> q;
        q.push(src);
        dist[src] = 0;
        par[src] = -1;

        int farthest = src;
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            if (dist[u] > dist[farthest]) farthest = u;

            for (int v : adj[u]) {
                if (dist[v] == -1) {
                    dist[v] = dist[u] + 1;
                    par[v] = u;
                    q.push(v);
                }
            }
        }
        return farthest;
    }

    // Returns {diameter_length, {endpoint_u, endpoint_v}}
    pair<int, pair<int, int>> get_diameter() {
        int u = bfs(1);
        int v = bfs(u);
        return {dist[v], {u, v}};
    }

    // Returns the sequence of vertices on the diameter path from
    //  $\hookrightarrow$  u to v
    vector<int> get_path() {
        auto [len, endpoints] = get_diameter();
        auto [u, v] = endpoints;
        vector<int> path;
        for (int curr = v; curr != -1; curr = par[curr]) {
            path.pb(curr);
        }
        reverse(all(path));
        return path;
    }
};
```

9.6 Euler Tour

9.6.1 EulerTour

Euler Tour Technique (Tree Flattening into 1D Array Intervals) - $O(V + E)$

```
// Subtree of node u corresponds to contiguous range [in_time[u],
  ↳ out_time[u]] in flat_tree
struct EulerTour {
    int n, timer = 0;
    vector<vector<int>> adj;
    vector<int> in_time, out_time, flat_tree;

    EulerTour(int n = 0) : n(n), adj(n + 1), in_time(n + 1),
  ↳ out_time(n + 1) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        adj[v].pb(u);
    }

    void dfs(int u, int p = -1) {
        in_time[u] = ++timer;
        flat_tree.pb(u);
        for (int v : adj[u]) {
            if (v != p) dfs(v, u);
        }
        out_time[u] = timer;
    }

    void build(int root = 1) {
        timer = 0;
        flat_tree.clear();
        dfs(root);
    }

    // Returns [L, R] 1-based index range for subtree queries on
  ↳ node u
    pair<int, int> get_subtree_range(int u) const {
        return {in_time[u], out_time[u]};
    }
};

/* THE EULER-TOUR TOOLBOX - four different tours, each answering a
  ↳ different query family.
CONVENTION HERE: in_time is 1-based (in_time[u] = ++timer) but
  ↳ flat_tree is a 0-based vector,
so flat_tree[in_time[u] - 1] == u. Keep that straight when you
  ↳ index a BIT.

1) SUBTREE = a contiguous range [in[v], out[v]].
   point update, subtree query -> BIT over in[]: add at
  ↳ in[u], query [in[v], out[v]]
   subtree update, point query -> DIFFERENCE BIT: +x at in[v],
  ↳ -x at out[v]+1,
                                     then the value at u is the
  ↳ prefix sum up to in[u]
2) PATH TO THE ROOT, using the SAME difference trick in the other
  ↳ direction:
   subtree update, path-to-root query <-> path update, point
  ↳ query (they are duals).
   PATH UPDATE u..v by +x, point query: +x at in[u], +x at
  ↳ in[v], -x at in[lca],
                                     -x at in[parent(lca)]
   PATH QUERY u..v, point update: query(in[u]) +
  ↳ query(in[v]) - 2*query(in[lca])
                                     (+ the LCA's own value
  ↳ if vertices carry weights)
3) THE 2n TOUR (+1 on entry, -1 on exit) turns ancestor tests and
  ↳ LCA into RMQ:
   u is an ancestor of v iff in[u] <= in[v] && out[v] <=
  ↳ out[u]
   LCA(u,v) = the vertex of minimum DEPTH in the tour range
  ↳ [in[u], in[v]] - and consecutive
   depths differ by exactly 1, which is what the  $O(n)/O(1)$  +-1
  ↳ RMQ exploits.
4) THE EDGE TOUR (each edge appears twice) is what Mo's-on-trees
  ↳ uses: a vertex appearing TWICE
   inside the range cancels, so a path becomes one contiguous
  ↳ range plus the LCA by hand.
   See 02 - Data Structures/06 - Square Root Decomposition/03 -
  ↳ MoOnPaths.cpp.

EDGE WEIGHTS instead of vertex weights: push each edge's weight
  ↳ onto its DEEPER endpoint, then
every path formula above drops the LCA term (the LCA's incoming
  ↳ edge is not on the path). */
```

9.7 Rerooting

9.7.1 Rerooting

REROOTING DP - compute $f(v)$ as root for EVERY v in $O(n)$.

```
// Four hooks: E() identity of merge * self(v) contribution of v
  ↳ itself *
//           merge(a,b) associative+commutative * lift(a,
  ↳ child) push a child's value
//           across one edge into the parent's frame.
// Shipped example: ans[v].first = sum of distances from v to
  ↳ every other node.
struct Reroot {
    typedef pair<ll, ll> T; // {sum of
  ↳ distances, #nodes}
    static T E() { return {0, 0}; }
    static T self(int v) { return {0, 1}; }
    static T merge(T a, T b) { return {a.first + b.first,
  ↳ a.second + b.second}; }
    static T lift(T a, int child) { return {a.first + a.second,
  ↳ a.second}; }

    int n;
    vector<vector<int>> adj;
    vector<T> down, up, ans;
    vector<int> par, ord;

    Reroot(int n) : n(n), adj(n), down(n, E()), up(n, E()),
  ↳ ans(n, E()), par(n, -1) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
  ↳ adj[v].push_back(u); }

    void build(int root = 0) {
        ord.clear(), ord.reserve(n);
        vector<int> st = {root};
        par[root] = -1;
        while (!st.empty()) { // iterative
  ↳ order: no recursion depth limit
            int u = st.back(); st.pop_back(); ord.push_back(u);
            for (int v : adj[u]) if (v != par[u]) par[v] = u,
  ↳ st.push_back(v);
        }
        for (int i = (int)ord.size() - 1; i >= 0; i--) { //
  ↳ pass 1: leaves -> root
            int u = ord[i];
            down[u] = self(u);
            for (int v : adj[u]) if (v != par[u]) down[u] =
  ↳ merge(down[u], lift(down[v], v));
            up[root] = E();
            for (int u : ord) { // pass 2: root
  ↳ -> leaves
                vector<int> ch;
                for (int v : adj[u]) if (v != par[u]) ch.push_back(v);
                int k = ch.size();
                vector<T> pre(k + 1, E()), suf(k + 1, E());
                for (int i = 0; i < k; i++) pre[i + 1] =
  ↳ merge(pre[i], lift(down[ch[i]], ch[i]));
                for (int i = k - 1; i >= 0; i--) suf[i] = merge(suf[i
  ↳ + 1], lift(down[ch[i]], ch[i]));
                ans[u] = merge(down[u], up[u]);
                for (int i = 0; i < k; i++)
                    up[ch[i]] = lift(merge(self(u), merge(up[u],
  ↳ merge(pre[i], suf[i + 1])), u);
            }
        }
};

// The prefix/suffix trick ("aggregate over all children EXCEPT
  ↳ one") is reusable on its own:
// use it whenever a DP needs "everything but this branch" and
  ↳ merge has no inverse.
// OTHER INSTANTIATIONS: max depth from each node (T = ll, lift =
  ↳ a+1, self = 0);
// count of nodes at even distance; product of subtree answers;
  ↳ tree diameter through each node.
```

9.8 Virtual Tree

9.8.1 VirtualTree

VIRTUAL (AUXILIARY) TREE - compress k marked vertices plus their pairwise LCAs into a tree

```
// with at most  $2k-1$  nodes, in  $O(k \log k)$ . Run the real DP on that
// → instead of on all  $n$  nodes.
// THE pattern for "q queries, each with  $k_i$  marked vertices, sum
// → of  $k_i$  bounded".
// Needs tin/tout from a DFS and an LCA (binary lifting or
// → Euler+sparse table).
struct VirtualTree {
    int n;
    vector<int> tin, tout, dep;
    vector<vector<int>> adj, vadj; // vadj =
    // → the compressed tree
    int timer = 0;

    VirtualTree(int n) : n(n), tin(n), tout(n), dep(n, 0),
    // → adj(n), vadj(n) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
    // → adj[v].push_back(u); }
    void dfs(int u, int p) {
        tin[u] = timer++;
        for (int v : adj[u]) if (v != p) dep[v] = dep[u] + 1,
    // → dfs(v, u);
        tout[u] = timer++;
    }
    bool isAnc(int u, int v) const { return tin[u] <= tin[v] &&
    // → tout[v] <= tout[u]; }

    // 'lca' must be a working LCA callable. Returns the roots'
    // → node list; edges land in vadj.
    template <class LCA>
    vector<int> build(vector<int> key, LCA lca) {
        if (key.empty()) return {};
        sort(all(key), [&](int a, int b) { return tin[a] <
    // → tin[b]; });
        key.erase(unique(all(key)), key.end());
        int k = key.size();
        for (int i = 0; i + 1 < k; i++) key.push_back(lca(key[i],
    // → key[i + 1]));
        sort(all(key), [&](int a, int b) { return tin[a] <
    // → tin[b]; });
        key.erase(unique(all(key)), key.end());
        for (int u : key) vadj[u].clear();
        vector<int> st = {key[0]};
        for (size_t i = 1; i < key.size(); i++) {
            int u = key[i];
            while (!isAnc(st.back(), u)) st.pop_back();
            vadj[st.back()].push_back(u); // parent
    // → -> child (directed is enough)
            st.push_back(u);
        }
        return key; // key[0]
    // → is the root of the virtual tree
    }
};
// ALWAYS clear vadj for exactly the nodes you used (done above) -
// → never clear all n per query,
// or the total cost becomes  $O(q*n)$  and defeats the whole point.
// The compressed edge  $u \rightarrow v$  represents the real path between
// → them; its "weight" is usually
//  $dep[v] - dep[u]$ , or a path aggregate you can query in  $O(1)$  with
// → prefix sums / HLD.
```

9.9 Isomorphism

9.9.1 Treelsomorphism

TREE ISOMORPHISM (AHU canonical form). $O(n \log n)$ with a map, $O(n)$ with per-level bucketing.

```
// Gives every rooted subtree a canonical ID: two subtrees are
// → isomorphic iff their IDs match.
// Exact - unlike random subtree hashing, which can collide.
struct TreeIso {
    map<vector<int>, int> id; //
    // → canonical child-multiset -> id
    // canonical id of the subtree rooted at u
    int rootedId(int u, int p, const vector<vector<int>>& adj) {
        vector<int> ch;
        for (int v : adj[u]) if (v != p) ch.push_back(rootedId(v,
    // → u, adj));
        sort(all(ch));
        auto it = id.find(ch);
        if (it != id.end()) return it->second;
        int nid = id.size();
        return id[ch] = nid;
    }

    // CENTERS of a tree: 1 or 2 vertices (Jordan). Root there to
    // → canonicalise an UNROOTED tree.
    vector<int> centers(int n, const vector<vector<int>>& adj) {
        if (n == 1) return {0};
        vector<int> deg(n), leaves;
        for (int i = 0; i < n; i++) if ((deg[i] = adj[i].size())
    // → <= 1) leaves.push_back(i);
        int removed = leaves.size();
        while (removed < n) {
            if (leaves.empty()) break; // guard:
    // → not a tree / disconnected
            vector<int> nxt;
            for (int u : leaves)
                for (int v : adj[u]) if (--deg[v] == 1)
    // → nxt.push_back(v);
            leaves = nxt, removed += leaves.size();
        }
        return leaves; // 1 or 2
    }
    // → vertices, adjacent if 2
    // Canonical id of an UNROOTED tree: min over its centers.
    int unrootedId(int n, const vector<vector<int>>& adj) {
        auto c = centers(n, adj);
        int best = rootedId(c[0], -1, adj);
        if (c.size() == 2) best = min(best, rootedId(c[1], -1,
    // → adj));
        return best;
    }
};
// Two unrooted trees are isomorphic iff unrootedId is equal (use
// → ONE shared TreeIso object so
// the id space is common). Two rooted trees: compare rootedId at
// → their roots.
// USES: "how many distinct tree shapes", dedupe trees, count
// → isomorphism classes, group
// identical subtrees for a DP over shapes.
// JORDAN: a tree has exactly one or two centers; if two, they are
// → adjacent.
```

9.10 Tree DP

9.10.1 TreeDP

TREE DP - the most common tree shape in a contest. Root the tree, compute a value per subtree

```
// bottom-up, and combine children. If the answer must be "for
    ↳ every root", see 07 - Rerooting.

// --- MAXIMUM WEIGHT INDEPENDENT SET / MINIMUM VERTEX COVER on a
    ↳ tree,  $O(n)$ . ---
//  $dp[v][0]$  = best for  $v$ 's subtree with  $v$  NOT taken,  $dp[v][1]$  =
    ↳ with  $v$  taken.
// A tree is bipartite, so  $\alpha = n - \nu$  and (Konig) min vertex
    ↳ cover = maximum matching - but
// this DP is shorter than building a matching, and it handles
    ↳ weights, which Konig does not.
pair<ll, ll> misTree(int v, int p, const vector<vector<int>>&
    ↳ adj, const vector<ll>& w) {
    ll take = w[v], skip = 0;
    for (int u : adj[v]) if (u != p) {
        auto [s, t] = misTree(u, v, adj, w);
        skip += max(s, t), take += s;
    }
    ↳ if  $v$  is taken, no child may be
    return {skip, take};
    ↳ answer = max(skip, take)
}

// MINIMUM VERTEX COVER (unweighted) =  $n - MIS$ . MAXIMUM MATCHING
    ↳ on a tree = the greedy "match
// every leaf to its parent, delete both, repeat" - and that
    ↳ greedy is optimal (exchange argument).

// --- TREE KNAPSACK,  $O(n^2)$  - and the one character that decides
    ↳  $n^2$  vs  $n^3$ . ---
//  $dp[v][j]$  = best value using exactly  $j$  vertices of  $v$ 's subtree.
    ↳ Merge child by child, capping
// BOTH loops by the ALREADY ACCUMULATED size, not by  $n$ . Each
    ↳ unordered pair  $(a, b)$  is then
// charged exactly once, at their LCA, giving  $O(n^2)$  total.
    ↳ Writing ' $u \leq n$ ' instead of
// ' $u \leq acc$ ' is  $O(n^3)$  and looks identical.
vector<ll> knapTree(int v, int p, const vector<vector<int>>& adj,
    ↳ const vector<ll>& w) {
    vector<ll> dp = {0, w[v]};
    ↳  $dp[0]$  = take nothing at all
    for (int u : adj[v]) if (u != p) {
        vector<ll> c = knapTree(u, v, adj, w);
        ↳  $c[0]$  = skip this child entirely
        vector<ll> nd(dp.size() + c.size() - 1, LLONG_MIN);
        nd[0] = 0;
        for (size_t i = 1; i < dp.size(); i++)
            ↳  $i \geq 1$ :  $v$  IS taken, so a child
            for (size_t j = 0; j < c.size(); j++)
                ↳ may attach and stay connected
                if (dp[i] != LLONG_MIN && c[j] != LLONG_MIN) nd[i
                    ↳ + j] = max(nd[i + j], dp[i] + c[j]);
                dp = nd;
    }
    return dp;
    ↳  $dp[j]$  = best CONNECTED set of  $j$ 
}
    ↳ vertices containing  $v$  ( $dp[0] = 0$ )
// Dropping the ' $i \geq 1$ ' restriction gives the UNCONSTRAINED
    ↳ version: best  $j$  vertices anywhere in
// the subtree, no connectivity - that is the
    ↳ group-knapsack-on-a-tree variant.
// WITH A CAP  $k$ : allocate  $\min(\text{size}, k) + 1$  slots instead, giving
    ↳  $O(n * k)$ .

// --- DIAMETER / LONGEST PATH by DP,  $O(n)$ . The version you reuse
    ↳ inside other tree DPs. ---
//  $down1[v]$ ,  $down2[v]$  = the two longest downward paths from  $v$ 
    ↳ through DIFFERENT children.
// The answer is max over  $v$  of  $down1[v] + down2[v]$ ; the two-BFS
    ↳ trick gives the same on an
// unweighted or POSITIVELY weighted tree, but only this DP
    ↳ survives negative weights.

/* MORE TREE-DP SHAPES, all  $O(n)$  unless noted
    COUNT SUBTREES / connected subgraphs containing  $v$ : prod over
    ↳ children of  $(1 + f(\text{child}))$ .
    COUNT MATCHINGS / independent sets: same  $[0/1]$  split, sum
    ↳ instead of max.
    SUM OF DISTANCES from every vertex: rerooting with (subtree
    ↳ size, subtree distance sum).
    K-COLOURINGS with adjacent-different:  $k * (k-1)^{(n-1)}$ . With a
    ↳ colour list per vertex, DP
     $dp[v][c]$  and take the product over children of (sum over  $c'$ 
    ↳  $!= c$  of  $dp[u][c']$ ) - keep the
```

total per child and subtract the forbidden term to stay $O(n * k)$.
 LONGEST PATH WITH AT MOST K EDGES / paths of length exactly L :
 ↳ $dp[v][d]$ merged small-to-large
 over the depth arrays, or long-path decomposition for $O(n)$.
 BINARY LIFTING ON A DP: when the transition is associative you
 ↳ can jump 2^k ancestors at once
 (max edge on a path, "the first ancestor with property P ").
 TREE + BITMASK: $dp[v][mask]$ for $k \leq 20$ marked vertices
 ↳ (Steiner tree on a tree).
 REROOTING is the answer to "compute $f(v)$ for EVERY root" - do
 ↳ not run the DP n times.
 AUXILIARY / VIRTUAL TREE (09 - Trees/08) collapses q marked
 ↳ vertices into an $O(q)$ tree, so a
 per-query tree DP costs $O(q \log q)$ instead of $O(n)$.
 DSU ON TREE (09 - Trees/02) is the answer when the per-subtree
 ↳ state is a MULTISSET you cannot
 merge algebraically (most frequent colour, k -th value), $O(n \log n)$.
 SEGMENT TREE MERGING (02 - DS/16) does the same in $O(n \log n)$
 ↳ while keeping the state
 queryable, at the cost of memory.
 RECURSION DEPTH: a path-shaped tree with $n = 2e5$ overflows the
 ↳ default stack once frames get
 fat. Move big locals out of the recursive function or convert
 ↳ to the iterative order used
 in 07 - Rerooting.cpp. */

9.11 Dynamic Trees

9.11.1 LinkCutTree

LINK-CUT TREE - a FOREST that changes shape, with path queries. $O(\log n)$ amortised per op.

```
// This is dynamic HLD: link, cut, re-root, and aggregate over the
// ↪ path u..v, all online.
// If the tree is STATIC use HLD (09 - Trees/03) - it is faster
// ↪ and far shorter. Reach for an LCT
// only when edges are added/removed, or when you need makeRoot.
// HOW IT WORKS: each vertex has at most one "preferred" child;
// ↪ preferred paths are stored as
// splay trees keyed by DEPTH; a non-preferred edge is a
// ↪ path-parent pointer the splay does not
// know about. access(v) makes root..v one preferred path, and the
// ↪ same heavy/light potential that
// bounds HLD bounds the number of preferred-child changes, hence
// ↪ the  $O(\log n)$  amortised.
struct LCT {
    struct Node { int ch[2] = {0, 0}, p = 0; ll val = 0, sum = 0;
        ↪ bool rev = 0; };
    vector<Node> t;
    LCT(int n) : t(n + 1) {}
    ↪ 1-indexed; node 0 is the null
    bool isRoot(int x) { return t[t[x].p].ch[0] != x &&
        ↪ t[t[x].p].ch[1] != x; }
    void pull(int x) { t[x].sum = t[t[x].ch[0]].sum ^ t[x].val ^
        ↪ t[t[x].ch[1]].sum; }
    void flip(int x) { if (x) swap(t[x].ch[0], t[x].ch[1]),
        ↪ t[x].rev ^= 1; }
    void push(int x) { if (t[x].rev) flip(t[x].ch[0]),
        ↪ flip(t[x].ch[1]), t[x].rev = 0; }
    void rot(int x) {
        int p = t[x].p, g = t[p].p, d = t[p].ch[1] == x;
        if (!isRoot(p)) t[g].ch[t[g].ch[1] == p] = x;
        t[x].p = g;
        t[p].ch[d] = t[x].ch[!d], t[t[x].ch[!d]].p = p;
        t[x].ch[!d] = p, t[p].p = x;
        pull(p), pull(x);
    }
    void splay(int x) {
        vector<int> st{x};
        for (int y = x; !isRoot(y); y = t[y].p)
        ↪ st.push_back(t[y].p);
        for (int i = st.size() - 1; i >= 0; i--) push(st[i]);
        while (!isRoot(x)) {
            int p = t[x].p, g = t[p].p;
            if (!isRoot(p)) rot((t[g].ch[1] == p) == (t[p].ch[1]
        ↪ == x) ? p : x);
            rot(x);
        }
        int access(int x) {
        ↪ root..x becomes one splay tree
            int last = 0;
            for (int y = x; y; y = t[y].p) splay(y), t[y].ch[1] =
        ↪ last, pull(y), last = y;
            splay(x);
            return last;
        ↪ = the LCA on the previous access
        }
        void makeRoot(int x) { access(x), flip(x), push(x); }
        int findRoot(int x) { access(x); while (t[x].ch[0]) push(x),
        ↪ x = t[x].ch[0]; splay(x); return x; }
        bool connected(int x, int y) { return findRoot(x) ==
        ↪ findRoot(y); }
        void link(int x, int y) { makeRoot(x); if (findRoot(y) != x)
        ↪ t[x].p = y; }
        void cut(int x, int y) {
        ↪ no-op if the edge is absent
            makeRoot(x);
            if (findRoot(y) == x && t[y].p == x && !t[y].ch[0])
        ↪ t[y].p = t[x].ch[1] = 0, pull(x);
        }
        ll query(int x, int y) { makeRoot(x), access(y); return
        ↪ t[y].sum; } // path aggregate
        void update(int x, ll v) { access(x), t[x].val = v, pull(x); }
        int lca(int r, int x, int y) { makeRoot(r), access(x); return
        ↪ access(y); }
};
// THE AGGREGATE is XOR here purely because it is self-inverse and
// ↪ needs no reversal handling.
// For SUM/MIN/MAX just change pull(); for a NON-COMMUTATIVE
// ↪ aggregate (matrix product, max
// prefix sum) you must store BOTH directions in the node and swap
// ↪ them inside flip().
// EDGE WEIGHTS: give each edge its own vertex (n + edgeId) with
// ↪ the weight, and link both ends
// to it. Vertex values then stay 0 and every path query still
```

```
↪ works.
// SUBTREE aggregates need "virtual subtree" bookkeeping: keep a
// ↪ second sum over the non-preferred
// children, updated in access() when you swap t[y].ch[1]. That
// ↪ doubles the code - if the tree is
// static, an Euler tour (09 - Trees/06) is  $O(\log n)$  and ten lines.
/* WHAT LCT UNLOCKS
    ONLINE DYNAMIC CONNECTIVITY ON A FOREST: connected() in  $O(\log
    ↪ n)$ . For general graphs (cycles)
    you need the offline segment-tree-on-time trick (02 - Data
    ↪ Structures/14) or Holm-de
    Lichtenberg-Thorup at  $O(\log^2 n)$ .
    DYNAMIC MST: to add edge (u,v,w), if u and v are connected find
    ↪ the MAXIMUM edge on the path;
    if it exceeds w, cut it and link the new edge.  $O(\log n)$  per
    ↪ update.
    MAXIMUM FLOW (Dinic with link-cut) improves the blocking-flow
    ↪ step to  $O(VE \log V)$  - almost
    never needed at contest sizes.
    OFFLINE "PAINT THE ROOT PATH": access() itself is the operation
    ↪ "recolour root..v with a new
    colour"; the number of colour changes is  $O(n \log n)$  total, so
    ↪ if each change costs a segment
    tree update you get  $O(n \log^2 n)$ . This is the standard trick
    ↪ for "count distinct substrings
    over all substrings of s" and for tree-path-colouring
    ↪ problems. */
```

10 Game Theory

10.1 GrundyAndNim

SPRAGUE-GRUNDY. Every FINITE, ACYCLIC (progressively bounded) IMPARTIAL game under normal

```

/* SPRAGUE-GRUNDY. Every FINITE, ACYCLIC (progressively bounded)
  ↳ IMPARTIAL game under normal
  play (last to move wins) is equivalent to a nim pile of size
  ↳ Grundy(position).
  ACYCLIC IS A HYPOTHESIS, NOT A DETAIL: on a graph with cycles
  ↳ the mex recursion is not
  well-founded and Grundy() below returns silently wrong values -
  ↳ use retrograde BFS instead. Grundy(p) = mex{ Grundy(q) : q
  ↳ reachable from p }.
  A position is LOSING for the player to move iff Grundy = 0.
  SUM of independent games: Grundy = XOR of the parts.
  ↳
  */
int mex(const vector<int>& v) {
    vector<bool> seen(v.size() + 1, false);
    for (int x : v) if (x >= 0 && x <= (int)v.size()) seen[x] =
    ↳ true;
    int m = 0;
    while (m < (int)seen.size() && seen[m]) m++;
    return m;
}

// Grundy over a DAG of at most n positions, memoised.
vector<int> g_;
int Grundy(int u, const vector<vector<int>>& nxt) {
    if (g_[u] != -1) return g_[u];
    g_[u] = 0;
    ↳ (DAG => no real cycles)
    vector<int> s;
    for (int v : nxt[u]) s.push_back(Grundy(v, nxt));
    return g_[u] = mex(s);
}

/* NIM AND FRIENDS
  NIM (take any number from one pile). Losing iff XOR of pile
  ↳ sizes == 0.
  Winning move: find a pile with  $(x \oplus X) < x$  where  $X$  is the
  ↳ total xor; reduce it to  $x \oplus X$ .
  MISERE NIM (last to move LOSES). If every pile is size 1: first
  ↳ player wins iff #piles is EVEN.
  Otherwise identical to normal nim (win iff xor != 0).
  NIM_k / MOORE'S NIM (may take from up to k piles). Losing iff,
  ↳ for every bit, the number of
  piles with that bit set is divisible by (k+1).
  STAIRCASE NIM. INDEX 0 IS THE GROUND - coins there are DEAD and
  ↳ have no move.
  Losing iff XOR of the counts at ODD indices == 0. (If
  ↳ instead a coin on index 0 may still
  slide OFF the board, the ground is a virtual index -1 and you
  ↳ must XOR the EVEN indices.
  Getting this backwards is wrong on 96 of the 256
  ↳ four-position states.)
  Recognise it as "move something toward a sink, and moving it
  ↳ to an even slot is reversible".
  SUBTRACTION GAME, allowed set S: Grundy is eventually PERIODIC.
  ↳ Compute a prefix, find the
  period, extrapolate - the standard route for n up to 1e9.
  S = {1..k}: Grundy(n) = n mod (k+1).
  WYTHOFF (two piles, take from one or equally from both). Losing
  ↳ positions are
  (floor(k*phi), floor(k*phi^2)), phi = (1+sqrt(5))/2, k >= 0.
  FIBONACCI NIM (first move any amount < pile; then at most 2x
  ↳ the previous move).
  First player loses iff the pile size is a Fibonacci number.
  TURNING / COIN GAMES: Grundy of a set of coins = XOR of the
  ↳ Grundy of each single coin.
  GREEN HACKENBUSH (all edges neutral): Grundy of a path of
  ↳ length n is n; FUSION principle -
  a cycle of odd length fuses to a single edge, an even cycle
  ↳ to nothing; sum branches by XOR.
  COLON PRINCIPLE (trees): Grundy of a node = XOR over children
  ↳ of (Grundy(child) + 1).
  */

// RETROGRADE ANALYSIS - when the game graph HAS CYCLES (draws
  ↳ possible) or the game is partisan,
// Grundy does not apply. BFS backwards from terminal positions
  ↳ counting out-degrees.
// win[u] = 1 first player to move wins, 0 loses, -1 draw.
vector<int> retrograde(int n, const vector<vector<int>>& radj,
  ↳ vector<int> deg, const vector<int>& terminallose) {
    vector<int> res(n, -1);
    queue<int> q;
    for (int u : terminallose) res[u] = 0, q.push(u);
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int u : radj[v]) {
            if (res[u] != -1) continue;

```

```

            if (res[v] == 0) res[u] = 1, q.push(u); // can
            ↳ move to a losing position
            else if (--deg[u] == 0) res[u] = 0, q.push(u); //
            ↳ all moves lead to wins
        }
    }
    return res;
    ↳ remaining -1 are draws
}

```

10.2 GameAlgorithms

Concrete game solvers. All impartial games under NORMAL play (last move wins) unless stated.

```
// SUBTRACTION GAME with allowed move set S: Grundy is eventually
// ↪ PERIODIC. Compute a prefix,
// ↪ detect the period, extrapolate to n up to 1e18. This is the
// ↪ standard route for huge n.
struct Periodic {
    vector<int> g;
    int start = -1, per = -1;
    Periodic(const vector<int>& S, int prefix = 2000) {
        g.assign(prefix, 0);
        for (int i = 1; i < prefix; i++) {
            vector<bool> seen(S.size() + 2, false);
            for (int s : S) if (s <= i && g[i - s] <
                ↪ (int)seen.size()) seen[g[i - s]] = true;
            while (g[i] < (int)seen.size() && seen[g[i]]) g[i]++;
            detect();
        }
        void detect() { //
            ↪ smallest (start, per) that fits
            int n = g.size();
            for (int p = 1; p <= n / 3; p++) {
                bool ok = true;
                for (int i = n / 3; i + p < n; i++) if (g[i] != g[i +
                ↪ p]) { ok = false; break; }
                if (!ok) continue;
                int st = 0; // the
                ↪ LAST break, not the first match:
                for (int i = n - p - 1; i >= 0; i--) //
                ↪ periodicity must hold from start ON
                    if (g[i] != g[i + p]) { st = i + 1; break; }
                if (n - p - st < 2 * p + 20) continue; //
                ↪ demand real evidence before believing
                start = st, per = p;
                return;
            }
            int at(11 n) const {
                if (n < (11)g.size()) return g[n];
                assert(per > 0); // no
                ↪ period found: do NOT extrapolate
                return g[start + (n - start) % per];
            }
        };
        // S = {1..k} => Grundy(n) = n mod (k+1) (no need to compute
        ↪ anything)

        // WYTHOFF'S GAME (two piles; take any amount from one pile, or
        ↪ the SAME amount from both).
        // Losing (P-)positions: (floor(k*phi), floor(k*phi^2)) for k >=
        ↪ 0, phi = (1+sqrt(5))/2.
        // Since phi^2 - phi = 1, the two coordinates differ by EXACTLY k,
        ↪ so k = b - a.
        // Tested exactly, no floating point: a == floor(k*phi) <=> d
        ↪ <= k*sqrt(5) < d+2, d = 2a-k.
        bool wythoffLosing(11 a, 11 b) {
            if (a > b) swap(a, b);
            11 k = b - a, d = 2 * a - k;
            return d >= 0 && (__int128)d * d <= (__int128)5 * k * k
                && (__int128)5 * k * k < (__int128)(d + 2) * (d
                ↪ + 2);
        }
        // P-positions: (0,0) (1,2) (3,5) (4,7) (6,10) (8,13) (9,15)
        ↪ (11,18) (12,20) (14,23) ...

        // MOORE'S NIM_k: you may take from up to k piles in one move.
        // Losing iff, for EVERY bit, the number of piles having that bit
        ↪ set is divisible by (k+1).
        bool mooreLosing(const vector<11>& piles, int k) {
            for (int b = 0; b < 63; b++) {
                int c = 0;
                for (11 x : piles) c += (x >> b) & 1;
                if (c % (k + 1)) return false;
            }
            return true;
        }

        // STAIRCASE NIM: coins on positions 0..n-1, a move slides some
        ↪ coins from i to i-1.
        // INDEX 0 IS THE GROUND: coins there are DEAD and cannot move.
        ↪ Then only ODD indices matter
        // (a move from an even index is reversible, so it is worth
        ↪ nothing).
        // If your problem instead lets a coin at index 0 slide OFF the
```

```
    ↪ board, the ground is a virtual
    // index -1 and you must XOR the EVEN indices. Brute force over 4
    ↪ positions: the wrong convention
    // is wrong on 96 of 256 states, e.g. (0,0,0,1).
    bool staircaseLosing(const vector<11>& cnt) {
        11 x = 0;
        for (size_t i = 1; i < cnt.size(); i += 2) x ^= cnt[i];
        return x == 0;
    }

    // GREEN HACKENBUSH (all edges neutral), Grundy of a rooted tree =
    ↪ COLON PRINCIPLE:
    // g(v) = XOR over children c of (g(c) + 1)
    int hackenbushTree(int u, int p, const vector<vector<int>>& adj) {
        int r = 0;
        for (int v : adj[u]) if (v != p) r ^= hackenbushTree(v, u,
        ↪ adj) + 1;
        return r;
    }

    // FUSION PRINCIPLE for graphs: all vertices of a cycle can be
    ↪ fused into one; an ODD cycle
    // leaves a single loop (= one edge, Grundy 1), an EVEN cycle
    ↪ leaves nothing (Grundy 0).

    // COIN TURNING GAMES: a position is a set of "heads". The Grundy
    ↪ value of the whole position is
    // the XOR of the Grundy values of each single head considered
    ↪ alone (Mock Turtles, Ruler, ...).

    // MISERE NIM (last to move LOSES): if EVERY pile is 1, the first
    ↪ player wins iff #piles is even;
    // otherwise the normal-play rule (xor != 0 => win) applies
    ↪ unchanged.
```


10.3 MinimaxAndMatching

Games that Sprague-Grundy does NOT cover: partisan games, scored games, draws.

```
// Needs: Kuhn (03 - Graph Theory/07 - Matching/02 -
    ↳ KuhnAndKonig.cpp) for the matching part.

// MINIMAX with ALPHA-BETA pruning. Returns the value in PLAYER
    ↳ 0's units (a fixed sign), not
// "the value for the player to move" - with maxing = false the
    ↳ result is still player-0 scored.
// 'moves(state)' lists successors; 'eval(state)' scores
    ↳ terminal/leaf states from player 0's view.
template <class S, class Moves, class Eval>
ll alphabeta(const S& s, int depth, ll alpha, ll beta, bool
    ↳ maxing, Moves moves, Eval eval) {
    auto nxt = moves(s);
    if (!depth || nxt.empty()) return eval(s);
    if (maxing) {
        ll best = LLONG_MIN;
        for (auto& t : nxt) {
            best = max(best, alphabeta(t, depth - 1, alpha, beta,
    ↳ false, moves, eval));
        alpha = max(alpha, best);
        if (beta <= alpha) break; // beta
    ↳ cutoff
        }
        return best;
    }
    ll best = LLONG_MAX;
    for (auto& t : nxt) {
        best = min(best, alphabeta(t, depth - 1, alpha, beta,
    ↳ true, moves, eval));
        beta = min(beta, best);
        if (beta <= alpha) break; //
    ↳ alpha cutoff
        }
        return best;
    }
// Order moves best-first: pruning is only good if you try strong
    ↳ moves early.
// Memoise on (state) when states repeat; use iterative deepening
    ↳ when depth is unbounded.

// MATCHING GAME ON A GRAPH: a token sits on vertex s; players
    ↳ alternately move it along an
// edge to an unvisited vertex; a player who cannot move loses.
// THEOREM: the FIRST player wins iff s is in EVERY maximum
    ↳ matching, i.e. iff removing s
// decreases the size of the maximum matching.
// For bipartite graphs use Kuhn (below). General graphs need
    ↳ Blossom, which this notebook does
// NOT carry - the helper below silently assumes the right side is
    ↳ numbered leftSize..n-1.
// Verified on 538 (graph, start) pairs over random graphs with n
    ↳ <= 7: 0 mismatches.
bool matchingGameFirstWins(int n, vector<vector<int>>& adj, int
    ↳ s, int leftSize) {
    auto solve = [&](int skip) {
        Kuhn K(leftSize, n - leftSize);
        for (int u = 0; u < leftSize; u++) if (u != skip)
            for (int v : adj[u]) if (v != skip) K.add_edge(u, v -
    ↳ leftSize);
        return K.maxMatching();
    };
    return solve(-1) > solve(s);
}
// GAMES WITH SCORES (not just win/lose): plain minimax DP over
    ↳ states,
// dp[state] = best over moves of (gain + (-dp[next])) for a
    ↳ zero-sum symmetric game,
// or keep two arrays (dp0, dp1) when the players' objectives
    ↳ differ.
// GAMES WITH DRAWS / CYCLES: retrograde BFS (see 01 -
    ↳ GrundyAndNim.cpp), never Grundy.
```

10.4 NamedGames

NAMED GAMES: the values you cannot re-derive during a contest =====

```
/* ===== NAMED GAMES: the values you cannot re-derive during a
    ↳ contest =====
Two different families live here: TAKE-AND-BREAK (octal) games,
    ↳ and COIN-TURNING games.
They have different theory - octal games are the splitting rule
    ↳ below, coin-turning games are
XOR of single-head values and, in 2-D, nim MULTIPLICATION. Do not
    ↳ mix them up.
Every value on this card was brute-force verified.
All are impartial, NORMAL play. Grundy of a sum = XOR of parts.

SPLITTING RULE (the general take-and-break move):
    g(v) = mex{ g(a1) XOR g(a2) XOR ... : v -> (a1, a2, ...) }

LASKER'S NIM (take any amount from a pile, OR split a pile of
    ↳ size >= 2 into two non-empty)
    g(4k+1) = 4k+1    g(4k+2) = 4k+2    g(4k+3) = 4k+4
    ↳ g(4k+4) = 4k+3

KAYLES (remove 1 or 2 adjacent pins, possibly splitting the row)
    Periodic with period 12 from n = 71 onward (Winning Ways says
    ↳ 72; 71 is tight).
    block for n >= 71: 7 4 1 2 8 1 4 7 2 1 8 2
    14 exceptions (n, G):
    ↳ (0,0) (3,3) (6,3) (9,4) (11,6) (15,7) (18,3) (21,4) (22,6) (28,5) (34,6)
        (39,3) (57,4) (70,6) last
    ↳ exception n = 70

DAWSON'S CHESS (octal game .137)
    Period 34 from n = 52 onward.
    block for n >= 52: 3 3 0 1 1 3 0 2 1 1 0 4 5 3 7 4 8 1 1 2 0
    ↳ 3 1 1 0 3 3 2 2 4 4 5 5 9
    7 exceptions (n, G):
    ↳ (0,0) (14,0) (16,2) (17,2) (31,2) (34,0) (51,2) last exception
    ↳ n = 51
DAWSON'S KAYLES (.07) period 34 from n = 53, same block, 8
    ↳ exceptions, last at n = 52.

GRUNDY'S GAME (split a pile into two UNEQUAL non-empty piles)
    Periodicity is an OPEN PROBLEM - more than 2*10^10 values
    ↳ computed and no period found.
    Do NOT assume a period here; compute the prefix you need.

GUY-SMITH PERIODICITY THEOREM (how to PROVE a period, not just
    ↳ guess it)
    For an octal game whose largest non-zero code digit has index
    ↳ k: if
        G(n + p) = G(n) for all n with n0 <= n < 2*n0 + p + k
    then the period holds for ALL n >= n0. So checking up to 2*n0 +
    ↳ p + k is a proof.

TURNING GAMES (a position is a set of heads; the whole
    ↳ position's Grundy value is the XOR of
the values of each single head considered alone)
    Twins (turn exactly 2 coins), 0-indexed: g(x) = x
    ↳ [plain nim]
    Mock Turtles (turn 1, 2 or 3 coins), 0-indexed: g(x) = the x-th
    ↳ ODDIOUS number
        (odious = odd popcount): 1, 2, 4, 7, 8, 11, 13, 14,
    ↳ 16, ...
        g(x) = 2x if 2x is odious, else 2x + 1
    Ruler (turn any number of CONSECUTIVE coins), 1-indexed:
        g(x) = largest power of two dividing x = x & (-x)
    ↳ -> 1,2,1,4,1,2,1,8,...
    TRAP: Ruler is 1-INDEXED while Twins and Mock Turtles are
    ↳ 0-INDEXED. Mixing them up gives a
    silently wrong answer.

NIM WITH ONE PASS (Guy's model: the pass may be used once, at any
    ↳ NON-terminal position)
    1 heap: G(0) = 0, G(n) = n+1 for n odd, G(n) = n-1 for n
    ↳ even >= 2.
    2 heaps: P-positions are exactly (0,0) and (a, a+1) with a ODD.
    3+ heaps: OPEN. No pattern; 34 P-positions with a<=b<c<=13 and
    ↳ neither xor==0 nor xor==1
    comes close. Never guess.
    If the pass is ALSO legal at a terminal position (a weaker
    ↳ rule), the game collapses:
        G = (XOR of piles) XOR 1, so P-positions are exactly XOR == 1.
=====
    ↳ */
bool odious(ll x) { return __builtin_parityll(x); }
ll mockTurtles(ll x) { return odious(2 * x) ? 2 * x : 2 * x + 1;
    ↳ } // 0-indexed
ll ruler(ll x) { return x & -x; }
```



```

    ↪ // 1-indexed
11 lasker(11 n) { // g(0) = 0,
    ↪ NOT -1
    11 r = n % 4;
    if (r == 0) return n ? n - 1 : 0;
    return r == 3 ? n + 1 : n;
}
// Grundy of a sum of turning-game heads = XOR of the single-head
↪ values.
11 turningGameValue(const vector<11>& heads, 11 (*g)(11)) {
    11 r = 0;
    for (11 h : heads) r ^= g(h);
    return r;
}

```

10.5 NimbersAndSums

NIMBERS, OCTAL GAMES, AND THE SUM VARIANTS. Everything on this page was brute-force verified.

```

// --- NIM PRODUCT.  $[0, 2^{(2^k)}]$  is a FIELD under (XOR, nimMul).
↪ ---
// Definition:  $a (*) b = \max\{a'(*)b \wedge a'(*)b' : a' < a, b' < b\}$ .
↪ Computed with the two identities: for a Fermat 2-power  $F = 2^{(2^k)}$ ,
//  $x (*) F = x * F$  for  $x < F$ , and  $F (*) F = 3F/2$ .
// USES: Turning Corners (turn the 4 corners of a rectangle) has
↪  $g(x,y) = x (*) y$ , and the TARTAN
// THEOREM says the 2-D product of two coin-turning games has
↪  $g(x,y) = g1(x) (*) g2(y)$ . Also
// CF 1310F, which asks you to solve  $a^*(*)x = b$  in the number
↪ field of size  $2^{64}$ .
map<pair<u64, u64>, u64> nmMemo;
u64 nimMul(u64 a, u64 b) {
    if (a < b) swap(a, b);
    if (b == 0) return 0;
    if (b == 1) return a;
    auto it = nmMemo.find({a, b});
    if (it != nmMemo.end()) return it->second;
    int L = 0;
    while (L < 6 && (a >> (1u << L))) L++;
    u64 F = 1ULL << (1u << (L - 1)); //
    ↪ largest Fermat 2-power <= a
    u64 ah = a / F, al = a % F, bh = b / F, bl = b % F;
    u64 A = nimMul(ah, bh), B = nimMul(ah, bl), C = nimMul(al,
    ↪ bh), D = nimMul(al, bl);
    return nmMemo[{a, b}] = ((A ^ B ^ C) * F) ^ D ^ nimMul(A, F /
    ↪ 2);
}

// --- OCTAL (TAKE-AND-BREAK) GAME GRUNDY GENERATOR. ---
// The code is ".d1d2d3...": digit d_k governs "remove exactly k
↪ tokens from one heap".
// bit 1 (value 1): you may take the WHOLE heap (i.e.
↪ leave zero heaps)
// bit 2 (value 2): you may leave ONE non-empty heap
// bit 4 (value 4): you may SPLIT the remainder into TWO
↪ non-empty heaps
// So .333... = Nim, .77 = Kayles, .07 = Dawson's Kayles, .137 =
↪ Dawson's chess.
// Being able to read a statement into a code string turns "invent
↪ a Grundy recurrence" into a
// function call.  $O(N^2 * \text{digits})$  because of the split term;  $O(N * \text{digits})$  without it.
vector<int> octal(const string& code, int N) { //
    ↪ code = the digits AFTER the dot
    vector<int> d;
    for (char c : code) d.push_back(isdigit(c) ? c - '0' : c -
    ↪ 'a' + 10);
    vector<int> g(N + 1, 0);
    for (int n = 1; n <= N; n++) {
        set<int> s;
        for (int i = 0; i < (int)d.size(); i++) {
            int k = i + 1;
            if (k > n) break;
            int rem = n - k;
            if ((d[i] & 1) && rem == 0) s.insert(0);
            if ((d[i] & 2) && rem >= 1) s.insert(g[rem]);
            if (d[i] & 4) for (int a = 1; a < rem; a++)
            ↪ s.insert(g[a] ^ g[rem - a]);
        }
        int m = 0;
        while (s.count(m)) m++;
        g[n] = m;
    }
    return g;
}

// GUY-SMITH PERIODICITY THEOREM turns a guess into a PROOF: with
↪ k = the index of the last
// non-zero code digit, if  $G(n + p) = G(n)$  for all  $n0 \leq n < 2n0$ 
↪ + p + k, the period holds for
// ALL  $n \geq n0$ . See 02 - GameAlgorithms.cpp for the (fixed) period
↪ detector.

// --- EUCLID'S GAME. (a, b); subtract a positive multiple of the
↪ smaller from the larger. ---
// FIRST PLAYER WINS iff  $a == b$  OR  $\max/\min > \phi$ . The usual
↪ phrasing " $\geq \phi$ " is WRONG on the
// whole diagonal: (a, a) is always an immediate win.
bool euclidWin(11 a, 11 b) {
    if (a < b) swap(a, b);
    if (b == 0) return false;

```

```

    if (a == b || a >= 2 * b) return true;
    return !euclidWin(b, a - b);
}

// --- ANTI-SG / the SJ theorem: MISERE play on a SUM of impartial
//    games. ---
// PRECONDITION, and it is not optional: in every component, SG(u)
//    == 0 must imply u is TERMINAL.
// (Without it this rule is wrong on ~1.3% of random impartial
//    games; with it, exact.)
// Misere nim satisfies it, which is why the classic misere-nim
//    rule looks so clean.
bool antiSGFirstWins(const vector<int>& g) {
    int x = 0, mx = 0;
    for (int v : g) x ^= v, mx = max(mx, v);
    return (x && mx > 1) || (!x && mx <= 1);
}

// --- EVERY-SG: you must move in EVERY unfinished component. Not
//    a function of the XOR at all. ---
// step(v) = 0                                if v is terminal
// step(v) = 1 + max{ step(u) : SG(u) == 0 }    if SG(v) > 0
//    (drag it out - you will win it)
// step(v) = 1 + min{ step(u) }                if SG(v) == 0
//    (end it fast - you will lose it)
// First player wins iff max over components of step(root) is ODD.
// The mirrored max/min assignment is WRONG; this orientation is
//    the verified one.
void everySGStep(const vector<vector<int>>& succ, const
    vector<int>& sg, vector<int>& step) {
    int n = succ.size();
    step.assign(n, 0);
    for (int u = 0; u < n; u++) {
        // u in REVERSE topological order
        if (succ[u].empty()) { step[u] = 0; continue; }
        if (sg[u] > 0) {
            int b = 0;
            for (int v : succ[u]) if (!sg[v]) b = max(b, step[v]
                + 1);
            step[u] = b;
        } else {
            int b = INT_MAX;
            for (int v : succ[u]) b = min(b, step[v] + 1);
            step[u] = b;
        }
    }
}

/* ===== VERIFIED CLOSED FORMS AND FACTS =====
COIN-TURNING FAMILY (a position is a set of heads; the whole
    position = XOR of single-head
values). THE INDEXING IS LOAD-BEARING - mixing 0- and 1-indexed
    values is silently wrong.
Twins          turn exactly 2          0-idx  g(x) = x
TurningTurtles turn 1 or 2             1-idx  g(x) = x
MockTurtles    turn 1, 2 or 3          0-idx  g(x) = 2x or
    2x+1, whichever is ODDIOUS
                                (= the x-th
    odious number: 1 2 4 7 8 11 13 )
Ruler          any CONSECUTIVE run     1-idx  g(x) = x & -x
Triplets       turn exactly 3          0-idx  g(x) = 0 for x
    < 2, MockTurtles(x-2) otherwise
Mogul          turn 1..5                NO closed form
    (1 2 4 8 16 31 32 64 103 ...)
TurningCorners turn 4 rectangle corners g(x, y) = x (*)
    y [nim product]
TARTAN THEOREM: for the 2-D product of two coin-turning games,
    g(x,y) = g1(x) (*) g2(y).

k-WYTHOFF (Fraenkel): take from one pile, or k1, k2 > 0 from both
    with |k1 - k2| < k.
P-positions: a_n = mex{a_i, b_i : i < n}, b_n = a_n + k*n.
Closed form: (floor(n*alpha), floor(n*beta)) with alpha = (2 -
    sqrt(k*k + 4)) / 2,
    beta = alpha + k. k = 1 is ordinary Wythoff.

SILVER DOLLAR (coins at p_1 < ... < p_k on a strip, slide one
    left onto an empty square, no
jumping): with gaps g_i = p_i - p_{i-1} - 1 and p_0 = -1, the
    GRUNDY VALUE is
    g_k XOR g_{k-2} XOR g_{k-4} XOR ... - staircase nim on
    alternate gaps counted FROM THE RIGHT.

CHOMP: every m x n board is a FIRST-PLAYER WIN except 1x1, by
    strategy stealing - and there is
NO known constructive winning move. Two-row Chomp: P-positions
    are exactly (b+1, b).
Strategy stealing is a pure EXISTENCE proof; if the problem

```

→ wants the move, it is useless.

BLUE-RED HACKENBUSH (partisan!): a string of edges from the
 ground has a NUMBER value, not a
 number. The first run of m equal colours contributes +m (Blue
 positive), and every later edge
 contributes +1/2, +1/4, ... halving. This is the cleanest
 concrete model of surreal numbers.
 Partisan values in general: 0 = second player wins, * = {0|0} =
 first player wins,
 up = {0|*} is positive but smaller than every positive number,
 a SWITCH {a|b} with a > b is
 "hot" (both players want to move there), and the TEMPERATURE is
 what it costs the opponent to
 let you move first. SIMPLICITY RULE: if some number lies
 strictly between the Left and Right
 options, the game IS the simplest such number. NUMBER
 AVOIDANCE: never move in a number
 component while a non-number component exists.

MISERE THEORY, honestly: misere nim's clean rule is an ACCIDENT.
 In general the misere outcome
 of a sum is NOT a function of the Grundy values; the correct
 invariant is the GENUS, and only
 "tame" games (genus matching nim's) obey nim-like rules.
 Anti-SG above is exactly the tame
 case, which is why it carries a precondition.

COMPLEXITY LANDSCAPE - when to stop looking for a formula:
 UNDIRECTED vertex geography is in P (the maximum-matching
 theorem in 03 - MinimaxAndMatching).
 DIRECTED vertex geography, Node Kayles, generalized geography
 and generalized Hex are all
 PSPACE-complete; generalized chess and Go are EXPTIME-complete.
 So a 1e5-sized instance of
 directed geography MUST have extra structure - find it instead
 of searching.

BOGUS NIM / REVERSIBLE MOVES: adding a move that the opponent can
 immediately undo does not
 change the Grundy value. That is precisely why the even indices
 in staircase nim are worth
 nothing, and it is the impartial case of the "bypass a reversible
 option" simplification. */

11 Geometry

11.1 Geometry 2D

11.1.1 Core

Point / vector core. $P < ll > =$ exact (use for all predicates), $P < ld > =$ real answers.

```
// cross(a,b) > 0 <=> b is counter-clockwise from a.
// a.cross(b,c) > 0 <=> c lies strictly LEFT of the directed line
//   a->b. [= orient(a,b,c)]
int sgn(ll x) { return (x > 0) - (x < 0); }
int sgn(ld x) { return (x > eps) - (x < -eps); }
int sgn(int x) { return (x > 0) - (x < 0); } // without
//   these two overloads a bare int or
int sgn(double x) { return sgn((ld)x); } // double
//   argument is AMBIGUOUS: hard error
ll gcdll(ll a, ll b) { return b ? gcdll(b, a % b) : a; } //
//   std::__gcd is libstdc++-only

template <class T>
struct P {
    typedef P Pt; T x = 0, y = 0;
    P(T x = 0, T y = 0) : x(x), y(y) {}
    Pt operator+(Pt p) const { return {x + p.x, y + p.y}; }
    Pt operator-(Pt p) const { return {x - p.x, y - p.y}; }
    Pt operator*(T d) const { return {x * d, y * d}; }
    Pt operator/(T d) const { return {x / d, y / d}; }
    T dot(Pt p) const { return x * p.x + y * p.y; }
    T cross(Pt p) const { return x * p.y - y * p.x; }
    T cross(Pt a, Pt b) const { return (a - *this).cross(b -
//   *this); } // orient
    T dist2() const { return x * x + y * y; }
    ld dist() const { return sqrtl((ld)dist2()); }
    ld angle() const { return atan2l(y, x); } // (-pi,
//   pi]
    Pt perp() const { return {-y, x}; } //
//   rotate +90 deg
    Pt unit() const { return *this / dist(); }
    Pt rot(ld a) const { return {x * cosl(a) - y * sinl(a), x *
//   sinl(a) + y * cosl(a)}; }
    bool operator<(Pt p) const { return tie(x, y) < tie(p.x,
//   p.y); }
    bool operator==(Pt p) const { return tie(x, y) == tie(p.x,
//   p.y); }
    friend istream& operator>>(istream& i, Pt& p) { return i >>
//   p.x >> p.y; }
    friend ostream& operator<<(ostream& o, Pt p) { return o <<
//   p.x << ' ' << p.y; }
};
typedef P<ll> Pi;
typedef P<ld> Pd;

// Signed angle from a to b in (-pi, pi]. Sign tells you the
//   rotation direction.
template <class T> ld angleTo(P<T> a, P<T> b) { return
//   atan2l(a.cross(b), a.dot(b)); }
// Angle a-0-b in [0, 2pi).
template <class T> ld angleAt(P<T> a, P<T> 0, P<T> b) {
    ld r = angleTo(a - 0, b - 0);
    return r < 0 ? r + 2 * PI : r;
}

// EXACT polar sort, no atan2. Orders by angle in [0, 2pi)
//   starting at +x axis.
// sort(all(v), angCmp<ll>); ties (same direction) keep input
//   order.
template <class T> bool half(P<T> p) { return p.y < 0 || (p.y ==
//   0 && p.x < 0); }
template <class T> bool angCmp(P<T> a, P<T> b) {
    return half(a) != half(b) ? half(a) < half(b) : a.cross(b) >
//   0;
}

// OVERFLOW: |coord| <= 1e9 => cross fits in ll (max ~4e18, ll
//   max 9.2e18).
// dist2 of a difference fits too. Anything squared
//   twice needs __int128.
```

11.1.2 Lines

Lines, segments, rays. "line ab" = infinite line through a and b.

```
// Predicates are exact on P<ll>; anything returning a point needs
//   P<ld>.

template <class P> ld lineDist(P p, P a, P b) { //
//   distance point -> line
    return fabs1((ld)(b - a).cross(p - a) / (b - a).dist());
}
template <class P> P proj(P p, P a, P b) { // foot
//   of perpendicular
    return a + (b - a) * ((ld)(p - a).dot(b - a) / (b -
//   a).dist2());
}
template <class P> P refl(P p, P a, P b) { return proj(p, a, b) *
//   2 - p; }

template <class P> bool onSeg(P p, P a, P b) { //
//   inclusive of endpoints
    return a.cross(b, p) == 0 && (a - p).dot(b - p) <= 0;
}
template <class P> ld segDist(P p, P a, P b) { //
//   distance point -> segment (exact-safe)
    if (a == b) return (p - a).dist();
    ld d = (ld)(b - a).dist2(), t = (ld)(p - a).dot(b - a);
    if (t <= 0) return (p - a).dist();
    if (t >= d) return (p - b).dist();
    return fabs1((ld)(b - a).cross(p - a) / sqrtl(d));
}
template <class P> P closestOnSeg(P p, P a, P b) {
    if ((p - a).dot(b - a) <= 0) return a;
    if ((p - b).dot(b - a) >= 0) return b;
    return proj(p, a, b);
}

// LINE x LINE. first = 1 unique, 0 parallel-disjoint, -1 same
//   line.
template <class P> pair<int, P> lineInter(P a, P b, P c, P d) {
    auto x = (b - a).cross(d - c);
    if (sgn(x) == 0) return {-a.cross(b, c) == 0, P()};
    auto p = c.cross(b, d), q = c.cross(d, a); // p +
//   q = x (affine weights)
    return {1, (a * p + b * q) / x}; // P
//   must be floating

// SEGMENT x SEGMENT. Returns 0, 1, or 2 points (2 = collinear
//   overlap endpoints).
// Exact on P<ll> when the answer has 0 or 2 points. The 1-point
//   case DIVIDES and the numerator
//   is ~2*C^3, so it is only exact for |coordinate| <= ~1.6e6.
// Above that use segCross for the
//   boolean, or Frac (01 - Miscellaneous/13) for an exact rational
//   point.
template <class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b), oc = a.cross(b,
//   c), od = a.cross(b, d);
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSeg(c, a, b)) s.insert(c);
    if (onSeg(d, a, b)) s.insert(d);
    if (onSeg(a, c, d)) s.insert(a);
    if (onSeg(b, c, d)) s.insert(b);
    return {all(s)};
}
// Just "do they touch?" - fully exact, NO division and no
//   allocation, so it is both safe at
//   |c| = 1e9 and ~30x faster than asking segInter. Use this
//   whenever you only need a bool.
template <class P> bool segCross(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b), oc = a.cross(b,
//   c), od = a.cross(b, d);
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0) return
//   true;
    return onSeg(c, a, b) || onSeg(d, a, b) || onSeg(a, c, d) ||
//   onSeg(b, c, d);
}

template <class P> ld segSegDist(P a, P b, P c, P d) {
    if (segCross(a, b, c, d)) return 0;
    return min({segDist(a, c, d), segDist(b, c, d), segDist(c, a,
//   b), segDist(d, a, b)});
}

// RAY from s through e.
template <class P> bool onRay(P p, P s, P e) { return s.cross(e,
```

```

    ↪ p) == 0 && (p - s).dot(e - s) >= 0; }
template <class P> ld rayDist(P p, P s, P e) { return (p -
    ↪ s).dot(e - s) <= 0 ? (p - s).dist() : lineDist(p, s, e); }

// Perpendicular bisector of ab, returned as two points on it.
// FLOATING P ONLY: (a+b)/2 truncates on P<ll> and returns a
    ↪ DIFFERENT line (for a=(0,0),
// b=(1,1) it gives y = -x instead of y = -x + 1). For an exact
    ↪ integer version work on the
// doubled lattice: the line through (a+b) and (a+b) +
    ↪ (b-a).perp().
template <class P> pair<P, P> perpBisector(P a, P b) {
    P m = (a + b) / 2;
    return {m, m + (b - a).perp()};
}

// Interior angle bisector direction at B for angle A-B-C (unit
    ↪ vectors summed).
template <class P> P bisectorDir(P a, P b, P c) { return (a -
    ↪ b).unit() + (c - b).unit(); }

// Line as a*x + b*y = c -> two points on it (needs floating P).
template <class P> pair<P, P> fromABC(ld A, ld B, ld C) {
    P n(A, B), p = n * (C / n.dist2());
    return {p, p + n.perp()};
}

```

11.1.3 Polygon

Polygons. Vertices in order (CW or CCW); convex routines want CCW, no collinear points.

```

template <class P> auto area2(const vector<P>& v) { //
    ↪ SIGNED 2*area. >0 <=> CCW.
    decltype(v[0].cross(v[0])) a = 0; //
    ↪ exact on P<ll>; safe on empty
    for (int i = 0, n = v.size(); i < n; i++) a +=
    ↪ v[i].cross(v[(i + 1) % n]);
    return a;
}

template <class P> ld area(const vector<P>& v) { return
    ↪ fabs1((ld)area2(v)) / 2; }
template <class P> ld perimeter(const vector<P>& v) {
    ld r = 0;
    for (int i = 0, n = v.size(); i < n; i++) r += (v[i] - v[(i +
    ↪ 1) % n]).dist();
    return r;
}

template <class P> P centroid(const vector<P>& v) { // area
    ↪ centroid (needs floating P)
    P r; ld A = 0;
    for (int i = 0, j = v.size() - 1, n = v.size(); i < n; j =
    ↪ i++) {
        auto c = v[j].cross(v[i]);
        r = r + (v[i] + v[j]) * c; A += c;
    }
    return r / A / 3;
}

// A completely collinear (zero-area) polygon reports TRUE here -
    ↪ guard with area2 != 0 if that
// matters. Bow-ties and spikes are correctly rejected.
template <class P> bool isConvex(const vector<P>& v) {
    int n = v.size(), pos = 0, neg = 0;
    for (int i = 0; i < n; i++) {
        int s = sgn(v[i].cross(v[(i + 1) % n], v[(i + 2) % n]));
        s > 0 ? pos++ : s < 0 ? neg++ : 0;
    }
    return !pos || !neg;
}

// Point in ANY simple polygon, O(n), exact. strict=true ->
    ↪ boundary counts as outside.
template <class P> bool inPoly(const vector<P>& v, P a, bool
    ↪ strict = true) {
    int cnt = 0, n = v.size();
    for (int i = 0; i < n; i++) {
        P q = v[(i + 1) % n];
        if (onSeg(a, v[i], q)) return !strict;
        cnt ^= ((a.y < v[i].y) - (a.y < q.y)) * a.cross(v[i], q)
    }
    ↪ > 0;
    return cnt;
}

// Convex hull, CCW, strictly convex (collinear points dropped).
    ↪ O(n log n).
// Change <= 0 to < 0 in the pop test to KEEP collinear points on
    ↪ the hull.
template <class P> vector<P> hull(vector<P> p) {
    if (p.size() <= 1) return p;
    sort(all(p));
    vector<P> h(p.size() + 1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(p)))
        for (P q : p) {
            while (t >= s + 2 && h[t - 2].cross(h[t - 1], q) <=
            ↪ 0) t--;
            h[t++] = q;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}

// Point in CONVEX polygon (CCW, strict), O(log n). 1 inside, 0 on
    ↪ boundary, -1 outside.
// h must be a strictly convex CCW hull with n >= 1. Returns 1
    ↪ inside, 0 on the boundary,
// -1 outside.
template <class P> int inConvex(const vector<P>& h, P p) {
    if (h.empty()) return -1;
    int n = h.size();
    if (n == 1) return p == h[0] ? 0 : -1;
    if (n == 2) return onSeg(p, h[0], h[1]) ? 0 : -1;
    if (h[0].cross(h[1], p) < 0 || h[0].cross(h[n - 1], p) > 0)
    ↪ return -1;
    int l = 1, r = n - 1;

```

```

    while (r - 1 > 1) { int m = (1 + r) / 2; (h[0].cross(h[m], p)
    ⇨ >= 0 ? 1 : r) = m; }
    int s = sgn(h[1].cross(h[1 + 1], p));
    if (s < 0) return -1;
    if (s == 0) return 0;
    return (h[0].cross(h[1], p) == 0 || h[0].cross(h[n - 1], p)
    ⇨ == 0) ? 0 : 1;
}

// Index of the hull vertex furthest in direction 'dir', O(log n).
⇨ CCW hull, no collinear.
// Use to get tangents from a far-away direction, or max/min dot
⇨ product.
#define _cmp(i, j) sgn(dir.perp().cross(h[(i) % n] - h[(j) % n]))
#define _ext(i) (_cmp(i + 1, i) >= 0 && _cmp(i, i - 1 + n) < 0)
template <class P> int extreme(const vector<P>& h, P dir) {
    int n = h.size(), lo = 0, hi = n;
    if (_ext(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (_ext(m)) return m;
        int ls = _cmp(lo + 1, lo), ms = _cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == _cmp(lo, m)) ? hi : lo) =
    ⇨ m;
    }
    return lo;
}

// Cut polygon by the line s->e, KEEP the part on the LEFT
⇨ (boundary included). O(n).
// Needs floating P. Boundary-inclusive matters: with a strict
⇨ test, vertices lying exactly on
// the line are dropped from BOTH halves and you lose area. Repeat
⇨ over many lines = half-plane
// intersection. cut(v,e,s) gives the other half; the two areas
⇨ sum to area(v).
// Keeps the part LEFT of the directed line s->e, boundary
⇨ INCLUSIVE. Vertices exactly on the
// line are kept, so a line through a vertex produces a DUPLICATE
⇨ point - fine for area, but
// dedup before feeding the result to hull / isConvex / minkowski
⇨ / diameter2.
template <class P> vector<P> cut(const vector<P>& v, P s, P e) {
    vector<P> r;
    for (int i = 0, n = v.size(); i < n; i++) {
        P cur = v[i], prv = v[(i + n - 1) % n];
        bool side = sgn(s.cross(e, cur)) >= 0;
        if (side != (sgn(s.cross(e, prv)) >= 0))
    ⇨ r.push_back(lineInter(s, e, cur, prv).second);
        if (side) r.push_back(cur);
    }
    return r;
}

// Minkowski sum of two CONVEX CCW polygons. O(n+m). Result is
⇨ convex CCW.
// BOTH polygons must be CONVEX and COUNTER-CLOCKWISE. On
⇨ clockwise input the merge below makes
// no progress and loops forever (an unexplained TLE/MLE, not a
⇨ wrong answer) - so normalise.
template <class P> vector<P> minkowski(vector<P> A, vector<P> B) {
    if (area2(A) < 0) reverse(all(A));
    if (area2(B) < 0) reverse(all(B));
    auto low = [](vector<P>& v) {
        rotate(v.begin(), min_element(all(v), [](P a, P b) {
    ⇨ return tie(a.y, a.x) < tie(b.y, b.x); }, v.end());
        });
    low(A); low(B);
    A.push_back(A[0]); A.push_back(A[1]);
    B.push_back(B[0]); B.push_back(B[1]);
    vector<P> r; size_t i = 0, j = 0;
    while (i + 2 < A.size() || j + 2 < B.size()) {
        r.push_back(A[i] + B[j]);
        auto c = (A[i + 1] - A[i]).cross(B[j + 1] - B[j]);
        if (c >= 0 && i + 2 < A.size()) i++;
        if (c <= 0 && j + 2 < B.size()) j++;
    }
    return r;
}

// Rotating calipers. Hull must be CCW strictly convex.
template <class P> auto diameter2(const vector<P>& h) { // max
    ⇨ squared distance
    int n = h.size(), j = n < 2 ? 0 : 1;
    decltype(h[0].dist2()) r = 0;
    for (int i = 0; i < j; i++)
        for (; j = (j + 1) % n) {
            r = max(r, (h[i] - h[j]).dist2());
            if ((h[(j + 1) % n] - h[j]).cross(h[i + 1] - h[i]) >=
    ⇨ 0) break;
        }
    return r;
}

template <class P> ld width(const vector<P>& h) { // min
    ⇨ distance between parallel supports
    int n = h.size(); if (n <= 2) return 0;
    ld r = 1e18; int j = 1;
    for (int i = 0; i < n; i++) {
        while ((h[(i + 1) % n] - h[i]).cross(h[(j + 1) % n] -
    ⇨ h[j]) >= 0) j = (j + 1) % n;
        r = min(r, lineDist(h[j], h[i], h[(i + 1) % n]));
    }
    return r;
}

// Minimum-area enclosing rectangle. One side always lies on a
⇨ hull edge (that's the theorem),
// so try every edge. O(n^2) on the HULL only - fine, and immune
⇨ to caliper tie bugs.
template <class P> ld minRectArea(const vector<P>& h) {
    int n = h.size(); if (n < 3) return 0;
    ld best = 1e18;
    for (int i = 0; i < n; i++) {
        P d = h[(i + 1) % n] - h[i];
        ld L = d.dist(), lo = 1e18, hi = -1e18, ht = 0;
        for (P q : h) {
            ld u = (ld)d.dot(q - h[i]) / L, v = (ld)d.cross(q -
    ⇨ h[i]) / L;
            lo = min(lo, u); hi = max(hi, u); ht = max(ht,
    ⇨ fabs(v));
        }
        best = min(best, (hi - lo) * ht);
    }
    return best;
}

// PICK'S THEOREM. Lattice polygon: A = I + B/2 - 1.
template <class P> ll boundaryPts(const vector<P>& v) { // B
    ll b = 0;
    for (int i = 0, n = v.size(); i < n; i++) {
        P d = v[(i + 1) % n] - v[i];
        b += gcdll(llabs(d.x), llabs(d.y));
    }
    return b;
}

// PICK'S THEOREM, and it needs a SIMPLE polygon with integer
⇨ vertices. On a self-intersecting
// or degenerate (zero-area) polygon the result is meaningless,
⇨ not merely approximate.
template <class P> ll interiorPts(const vector<P>& v) { // I =
    ⇨ A - B/2 + 1
    return (llabs(area2(v)) - boundaryPts(v)) / 2 + 1;
}

```


11.1.4 Circles

Circles. All of these want a floating P (Pd).

```
// --- circle x line: 0, 1 (tangent) or 2 points on segment/line
↳ ab ---
template <class P> vector<P> circleLine(P o, ld r, P a, P b) {
    P p = proj(o, a, b);
    ld h2 = r * r - (p - o).dist2();
    if (h2 < -eps) return {};
    if (h2 < eps) return {p};
    P h = (b - a).unit() * sqrt1(h2);
    return {p - h, p + h};
}

template <class P> vector<P> circleSeg(P o, ld r, P a, P b) {
    vector<P> out;
    for (P p : circleLine(o, r, a, b)) if (segDist(p, a, b) <
↳ eps) out.push_back(p);
    return out;
}

// --- circle x circle: the 0/1/2 intersection points ---
template <class P> vector<P> circleCircle(P a, ld r1, P b, ld r2)
↳ {
    P d = b - a; ld d2 = d.dist2();
    if (d2 < eps) return {}; //
    ↳ concentric (equal => infinite)
    ld p = (d2 + r1 * r1 - r2 * r2) / (2 * d2), h2 = r1 * r1 - p
↳ * p * d2;
    if (r1 + r2 < sqrt1(d2) - eps || fabs1(r1 - r2) > sqrt1(d2) +
↳ eps) return {};
    P mid = a + d * p, per = d.perp() * sqrt1(max((ld)0, h2) /
↳ d2);
    if (h2 < eps) return {mid};
    return {mid + per, mid - per};
}

// --- AREA of the lens (intersection region) of two circles ---
template <class P> ld circleInterArea(P a, ld r1, P b, ld r2) {
    ld d = (a - b).dist();
    if (d >= r1 + r2) return 0;
    if (d + min(r1, r2) <= max(r1, r2)) return PI * min(r1, r2) *
↳ min(r1, r2);
    ld A = acos1((d * d + r1 * r1 - r2 * r2) / (2 * d * r1));
    ld B = acos1((d * d + r2 * r2 - r1 * r1) / (2 * d * r2));
    return r1 * r1 * (A - sin1(2 * A) / 2) + r2 * r2 * (B -
↳ sin1(2 * B) / 2);
}

// --- TANGENTS. Each result is a pair (touch point on c1, touch
↳ point on c2).
// inner=false -> the 2 external tangents; inner=true -> the 2
↳ internal (crossing) ones.
// Tangent lines from a POINT p to circle (c,r): tangents(p, 0,
↳ c, r, false).
template <class P> vector<pair<P, P>> tangents(P c1, ld r1, P c2,
↳ ld r2, bool inner) {
    if (inner) r2 = -r2;
    P d = c2 - c1;
    ld dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 < eps || h2 < -eps) return {};
    vector<pair<P, P>> out;
    for (ld s : {-1.0L, 1.0L}) {
        P v = (d * dr + d.perp() * sqrt1(max((ld)0, h2)) * s) /
↳ d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 < eps) out.pop_back();
    return out;
}

// --- circumcircle / incircle of a triangle ---
template <class P> P circumcenter(P a, P b, P c) {
    P B = c - a, C = b - a;
    return a + (B * C.dist2() - C * B.dist2()).perp() /
↳ B.cross(C) / 2;
}

template <class P> ld circumradius(P a, P b, P c) {
    return (b - a).dist() * (c - b).dist() * (a - c).dist() / (2
↳ * fabs1((ld)(b - a).cross(c - a)));
}

template <class P> P incenter(P a, P b, P c) {
    ld A = (b - c).dist(), B = (c - a).dist(), C = (a - b).dist();
    return (a * A + b * B + c * C) / (A + B + C);
}

template <class P> ld inradius(P a, P b, P c) {
    ld s = ((b - a).dist() + (c - b).dist() + (a - c).dist()) / 2;
    return fabs1((ld)(b - a).cross(c - a)) / 2 / s;
}
```

```
}
template <class P> P orthocenter(P a, P b, P c) { return a + b +
↳ c - circumcenter(a, b, c) * 2; }

// --- MINIMUM ENCLOSING CIRCLE, expected O(n) (Welzl,
↳ incremental) ---
template <class P> pair<P, ld> mec(vector<P> p) {
    if (p.empty()) return {P(), 0};
    shuffle(all(p),
↳ mt19937(chrono::steady_clock::now().time_since_epoch().count()));
    P o = p[0]; ld r = 0, E = 1 + 1e-9;
    for (int i = 0; i < (int)p.size(); i++) if ((o - p[i]).dist()
↳ > r * E) {
        o = p[i], r = 0;
        for (int j = 0; j < i; j++) if ((o - p[j]).dist() > r *
↳ E) {
            o = (p[i] + p[j]) / 2, r = (o - p[i]).dist();
            for (int k = 0; k < j; k++) if ((o - p[k]).dist() > r
↳ * E)
                o = circumcenter(p[i], p[j], p[k]), r = (o -
↳ p[i]).dist();
        }
    }
    return {o, r};
}

// --- AREA of (circle centered at c, radius r) INTERSECT polygon
↳ ps ---
template <class P> ld circlePolyArea(P c, ld r, vector<P> ps) {
    auto arg = [](P p, P q) { return atan21(p.cross(q),
↳ p.dot(q)); };
    auto tri = [&](P p, P q) -> ld {
        ld r2 = r * r / 2;
        P d = q - p;
        ld a = d.dot(p) / d.dist2(), b = (p.dist2() - r * r) /
↳ d.dist2(), det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        ld s = max((ld)0, -a - sqrt1(det)), t = min((ld)1, -a +
↳ sqrt1(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = p + d * t;
        return arg(p, u) * r2 + u.cross(v) / 2 + arg(v, q) * r2;
    };
    ld sum = 0;
    for (int i = 0, n = ps.size(); i < n; i++) sum += tri(ps[i] -
↳ c, ps[(i + 1) % n] - c);
    return fabs1(sum);
}

// --- power of a point / radical axis ---
// pow(p) = |p-c|^2 - r^2 : <0 inside, 0 on circle, >0 outside;
↳ equals (tangent length)^2.
template <class P> ld power(P p, P c, ld r) { return (p -
↳ c).dist2() - r * r; }

// Radical axis of two non-concentric circles, as two points on it.
template <class P> pair<P, P> radicalAxis(P a, ld r1, P b, ld r2)
↳ {
    P d = b - a; ld t = ((r1 * r1 - r2 * r2) / d.dist2() + 1) / 2;
    P m = a + d * t;
    return {m, m + d.perp()};
}
```


11.1.5 Sweep

Global / sweep-line geometry: problems about a whole SET of objects.

```
// --- CLOSEST PAIR of points,  $O(n \log n)$ . Returns the two points.
//  $\hookrightarrow$  Exact on  $P<11>$ . ---
template <class P> pair<P, P> closestPair(vector<P> v) {
    assert(v.size() > 1); // n <
    // 2 has no pair to return
    set<P> S; //
    // ordered by (x, y)
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> best{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d{1 + (ll)sqrtl((ld)best.first), 0};
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo) best = min(best, {(ll)lo -
        // p).dist2(), {*lo, p}});
        S.insert(p);
    }
    return best.second;
}

// --- AREA OF UNION OF AXIS-ALIGNED RECTANGLES,  $O(n \log n)$ . ---
// rects: {x1, y1, x2, y2} half-open [x1,x2) x [y1,y2). Sweep x,
//  $\hookrightarrow$  segment tree counts covered y.
struct CoverTree { //
    // "count>0 length" segment tree
    int n; vector<int> cnt; vector<ll> len; vector<ll> ys;
    CoverTree(vector<ll> y) : ys(y) { n = ys.size() - 1;
    // cnt.assign(4 * n, 0); len.assign(4 * n, 0); }
    void upd(int v, int l, int r, int ql, int qr, int d) {
        if (qr <= l || r <= ql) return;
        if (ql <= l && r <= qr) cnt[v] += d;
        else {
            int m = (l + r) / 2;
            upd(2 * v + 1, m, r, ql, qr, d); upd(2 * v, l, m, ql, qr, d);
        }
        len[v] = cnt[v] ? ys[r] - ys[l] : (r - l == 1 ? 0 : len[2
        // * v] + len[2 * v + 1]);
        void upd(int l, int r, int d) { upd(1, 0, n, l, r, d); }
        ll get() { return len[1]; }
};

ll rectUnionArea(vector<array<ll, 4>> r) {
    vector<ll> ys;
    for (auto& q : r) ys.push_back(q[1]), ys.push_back(q[3]);
    sort(all(ys)); ys.erase(unique(all(ys)), ys.end());
    if (ys.size() < 2) return 0;
    auto id = [&](ll y) { return (int)(lower_bound(all(ys), y) -
    // ys.begin()); };
    vector<array<ll, 4>> ev; // {x,
    // +1, y1, y2}
    for (auto& q : r) ev.push_back({q[0], 1, q[1], q[3]}),
    // ev.push_back({q[2], -1, q[1], q[3]});
    sort(all(ev));
    CoverTree T(ys);
    ll area = 0, px = ev.empty() ? 0 : ev[0][0];
    for (auto& e : ev) {
        area += T.get() * (e[0] - px); px = e[0];
        T.upd(id(e[2]), id(e[3]), (int)e[1]);
    }
    return area;
}

// PERIMETER of the union: run the same sweep twice (once per
//  $\hookrightarrow$  axis) accumulating
// |len(after) - len(before)| at each event x, and add the
//  $\hookrightarrow$  vertical contributions.

// --- MAX POINTS ON A LINE,  $O(n^2 \log n)$  ---
template <class P> int maxOnLine(vector<P> p) {
    int n = p.size(), best = min(n, 1);
    for (int i = 0; i < n; i++) {
        map<pair<ll, ll>, int> c; int dup = 0;
        for (int j = 0; j < n; j++) if (j != i) {
            ll dx = p[j].x - p[i].x, dy = p[j].y - p[i].y;
            if (!dx && !dy) { dup++; continue; } //
            // duplicate points
            ll g = gcdll(llabs(dx), llabs(dy)); dx /= g; dy /= g;
            if (dx < 0 || (dx == 0 && dy < 0)) dx = -dx, dy = -dy;
            c[{dx, dy}]++;
        }
        best = max(best, dup + 1);
        for (auto& [k, v] : c) best = max(best, v + dup + 1);
    }
}
```

```
return best;
}

// --- COUNT PAIRS OF INTERSECTING SEGMENTS,  $O(n^2)$  exact. ---
// For  $O((n+k) \log n)$  use Bentley-Ottmann; almost never needed in
//  $\hookrightarrow$  contests.
template <class P> ll countSegInter(vector<pair<P, P>> s) {
    ll c = 0;
    for (int i = 0; i < (int)s.size(); i++)
        for (int j = i + 1; j < (int)s.size(); j++)
            c += segCross(s[i].first, s[i].second, s[j].first,
            // s[j].second);
    return c;
}

// --- CONVEX LAYERS (onion peeling): repeatedly strip the hull.
//  $\hookrightarrow O(n^2 \log n)$  naive. ---
template <class P> vector<vector<P>> onion(vector<P> p) {
    vector<vector<P>> L;
    while (!p.empty()) {
        auto h = hull(p);
        L.push_back(h);
        for (P q : h) p.erase(find(all(p), q));
    }
    return L;
}
```

11.1.6 HalfPlaneIntersection

HALF-PLANE INTERSECTION - $O(n \log n)$. A half-plane is "everything LEFT of the directed

```
// line a -> b". Returns the CCW vertices of the intersection, or
//  $\hookrightarrow$  {} if it is empty.
// A big bounding box is added, so the result is always bounded.
// 2D LINEAR PROGRAMMING is exactly this: constraints ->
//  $\hookrightarrow$  half-planes, then optimise over the
// resulting convex polygon (ternary search on the boundary, or
//  $\hookrightarrow$  check all vertices).
struct HP {
    Pd p, d; // keep the side
    // LEFT of p -> p + d
    HP() {}
    HP(Pd a, Pd b) : p(a), d(b - a) {}
    bool out(Pd r) const { return d.cross(r - p) < -eps; }
};

Pd hpInter(const HP& a, const HP& b) { // assumes a.d and
    // b.d are not parallel
    return a.p + a.d * ((b.p - a.p).cross(b.d) / a.d.cross(b.d));
}

vector<Pd> halfPlaneInter(vector<HP> h, ld BOX = 1e9) {
    Pd box[4] = {Pd(-BOX, -BOX), Pd(BOX, -BOX), Pd(BOX, BOX),
    // Pd(-BOX, BOX)};
    for (int i = 0; i < 4; i++) h.push_back(HP(box[i], box[(i +
    // 1) % 4]));
    sort(all(h), [](const HP& a, const HP& b) {
        if (half(a.d) != half(b.d)) return half(a.d) < half(b.d);
        ld c = a.d.cross(b.d);
        if (fabsl(c) > eps) return c > 0;
        return a.out(b.p); // same
    });
    deque<HP> dq;
    for (auto& x : h) {
        if (!dq.empty() && fabsl(dq.back().d.cross(x.d)) <= eps
        // && dq.back().d.dot(x.d) > 0) continue;
        while (dq.size() > 1 && x.out(hpInter(dq[dq.size() - 1],
        // dq[dq.size() - 2]))) dq.pop_back();
        while (dq.size() > 1 && x.out(hpInter(dq[0], dq[1])))
        // dq.pop_front();
        dq.push_back(x);
    }
    while (dq.size() > 2 && dq[0].out(hpInter(dq[dq.size() - 1],
    // dq[dq.size() - 2]))) dq.pop_back();
    while (dq.size() > 2 && dq.back().out(hpInter(dq[0], dq[1])))
    // dq.pop_front();
    if (dq.size() < 3) return {};
    vector<Pd> r;
    for (size_t i = 0; i < dq.size(); i++)
        // r.push_back(hpInter(dq[i], dq[(i + 1) % dq.size()]));
    return r;
}

// If you only have a handful of half-planes, repeated cut() (03 -
//  $\hookrightarrow$  Polygon.cpp) on a big box is
// shorter to retype and just as correct -  $O(n^2)$  but n is small.
```

11.1.7 Unions

AREA OF UNIONS. The three shapes that actually show up.

```
// --- UNION OF POLYGONS (any simple polygons, may overlap),  $O(n^2 \log n)$ . ---
// Works by, for every edge, measuring how much of it is "on the
// boundary of the union" and
// accumulating the shoelace contribution with the right
// multiplicity.
ld edgeRatio(Pd a, Pd b) { return sgn(b.x) ? a.x / b.x : a.y /
    b.y; }
ld polyUnion(vector<vector<Pd>>& ps) { //
    // every polygon must be CCW
    ld ret = 0;
    for (size_t i = 0; i < ps.size(); i++)
        for (size_t v = 0; v < ps[i].size(); v++) {
            Pd A = ps[i][v], B = ps[i][(v + 1) % ps[i].size()];
            vector<pair<ld, int>> seg = {{0, 0}, {1, 0}};
            for (size_t j = 0; j < ps.size(); j++) {
                if (i == j) continue;
                for (size_t u = 0; u < ps[j].size(); u++) {
                    Pd C = ps[j][u], D = ps[j][(u + 1) %
                        ps[j].size()];
                    int sc = sgn(A.cross(B, C)), sd =
                        sgn(A.cross(B, D));
                    if (sc != sd) {
                        ld sa = C.cross(D, A), sb = C.cross(D, B);
                        if (min(sc, sd) < 0) seg.push_back({sa /
                            (sa - sb), sc > sd ? 1 : -1});
                    } else if (!sc && !sd && j < i && sgn((B -
                        A).dot(D - C)) > 0) {
                        seg.push_back({edgeRatio(C - A, B - A),
                            1});
                        seg.push_back({edgeRatio(D - A, B - A),
                            -1});
                    }
                }
            }
            sort(all(seg));
            for (auto& s : seg) s.first = min(max(s.first,
                (ld)0), (ld)1);
            ld sum = 0;
            int cnt = seg[0].second;
            for (size_t j = 1; j < seg.size(); j++) { //
                // fraction of AB on the union boundary
                if (!cnt) sum += seg[j].first - seg[j - 1].first;
                cnt += seg[j].second;
            }
            ret += A.cross(B) * sum;
        }
    return ret / 2;
}

// --- UNION OF CIRCLES,  $O(n^2 \log n)$  by integrating each arc that
// is on the outer boundary. ---
ld circleUnion(vector<Pd> c, vector<ld> r) {
    int n = c.size();
    ld total = 0;
    for (int i = 0; i < n; i++) {
        vector<pair<ld, int>> ev;
        int cover = 0;
        for (int j = 0; j < n; j++) {
            if (i == j) continue;
            ld d = (c[i] - c[j]).dist();
            if (d + r[i] <= r[j] + eps && r[i] < r[j] ||
                (fabs1(r[i] - r[j]) < eps && j < i)) { cover = -1; break; }
            if (d >= r[i] + r[j] - eps || d + r[j] <= r[i] + eps)
                continue;
            ld a = (c[j] - c[i]).angle();
            ld w = acos1((d * d + r[i] * r[i] - r[j] * r[j]) / (2
                * d * r[i]));
            ld L = a - w, R = a + w;
            if (L < -PI) L += 2 * PI;
            if (R > PI) R -= 2 * PI;
            if (L > R) ev.push_back({L, 1}), ev.push_back({PI,
                -1}), ev.push_back({-PI, 1}), ev.push_back({R, -1});
            else ev.push_back({L, 1}), ev.push_back({R, -1});
        }
        if (cover < 0) continue;
        ev.push_back({-PI, 0}), ev.push_back({PI, 0});
        sort(all(ev));
        int cnt = 0; ld last = -PI;
        for (auto [a, t] : ev) {
            if (!cnt && a > last) {
                // arc [last, a] is on the boundary
                Pd p1 = c[i] + Pd(cos1(last), sin1(last)) * r[i];
                Pd p2 = c[i] + Pd(cos1(a), sin1(a)) * r[i];
                total += p1.cross(p2) / 2 + r[i] * r[i] * ((a -
                    last) - sin1(a - last)) / 2;
            }
            cnt += t;
            last = max(last, a);
        }
    }
    return total;
}

// --- PERIMETER OF A UNION OF AXIS-ALIGNED RECTANGLES ---
// Run the rectUnionArea sweep TWICE (once per axis). At each
// event x, the vertical contribution
// is |coveredLen(after) - coveredLen(before)|; sum those, then
// repeat with x/y swapped.
```

11.1.8 Transforms

COORDINATE TRANSFORMS - the tricks that turn a hard metric into an easy one.

```
// CHEBYSHEV <-> MANHATTAN. rot45(p) = (x+y, x-y).
// Manhattan distance in the original = Chebyshev distance after
//   ↪ rot45 (and vice versa).
// Use it for: "max Manhattan distance over a set" (becomes max
//   ↪ over 4 coordinate extremes),
// "count pairs within Manhattan distance k", chessboard-king
//   ↪ problems.
template <class P> P rot45(P p) { return P(p.x + p.y, p.x - p.y);
//   ↪ }
template <class P> P unrot45(P p) { return P((p.x + p.y) / 2,
//   ↪ (p.x - p.y) / 2); }
// Max Manhattan distance between any two of the points, O(n).
ll maxManhattan(const vector<Pi>& v) {
    ll best = 0;
    for (int s = 0; s < 2; s++) { // the
//   ↪ 2^1 sign patterns for (x +- y)
        ll mn = LLONG_MAX, mx = LLONG_MIN;
        for (auto& p : v) { ll t = s ? p.x - p.y : p.x + p.y; mn
//   ↪ = min(mn, t), mx = max(mx, t); }
            best = max(best, mx - mn);
        }
        return best;
    }
// In d dimensions the same idea uses all 2^(d-1) sign patterns.

// MANHATTAN MST - the MST under L1 distance, O(n log n) instead
//   ↪ of O(n^2).
// Key fact: every point only needs edges to its nearest neighbour
//   ↪ in each of 8 octants; by
// symmetry 4 sweeps suffice (rotate/reflect the plane between
//   ↪ sweeps).
vector<array<ll, 3>> manhattanEdges(vector<Pi> ps) { // {w,
//   ↪ i, j}; indices stay ORIGINAL
    int n = ps.size();
    vector<int> id(n);
    iota(all(id), 0);
    vector<array<ll, 3>> e;
    for (int s = 0; s < 4; s++) {
        sort(all(id), [&](int a, int b) { return ps[a].x +
//   ↪ ps[a].y < ps[b].x + ps[b].y; });
        map<ll, int> act; // key =
//   ↪ -y, value = index
        for (int i : id) {
            for (auto it = act.lower_bound(-ps[i].y); it !=
//   ↪ act.end(); act.erase(it++)) {
                int j = it->second;
                Pi d = ps[i] - ps[j];
                if (d.y > d.x) break;
                e.push_back({d.x + d.y, (ll)i, (ll)j});
            }
            act[-ps[i].y] = i;
        }
        for (auto& p : ps) { if (s & 1) p.x = -p.x; else
//   ↪ swap(p.x, p.y); } // next octant pair
    }
    return e; // feed
}

// LINEAR TRANSFORMATION mapping p0->q0 and p1->q1 (rotation +
//   ↪ scaling + translation).
Pd linTrans(Pd p0, Pd p1, Pd q0, Pd q1, Pd r) {
    Pd dp = p1 - p0, dq = q1 - q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + Pd((r - p0).cross(num), (r - p0).dot(num)) /
//   ↪ dp.dist2();
}

// POINT <-> LINE DUALITY: point (a, b) <-> line y = a*x - b.
// "max points on a common line" becomes "max lines through a
//   ↪ common point"
// convex hull of points <-> upper/lower envelope of lines
//   ↪ (this IS the CHT correspondence)
// incidences are preserved, above/below is preserved.
```

11.1.9 KDTree

KD-TREE over 2D points: nearest-neighbour and "count points in a rectangle".

```
// Build O(n log n); NN query O(log n) expected, O(n) worst for
//   ↪ NN; the O(sqrt n) bound is for RECTANGLE queries; rectangle
//   ↪ count O(sqrt n + k).
// Use when coordinates are large/sparse and a grid or sort-sweep
//   ↪ will not do.
struct KD {
    struct Node { // the
        Pi p; // splitting point
        ll x0 = LLONG_MAX, x1 = LLONG_MIN, y0 = LLONG_MAX, y1 =
//   ↪ LLONG_MIN; // bounding box
        int l = -1, r = -1, cnt = 0;
    };
    vector<Node> t;
    vector<Pi> pts;

    KD(vector<Pi> v) : pts(v) { if (!v.empty()) build(0,
//   ↪ v.size(), false); }

    int build(int lo, int hi, bool divX) {
        if (lo >= hi) return -1;
        int id = t.size();
        t.push_back({});
        int mid = (lo + hi) / 2;
        nth_element(pts.begin() + lo, pts.begin() + mid,
//   ↪ pts.begin() + hi,
//   ↪ [&](Pi a, Pi b) { return divX ? a.x < b.x :
//   ↪ a.y < b.y; });
        Node nd;
        nd.p = pts[mid], nd.cnt = hi - lo;
        for (int i = lo; i < hi; i++)
            nd.x0 = min(nd.x0, pts[i].x), nd.x1 = max(nd.x1,
//   ↪ pts[i].x),
            nd.y0 = min(nd.y0, pts[i].y), nd.y1 = max(nd.y1,
//   ↪ pts[i].y);
        t[id] = nd;
        int L = build(lo, mid, !divX), R = build(mid + 1, hi,
//   ↪ !divX);
        t[id].l = L, t[id].r = R;
        return id;
    }
    ll boxDist(int i, Pi q) const { //
//   ↪ squared distance from q to the box
        const Node& n = t[i];
        ll dx = max({(ll)0, n.x0 - q.x, q.x - n.x1}), dy =
//   ↪ max({(ll)0, n.y0 - q.y, q.y - n.y1});
        return dx * dx + dy * dy;
    }
    void nn(int i, Pi q, ll& best, Pi& bp, bool skipSelf) const {
        if (i < 0 || boxDist(i, q) >= best) return;
        ll d = (t[i].p - q).dist2();
        if (d < best && !(skipSelf && d == 0)) best = d, bp =
//   ↪ t[i].p;
        int a = t[i].l, b = t[i].r;
        if (a >= 0 && b >= 0 && boxDist(b, q) < boxDist(a, q))
//   ↪ swap(a, b);
        nn(a, q, best, bp, skipSelf), nn(b, q, best, bp,
//   ↪ skipSelf);
    }
    // skipSelf skips EVERY point at distance 0, not just q itself
    // - with duplicate input points
    // that is wrong for "nearest OTHER point". If duplicates are
    //   ↪ possible, count multiplicity
    // separately and treat a repeated point as its own nearest
    //   ↪ neighbour at distance 0.
    pair<ll, Pi> nearest(Pi q, bool skipSelf = false) const {
        if (t.empty()) return {llinf, Pi()};
        ll best = LLONG_MAX; Pi bp;
        nn(0, q, best, bp, skipSelf);
        return {best, bp}; //
    }
    // {squared distance, point}
    int count(int i, ll x0, ll y0, ll x1, ll y1) const { //
//   ↪ #points in [x0,x1] x [y0,y1]
        if (i < 0 || t[i].x1 < x0 || t[i].x0 > x1 || t[i].y1 < y0
//   ↪ || t[i].y0 > y1) return 0;
        if (x0 <= t[i].x0 && t[i].x1 <= x1 && y0 <= t[i].y0 &&
//   ↪ t[i].y1 <= y1) return t[i].cnt;
        int r = (x0 <= t[i].p.x && t[i].p.x <= x1 && y0 <=
//   ↪ t[i].p.y && t[i].p.y <= y1);
        return r + count(t[i].l, x0, y0, x1, y1) + count(t[i].r,
//   ↪ x0, y0, x1, y1);
    }
    int count(ll x0, ll y0, ll x1, ll y1) const { return
//   ↪ t.empty() ? 0 : count(0, x0, y0, x1, y1); }
```

};

11.1.10 Triangulation

EAR CLIPPING - triangulates any SIMPLE polygon (convex or not, no self-intersections, no holes)

```
// Needs: Core.cpp, Polygon.cpp (area2).
// into n-2 triangles,  $O(n^2)$ . Exact with  $P<ll>$ . Returns index
//   ↳ triples into the ORIGINAL vector.
// An "ear" is a convex vertex whose triangle contains no other
//   ↳ vertex of the polygon.
// USES: area/integral of anything over a polygon, rendering,
//   ↳ splitting a region for a sweep,
// point-location preprocessing, computing a polygon's centre of
//   ↳ mass with a weight function.
template <class P>
vector<array<int, 3>> triangulate(const vector<P>& poly) {
    int n = poly.size();
    vector<array<int, 3>> tri;
    if (n < 3) return tri;
    vector<int> id(n);
    iota(all(id), 0);
    if (area2(poly) < 0) reverse(all(id));
    ↳ work counter-clockwise
    auto inTri = [&](int a, int b, int c, int q) {
        ↳ closed triangle
        return poly[a].cross(poly[b], poly[q]) >= 0 &&
        poly[b].cross(poly[c], poly[q]) >= 0 &&
        poly[c].cross(poly[a], poly[q]) >= 0;
    };
    while (id.size() > 2) {
        int m = id.size(), cut = -1, flat = -1;
        for (int i = 0; i < m && cut < 0; i++) {
            int a = id[(i + m - 1) % m], b = id[i], c = id[(i +
            ↳ 1) % m];
            auto o = poly[a].cross(poly[b], poly[c]);
            if (o == 0) { flat = i; continue; }
            ↳ // collinear vertex: just drop it
            if (o < 0) continue;
            ↳ // reflex
            bool ok = true;
            for (int j = 0; j < m && ok; j++)
                if (id[j] != a && id[j] != b && id[j] != c &&
                ↳ inTri(a, b, c, id[j])) ok = false;
            if (ok) cut = i, tri.push_back({a, b, c});
        }
        if (cut < 0) cut = flat;
        if (cut < 0) break;
        ↳ // not a simple polygon
        id.erase(id.begin() + cut);
    }
    return tri;
}
// FASTER: monotone decomposition (sweep,  $O(n \log n)$ ) then
//   ↳ triangulate each monotone piece in  $O(n)$ .
// Only worth writing for n beyond ~2000; ear clipping handles
//   ↳ everything a contest throws at you.
// DELAUNAY TRIANGULATION maximises the minimum angle. Cheap way
//   ↳ to get one: lift each point to the
// paraboloid  $(x, y, x^2+y^2)$  and take the LOWER hull of the 3D
//   ↳ convex hull - its faces project to
// the Delaunay triangles. The dual graph is the VORONOI DIAGRAM.
// Delaunay contains: the Euclidean minimum spanning tree, the
//   ↳ nearest-neighbour graph, and the
// closest pair. So "EMST of points in the plane" = Delaunay
//   ↳ ( $O(n \log n)$  edges) + Kruskal.
// In-circle test (a,b,c counter-clockwise): d is strictly
//   ↳ inside the circumcircle of a,b,c iff
// the  $3 \times 3$  determinant of rows  $(a-d, |a-d|^2)$ ,  $(b-d, |b-d|^2)$ ,
//   ↳  $(c-d, |c-d|^2)$  is  $> 0$ . [__int128]
// ART GALLERY THEOREM: floor(n/3) guards always suffice for a
//   ↳ simple n-gon, and are sometimes
// necessary (comb polygon). Proof = triangulate, 3-colour the
//   ↳ triangulation graph, take the
// smallest colour class. The triangulation above is exactly what
//   ↳ that construction needs.
// POLYGON WITH HOLES: connect each hole to the outer boundary
//   ↳ with a "bridge" edge (pick the hole's
// rightmost vertex, shoot a ray right, split the hit edge) to get
//   ↳ one simple polygon, then clip.
```

11.1.11 PolygonOps

POLYGON AGAINST POLYGON: clipping, intersection, containment, distance, and the $O(\log n)$

```
// Needs: Core.cpp, Lines.cpp, Polygon.cpp.
// queries you get once one side is a convex hull.

// --- IS THIS POLYGON SIMPLE?  $O(n^2)$ , exact on  $P<ll>$ . ---
// Every polygon routine (inPoly, area, triangulate, Pick,
//   ↳ polyUnion) silently returns nonsense on
// a self-intersecting polygon, so this is also the oracle you
//   ↳ stress-test them with.
template <class P> bool isSimple(const vector<P>& v) {
    int n = v.size();
    if (n < 3) return false;
    for (int i = 0; i < n; i++) {
        if (v[i] == v[(i + 1) % n]) return false;
        ↳ // zero-length edge
        for (int j = i + 1; j < n; j++) {
            int a = i, b = (i + 1) % n, c = j, d = (j + 1) % n;
            if (a == c || a == d || b == c || b == d) {
                ↳ // adjacent edges: they may only
                if (a == d || b == c) {
                    ↳ // share the one common vertex
                    P sh = (a == d) ? v[a] : v[b];
                    for (P q : {v[a], v[b], v[c], v[d]})
                        if (!q == sh) && onSeg(q, v[a], v[b]) &&
                        ↳ onSeg(q, v[c], v[d]) return false;
                }
                continue;
            }
            if (segCross(v[a], v[b], v[c], v[d])) return false;
        }
    }
    return true;
}
// --- WINDING NUMBER. inPoly uses the EVEN-ODD rule, which is
//   ↳ wrong for a self-intersecting
// outline and for "polygon with holes given as one vertex list".
// The nonzero rule uses this. ---
template <class P> int windingNumber(const vector<P>& v, P p) {
    ↳ // 0 = outside (nonzero rule)
    int n = v.size(), w = 0;
    for (int i = 0; i < n; i++) {
        P a = v[i], b = v[(i + 1) % n];
        if (onSeg(p, a, b)) return 0;
        ↳ // on the boundary: caller decides
        if (a.y <= p.y) { if (b.y > p.y && a.cross(b, p) > 0)
        ↳ w++; }
        else if (b.y <= p.y && a.cross(b, p) < 0) w--;
    }
    return w;
}
// --- SUTHERLAND-HODGMAN: clip any polygon by a CONVEX window,
//   ↳  $O(n * m)$ . ---
// The window must be convex and counter-clockwise; the subject
//   ↳ may be concave.
// This is the general "intersect with a convex region" and it is
//   ↳ just 'cut' in a loop.
template <class P> vector<P> clipByConvex(vector<P> subject,
    ↳ const vector<P>& win) {
    int m = win.size();
    for (int i = 0; i < m && !subject.empty(); i++)
        subject = cut(subject, win[i], win[(i + 1) % m]);
    return subject;
}
// --- CONVEX intersect CONVEX in  $O(n + m)$  is the classic O'Rourke
//   ↳ walk, but clipByConvex is  $O(nm)$  and
// fits on one line - at contest sizes (n, m <= a few thousand)
//   ↳ prefer it. Use halfPlaneInter
// (06) when you have the polygons as half-plane constraints
//   ↳ instead of vertex lists.

// --- MINIMUM DISTANCE BETWEEN TWO CONVEX POLYGONS,  $O(n * m)$  as
//   ↳ written. ---
// 0 if they touch or overlap. The  $O(n+m)$  rotating-calipers
//   ↳ version exists but this is 6 lines and
// the distance is a floating answer anyway. Equivalent trick:
//   ↳ dist(origin, minkowski(A, -B)).
template <class P> ld convexDist(const vector<P>& A, const
    ↳ vector<P>& B) {
    for (P p : A) if (inPoly(B, p, false)) return 0;
    for (P p : B) if (inPoly(A, p, false)) return 0;
    ld best = 1e18;
    int n = A.size(), m = B.size();
    for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) {
        P a = A[i], b = A[(i + 1) % n], c = B[j], d = B[(j + 1) %
        ↳ m];
```

```

    if (segCross(a, b, c, d)) return 0;
    best = min(best, segSegDist(a, b, c, d));
}
return best;
}
// --- TANGENTS FROM A POINT TO A CONVEX POLYGON, O(n) here (O(log
    ↪ n) with a binary search). ---
// Returns the two vertex indices l, r such that the whole polygon
    ↪ lies to one side of p->h[l]
// and of p->h[r]: the visible arc is h[r], h[r+1], ..., h[l]. p
    ↪ must be strictly OUTSIDE.
// USES: visibility ("how much of the wall can I see"),
    ↪ shadow/silhouette, and "add p to the hull".
template <class P> pair<int, int> pointTangents(const vector<P>&
    ↪ h, P p) {
    int n = h.size(), l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (p.cross(h[i], h[l]) > 0) l = i;
        ↪ // h[l] is the CCW-most
        if (p.cross(h[i], h[r]) < 0) r = i;
        ↪ // h[r] is the CW-most
    }
    return {l, r};
}
// --- LINE x CONVEX POLYGON in O(log n): which two edges does the
    ↪ line a->b cross? ---
// Returns {-1,-1} for no hit; {i,-1} for a single touched corner
    ↪ i; {i,j} when the line crosses
// the edges (i,i+1) and (j,j+1). h must be a strictly convex CCW
    ↪ hull (no collinear vertices).
// USES: "length of the segment inside the polygon" per query, in
    ↪ O(log n) instead of O(n).
template <class P> pair<int, int> lineHull(const vector<P>& h, P
    ↪ a, P b) {
    int n = h.size();
    int endA = extreme(h, (a - b).perp()), endB = extreme(h, (b -
    ↪ a).perp());
    auto cmpL = [&](int i) { return sgn(a.cross(h[i], b)); };
    if (cmpL(endA) < 0 || cmpL(endB) > 0) return {-1, -1};
    ↪ // whole hull on one side
    auto bs = [&](int lo, int hi) {
        ↪ // last index still >= 0
        int step = (hi - lo + n) % n;
        for (int s = step; s > 0; s /= 2)
            while (s && cmpL((lo + s) % n) >= 0) lo = (lo + s) %
        ↪ n, step -= s;
        return lo;
    };
    int i = bs(endB, endA), j = bs(endA, endB);
    if (i == j) return {i, -1};
    return {i, j};
}
/* MORE POLYGON FACTS AND RECIPES
AREA OF INTERSECTION of two CONVEX polygons: clipByConvex then
    ↪ area(). For two ARBITRARY simple
polygons, triangulate one (10 - Triangulation.cpp) and clip
    ↪ each triangle by the other.
AREA OF THE UNION: |A| + |B| - |A n B|, or polyUnion (07 -
    ↪ Unions.cpp) for many polygons at once.
SYMMETRIC DIFFERENCE: |A| + |B| - 2|A n B|.
POINT IN A POLYGON WITH HOLES: outer boundary CCW, every hole CW,
    ↪ then the sign of the total
winding number is the answer (this is exactly why windingNumber
    ↪ exists).
CONVEX POLYGON CONTAINS CONVEX POLYGON: every vertex of B is in A
    ↪ (inConvex, O(m log n)).
CENTROID OF THE PERIMETER (not of the area): sum of edge
    ↪ midpoints weighted by edge LENGTH.
The two differ - "balance the wire frame" vs "balance the
    ↪ plate".
POLYGON DIAMETER / WIDTH / MIN RECTANGLE: 03 - Polygon.cpp
    ↪ (rotating calipers).
REGULAR n-GON of circumradius R centred at c: c + R*(cos(2*pi*k/n
    ↪ + t), sin(2*pi*k/n + t)).
Its area is (n/2) R^2 sin(2 pi/n); with INRADIUS r it is n r^2
    ↪ tan(pi/n).
SPLIT A POLYGON BY A LINE into both halves: cut(v, s, e) and
    ↪ cut(v, e, s).
LARGEST INSCRIBED / SMALLEST ENCLOSING: smallest enclosing circle
    ↪ is mec (04 - Circles.cpp);
smallest enclosing rectangle is minRectArea; largest inscribed
    ↪ CIRCLE in a convex polygon is a
binary search on the radius plus a half-plane intersection of
    ↪ the inward-offset edges;
largest inscribed AXIS-ALIGNED rectangle needs a sweep and is
    ↪ genuinely hard - do not improvise.
OFFSET / BUFFER a convex polygon by d: push every edge outward by

```

↪ d along its normal and
 intersect the half-planes (06 - HalfPlaneIntersection.cpp). The
 ↪ exact offset of a NON-convex
 polygon has circular arcs at the reflex corners - approximate
 ↪ them.
 TRIANGLE CONTAINS POINT, exact: the three orientations all have
 ↪ the same sign (or one is 0 for
 the boundary). Faster than inPoly for n = 3 and it is the inner
 ↪ loop of triangulate. */

11.1.12 Primitives

THE SMALL-FUNCTION LAYER. Individually trivial, collectively the thing you actually spend

```
// Needs: Core.cpp, Lines.cpp, Circles.cpp.
// contest minutes re-deriving. Everything here is one to six
// ↪ lines.

// --- ANGLE BISECTORS OF TWO LINES. Two lines have TWO bisectors,
// ↪ perpendicular to each other.
// Direction of the internal one = the sum of the unit direction
// ↪ vectors; external = difference.
// (Take the pair whose sign matches the side you want; if the
// ↪ lines are parallel the bisector is
// the parallel line halfway between them.)
template <class P> pair<P, P> lineBisectors(P d1, P d2) {
    // ↪ // returns the two DIRECTIONS
    P u = d1.unit(), v = d2.unit();
    return {u + v, u - v};
}

// Internal bisector of the angle at b in triangle a-b-c, as a
// ↪ direction from b:
template <class P> P angleBisector(P a, P b, P c) { return (a -
    // ↪ b).unit() + (c - b).unit(); }
// The internal bisector from A meets BC at the point dividing it
// ↪ in ratio AB : AC.
template <class P> P bisectorFoot(P a, P b, P c) {
    ld x = (a - b).dist(), y = (a - c).dist();
    return (b * y + c * x) / (x + y);
}

// --- ROTATE / SCALE ABOUT AN ARBITRARY CENTRE (rot() in Core is
// ↪ about the origin). ---
template <class P> P rotAbout(P p, P c, ld a) { return c + (p -
    // ↪ c).rot(a); }
template <class P> P scaleAbout(P p, P c, ld k) { return c + (p -
    // ↪ c) * k; }
// Reflect p across the POINT c (a 180-degree rotation):
template <class P> P reflPoint(P p, P c) { return c * 2 - p; }
// --- SIGNED AREA OF A TRIANGLE, and the barycentric coordinates
// ↪ of p in a-b-c. ---
template <class P> auto triArea2(P a, P b, P c) { return
    // ↪ a.cross(b, c); } // 2 * signed
template <class P> array<ld, 3> barycentric(P a, P b, P c, P p) {
    ld s = a.cross(b, c);
    return {(ld)p.cross(b, c) / s, (ld)a.cross(p, c) / s,
    // ↪ (ld)a.cross(b, p) / s};
}
// p is inside the triangle iff all three coordinates are >= 0
// ↪ (they always sum to 1).
template <class P> bool inTriangle(P a, P b, P c, P p) {
    // ↪ // exact, boundary included
    int s1 = sgn(a.cross(b, p)), s2 = sgn(b.cross(c, p)), s3 =
    // ↪ sgn(c.cross(a, p));
    return (s1 >= 0 && s2 >= 0 && s3 >= 0) || (s1 <= 0 && s2 <= 0
    // ↪ && s3 <= 0);
}

// --- SEGMENT x CIRCLE: clip the segment a-b to the disc,
// ↪ returning the surviving piece. ---
template <class P> vector<P> segCircle(P a, P b, P c, ld r) {
    auto v = circleLine(c, r, a, b);
    // ↪ // 0, 1 or 2 points
    vector<P> res;
    for (P p : v) if (onSeg(p, a, b)) res.push_back(p);
    return res;
}

// --- CIRCLE THROUGH THREE POINTS = circumcenter (O4 -
// ↪ Circles.cpp); its radius = circumradius.
// CIRCLE THROUGH TWO POINTS WITH A GIVEN RADIUS: the two centres
// ↪ are m +- h * (b-a).perp().unit()
// where m is the midpoint and h = sqrt(r^2 - |b-a|^2/4). None if
// ↪ 2r < |b-a|.
template <class P> vector<P> circlesThrough(P a, P b, ld r) {
    ld d2 = (b - a).dist2(), h2 = r * r - d2 / 4;
    if (d2 < eps || h2 < 0) return {};
    // ↪ // a = b: infinitely many
    P m = (a + b) / 2, dir = (b - a).perp().unit() * sqrt1(h2);
    return h2 < eps ? vector<P>{m} : vector<P>{m - dir, m + dir};
}

// --- APOLLONIUS CIRCLE: the locus of p with |pa| / |pb| = k (k
// ↪ != 1) is a circle whose diameter
// endpoints are the two points dividing ab internally and
// ↪ externally in ratio k : 1.
template <class P> pair<P, ld> apollonius(P a, P b, ld k) {
    P u = (b * k + a) / (k + 1), v = (b * k - a) / (k - 1);
    // ↪ // internal, external
    return {(u + v) / 2, (u - v).dist() / 2};
}

// --- IS THE QUADRILATERAL a,b,c,d CONCYCLIC? Exact on P<ld> with
```

```
    // ↪ ↪ __int128 (in-circle test). ---
    // > 0: d strictly inside the circumcircle of a CCW triangle
    // ↪ a,b,c; == 0: concyclic.
template <class P> int inCircle(P a, P b, P c, P d) {
    auto f = [&](P p) { return (__int128)(p - d).dist2(); };
    __int128 det = (__int128)(a.x - d.x) * ((b.y - d.y) * f(c) -
    // ↪ (c.y - d.y) * f(b))
    // ↪ - (__int128)(a.y - d.y) * ((b.x - d.x) * f(c) -
    // ↪ (c.x - d.x) * f(b))
    // ↪ + f(a) * ((__int128)(b.x - d.x) * (c.y - d.y) -
    // ↪ (__int128)(c.x - d.x) * (b.y - d.y));
    return (det > 0) - (det < 0);
}

// --- MAXIMUM POINTS COVERED BY A CIRCLE OF RADIUS r, O(n^2 log
// ↪ n). ---
// For each i, every j within 2r gives an ARC of centre-angles
// ↪ that cover both; sweep the arcs.
template <class P> int maxCovered(const vector<P>& p, ld r) {
    int n = p.size(), best = 1;
    for (int i = 0; i < n; i++) {
        vector<pair<ld, int>> ev;
        int base = 1;
        // ↪ // p[i], plus its duplicates
        for (int j = 0; j < n; j++) if (j != i) {
            ld d = (p[j] - p[i]).dist();
            if (d > 2 * r + eps) continue;
            if (d < eps) { base++; continue; }
            // ↪ // DUPLICATE of p[i]: the
            // ↪ // division below would be 0/0
            // ↪ // and it is always covered
            ld th = (p[j] - p[i]).angle(), ph = acos1(min((ld)1,
            // ↪ max((ld)-1, d / (2 * r))));
            ld s = th - ph, e = th + ph;
            // ↪ // the arc of valid centres
            while (s < -PI) s += 2 * PI;
            // ↪ // normalise, then SPLIT the
            e = s + 2 * ph;
            // ↪ // arcs that wrap past +pi -
            // ↪ // 0 = arc opens, 1 = arc closes. Pair-sorting then
            // ↪ puts the OPEN first at an equal
            // ↪ // angle, which is what makes a TANGENTIAL arc (d ==
            // ↪ 2r, so the arc is a single point)
            // ↪ // still count - with {-1}/{-1} the close sorts first
            // ↪ and the point is lost.
            if (e <= PI) ev.push_back({s, 0}), ev.push_back({e,
            // ↪ 1});
            else {
                ev.push_back({s, 0}), ev.push_back({PI, 1});
                ev.push_back({-PI, 0}), ev.push_back({e - 2 * PI,
            // ↪ 1});
            }
        }
        sort(all(ev));
        int cur = base;
        best = max(best, cur);
        for (auto& [a, t] : ev) best = max(best, cur += t ? -1 :
            // ↪ 1);
        }
        return best;
    }
    // ↪ // the circle passes through p[i]
}

// --- GEOMETRIC MEDIAN (minimise the SUM of distances). Nested
// ↪ ternary search - the objective is
// convex, so this is exact to any eps and cannot get stuck.
// ↪ Weiszfeld iteration is faster but
// breaks when the iterate lands exactly on a sample point.
template <class P> P geoMedian(const vector<P>& p) {
    auto f = [&](ld x, ld y) { ld s = 0; for (auto& q : p) s +=
    // ↪ (P{x, y} - q).dist(); return s; };
    auto fy = [&](ld x) { ld lo = -2e9, hi = 2e9;
        for (int it = 0; it < 100; it++) { ld a = lo + (hi - lo)
            // ↪ / 3, b = hi - (hi - lo) / 3;
            // ↪ f(x, a) < f(x, b) ? hi = b : lo = a; } return (lo +
            // ↪ hi) / 2; };
    ld lo = -2e9, hi = 2e9;
    for (int it = 0; it < 100; it++) {
        ld a = lo + (hi - lo) / 3, b = hi - (hi - lo) / 3;
        f(a, fy(a)) < f(b, fy(b)) ? hi = b : lo = a;
    }
    ld x = (lo + hi) / 2;
    return {x, fy(x)};
}

/* THE REST OF THE SMALL LAYER, as one-liners you can reconstruct
// ↪ instantly
COLLINEAR a, b, c a.cross(b, c) == 0
```


PARALLEL / PERPENDICULAR $d1.cross(d2) == 0$ / $d1.dot(d2) \rightarrow == 0$

SAME SIDE OF A LINE $sgn(a.cross(b, p)) == sgn(a.cross(b, q))$

POINT ON A RAY / SEGMENT $onRay, onSeg$ (02 - Lines.cpp)

DISTANCE point-line/segment $lineDist, segDist$ ray: $rayDist$
 $\rightarrow seg-seg: segSegDist$

PROJECT / REFLECT over a line $proj, refl$ (02 - Lines.cpp)

ANGLE between two vectors $atan2(cross, dot) \rightarrow$ signed, in $(-\pi, \pi]$. NEVER $acos(dot/|a||b|)$:
 $\rightarrow$ it loses all precision near 0 and π
 $\rightarrow$ and cannot give you a sign.

ROTATE 90 DEGREES $p.perp() = (-y, x)$. Twice = negation. This is the whole of "turn left"; you almost never need a general $rot()$ for a right angle.

SORT BY ANGLE $angCmp$ (01 - Core.cpp), NOT $atan2$ - $\rightarrow$ exact and 5x faster.

AREA OF A TRIANGLE $|cross| / 2$. Heron only if you are given the three SIDES, and then use the sorted stable form: sort $a \geq b \geq c$ and
 $A = \sqrt{(a+(b+c))(c-(a-b))(c+(a-b))(a+(b-c))} / 4$.

TRIANGLE CENTRES centroid $(a+b+c)/3$, circumcenter, incenter, orthocenter (04).
 EULER LINE: O, G, N, H are collinear
 $\rightarrow$ with $OG:GN:NH = 2:1:3$, and $H = A + B + C - 2O$. Nine-point centre = midpoint of OH, radius $R/2$.

TRIANGLE IDENTITIES $A = rs = abc/(4R)$; $R = abc/(4A)$; $r = A/s$; $r_a = A/(s-a)$
 law of cosines $c^2 = a^2 + b^2 - 2ab \cos C$
 law of sines $a/\sin A = 2R$
 $OI^2 = R(R - 2r) \Rightarrow$ EULER'S
 $\rightarrow$ INEQUALITY $R \geq 2r$
 Stewart (cevian d to BC split $m + n$): $b^2 m + c^2 n = a(d^2 + mn)$
 median $m_a = \sqrt{2b^2 + 2c^2 - a^2}/2$

CYCLIC QUADRILATERAL PTOLEMY: $AC*BD = AB*CD + BC*AD$ (in cyclic order). For ANY four points $AC*BD \leq AB*CD + BC*AD$, equality iff concyclic in that order.
 BRAHMAGUPTA area = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
 General quadrilateral (BRETSCHNEIDER, angle-free):
 $A = \sqrt{4p^2 q^2 - (b^2 + d^2 - a^2 - c^2)^2}/4$ with p, q the diagonals.

POWER OF A POINT $pow(p) = |pc|^2 - r^2$; for ANY line through p meeting the circle at A and B the signed product $PA*PB$ equals it. Outside: = tangent².
 RADICAL AXIS = equal power, a line perpendicular to the centre line.
 RADICAL CENTRE of three circles with non-collinear centres: the three pairwise axes are concurrent.

DESCARTES CIRCLE THEOREM four mutually tangent circles, curvatures $k_i = \pm 1/r_i$ (negative for the enclosing one):
 $(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$,
 so $k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1 k_2 + k_2 k_3 + k_3 k_1}$.

LATTICE POINTS on a segment: $\gcd(|dx|, |dy|) + 1$
 $\rightarrow$ endpoints included.
 inside a polygon: PICK, $I = A - B/2 + 1$ (simple polygons only).
 under a line, in $O(\log)$: floorSum (04 - Number Theory/01/03).

HELLY (2D) if every THREE of a family of convex sets meet, all of them meet.

RADON (2D) any FOUR points split into two sets whose hulls intersect.

CARATHEODORY (2D) a point in the hull of S is in the hull of at most THREE points of S - which is exactly why the minimum enclosing circle needs only 3.

ART GALLERY floor($n/3$) guards always suffice for a simple n -gon (triangulate, 3-colour, take the smallest class).
 $\rightarrow$ Orthogonal polygon: floor($n/4$).

EULER, PLANAR $V - E + F = 1 + C$. n lines in general position cut the plane into $1 + n + C(n, 2)$ regions, $C(n-1, 2)$ of

$\rightarrow$ them bounded; n circles in general position give $n^2 - n + 2$

$\rightarrow$ regions. point $(a, b) \leftrightarrow$ line $y = ax - b$.
 $\rightarrow$ Incidence and above/below are preserved, so "max collinear points" becomes "max concurrent lines"
 $\rightarrow$ envelope. */
 and a convex hull becomes an upper

11.2 Geometry 3D

Problem Solving

11.2.1 Geometry3D

3D point / vector, planes, and the few 3D things that actually appear.

```
template <class T>
struct P3 {
    typedef P3 Pt; T x = 0, y = 0, z = 0;
    P3(T x = 0, T y = 0, T z = 0) : x(x), y(y), z(z) {}
    Pt operator+(Pt p) const { return {x + p.x, y + p.y, z +
    ↪ p.z}; }
    Pt operator-(Pt p) const { return {x - p.x, y - p.y, z -
    ↪ p.z}; }
    Pt operator*(T d) const { return {x * d, y * d, z * d}; }
    Pt operator/(T d) const { return {x / d, y / d, z / d}; }
    T dot(Pt p) const { return x * p.x + y * p.y + z * p.z; }
    Pt cross(Pt p) const { return {y * p.z - z * p.y, z * p.x - x
    ↪ * p.z, x * p.y - y * p.x}; }
    T dist2() const { return x * x + y * y + z * z; }
    ld dist() const { return sqrt1((ld)dist2()); }
    Pt unit() const { return *this / dist(); }
    ld phi() const { return atan2l(y, x); }
    ↪ // longitude,  $(-\pi, \pi]$ 
    ld theta() const { return atan2l(sqrt1((ld)(x * x + y * y)),
    ↪ z); } // colatitude,  $[0, \pi]$ 
    Pt rot(ld a, Pt ax) const {
    ↪ // Rodrigues, ccw around ax
        Pt u = ax.unit(); ld s = sinl(a), c = cosl(a);
        return u * dot(u) * (1 - c) + (*this) * c +
    ↪ u.cross(*this) * s;
    }
    bool operator<(Pt p) const { return tie(x, y, z) < tie(p.x,
    ↪ p.y, p.z); }
    bool operator==(Pt p) const { return tie(x, y, z) == tie(p.x,
    ↪ p.y, p.z); }
    friend istream& operator>>(istream& i, Pt& p) { return i >>
    ↪ p.x >> p.y >> p.z; }
    friend ostream& operator<<(ostream& o, Pt p) { return o <<
    ↪ p.x << ' ' << p.y << ' ' << p.z; }
};

typedef P3<ll> Pi3;
typedef P3<ld> Pd3;

// PLANE  $n \cdot \text{dot}(X) = d$ . Build from 3 points, or from a normal
    ↪ and a point.
template <class P> struct Plane {
    P n; ld d;
    Plane(P n = P(), ld d = 0) : n(n), d(d) {}
    Plane(P a, P b, P c) : n((b - a).cross(c - a)), d(n.dot(a)) {}
    ld side(P p) const { return n.dot(p) - d; }
    ↪ // sign = which side
    ld dist(P p) const { return fabs1(side(p)) / n.dist(); }
    P proj(P p) const { return p - n * (side(p) / n.dist2()); }
    P refl(P p) const { return p - n * (2 * side(p) / n.dist2()); }
    ↪ }
};

// LINE  $a \rightarrow b$  vs plane. {0 parallel-disjoint, 1 unique, -1 line
    ↪ lies in plane}
template <class P> pair<int, P> planeLine(const Plane<P>& pl, P
    ↪ a, P b) {
    ld da = pl.side(a), db = pl.side(b);
    if (fabs1(da - db) < eps) return {fabs1(da) < eps ? -1 : 0,
    ↪ P()};
    return {1, (b * da - a * db) / (da - db)};
}

// PLANE  $\times$  PLANE  $\rightarrow$  a line, given as {point on it, direction}.
    ↪ dir==0 means parallel.
template <class P> pair<P, P> planePlane(const Plane<P>& A, const
    ↪ Plane<P>& B) {
    P dir = A.n.cross(B.n);
    if (dir.dist2() < eps) return {P(), P()};
    P p = (B.n * A.d - A.n * B.d).cross(dir) / dir.dist2();
    return {p, dir};
}

// VOLUME of the tetrahedron abcd (signed;  $x6$  form is exact on
    ↪  $P3<ll>$  only up to  $|coordinate| \sim 5.7e5$  ( $|vol6| \leq 48 \cdot C^3$ );
// above that make the return type --int128).
template <class P> auto vol6(P a, P b, P c, P d) { return (b -
    ↪ a).cross(c - a).dot(d - a); }
template <class P> ld tetraVolume(P a, P b, P c, P d) { return
    ↪ fabs1((ld)vol6(a, b, c, d)) / 6; }

// SPHERE  $\times$  LINE: points where line  $a \rightarrow b$  meets the sphere  $(o, r)$ .
template <class P> vector<P> sphereLine(P o, ld r, P a, P b) {
    P d = b - a, f = a - o;
    ld A = d.dist2(), B = 2 * f.dot(d), C = f.dist2() - r * r, D
    ↪ = B * B - 4 * A * C;
    if (D < -eps) return {};
```

```
D = sqrt1(max((ld)0, D));
if (D < eps) return {a + d * (-B / (2 * A))};
return {a + d * ((-B - D) / (2 * A)), a + d * ((-B + D) / (2
    ↪ * A))};
}

// GREAT-CIRCLE distance on a sphere of radius R, inputs in
    ↪ DEGREES (haversine).
ld greatCircle(ld lat1, ld lon1, ld lat2, ld lon2, ld R) {
    auto rad = [] (ld d) { return d * PI / 180; };
    ld dl = rad(lat2 - lat1) / 2, dg = rad(lon2 - lon1) / 2;
    ld h = sinl(dl) * sinl(dl) + cosl(rad(lat1)) *
    ↪ cosl(rad(lat2)) * sinl(dg) * sinl(dg);
    return 2 * R * asinl(sqrt1(min((ld)1, h)));
}
```

12 Problem Solving

12.1 Reframing

REFRAMING: CHANGE THE PROBLEM, NOT THE EFFORT

```
/* ===== REFRAMING: CHANGE THE PROBLEM, NOT THE EFFORT
   =====
```

This chapter has no algorithms - it has the moves you make BEFORE
 ↳ you know which template to
 paste. Stuck for 10 minutes? Do not re-read the statement a
 ↳ fourth time. Walk the six questions,
 then walk the TRIGGERS below and try each against your problem.

THE SIX QUESTIONS

1. What object am I counting / optimising over? Say it in one sentence with no input format.
2. What is INVARIANT under the allowed operation?
3. What does the ANSWER look like (a prefix? one of the inputs? a threshold? always ≤ 2 ?)
4. Can I CHECK a candidate faster than I can find one?
 ↳ -> binary search the answer
5. What do the constraints forbid, and what does the leftover
 ↳ budget exactly equal?
6. What would the brute force print for $n = 1..8$?

--- REVERSE THE PROCESS ---

Run time backwards: deletions become insertions, splits become
 ↳ merges. Every structure is
 add-friendly and delete-hostile (DSU above all), so reversing
 ↳ turns impossible into trivial.
 TRIGGER: things are removed one at a time and you must answer
 ↳ after each removal.
 EXAMPLE: CF 722C Destroying Array - process destructions in
 ↳ reverse, union with restored
 neighbours. Also CSES 1163 Traffic Lights.
 TRAP: emit answers in reverse too, and check the off-by-one at
 ↳ both ends with $n = 1$.

--- WORK BACKWARDS FROM THE GOAL ---

Define the answer at terminal states and propagate backwards.
 ↳ Standard for games and for
 "what must have been true one step earlier".
 TRIGGER: few terminal states, huge forward branching, small
 ↳ backward branching.
 TRAP: on a CYCLIC game graph this is NOT a plain DP - you need
 ↳ retrograde BFS with out-degree
 counters (a position is losing only once ALL successors are
 ↳ known winning). See 10 - Game
 Theory. Writing win[v] = OR over children on a cyclic graph
 ↳ silently mislabels draws.

--- COMPLEMENTARY COUNTING ---

Count the bad ones and subtract; count $P(X \leq x)$ instead of $P(X = x)$.
 TRIGGER: "at least one", "there exists", "not all equal", "some
 ↳ pair".
 EXAMPLE: $E[\max] = \sum \text{over } x \text{ of } P(\max \geq x)$, with $P(\max \leq x) =$
 ↳ $(x/n)^k$.
 TRAP: the complement is easy for AT MOST / AT LEAST, not for
 ↳ EXACTLY. Getting from one to the
 other is binomial inversion: $\text{exactly_}k = \sum_{j \geq k}$
 ↳ $(-1)^{j-k} C(j, k) \text{atleast_}j$.

--- THE CONTRIBUTION TECHNIQUE ---

Instead of "for each subarray, compute f", ask "for each
 ↳ element, in how many objects does it
 contribute, and how much". Turns $O(n^2)$ enumeration into $O(n)$.
 TRIGGER: "sum over all subarrays / pairs / subsets of [min |
 ↳ $\max$ | gcd | #distinct | bottleneck]".
 EXAMPLE: sum over subarrays of the minimum -> monotonic stack
 ↳ gives each element's span.
 Sum over pairs of the bottleneck -> Kruskal, each merge
 ↳ contributes $w * |A| * |B|$ pairs.
 TRAP: ties. With equal values the span must be strict on ONE
 ↳ side and non-strict on the other,
 or every duplicated minimum is counted twice. Decide which
 ↳ side BEFORE coding.

--- SWAP THE ORDER OF SUMMATION ---

$\sum_i \sum_{d | i} f(d) = \sum_d f(d) * \text{floor}(n/d)$. Move the
 ↳ outer loop to the variable whose
 inner count has a closed form.
 TRIGGER: a double sum whose inner range depends on the outer
 ↳ index; divisors, multiples,
 ancestors, intervals containing a point.
 EXAMPLE: sum of $\sigma(i)$ for $i \leq n = \sum_d d * \text{floor}(n/d)$,
 ↳ then divisor blocks -> $O(\sqrt{n})$.

--- LINEARITY OF EXPECTATION ---

$E[\sum X_i] = \sum E[X_i]$, with NO independence needed. Decompose
 ↳ into indicators.
 TRIGGER: "expected number of ...", or an expectation over a
 ↳ process you cannot untangle.
 TRAP: linearity does not survive $\max$, $\min$, or division. $E[\max]$
 ↳ $\neq \max E$, $E[1/X] \neq 1/E[X]$.
 For an expected maximum use the complementary-counting tail
 ↳ sum above.

--- FIX THE EXTREMUM ---

Add an artificial outer loop "for each choice of the maximum /
 ↳ the leftmost / the last move".
 Everything else becomes bounded relative to the fixed thing.
 TRIGGER: the statement mentions $\max$ or $\min$ over a range; a
 ↳ Cartesian-tree recursion is available.
 EXAMPLE: CF 1156E - fix the position of the maximum, then
 ↳ enumerate the SMALLER side.
 TRAP: always iterate the smaller side (small-to-large, $O(n \log$
 ↳ $n)$ total). Iterating the left
 side unconditionally is $O(n^2)$ on a sorted permutation - and
 ↳ it passes the samples.

--- DUALIZE ---

$\max\text{-flow} = \min\text{-cut}$; $\max$ bipartite matching = $\min$ vertex cover
 ↳ (Konig); $\min$ path cover of a DAG
 = $n - \text{matching}$; longest antichain = $\min$ chain cover (Dilworth);
 ↳ "a perfect matching exists" =
 "no Hall violator".
 TRIGGER: the statement asks for a MINIMUM removal / blocking
 ↳ set / number of chains - that is
 almost always the dual of a maximum you can compute.
 TRAP: Konig gives the SIZE for free, but recovering the cover
 ↳ needs the alternating-BFS
 construction $(L \setminus Z) \cup (R \text{ and } Z)$. And Konig is BIPARTITE
 ↳ ONLY.

--- MODEL IT AS A GRAPH ---

Objects -> nodes, relations -> edges. States can be tuples:
 ↳ (cell, fuel), (vertex, parity),
 (position, mask).
 TRIGGER: "can I get from A to B", "minimum number of
 ↳ operations", "are these constraints
 consistent", "each x points at exactly one y " (functional
 ↳ graph).
 TRAP: compute $|V| * |E|$ explicitly before coding. In a layered
 ↳ graph the answer is usually
 "any layer", not layer 0.

--- MODEL IT AS INTERVALS / A TIMELINE ---

Recast objects as [start, end] and use a sweep,
 ↳ greedy-by-endpoint, or a difference array.
 Tree subtrees become intervals via the Euler tour.
 TRIGGER: "from time a to time b", "covers l..r", "is available
 ↳ during", subtree aggregates.
 TRAP: decide ONCE whether [1,3] and [3,5] overlap, and encode
 ↳ it in exactly one place.

--- MODEL IT AS A POLYNOMIAL ---

"For every possible sum, count the ways" is a convolution.
 ↳ Multiplying GFs = combining
 independent choices; x^k = "used k of the budget".
 TRIGGER: modulus 998244353 (NTT is intended); independent parts
 ↳ whose sizes add.
 TRAP: convolving an array with itself double-counts $i \neq j$ and
 ↳ mis-counts $i = j$.

--- MODEL IT AS LINEAR ALGEBRA OVER GF(2) ---

Each element is a bit-vector; "can I make X" = "is X in the
 ↳ span"; the number of subsets giving
 a representable Y is $2^{(n - \text{rank})}$.
 TRIGGER: XOR anywhere; "product is a perfect square" (exponent
 ↳ parity vector over primes);
 switch-toggling puzzles; "each constraint satisfied an even
 ↳ number of times".
 TRAP: $2^{(n - \text{rank})}$ counts solutions only if the system is
 ↳ CONSISTENT - check first. And subtract
 the empty subset when the problem wants a non-empty one.

--- MODEL IT AS THE SUBSET LATTICE ---

Bitmask DP plus zeta/Mobius (SOS) to aggregate over submasks in
 ↳ $O(2^n n)$ instead of $O(3^n n)$.
 TRIGGER: $n \leq 20$ with a "set of chosen items" state, or a $\leq$
 ↳ 22-bit value domain with queries
 "over all y contained in x".
 TRAP: the bit loop goes OUTSIDE the mask loop. Swapping them
 ↳ gives a wrong-but-plausible array.
 Submask enumeration must be for $(s = m; ; s = (s-1) \& m) \{$

→ ...; if (!s) break; }.

--- THINK ABOUT WHAT THE ANSWER LOOKS LIKE ---

Characterise the optimum's SHAPE first: it is a prefix; it is
 → one of the inputs; it is at most
 2; it has a threshold structure. Then search only that shape.
 TRIGGER: the output is one small number regardless of n; the
 → problem screams "constructive".
 TRAP: "the answer is at most k" needs a matching lower-bound
 → example, or you print k-1.

--- CHANGE OF VARIABLES ---

Difference array (range add → two point updates; "make all
 → equal" → zero the differences),
 prefix sums (subarray → pair of points), Chebyshev ↔
 → Manhattan via $(x+y, x-y)$, logs
 (products → sums).
 TRIGGER: operations act on ranges; the objective is
 → $\max(|dx|, |dy|)$ or $|dx|+|dy|$.
 TRAP: C++ % is negative for negative operands - use $((x \% n) +$
 → $n) \% n$. Seed the empty prefix.

--- PER-BIT / PER-PRIME INDEPENDENCE ---

If the objective decomposes across bits (XOR/AND/OR sums,
 → gcd/lcm exponent-wise), solve 30 tiny problems and recombine.
 EXAMPLE: sum over pairs of $(a_i \text{ XOR } a_j) = \text{sum_b } 2^b * \text{cnt0_b} * \text{cnt1_b}$.
 TRAP: bit-independence holds for SUMS, not for COMPARISONS.
 → "Count pairs with $a_i \wedge a_j < k$ " is
 not separable - that needs a binary trie walked high bit
 → first.

--- REDUCE TO SOMETHING YOU ALREADY HAVE ---

"minimum adjacent swaps to sort" = inversions; "maximum
 → pairwise-compatible set" = matching;
 "minimise the maximum edge on a path" = MST bottleneck /
 → Kruskal reconstruction tree; "longest
 chain of pairs" = LIS; "pairwise gap constraints" = difference
 → constraints + Bellman-Ford;
 "each person points at one other" = functional graph.
 TRIGGER: you can state the problem without the story. Do that,
 → then match it.
 TRAP: the disguise is usually ALMOST the classic - one extra
 → constraint breaks the reduction.
 Check the reduction on the samples by hand before trusting it.

→ */

12.2 SearchAndProof

SEARCH: TURN "FIND" INTO "CHECK" =====

/* ===== SEARCH: TURN "FIND" INTO "CHECK" =====

BINARY SEARCH ON THE ANSWER. If feasible(x) is monotone, stop
 → optimising and find the boundary.
 TRIGGER: "minimise the maximum", "maximise the minimum",
 → "minimum time such that", "possible
 within k". Also: the answer is an integer in a huge range and
 → no formula presents itself.
 TRAPS: (1) the check itself overflows - break early once a
 → running count passes the target;
 (2) 'hi' must be PROVABLY feasible - seed it with the trivial
 → answer, not 1e9;
 (3) if the predicate is not monotone you get a plausible
 → wrong answer and no crash.

BINARY SEARCH THE VALUE, COUNT WITH A SECOND SEARCH. For "k-th
 → smallest X" over a set you cannot
 materialise (pairs, products, distances, subarray sums): binary
 → search x, count how many are
 ≤ x. The counting routine is the real problem.
 TRAP: use firstTrue(count(x) ≥ k) so the returned x actually
 → occurs.

PARAMETRIC SEARCH / LAGRANGIAN (Aliens, wqs). Replace "exactly k"
 → by a penalty per item and
 binary search the penalty. Requires the optimum to be CONVEX in
 → k. See 08 - DP/01/07.
 TRAP: verify convexity by brute-forcing f(k) for small n; and
 → the inner DP must break ties
 toward the largest count or the final subtraction is off by
 → an arbitrary amount.

TERNARY SEARCH. Unimodal f: cut a third each step. Reals: fixed
 → ~200 iterations (never an eps
 loop). Integers: while (hi - lo > 2), then scan the survivors.
 TRAP: PLATEAUS KILL IT. $f(m1) == f(m2)$ on a flat region can
 → discard the true optimum. For any
 discretely convex f use binary search on the difference
 → instead:
 $\text{argmin} = \text{firstTrue}(lo, hi-1, [\&](ll\ x)\{ \text{return } f(x+1) \geq$
 → $f(x); \})$;

TWO POINTERS - and why it is legal. Legal precisely when validity
 is MONOTONE IN l for fixed r:

if [l, r] is valid then so is [l', r] for every l' ≥ l.
 TRAP: "subarray with sum ≤ S" is two-pointerable with positive
 → values and WRONG with negative
 ones. Every time you reach for two pointers, say out loud
 → "shrinking can only help". If it
 cannot, stop.

SWEEP LINE. Sort events by one coordinate, maintain a structure
 → over the other.
 TRAP: event ordering at TIES is the whole problem - for closed
 → intervals process opens first;
 for touching rectangles process closes first. Getting it
 → wrong is off-by-exactly-the-ties.

COORDINATE COMPRESSION. Only the order matters; map ≤ 2n values
 → to 0..m-1.
 TRAP: if the problem cares about GAPS (lengths, areas, "how
 → many integers strictly between"),
 keep the original values to weight each compressed cell.
 → Insert both l and r+1.

DIVIDE AND CONQUER ON TIME. Put each object's lifetime on a
 → segment tree over the timeline, DFS
 it with a ROLLBACK structure, answer at the leaves. This is how
 → you delete from a DSU.
 TRAP: rollback DSU must use union by size only - path
 → compression makes undo impossible, so
 find is $O(\log n)$, not alpha. Budget accordingly.

MEET IN THE MIDDLE. Split in half, enumerate $2^{(n/2)}$ per side,
 → recombine by sorting + binary
 search / two pointers / hash map.
 TRIGGER: n ≤ 40..44 with a subset-flavoured question.
 TRAP: memory ($2^{20} \cdot 11 = 8 \text{ MB}$ per array); and in "mod m"
 → variants you must also consider the
 wrap-around pair. Put the extra element of an odd n in the
 → CHEAPER half.

SQRT DECOMPOSITION OF THE PROBLEM (not the data structure). Two
 → complementary bad algorithms,
 $O(x)$ and $O(n/x)$: threshold at sqrt and use each where it wins.

→ Heavy vs light vertices, small vs large divisors, big vs small queries, "sum $a_i \leq 1e5$ so at most $\sqrt{q}$ distinct values".
 TRAP: tune the threshold to the ACTUAL costs; if updates are
 → 10α cheaper the optimum is far from $\sqrt{q}$. Measure.

OFFLINE REORDERING. If nothing forces you online, sort the
 → queries: by right endpoint (+ BIT), by threshold (+ DSU), by Mo's ordering, by time. The
 → highest-yield contest move for "q queries on a static array".
 TRIGGER: no "xor with the previous answer" marker; all queries
 → readable up front.
 TRAP: store answers by the ORIGINAL query id and print in input
 → order.

SMALL TO LARGE. Always insert the smaller container into the
 → larger: each element moves $O(\log n)$ times. SWAP THE HANDLES, never copy - copying the big
 → one in is $O(n^2)$ and looks identical.

AMORTIZATION / POTENTIAL. An occasionally-expensive operation is
 → cheap on average if each expensive step DESTROYS something expensive to create.
 → Justifies monotonic stacks, ODT, beats.
 TRAP: ODT's amortization needs assign-heavy input; it is $O(n)$
 → per op in the worst case.
 Monotonic-stack amortization dies if you ever re-push an
 → index.

ADD ONE ELEMENT AT A TIME. Solve for prefixes and maintain the
 → answer incrementally. Turns "for every prefix, output X" from $O(n^2)$ into $O(n \log n)$ and usually
 → reveals the structure.
 TRAP: LIS - lower_bound gives strictly increasing, upper_bound
 → gives non-decreasing. The tails array is NOT the subsequence; reconstruct with parent
 → pointers.

PARALLEL BINARY SEARCH. q independent binary searches over one
 → shared timeline: bucket queries by their current mid, sweep the timeline once, resolve one bit
 → for every query.
 TRAP: reset or rebuild the shared structure between LEVELS.
 → Forgetting produces monotone-looking garbage.

===== PROOF TOOLS: THE TEN MINUTES THAT SAVE THE SUBMIT
 → =====

EXCHANGE ARGUMENT (for greedy). Take any optimum; show that
 → swapping two ADJACENT elements toward your greedy order never makes it worse. For a sorting
 → greedy this DERIVES the comparator: compare $\text{cost}(a \text{ then } b)$ against $\text{cost}(b \text{ then } a)$.
 TRAP: the derived comparator must be a STRICT WEAK ORDERING.
 → Ties returning true, or an overflowing cross-multiplication, makes std::sort run off the
 → array - an RE that looks like a logic bug. Test with all-equal input, and cross-multiply in
 → $_int128$.

INVARIANTS. A quantity unchanged by every legal operation. If
 → start and target differ on it the answer is "impossible" - and you have a proof instead of a
 → guess.
 TRAP: an invariant proves IMPOSSIBILITY only. You still owe a
 → construction for the other side.

MONOVARIANTS. A quantity that strictly decreases and is bounded:
 → proves termination and bounds the number of operations. (Euclid: $a+b$ strictly decreases.)
 TRAP: on reals a monovariant proves nothing - you need
 → integrality or a floor.

PARITY AND COLOURING. Colour the board so every legal move
 → changes the colour counts in a fixed way, then count. Try in order: checkerboard, columns mod k,
 → $(i+j) \bmod 3$, a weight function.
 EXAMPLE: the mutilated chessboard has 32/30 of the two colours,
 → so no domino tiling exists.
 TRAP: if no colouring produces an imbalance in five minutes,
 → the obstruction is not parity.

THE EXTREMAL PRINCIPLE. Argue about the largest / smallest /

→ deepest object - extremes have neighbours on one side only, which collapses the cases.
 EXAMPLE: two-BFS tree diameter is correct because the farthest
 → vertex from ANY start is an endpoint of SOME diameter.
 TRAP: that theorem is TREE-ONLY. It is false on general graphs
 → and on negative weights.

PIGEONHOLE. $n+1$ objects in n boxes. In practice: $n+1$ prefix sums
 → but only n residues, so some subarray sum is divisible by n .
 TRAP: the constructive version needs the FIRST repeat - store
 → the earliest index per residue.

SYMMETRY AND WLOG. If the problem is invariant under swapping two
 → roles, assume an order and divide by the symmetry factor at the end.
 TRAP: handle the diagonal $i == j$ separately: (total -
 → diagonal)/2 + diagonal. Under a modulus, "divide by 2" means multiply by $\text{inv}2$.

STRATEGY STEALING. In a symmetric game where an extra move never
 → hurts, the first player wins or draws: if the second player had a winning strategy the first
 → could steal it.
 TRAP: pure EXISTENCE. It gives you no winning move, and it
 → fails under misere play.

ADVERSARY ARGUMENTS / QUERY LOWER BOUNDS. q binary queries
 → distinguish at most 2^q inputs, so $q \geq \log_2(\#\text{inputs})$; comparison sorting needs $\log_2(n!) \sim n \log n$.
 TRAP: assume the interactor is ADAPTIVE unless told otherwise -
 → it fixes nothing in advance, only stays consistent. Strategies relying on "the answer was
 → fixed before I started" lose.

===== CONSTRAINTS ARE A MESSAGE FROM THE SETTER

→ =====
 $n \leq 10$ permutations $n!$, or backtracking with pruning
 $n \leq 20$ 2^n bitmask DP, SOS, 3^n submask partition
 $n \leq 24..28$ 2^n with a small factor
 $n \leq 40..44$ MEET IN THE MIDDLE, $2^{(n/2)}$
 $n \leq 100$ $O(n^3)/O(n^4)$: Floyd, Gauss, interval DP,
 → min-cost flow, Hungarian
 $n \leq 500$ $O(n^3)$ or $O(n^2 \log n)$; flows on n^2 edges
 $n \leq 5000$ $O(n^2)$: LCS, tree knapsack, or $O(n^2/64)$ with a
 → bitset
 $n \leq 1e5$, $\log^2$ slack segtree of segtrees, merge-sort tree,
 → parallel binary search, HLD
 $n \leq 2e5$ $O(n \log n)$. This is the default and it means
 → "find the $n \log n$ "
 $n \leq 1e6$ $O(n)$ or $O(n \log n)$ with a SMALL constant: no
 → map, no set, fast IO mandatory
 $n, q \leq 1e5$, offline Mo's ($n \sqrt{q}$) is in budget, $\sim 3e7$
 → pointer moves
 value $\leq 1e9$, n small coordinate compression, or binary
 → search the value
 $a_i \leq 1e18$ Miller-Rabin + Pollard rho, $_int128$, no sieve
 sum $n \leq 2e5$ over tests per-test $O(n \log n)$; NEVER memset a
 → MAXN global per test
 $t \leq 1e4$, no sum bound the intended solution is $O(1)$ or $O(\log)$
 → per test
 mod 998244353 NTT / polynomial methods intended
 mod $1e9+7$ counting DP / combinatorics, NOT NTT
 real output, $1e-6$ geometry, ternary/binary search on
 → reals, or expectation
 $\leq 20-30$ oracle queries binary search ($2^{20} > 1e6$) or
 → bit-by-bit determination
 TL 3-5 s at $n \leq 1e5$ the intended solution really is $n \sqrt{q}$
 → n or $n \log^2 n$
 ML 32-64 MB offline/streaming, rolling DP arrays, no
 → persistence

THE /64 ARGUMENT. An "impossible" $O(n^2)$ or $O(n^3)$ over BOOLEANS
 → becomes $n^2/64$ - fine up to $\sim 1e10$ raw ops. Reachability, subset sum, LCS, string matching,
 → set intersection.
 TRAP: memory. n bitsets of n bits is $n^2/8$ bytes: at $n = 5e4$
 → that is 312 MB. Process in blocks of ~ 1000 and reuse. bitset needs a compile-time size;
 → a vector<u64> hand-roll is $\sim 30\%$ slower and always available.

THE "ANSWER IS SMALL" ARGUMENT. Prove the answer is $\leq$ a tiny
 → constant, then brute force over it. Each operation halves something / clears a bit / merges two
 → components; "OR of a range

equals its SUM" means the values are bit-disjoint so at most 30
 ↳ are nonzero; $\sum a_i \leq 1e5$
 means at most ~ 632 distinct values; $a_i \leq 1e6$ means at most 7
 ↳ distinct primes, 240 divisors;
 gcd over an expanding range takes only $O(\log V)$ distinct values.
 TRAP: prove the bound, do not observe it on samples, and handle
 ↳ "impossible" explicitly.

↳ */

12.3 ContestCraft

CONSTRUCTIVE, INTERACTIVE, RANDOMISED, EMPIRICAL

/* ===== CONSTRUCTIVE, INTERACTIVE, RANDOMISED, EMPIRICAL
 ↳ =====

BUILD GREEDILY AND PROVE AS YOU GO. Emit the answer one decision
 ↳ at a time, always taking the best local choice that **KEEPS THE REMAINDER FEASIBLE** - that
 ↳ check is the whole problem.
 TRAP: for **LEXICOGRAPHICALLY SMALLEST**, greedy alone is wrong.
 ↳ You need "pick the smallest candidate such that the rest is still completable", and the
 ↳ completeness test must be exact. If you cannot write that test, you cannot do
 ↳ lexicographic greedy.

BUILD RECURSIVELY. Solve n from $n/2$ (or $n-1$, or four quadrants)
 ↳ and glue with a fixed pattern.
 TRIGGER: n is a power of two, or the object is self-similar
 ↳ (Gray code, Hanoi, de Bruijn).
 TRAP: base cases are usually $n = 1$ AND $n = 2$. Recursive output
 ↳ of 2^n lines needs buffered output - endl will TLE.

EXTEND FROM A KNOWN SMALL CASE. Brute force $n = 1..8$, find the
 ↳ pattern of CONSTRUCTION (not just the count), then prove the inductive step.
 TRAP: the small cases are exactly where the pattern fails. Find
 ↳ the smallest n where it first works and special-case everything below. Nearly every
 ↳ constructive WA is a small- n exception.

CERTIFICATE CONSTRUCTION (prove the NO). Output the witness: the
 ↳ negative cycle, the odd cycle, the Hall violator, the parity mismatch. Producing it also
 ↳ validates your YES logic.
 TRAP: after n Bellman-Ford rounds you have a vertex **AFFECTED BY**
 ↳ a negative cycle, not one ON it. Walk n parent pointers back first, then follow parents
 ↳ until a vertex repeats.

INTERACTIVE: BUDGET FIRST, ALGORITHM SECOND. Compare 2^Q against
 ↳ the number of possibilities.
 $Q \sim \log n \rightarrow$ binary search; $Q \sim n \rightarrow$ one query per element; Q
 ↳ $\sim n \log n \rightarrow$ sorting/merging.
 TRAPS: (1) **FLUSH** after every query (endl, or fflush) or you get
 ↳ "idleness limit exceeded", which looks nothing like the real bug; (2) count your queries
 ↳ locally and assert the limit; (3) assume the interactor is **ADAPTIVE** - it only has to stay
 ↳ consistent with past answers.

RANDOM SHUFFLE to destroy adversarial order: Kuhn's matching,
 ↳ quickselect, incremental convex hull, minimum enclosing circle (which is $O(n)$ only after a
 ↳ shuffle), KD-tree splits.
 TRAP: seed from the clock (mt19937_64 with steady_clock), never
 ↳ srand(time(0)) + rand().
 unordered_map needs the splitmix64 custom hash or it is
 ↳ $O(n^2)$ on a crafted test.

RANDOM SAMPLING. If a target occupies more than a $1/c$ fraction, k
 ↳ samples miss it with probability $(1 - 1/c)^k$.
 TRIGGER: "majority element of a range", "a value shared by at
 ↳ least half".
 TRAP: sampling gives a **CANDIDATE** - you must verify it exactly.
 ↳ And the failure probability compounds over q queries: with $q = 1e5$ you need per-query
 ↳ failure below $1e-12$, so ~ 40 samples, not 20.

RANDOM WEIGHTS FOR UNIQUENESS (Zobrist / XOR hashing). Give each
 ↳ value a random 64-bit label; then multiset equality, "every value appears a multiple of k
 ↳ times", and "this subtree shape occurred before" become single-number comparisons.
 TRAP: one random assignment has a real false-positive rate -
 ↳ repeat ~ 30 times with **FRESH** randoms. A fixed small set of randoms is breakable by
 ↳ Gaussian elimination.

HASHING AS $O(1)$ EQUALITY. Polynomial hashing for substrings, $2D$
 ↳ for submatrices.
 TRAP: **RANDOM** base, and either double-mod or one 61-bit Mersenne
 ↳ mod. Fixed base 31 with mod

1e9+7 is anti-hashed by a Thue-Morse construction. Birthday
 ↳ bound: comparing m hashes
 pairwise collides with probability $\sim m^2/(2*MOD)$, so $m = 1e6$
 ↳ against a single $1e9+7 \bmod$ is a coin flip.

THE REPETITION MATH. k independent trials fail with p^k . Budget
 ↳ PER QUERY, not per test, then
 multiply by 100 for safety. For primality use the DETERMINISTIC
 ↳ witness set
 {2,3,5,7,11,13,17,19,23,29,31,37} (valid below $3.3e24$) - zero
 ↳ failure probability, same cost.

STRESS TESTING AS A THINKING TOOL. gen + brute + a diff loop;
 ↳ then SHRINK the counterexample
 until it fits on one line and stare at it.
 TRAPS: (1) the generator must produce DEGENERATE cases - all
 ↳ equal, $n = 1$, all zero, maximum
 values, duplicate coordinates. A uniform generator over [1,
 ↳ 1e9] never produces a tie and
 will never find your tie bug. (2) If the problem says "print
 ↳ any", diffing is useless -
 write a CHECKER. (3) Stress the brute force against the
 ↳ samples first.

BRUTE FORCE SMALL CASES, THEN LOOK IT UP. Compute $n = 1..12$
 ↳ exactly, then search OEIS, or feed
 the sequence to Berlekamp-Massey, or fit a polynomial by
 ↳ Lagrange interpolation.
 TRAP: an OEIS match on 4 terms is noise - get 8, then confirm
 ↳ the 9th. BM needs terms computed
 under the SAME prime you will use, and at least $2x$ the order.

PRINT THE DP TABLE AND LOOK FOR STRUCTURE. Dump the $n \times n$ table
 ↳ (and the argmin table) for
 $n = 20$ and look for monotone rows, monotone $opt[i][j]$, convex
 ↳ differences, sparsity.
 That is how you decide between Knuth, divide & conquer, and
 ↳ Aliens.
 TRAP: monotonicity that holds at $n = 20$ can fail at $n = 200$.
 ↳ Also verify the quadrangle
 inequality numerically on random quadruples.

INSTRUMENT, DO NOT GUESS (for TLE). Print a counter of inner-loop
 ↳ iterations on your largest
 local test and compare against $1e8-1e9$. That distinguishes
 ↳ "wrong complexity" from "wrong
 constant", which need completely different fixes. Compile
 ↳ locally with -O2.

===== CONTEST CRAFT =====

READING THE SET

Minute 0-15: all three read DIFFERENT problems, front to back.
 ↳ Nobody codes. On the board, per
 problem: a one-line restatement, n , and a guess at the family.
 Order by (confidence $\propto 1/\text{effort}$), not by letter. Setters
 ↳ mis-order ~30% of the time.
 Use the scoreboard from minute 30 - it is the best difficulty
 ↳ signal you have.
 ONE coder at the keyboard, always. A problem is not "started"
 ↳ until it has a written plan with
 its complexity and its main data structure.

WHEN TO ABANDON - when any TWO hold:

- * 25+ minutes with no NEW idea (the idea count stopped growing,
 ↳ not the code).
- * The complexity is $10x$ over budget with no plan to close it.
- * Two teammates failed to find the bug and 5 minutes of stress
 ↳ testing found nothing
 => you are testing the wrong thing or misread the statement.
 ↳ Re-read it OUT LOUD.
- * Another unattempted problem has more solves.
 Write the state on paper before switching, and set a hard
 ↳ return time.

DEBUGGING UNDER TIME PRESSURE - in this order:

1. RE-READ THE STATEMENT: output format, the modulus, 1-based
 ↳ vs 0-based, multi-test.
 About a quarter of "impossible" bugs are misreads.
2. Re-read your input parsing (n and m swapped; looping to n
 ↳ over an array sized m).
3. Run ALL the samples, including the one you skipped.
4. Run $n = 1$, $n = 2$, and the all-equal case.
5. Stress test. This beats reading code whenever you have a
 ↳ brute force.
6. Only then read the code - out loud, to a teammate, for

- ↳ INTENT. Silent re-reading just
 re-hallucinates the same thing.
- 7. Nothing after 15 minutes: REWRITE the suspicious function
 ↳ from scratch.

PRE-SUBMIT CHECKLIST (40 seconds, every submit)

- [] Overflow: every product of two inputs; inside comparators;
 ↳ inside prefix sums.
- [] Modulus: right constant, +MOD after subtraction, inverse
 ↳ instead of division, non-negative.
- [] $n = 1$, $m = 1$, $k = 0$, single-element array.
- [] Degenerate input: empty string, zero edges, one-node tree,
 ↳ all equal, all zero, negatives.
- [] STATE RESET between test cases: every global, vector,
 ↳ counter, visited array, the answer
 variable, and any function-local static. Clear only what
 ↳ you touched if sum n is bounded.
- [] Right bound read: MAXN vs the actual limit; the sum over
 ↳ queries can exceed the per-test n .
- [] Uninitialised: dp arrays that need -INF not 0; DSU
 ↳ $parent[i] = i$.
- [] Recursion depth: DFS on a path with $n = 2e5$ is $2e5$ frames -
 ↳ move big locals out.
- [] Output format: newline vs space, YES vs Yes, -1 vs
 ↳ IMPOSSIBLE, setprecision.
- [] endl removed from any loop running more than $\sim 1e4$ times;
 ↳ debug output removed.
- [] The 'cin >> tc' line matches whether the problem is
 ↳ multi-test.

VERDICT TRIAGE

- WA -> which test? Test 1 means a misread or a format error; a
 ↳ late test means an edge case or
 overflow. Then: modulus, overflow, ties/duplicates
- ↳ (strict vs non-strict), stress test,
 unreset state, floating-point comparison.
- TLE -> is the complexity really what you think (count the loops
 ↳ on paper)? endl in a loop /
 missing sync_with_stdio. map/set in a hot loop.
- ↳ Reallocation and pass-by-value. memset
 of a huge global per test. Infinite loop: a binary
 ↳ search with $l = \text{mid}$.
 Only then change algorithm. A /64 win never fixes a
 ↳ wrong complexity class.
- RE -> array bounds first: rebuild with
 ↳ -fsanitize=address,undefined -D_GLIBCXX_DEBUG and run
 the samples; it finds it in seconds. Then stack
- ↳ overflow, division by zero, a comparator
 that is not a strict weak ordering (crashes INSIDE
- ↳ std::sort), .back()/top() on an
 empty container, 0- vs 1-indexed access.
- MLE -> $n \times n$ arrays (5000^2 ints = 100 MB); vector<vector<int>>
 ↳ for a graph (5-8x the raw
 data - use CSR); persistent structures (count the nodes
- ↳ before allocating); recursion
 frames; go offline and stream instead of storing
- ↳ versions.
 Idleness on an interactive -> a missing flush, or debug output
 ↳ on stdout.

TEAM PROTOCOL

- Never submit without the checklist: a rejection costs 20
 ↳ penalty minutes plus morale.
- One person owns the printed notebook and fetches pages; the
 ↳ coder does not browse.
- Paste, adapt, then read once - the #1 template bug is an
 ↳ index-convention mismatch.
- Two solutions disagree? Trust the brute force. Always.
- Last 30 minutes: stop starting new problems; finish the nearest
 ↳ one and re-check the submits.

THE TEN MOST EXPENSIVE RECURRING BUGS (by contest-time cost)

1. Overflow in an intermediate product, invisible because the
 ↳ final answer is small.
2. State not reset between test cases - passes locally, fails
 ↳ on test 2.
3. Wrong strictness ($<$ vs $\leq$) in a comparator, a binary-search
 ↳ predicate, or a tie-break.
4. Two pointers on a non-monotone predicate: plausible output,
 ↳ silently wrong.
5. Binary search bounds: hi not provably feasible, or $l = \text{mid}$
 ↳ causing an infinite loop.
6. 0- vs 1-based off-by-one when pasting a template.
7. endl in an output loop, or a missing sync_with_stdio, at $n =$
 ↳ $1e6$.
8. Negative modulo in C++ ($-3 \% 5 == -3$).
9. A greedy that was never proved and passes the samples.

10. Recursion depth on a path-shaped tree with $n = 2e5$.

=====

↪ */

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